

SSerxhs 的 ICPC 模板

SSerxhs

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1 前言

此模板的初衷是个人使用，因此已有的模板可能未列出。建议结合 Heltion 模板和 HDU 模板使用。

模板需要的版本为 cpp17 或 cpp20。

optional 的用法：一个 optional 变量 r 可以用 if (r) 判断其是否有值。取出值的方法是 *r。常见于包含无解又包含空集解的代码中，便于区分无解和空集解。

rev 宏通常用于数据结构中，正向信息是否与反向信息相同，若定义了宏则表示不同。例如矩阵乘法需要定义，而区间求和则不需要。定义后额外统计信息，会慢一点点。

常见的被漏掉的初始代码：

```
#define all(x) (x).begin(),(x).end()
template<class T1, class T2> bool cmin(T1 &x, const T2 &y) { if (y < x) { x = y; return 1; }
    return 0; }
template<class T1, class T2> bool cmax(T1 &x, const T2 &y) { if (x < y) { x = y; return 1; }
    return 0; }
```

常见的缺漏算法：

回文自动机，最小树形图。

2 数据结构

2.1 树状数组

支持单点修改、求前缀和、二分前缀和大于等于 x 的第一个位置。

`lower_bound` 要求前缀和单调，常见用法是所有单点值非负。

```
template<class T> struct bit
{
    vector<T> a;
    int n;
    bit() { }
    bit(int nn) : n(nn), a(nn + 1) { }
    template<class TT> bit(int nn, TT *b) : n(nn), a(nn + 1)
    {
        for (int i = 1; i <= n; i++) a[i] = b[i];
        for (int i = 1; i <= n; i++) if (i + (i & -i) <= n) a[i + (i & -i)] += a[i];
    }
    void add(int x, T y)
    {
        //cerr<<"add "<<x<<" by "<<y<<endl;
        assert(1 <= x && x <= n);
        a[x] += y;
        while ((x += x & -x) <= n) a[x] += y;
    }
    T sum(int x)
    {
        //cerr<<"sum "<<x;
        assert(0 <= x && x <= n);
        T r = a[x];
        while (x ^= x & -x) r += a[x];
        //cerr<<"= "<<r<<endl;
        return r;
    }
    T sum(int x, int y)
    {
        return sum(y) - sum(x - 1);
    }
    int lower_bound(T x) //要求所有单点值非负，从而前缀和单调
    {
        if (n == 0 || x <= 0) return 0;
        int i = __lg(n), j = 0;
        for (; i >= 0; i--) if ((1 << i | j) <= n && a[1 << i | j] < x) j |= 1 << i, x -= a[j];
        return j + 1;
    }
};
```

2.2 线段树

包含标记的线段树，支持线段树上二分，采用左闭右闭。但只支持求左侧第一个符合条件的下标。

要求：具有 `info+info`, `info+=tag`, `tag+=tag`。 `info`, `tag` 需要有默认构造，但不必有正确的值。

示例的 `tag` 和 `info` 是区间加区间覆盖区间历史最大值。

对于线段树上二分，如果你要 `check` 的是前缀，可以在 `false` 时加入信息。

例如查找前缀和大于等于 s 的第一个下标，可以写成

```
[&](const int &x) {
    if (x >= s) return true;
    s -= x;
    return false; }
```

```
template<class info, class tag> struct sgt
{
    int n, shift;
    info *a;
    info tmp;
    vector<info> s;
    vector<tag> tg;
    vector<int> lz;
    bool flg;
    void build(int x, int l, int r)
    {
        if (l == r)
        {
            s[x] = (flg ? tmp : a[l]);
            return;
        }
        int c = x * 2, m = l + r >> 1;
        build(c, l, m); build(c + 1, m + 1, r);
        s[x] = s[c] + s[c + 1];
    }
    sgt(info *b, int L, int R) :n(R - L + 1), shift(L - 1), a(b + L - 1), s(R - L + 1 << 2), tg(R - L + 1 << 2), lz(R - L + 1 << 2)
    {
        flg = 0;
        build(1, 1, n);
    } // [L,R]
    sgt(info b, int L, int R) :n(R - L + 1), shift(L - 1), s(R - L + 1 << 2), tg(R - L + 1 << 2), lz(R - L + 1 << 2)
    {
        tmp = b;
        flg = 1;
        build(1, 1, n);
    } // [L,R]
    int z, y;
    info res;
    tag dt;
    bool fir;
private:
    void _modify(int x, int l, int r)
    {
        if (z <= l && r <= y)
        {
            s[x] += dt;
            if (lz[x]) tg[x] += dt; else tg[x] = dt;
            lz[x] = 1;
            return;
        }
        int c = x * 2, m = l + r >> 1;
        if (lz[x])
        {
            if (lz[c]) tg[c] += tg[x]; else tg[c] = tg[x];
```



```

        lz[c] = 1; s[c] += tg[x]; c ^= 1;
        if (lz[c]) tg[c] += tg[x]; else tg[c] = tg[x];
        lz[c] = 1; s[c] += tg[x]; c ^= 1;
        lz[x] = 0;
    }
    if (z <= m) _modify(c, l, m);
    if (m < y) _modify(c + 1, m + 1, r);
    s[x] = s[c] + s[c + 1];
}
void ask(int x, int l, int r)
{
    if (z <= l && r <= y)
    {
        res = fir ? s[x] : res + s[x];
        fir = 0;
        return;
    }
    int c = x * 2, m = l + r >> 1;
    if (lz[x])
    {
        if (lz[c]) tg[c] += tg[x]; else tg[c] = tg[x];
        lz[c] = 1; s[c] += tg[x]; c ^= 1;
        if (lz[c]) tg[c] += tg[x]; else tg[c] = tg[x];
        lz[c] = 1; s[c] += tg[x]; c ^= 1;
        lz[x] = 0;
    }
    if (z <= m) ask(c, l, m);
    if (m < y) ask(c + 1, m + 1, r);
}
function<bool(info)> check;
void find_left_most(int x, int l, int r)
{
    if (r < z || z <= l && !check(s[x])) return;
    if (l == r) { y = l; res = s[x]; return; }
    int c = x * 2, m = l + r >> 1;
    if (lz[x])
    {
        if (lz[c]) tg[c] += tg[x]; else tg[c] = tg[x];
        lz[c] = 1; s[c] += tg[x]; c ^= 1;
        if (lz[c]) tg[c] += tg[x]; else tg[c] = tg[x];
        lz[c] = 1; s[c] += tg[x]; c ^= 1;
        lz[x] = 0;
    }
    find_left_most(c, l, m);
    if (y == n + 1) find_left_most(c + 1, m + 1, r);
}
void find_right_most(int x, int l, int r)
{
    if (l > y || r <= y && !check(s[x])) return;
    if (l == r) { z = l; res = s[x]; return; }
    int c = x * 2, m = l + r >> 1;
    if (lz[x])
    {
        if (lz[c]) tg[c] += tg[x]; else tg[c] = tg[x];
        lz[c] = 1; s[c] += tg[x]; c ^= 1;
        if (lz[c]) tg[c] += tg[x]; else tg[c] = tg[x];
        lz[c] = 1; s[c] += tg[x]; c ^= 1;
    }

```

```

        lz[x] = 0;
    }
    find_right_most(c + 1, m + 1, r);
    if (z == 0) find_right_most(c, l, m);
}
public:
    void modify(int l, int r, const tag &x)//[l,r]
    {
        z = l - shift; y = r - shift; dt = x;
        // cerr<<"modify ["<<l<<','<<r<<" "<<'\\n';
        assert(1 <= z && z <= y && y <= n);
        _modify(1, 1, n);
    }
    void modify(int pos, const info &o)
    {
        pos -= shift;
        int l = 1, r = n, m, c, x = 1;
        while (l < r)
        {
            c = x * 2; m = l + r >> 1;
            if (lz[x])
            {
                if (lz[c]) tg[c] += tg[x]; else tg[c] = tg[x];
                lz[c] = 1; s[c] += tg[x]; c ^= 1;
                if (lz[c]) tg[c] += tg[x]; else tg[c] = tg[x];
                lz[c] = 1; s[c] += tg[x]; c ^= 1;
                lz[x] = 0;
            }
            if (pos <= m) x = c, r = m; else x = c + 1, l = m + 1;
        }
        s[x] = o;
        while (x >= 1) s[x] = s[x * 2] + s[x * 2 + 1];
    }
    info ask(int l, int r)//[l,r]
    {
        z = l - shift; y = r - shift; fir = 1;
        // cerr<<"ask ["<<l<<','<<r<<" "<<'\\n';
        assert(1 <= z && z <= y && y <= n);
        ask(1, 1, n);
        return res;
    }
}
pair<int, info> find_left_most(int l, const function<bool(info)> &_check)//y=n+1 第二个参数是乱给的
{
    check = _check;
    z = l - shift; y = n + 1;
    assert(1 <= z && z <= n + 1);
    find_left_most(1, 1, n);
    return {y + shift, res};
}
pair<int, info> find_right_most(int r, const function<bool(info)> &_check)//z=0 第二个参数是乱给的
{
    check = _check;
    z = 0; y = r - shift;
    assert(0 <= y && y <= n);
    find_right_most(1, 1, n);
}

```

```

        return {z + shift, res};
    }
};

struct tag
{
    ll hadd, hc, now;
    bool t;
    void operator+=(const tag &o)
    {
        if (t == 0)
        {
            cmax(hc, o.hc);
            cmax(hadd, now + o.hadd);
        }
        else
            hc = max({hc, now + o.hadd, o.hc});
        if (o.t) now = o.now, t = 1;
        else now += o.now;
    }
};

struct info
{
    ll mx, hmx;
    info operator+(const info &o) const { return {max(mx, o.mx), max(hmx, o.hmx)}; }
    void operator+=(const tag &o)
    {
        hmx = max({hmx, mx + o.hadd, o.hc});
        if (o.t) mx = o.now; else mx += o.now;
    }
};

```

2.3 珂朵莉树

支持区间赋值、单点访问。维护每个连续段的范围和值。

如果希望维护所有连续段的整体信息（如长度的最大值），修改 `add` 和 `del` 函数即可，分别表示连续段被加入和被删去。

特别注意一开始 `insert` 的不会触发 `add`，只有 `modify` 会触发。

```

namespace chtholly_tree
{
    using T = int; // 可以把 T 修改为任意想要的类型。
    struct node
    {
        int l;
        mutable int r;
        mutable T v;
        int len() const { return r - l + 1; }
        bool operator<(const node &x) const { return l < x.l; }
    };
    void add(const node &a) { }
    void del(const node &a) { }
    class odt : public set<node>
    {
    public:
        typedef odt::iterator iter;
    };
}

```

```

iter split(int x)
{
    iter it = lower_bound({x});
    if (it != end() && it->l == x) return it;
    node t = *--it, a = {t.l, x - 1, t.v}, b = {x, t.r, t.v};
    del(*it); add(a); add(b);
    erase(it); insert(a);
    return insert(b).first;
}

iter modify(int l, int r, T v) //[l,r]
{
    iter lt, rt, it;
    rt = r == rbegin()->r ? end() : split(r + 1); lt = split(l); //[lt,rt)
    while (lt != begin() && (it = prev(lt))->v == v) l = (lt = it)->l;
    while (rt != end() && rt->v == v) r = (rt++)->r;
    for (it = lt; it != rt; it++) del(*it);
    add({l, r, v});
    erase(lt, rt);
    return insert({l, r, v}).first;
}

T operator[](const int x) const { return prev(upper_bound({x}))->v; } //直接访问单点
iter find(int x) const { return prev(upper_bound({x})); } //找到对应的线段
};
}

using chtholly_tree::node, chtholly_tree::odt;
typedef odt::iterator iter;
int main()
{
    odt s;
    s.insert({0, 5, 1}); // 先 insert({L,R,x}) 表示整个下标范围和初始值。 左闭右闭。
    // s={1,1,1,1,1,1}
    s.modify(2, 3, 2); // 左闭右闭。 s={1,1,2,2,1,1}
    for (auto [l, r, v] : s)
    {
        //(l,r,v)=(0,1,1)
        //(l,r,v)=(2,3,2)
        //(l,r,v)=(4,5,1)
    }
}

```

2.4 带删堆

本质是额外维护一个堆 q 表示要被删除的元素，当 p 的最值和 q 一样时删除。

需要保证每次 pop 的元素都存在于堆中。

本代码的用法和 `priority_queue` 一致。

```

template<class T, class T1 = vector<T>, class T2 = less<T>> struct heap
{
private:
    priority_queue<T, T1, T2> p, q;
public:
    void push(const T &x)
    {
        if (!q.empty() && q.top() == x)
        {
            q.pop();
        }
    }
}

```

```

        while (!q.empty() && q.top() == p.top()) p.pop(), q.pop();
    }
    else p.push(x);
}
void pop()
{
    p.pop();
    while (!q.empty() && p.top() == q.top()) p.pop(), q.pop();
}
void pop(const T &x)
{
    if (p.top() == x)
    {
        p.pop();
        while (!q.empty() && p.top() == q.top()) p.pop(), q.pop();
    }
    else q.push(x);
}
T top() const { return p.top(); }
int size() const { return p.size() - q.size(); }
bool empty() const { return p.empty(); }
vector<T> to_vector() const
{
    vector<T> a;
    auto P = p, Q = q;
    while (P.size())
    {
        a.push_back(P.top()); P.pop();
        while (Q.size() && P.top() == Q.top()) P.pop(), Q.pop();
    }
    return a;
}
};

```

2.5 前 k 大的和

本质是用小根堆维护前 k 大的数，用大根堆维护其余数。

如果需要支持删除，结合前面一个使用，或者直接用 `multiset` 进行 `extract`。

为了方便起见，直接给出支持删除的版本，并且使用 `long long`。如果不需要支持删除，类型改为优先队列并去掉 `pop` 函数即可。

注意：复杂度为 $O(|k - k'|)$ ，其中 k' 是上一次询问的 k 。也就是说，多组询问时询问的 k 的差值应该尽可能小。

其用法与 `priority_queue` 保持一致，可以用同样的方法改写成前 k 小。

```

using ll = long long;
template<class T, class T1 = vector<T>, class T2 = less<T>> struct ksum_pop
{
private:
    struct __cmp
    {
        bool operator()(const T &x, const T &y) const
        {
            return x != y && !T2()(x, y);
        }
    };
};

```

```

    heap<T, T1, __cmp> p;
    heap<T, T1, T2> q;
    ll cur;
public:
    ksum_pop() : cur(0) { }
    void push(const T &x)
    {
        if (!q.size() || !T2()(x, q.top())) p.push(x), cur += x; else q.push(x);
    }
    int size() const { return p.size() + q.size(); }
    void pop(const T &x)
    {
        if (q.size() && !T2()(q.top(), x)) q.pop(x);
        else p.pop(x), cur -= x;
    }
    ll sum(int k)
    {
        while (p.size() < k)
        {
            cur += q.top();
            p.push(q.top());
            q.pop();
        }
        while (p.size() > k)
        {
            cur -= p.top();
            q.push(p.top());
            p.pop();
        }
        return cur;
    }
};

```

2.6 左偏树/可并堆

建议不要使用。pb_ds 可以替代这个功能。我完全没有使用过这个板子。

$O((n+q)\log n)$, $O(n)$ 。

```

struct left_tree//小根堆，大根堆需要改的地方注释了
{
    int jl[N], v[N], f[N], c[N][2], tf[N], n; //tf只有删非堆顶才用
    bool ed[N];
    void init(const int nn, const int *a)
    {
        jl[0] = -1; n = nn;
        memset(jl + 1, 0, n << 2);
        memset(tf + 1, 0, n << 2); //同上
        memset(c + 1, 0, n << 3);
        memset(ed + 1, 0, n);
        for (int i = 1; i <= n; i++) v[f[i] = i] = a[i];
    }
    int mg(int x, int y)
    {
        if (!(x && y)) return x | y;
        if (v[x] > v[y] || v[x] == v[y] && x > y) swap(x, y); //改
        tf[c[x][1] = mg(c[x][1], y)] = x; //同上
    }
};

```

```

        if (j1[c[x][0]] < j1[c[x][1]]) swap(c[x][0], c[x][1]);
        j1[x] = j1[c[x][1]] + 1;
        return x;
    }
    int getf(int x)
    {
        if (f[x] == x) return x;
        return f[x] = getf(f[x]);
    }
    int merge(int x, int y)
    {
        if (ed[x] || ed[y] || (x = getf(x)) == (y = getf(y))) return x;
        int z = mg(x, y); return f[x] = f[y] = z;
    }
    int getv(int x) //需要自行判断是否存在
    {
        return v[getf(x)];
    }
    int del(int x) //删除堆内最值
    {
        tf[c[x][0]] = tf[c[x][1]] = 0;
        f[c[x][0]] = f[c[x][1]] = f[x] = mg(c[x][0], c[x][1]);
        ed[x] = 1; c[x][0] = c[x][1] = tf[x] = 0; return f[x];
    }
    int del_all(int x) //删除堆内非最值 (没验证过)
    {
        int fa = tf[x];
        if (f[c[x][0]] == x) f[c[x][0]] = getf(tf[x]);
        if (f[c[x][1]] == x) f[c[x][1]] = f[tf[x]];
        tf[x] = tf[c[x][0]] = tf[c[x][1]] = 0;
        tf[c[fa][c[fa][1]] == x] = mg(c[x][0], c[x][1]) = fa;
        c[x][0] = c[x][1] = 0;
        while (j1[c[fa][0]] < j1[c[fa][1]])
        {
            swap(c[fa][0], c[fa][1]);
            j1[fa] = j1[c[fa][1]] + 1;
            fa = tf[fa];
        }
    }
    void out(int n)
    {
        for (int i = 1; i <= n; i++) printf("%d: %d %d %d %d\n", i, c[i][0], c[i][1], f[i], v[i]);
    }
};

```

2.7 树状数组区间加区间求和

本质: a_n 区间加等价于差分数组 d_n 的单点加。

$$\sum_{i=1}^m a_i = \sum_{i=1}^m \sum_{j=1}^i d_j = \sum_{j=1}^m d_j (m - j + 1) = ((m + 1) \sum_{j=1}^m d_j) - (\sum_{j=1}^m j d_j)。$$

分别维护 d_j 和 $j d_j$ 的前缀和。

$O(n) \sim O(q \log n)$, $O(n)$ 。

```
template<class T> struct bit
```

```

{
    vector<T> a, b;
    int n;
    template<class TT> bit(int n, TT *c) :n(n), a(n + 1), b(n + 1)
    {
        for (int i = 1; i <= n; i++) b[i] = -c[i];
        for (int i = 1; i <= n; i++) if (i + (i & -i) <= n) b[i + (i & -i)] += b[i];
    }
    void add(int l, int r, T d)
    {
        T x;
        int i;
        for (i = l, x = d * i; i <= n; i += i & -i) a[i] += d, b[i] += x;
        for (i = r + 1, x = d * i; i <= n; i += i & -i) a[i] -= d, b[i] -= x;
    }
    void add(int x, T d)
    {
        for (int i = x; i <= n; i += i & -i) b[i] -= d;
    }
    T sum(int x)
    {
        T r1 = 0, r2 = 0;
        for (int i = x; i; i ^= i & -i) r1 += a[i], r2 += b[i];
        return r1 * (x + 1) - r2;
    }
    T sum(int l, int r)
    {
        return sum(r) - sum(l - 1);
    }
};

```

2.8 二维树状数组矩形加矩形求和

本质还是差分，只不过这次要维护 $d_{i,j}, d_{i,j}i, d_{i,j}j, d_{i,j}ij$ 。

$O(n^2 + q \log^2 n), O(n^2)$

```

template <class T> struct bit
{
    int n, m;
    vector<vector<T>> a, b, c, d;
private:
    void modify(vector<vector<T>> &a, int x, int y, T z)
    {
        for (int i = x; i <= n; i += i & -i) for (int j = y; j <= m; j += j & -j) a[i][j] += z;
    }
    T ask(const vector<vector<T>> &a, int x, int y) const
    {
        T res = 0; --x; --y;
        for (int i = x; i; i ^= i & -i) for (int j = y; j; j ^= j & -j) res += a[i][j];
        return res;
    }
    void cg(int x, int y, T t)
    {
        if (x > n || y > m) return;
        modify(a, x, y, t);
        modify(b, x, y, x * t);
    }
};

```



```

        modify(c, x, y, y * t);
        modify(d, x, y, x * y * t);
    }
public:
    bit(int n, int m) : n(n), m(m), a(n + 1, vector<T>(m + 1)), b(a), c(a), d(a) { }
    void add(int x1, int y1, int x2, int y2, T t)
    {
        ++x2, ++y2;
        cg(x1, y1, t); cg(x2, y2, t);
        cg(x1, y2, -t); cg(x2, y1, -t);
    }
    T sum(int x, int y) const
    {
        if (x <= 0 || y <= 0) return 0;
        ++x; ++y;
        return ask(a, x, y) * x * y + ask(d, x, y) - ask(b, x, y) * y - ask(c, x, y) * x;
    }
    T sum(int x1, int y1, int x2, int y2) const
    {
        --x1; --y1;
        return sum(x2, y2) + sum(x1, y1) - sum(x2, y1) - sum(x1, y2);
    }
};

```

2.9 带修莫队

按照 $n^{\frac{2}{3}}$ 分块，排序关键字是 l, r, t 所在的块 (t 是版本号，每次修改都会增加一个版本)，可以奇偶分块优化。

相比于传统莫队多了一个 `modify`。这里只给出参考代码，功能是带修区间数颜色。

$O(n^{\frac{5}{3}})$, $O(n)$ 。

```

#include "bits/stdc++.h"
using namespace std;
typedef long long ll;
#define all(x) (x).begin(), (x).end()
const int N = 1.4e5, M = 1e6 + 2;
int a[N], ans[N], bel[N], cnt[M], sum, z, y, cur;
struct P
{
    int p, v;
};
struct Q
{
    int l, r, t, p;
    bool operator<(const Q &o) const
    {
        if (bel[l] != bel[o.l]) return bel[l] < bel[o.l];
        if (bel[r] != bel[o.r]) return (bel[l] & 1) ^ bel[r] < bel[o.r];
        return (bel[r] & 1) ? t < o.t : t > o.t;
    }
};
Q b[N];
P d[N];
void add(const int &x) { sum += !(cnt[a[x]]++); }
void del(const int &x) { sum -= !(--cnt[a[x]]); }
void mdf(const int &x)

```

```

{
    auto &[p, v] = d[x];
    if (z <= p && p <= y) del(p);
    swap(a[p], v);
    if (z <= p && p <= y) add(p);
}
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    int n, m, q1 = 0, q2 = 0, i, ksiz;
    cin >> n >> m;
    for (i = 1; i <= n; i++) cin >> a[i];
    for (i = 1; i <= m; i++)
    {
        char c;
        int l, r;
        cin >> c >> l >> r;
        if (c == 'Q') ++q1, b[q1] = {l, r, q2, q1};
        else d[++q2] = {l, r};
    }
    ksiz = max(1.0, round(cbrt((1ll)n * n)));
    for (i = 1; i <= n; i++) bel[i] = i / ksiz;
    sort(b + 1, b + q1 + 1);
    z = b[1].l; y = z - 1; cur = 0;
    for (i = 1; i <= q1; i++)
    {
        auto [l, r, t, p] = b[i];
        while (z > l) add(--z);
        while (y < r) add(++y);
        while (z < l) del(z++);
        while (y > r) del(y--);
        while (cur < t) mdf(++cur);
        while (cur > t) mdf(cur--);
        ans[p] = sum;
    }
    for (i = 1; i <= q1; i++) cout << ans[i] << '\n';
}

```

2.10 二次离线莫队

直接摘录题解，用途不大。

$O(n\sqrt{n})$, $O(n)$ 。

珂朵莉给了你一个序列 a ，每次查询给一个区间 $[l, r]$ ，查询 $l \leq i < j \leq r$ ，且 $a_i \oplus a_j$ 的二进制表示下有 k 个 1 的二元组 (i, j) 的个数。 $\oplus$ 是指按位异或。

二次离线莫队，通过扫描线，再次将更新答案的过程离线处理，降低时间复杂度。假设更新答案的复杂度为 $O(k)$ ，它将莫队的复杂度从 $O(nk\sqrt{n})$ 降到了 $O(nk + n\sqrt{n})$ ，大大简化了计算。设 x 对区间 $[l, r]$ 的贡献为 $f(x, [l, r])$ ，我们考虑区间端点变化对答案的影响：以 $[l..r]$ 变成 $[l..(r+k)]$ 为例， $\forall x \in [r+1, r+k]$ 求 $f(x, [l, x-1])$ 。我们可以进行差分： $f(x, [l, x-1]) = f(x, [1, x-1]) - f(x, [1, l-1])$ ，这样转化为了一个数对一个前缀的贡献。保存下来所有这样的询问，从左到右扫描数组计算就可以了。但是这样做，空间是 $O(n\sqrt{n})$ 的，不太优秀，而且时间常数巨大。。这样的贡献分为两类：

1. 减号左边的贡献永远是一个前缀和它后面一个数的贡献。这可以预处理出来。2. 减号右边的贡献对于一次移动中所有的 x 来说，都是不变的。我们打标记的时候，可以只标记左右端点。

这样，减小时间常数的同时，空间降为了 $O(n)$ 级别。是一个很优秀的算法了。处理前缀询问

的时候，我们利用异或运算的交换律，即 $a \text{ xor } b = c \iff a \text{ xor } c = b$ 开一个桶 t ， $t[i]$ 表示当前前缀中与 i 异或有 k 个数位为 1 的数有多少个。则每加入一个数 $a[i]$ ，对于所有 $\text{popcount}(x) = k$ 的 x ， $t[a[i] \text{ xor } x] \leftarrow t[a[i] \text{ xor } x] + 1$ 即可。

```
const int N = 1e5 + 2, M = 1 << 14;
ll f[N], ans[N], ta[N];
int a[N], cnt[M], bel[N], pc[M], st[N];
int n, m, ksiz;
struct Q
{
    int z, y, wz;
    bool operator<(const Q &x) const { return (bel[z] < bel[x.z]) || (bel[z] == bel[x.z]) && ((y < x.y) && (bel[z] & 1) || (y > x.y) && (1 ^ bel[z] & 1)); }
};
Q mq(const int x, const int y, const int z)
{
    Q a;
    a.z = x; a.y = y; a.wz = z;
    return a;
}
Q q[N];
vector<Q> b[N];
int main()
{
    ios::sync_with_stdio(false);
    cin.tie(0);
    int i, j, k, l = 1, r = 0, tp = 0, x, na;
    cin >> n >> m >> k; ksiz = sqrt(n);
    for (i = 1; i <= n; i++) { cin >> a[i]; bel[i] = (i - 1) / ksiz + 1; }
    if (k == 0) st[++tp] = 0;
    for (i = 1; i < 16384; i++)
    {
        if (i & 1) pc[i] = pc[i >> 1] + 1; else pc[i] = pc[i >> 1];
        if (pc[i] == k) st[++tp] = i;
    }
    for (i = 1; i <= n; i++)
    {
        j = tp + 1; f[i] = f[i - 1];
        while (--j) f[i] += cnt[st[j] ^ a[i]];
        ++cnt[a[i]];
    }
    for (i = 1; i <= m; i++) { cin >> q[i].z >> q[q[i].wz = i].y; }
    sort(q + 1, q + m + 1);
    for (i = 1; i <= m; i++)
    {
        ans[i] = f[q[i].y] - f[r] + f[q[i].z - 1] - f[l - 1];
        if (k == 0) ans[i] += q[i].z - 1;
        if (r < q[i].y)
        {
            b[l - 1].push_back(mq(r + 1, q[i].y, -i));
            r = q[i].y;
        }
        if (l > q[i].z)
        {
            b[r].push_back(mq(q[i].z, l - 1, i));
            l = q[i].z;
        }
    }
}
```

```

    if (r > q[i].y)
    {
        b[l - 1].push_back(mq(q[i].y + 1, r, i));
        r = q[i].y;
    }
    if (l < q[i].z)
    {
        b[r].push_back(mq(l, q[i].z - 1, -i));
        l = q[i].z;
    }
}
memset(cnt, 0, sizeof(cnt));
for (i = 1; i <= n; i++)
{
    j = tp + 1; x = a[i];
    while (--j) ++cnt[x ^ st[j]];
    for (j = 0; j < b[i].size(); j++)
    {
        na = 0; l = b[i][j].z; r = b[i][j].y;
        for (k = l; k <= r; k++) na += cnt[a[k]];
        if (b[i][j].wz > 0) ans[b[i][j].wz] += na; else ans[-b[i][j].wz] -= na;
    }
}
for (i = 2; i <= m; i++) ans[i] += ans[i - 1];
for (i = 1; i <= m; i++) ta[q[i].wz] = ans[i];
for (i = 1; i <= m; i++) printf("%lld\n", ta[i]);
}

```

2.11 回滚莫队

不删除的莫队，比如求 $\max$ 。

做法：块内询问暴力。对于 l 所在块相同的询问，按照 r 升序排序，并且将左指针固定在 l 所在块的最右侧。（由于块内询问暴力，这不会导致左指针更大）

回答每个询问的时候，先右端点右移到 r ，然后左端点左移到 l 。询问完成后，把左端点移回去。移回去的过程虽然涉及删除，但不需要维护答案变成什么了（因为在左端点左移之前已经求过了）。换句话说，相当于“撤销”而不是删除，完全可以记录移动过程中的所有变化来撤销。

$O(n\sqrt{n})$, $O(n)$ 。

```

#include "bits/stdc++.h"
using namespace std;
const int N = 2e5 + 2;
int a[N], z[N], y[N], wz[N], b[N], d[N], bel[N], ans[N], st[N][2], pos[N][2];
void qs(int l, int r)
{
    int i = l, j = r, m = bel[z[l + r >> 1]], mm = y[l + r >> 1];
    while (i <= j)
    {
        while ((bel[z[i]] < m) || (bel[z[i]] == m) && (y[i] < mm)) ++i;
        while ((bel[z[j]] > m) || (bel[z[j]] == m) && (y[j] > mm)) --j;
        if (i <= j)
        {
            swap(wz[i], wz[j]);
            swap(z[i], z[j]);
            swap(y[i++], y[j--]);
        }
    }
}

```

```

    }
    if (i < r) qs(i, r);
    if (l < j) qs(l, j);
}
int main()
{
    ios::sync_with_stdio(false);
    cin.tie(0);
    cin >> n;
    ksiz = sqrt(n);
    for (i = 1; i <= n; i++) { cin >> a[i]; b[i] = a[i]; bel[i] = (i - 1) / ksiz + 1; }
    sort(b + 1, b + n + 1);
    d[gs = 1] = b[1];
    for (i = 2; i <= n; i++) if (b[i] != b[i - 1]) d[++gs] = b[i];
    for (i = 1; i <= n; i++) a[i] = lower_bound(d + 1, d + gs + 1, a[i]) - d;
    cin >> m; assert(int(n / sqrt(m)));
    for (i = 1; i <= m; i++) cin >> z[i] >> y[wz[i] = i];
    qs(1, m);
    for (i = 1; i <= m; i++)
    {
        if (bel[z[i]] > bel[z[i - 1]])
        {
            while (l <= r) { pos[a[l]][0] = pos[a[l]][1] = 0; ++l; } na = 0;
            if (bel[z[i]] == bel[y[i]])
            {
                for (j = z[i]; j <= y[i]; j++) if (pos[a[j]][0]) na = max(na, j - pos[a[j]][0]);
            }
            else pos[a[j]][0] = j;
            ans[wz[i]] = na; for (j = z[i]; j <= y[i]; j++) pos[a[j]][0] = 0; na = 0; l =
            ksiz * bel[z[i]]; r = l - 1;
            continue;
        }
        l = ksiz * bel[z[i]]; r = l - 1; na = 0;
    }
    if (bel[z[i]] == bel[y[i]])
    {
        while (l <= r) { pos[a[l]][0] = pos[a[l]][1] = 0; ++l; } na = 0;
        for (j = z[i]; j <= y[i]; j++) if (pos[a[j]][0]) na = max(na, j - pos[a[j]][0]); else
        pos[a[j]][0] = j;
        ans[wz[i]] = na; for (j = z[i]; j <= y[i]; j++) pos[a[j]][0] = 0;
        l = ksiz * bel[z[i]]; r = l - 1; na = 0;
        continue;
    }
    while (r < y[i])
    {
        x = a[++r]; pos[x][1] = r;
        if (!pos[x][0]) pos[x][0] = r; else na = max(na, r - pos[x][0]);
    } c = na;
    while (l > z[i])
    {
        x = a[--l]; st[++tp][0] = x; st[tp][1] = pos[x][0];
        pos[x][0] = 1;
        if (!pos[x][1])
        {
            st[++tp][0] = x + n; st[tp][1] = 0;
            pos[x][1] = 1;
        }
        else na = max(na, pos[x][1] - 1);
    }
}

```

```

    }
    ans[wz[i]] = na; na = c; ++tp; l = ksiz * bel[z[i]];
    while (--tp) if (st[tp][0] <= n) pos[st[tp][0]][0] = st[tp][1]; else pos[st[tp][0] - n][1] = st[tp][1];
    }
    for (i = 1; i <= m; i++) cout << ans[i] << "\n";
}

```

2.12 李超树

题意：插入线段，查询某个 x 的最大 y （输出最小编号）

算法核心：修改时，线段树每个点只维护在中点取值最大的线段，中点取值较小的线段只会在至多一侧有用，递归下去插入，复杂度 $O(\log^2)$ 。查询时询问线段树上 $\log$ 个点的线段中最大的。

```

struct Q
{
    int x0, y0, dx, dy, id;
    Q() :x0(0), y0(-1), dx(1), dy(0), id(-1) { } //y>=0
    Q(int a, int b, int c, int d, int e) :x0(a), y0(b), dx(c), dy(d), id(e) { }
    bool contains(const int &x) const { return x0 <= x && x <= x0 + dx; }
};

bool cmp(const Q &a, const Q &b, int x) //小心数值爆炸
{
    ll A = ((ll)a.y0 * a.dx + (ll)(x - a.x0) * a.dy) * b.dx, B = ((ll)b.y0 * b.dx + (ll)(x - b.x0) * b.dy) * a.dx;
    if (A != B) return A < B;
    return a.id > b.id;
}

bool cmp2(const Q &a, const Q &b)
{
    if (a.y0 + a.dy != b.y0 + b.dy) return a.y0 + a.dy < b.y0 + b.dy;
    return a.id > b.id;
}

const int inf = 1e9;
int ans;
namespace seg
{
    const int N = 4e4 + 2, M = N * 4;
    Q s[M], X[N];
    int n, z, y;
    void init(int nn)
    {
        n = nn;
        for (int i = 1; i <= n * 4; i++) s[i] = Q();
        for (int i = 1; i <= n; i++) X[i] = Q();
    }
    void insert(int x, int l, int r, Q dt)
    {
        int c = x * 2, m = l + r >> 1;
        if (z <= l && r <= y)
        {
            if (cmp(s[x], dt, m)) swap(s[x], dt);
            if (l == r) return;
            if (cmp(s[x], dt, l)) insert(c, l, m, dt);
            else if (cmp(s[x], dt, r)) insert(c + 1, m + 1, r, dt);
            return;
        }
    }
}

```

```

    }
    if (z <= m) insert(c, l, m, dt);
    if (y > m) insert(c + 1, m + 1, r, dt);
}
void insert(const Q &o)
{
    z = o.x0; y = z + o.dx;
    assert(1 <= z && z <= y && y <= n);
    if (z == y)
    {
        if (cmp2(X[z], o)) X[z] = o;
        return;
    }
    insert(1, 1, n, o);
}
Q askmax(int p)
{
    Q ans = s[1].contains(p) ? s[1] : Q();
    int x = 1, l = 1, r = n, c, m;
    while (l < r)
    {
        c = x * 2, m = l + r >> 1;
        if (p <= m) x = c, r = m; else x = c + 1, l = m + 1;
        if (s[x].contains(p) && cmp(ans, s[x], p)) ans = s[x];
    }
    Q o(X[p].x0, X[p].y0 + X[p].dy, 1, 0, X[p].id);
    return cmp(ans, o, p) ? X[p] : ans;
}
}
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cout << setiosflags(ios::fixed) << setprecision(15);
    int n = 4e4, m, i;
    seg::init(n);
    cin >> m;
    while (m--)
    {
        int op;
        cin >> op;
        if (op)
        {
            int x[2], y[2];
            cin >> x[0] >> y[0] >> x[1] >> y[1];
            for (int &v : x) v = (v + ans - 1) % 39989 + 1;
            for (int &v : y) v = (v + ans - 1) % inf + 1;
            if (x[0] > x[1] || x[0] == x[1] && y[0] > y[1]) swap(x[0], x[1]), swap(y[0], y[1]);
            static int id;
            seg::insert({x[0], y[0], x[1] - x[0], y[1] - y[0], ++id});
        }
        else
        {
            int x;
            cin >> x;
            x = (x + ans - 1) % 39989 + 1;
            cout << (ans = max(0, seg::askmax(x).id)) << '\n';
        }
    }
}

```

```

    }
}

```

2.13 李超树（动态开点）

```

struct Q
{
    int k;
    ll b;
    ll y(const int &x) const { return (ll)k * x + b; }
};

const int inf = 1e9;
const ll INF = 1e18;
struct seg//可以析构，不能并行
{
    const static int N = 4e5 + 2, M = N * 8 * 8 + (1 << 23);
    const static ll npos = 9e18;
    static Q s[M];
    static int c[M][2], id;
    int z, y, L, R;
    seg(int l, int r)
    {
        L = l; R = r; id = 1;
        s[1] = {0, npos};
        assert(L <= R && (ll)R - L < 1ll << 32);
    }
private:
    void insert(int &x, int l, int r, Q o)
    {
        if (!x)
        {
            x = ++id;
            assert(id < M);
            s[x] = {0, npos};
        }
        int m = l + (r - l >> 1);
        if (z <= l && r <= y)
        {
            if (s[x].y(m) > o.y(m)) swap(s[x], o);
            if (s[x].y(l) > o.y(l)) insert(c[x][0], l, m, o);
            else if (s[x].y(r) > o.y(r)) insert(c[x][1], m + 1, r, o);
            return;
        }
        if (z <= m) insert(c[x][0], l, m, o);
        if (y > m) insert(c[x][1], m + 1, r, o);
    }
public:
    void insert(const Q &x, const int &l, const int &r)//[l,r]
    {
        z = l; y = r; int tmp = 1;
        insert(tmp, L, R, x);
        assert(tmp == 1);
    }
    ll askmin(const int &p)
    {
        ll res = s[1].y(p);
    }
}

```



```

    int l = L, r = R, m, x = 1;
    while (l < r)
    {
        m = l + (r - l >> 1);
        if (p <= m) x = c[x][0], r = m; else x = c[x][1], l = m + 1;
        if (!x) return res;
        res = min(res, s[x].y(p));
    }
    return res;
}
~seg()
{
    ++id;
    while (--id) c[id][0] = c[id][1] = 0;
}
};
Q seg::s[seg::M];
int seg::c[seg::M][2], seg::id;

```

2.14 线性基

$O((n + q) \log a)$, $O(n \log a)$ 。

```

template<class T, int M = sizeof(T) * 8> struct base//线性基
{
    array<T, M> a;
    int num, dim;
    base() :a{ }, num(0), dim(0) { }
    bool insert(T x)//线性基插入
    {
        ++num;
        if (x == 0) return 0;
        for (int i = __lg(x); x; i = __lg(x))
        {
            if (!a[i]) return ++dim, a[i] = x;
            x ^= a[i];
        }
        return 0;
    }
    bool contains(T x, bool empty_allowed = true) const//查询是否能 xor 出 x
    {
        if (x == 0) return empty_allowed || dim != num;
        for (int i = __lg(x); x; i = __lg(x))
        {
            if (!a[i]) return 0;
            x ^= a[i];
        }
        return 1;
    }
}
T max(T x = 0, T k = 0, bool empty_allowed = true) const//查询子集 xor 的 k+1 大。若有传入参数
x, 表示子集 xor x 的 k+1 大。
{
    assert(M <= 64);
    bool zero = empty_allowed || num != dim;
    if (dim < 64 && k >= (1ull << dim) - !zero || k == (-1ull) && !zero)
    {
        assert(M != sizeof(T) * 8);
    }
}

```

```

        return -1;
    }
    if (!zero)
    {
        ull z = 0;
        int d = dim;
        for (int i = M - 1; i >= 0; i--) if (a[i]) z |= (1ull ^ (x >> i & 1)) << --d;
        if (k >= z) ++k;
    }
    int d = dim;
    for (int i = M - 1; i >= 0; i--)
        if (a[i] && (1 ^ (k >> --d ^ x >> i) & 1))
            x ^= a[i];
    return x;
}

```

T min(T x = 0, T k = 0, bool empty_allowed = true) const//查询子集 xor 的 k+1 大。若有传入参数 x, 表示子集 xor x 的 k+1 大。

```

{
    assert(M <= 64);
    bool zero = empty_allowed || num != dim;
    if (dim == 0 && k >= zero || dim < 64 && k >= (1ull << dim) - !zero || k == (-1ull) && !
zero)
    {
        assert(M != sizeof(T) * 8);
        return -1;
    }
    if (!zero)
    {
        ull z = 0;
        int d = dim;
        for (int i = M - 1; i >= 0; i--) if (a[i]) z |= (0ull + (x >> i & 1)) << --d;
        if (k >= z) ++k;
    }
    int d = dim;
    for (int i = M - 1; i >= 0; i--)
        if (a[i] && ((k >> --d ^ x >> i) & 1))
            x ^= a[i];
    return x;
}

```

base &operator|=(const base &o)//合并线性基

```

{
    int t = num;
    for (T x : o.a) if (x) insert(x);
    num = t + o.num;
    return *this;
}
base operator|(base o) const { return o += *this; }//合并线性基
base operator&(base o) const
{
    array<T, M> g1{ }, g2{ }, val{ };
    int i, j, k;
    for (i = 0; i < M; i++)
        for (j = 0; j < M; j++)
            g1[i] |= (a[j] >> i & 1) << j,
            g2[i] |= (o.a[j] >> i & 1) << j;

    T mask1 = 0, mask2 = 0;
}

```

```

    for (i = M - 1; i >= 0; i--) if (g1[i] || g2[i])
    {
        auto &x = (g1[i] ? g1 : g2);
        int g = __lg(x[i]);
        (g1[i] ? mask1 : mask2) |= (T)1 << g;
        if (g1[i]) val[i] = a[g];
        for (j = 0; j < M; j++)
            if (i != j && (x[j] >> g & 1))
                g1[j] ^= g1[i], g2[j] ^= g2[i];
    }
    base<T, M> res;
    for (i = 0; i < M; i++) if (a[i] && !(mask1 >> i & 1))
    {
        T v = a[i];
        for (j = 0; j < M; j++)
            if (g1[j] >> i & 1)
                v ^= val[j];
        res.insert(v);
    }
    for (i = 0; i < M; i++) if (o.a[i] && !(mask2 >> i & 1))
    {
        T v = 0;
        for (j = 0; j < M; j++)
            if (g2[j] >> i & 1)
                v ^= val[j];
        res.insert(v);
    }
    return res;
}

base &operator&=(const base &o) { return *this = *this & o; }

};

template<class T = ll, int M = sizeof(T) * 8> struct rangebase//[0,...)
{
    vector<array<pair<T, int>, M>> a;
    rangebase() :a{{ }} { }
    rangebase(const vector<T> &b) :a{{ }} { for (T x : b) push_back(x); }//直接用一个 vector 构造
    void push_back(T x)//在最后插入 x
    {
        int n = a.size() - 1;
        a.push_back(a.back());
        if (x == 0) return;
        for (int i = __lg(x); x; i = __lg(x))
        {
            auto &[v, p] = a.back()[i];
            if (v)
            {
                if (n > p)
                {
                    swap(x, v);
                    swap(n, p);
                }
                x ^= v;
            }
            else
            {
                v = x;
            }
        }
    }
};

```

```

        p = n;
        return;
    }
}
}
base<T, M> ask(int l, int r)//查询  $[l,r]$  元素构成的线性基。下标从 0 开始 (同 vector)
{
    assert(0 <= l && l <= r && r <= a.size());
    base<T, M> res;
    res.num = r - l;
    for (int i = 0; i < M; i++)
    {
        auto [v, p] = a[r][i];
        if (v && p >= l) res.a[i] = v, ++res.dim;
    }
    return res;
}
};

```

2.15 splay

指针版。

$O(n)$, $O((n+q)\log n)$ 。

```

template<class info, class tag> struct splay
{
#define _rev
    struct node
    {
        node *c[2], *f;
        int siz;
        info s, v;
        tag t;
        node() :c{ }, f(0), siz(1), s(), v(), t() { }
        node(info x) :c{ }, f(0), siz(1), s(x), v(x), t() { }
        void operator+=(const tag &o)
        {
            s += o; v += o; t += o;
#ifdef _rev
            if (o.rev) swap(c[0], c[1]);
#endif
        }
        void pushup()
        {
            if (c[0]) s = c[0]->s + v, siz = c[0]->siz + 1; else s = v, siz = 1;
            if (c[1]) s = s + c[1]->s, siz += c[1]->siz;
        }
        void pushdown()
        {
            for (auto x : c) if (x) *x += t;
            t = { };
        }
        void zigzag()
        {
            node *y = f, *z = y->f;

```

```

        int typ = y->c[0] == this;
        if (z) z->c[z->c[1] == y] = this;
        f = z; y->f = this;
        y->c[typ ^ 1] = c[typ];
        if (c[typ]) c[typ]->f = y;
        c[typ] = y;
        y->pushup();
    }
    void splay(node *tar)//不要在 makeroot 以外调用
    {
        for (node *y = f; y != tar; zigzag(), y = f) if (node *z = y->f; z != tar) (z->c[1]
== y ^ y->c[1] == this ? this : y)->zigzag();
        pushup();
    }
    void clear()
    {
        for (node *x : c) if (x) x->clear();
        delete this;
    }
};
node *rt;
void debug()
{
    map<node *, int> id;
    id[0] = 0; id[rt] = 1;
    int cnt = 1;
    function<void(node *)> out = [&](node *x) {
        if (!x) return;
        for (auto y : x->c) if (!id.count(y)) id[y] = ++cnt;
        cerr << id[x] << '␣' << id[x->c[0]] << '␣' << id[x->c[1]] << '␣' << id[x->f] << '␣'
<< x->siz << '\n';
        for (auto y : x->c) out(y);
    };
    out(rt);
}
node *build(info *a, int n)
{
    if (n == 0) return 0;
    int m = n - 1 >> 1;
    node *x = new node(a[m]);
    x->c[0] = build(a, m);
    x->c[1] = build(a + m + 1, n - 1 - m);
    for (node *y : x->c) if (y) y->f = x;
    x->pushup();
    return x;
}
splay()
{
    shift = 0;
    rt = new node;
    rt->c[1] = new node;
    rt->c[1]->f = rt;
    rt->siz = 2;
}
int shift;
splay(info *a, int l, int r)//[l,r)
{

```

```

    shift = l - 1;
    rt = new node;
    rt->c[1] = new node;
    rt->c[1]->f = rt;
    if (l < r)
    {
        rt->c[1]->c[0] = build(a + 1, r - 1);
        rt->c[1]->c[0]->f = rt->c[1];
    }
    rt->c[1]->pushup();
    rt->pushup();
}
void makeroot(node *u, node *tar)
{
    if (!tar) rt = u;
    u->splay();
}
void findnth(int k, node *tar)
{
    node *x = rt;
    while (1)
    {
        x->pushdown();
        int v = x->c[0] ? x->c[0]->siz : 0;
        if (v + 1 == k) { x->splay(tar); if (!tar) rt = x; return; }
        if (v >= k) x = x->c[0]; else x = x->c[1], k -= v + 1;
    }
}
void split(int l, int r)
{
    assert(1 <= l && r <= rt->siz - 2 && l - 1 <= r);
    findnth(l, 0);
    findnth(r + 2, rt);
}
#ifdef _rev
void reverse(int l, int r)
{
    l -= shift; r -= shift + 1;
    if (l - 1 == r) return;
    assert(1 <= l && l <= r && r <= rt->siz - 2);
    split(l, r);
    *(rt->c[1]->c[0]) += tag(1);
}
#endif
void insert(int pos, info x)//insert before pos
{
    pos -= shift;
    assert(1 <= pos && pos <= rt->siz - 1);
    split(pos, pos - 1);
    rt->c[1]->c[0] = new node(x);
    rt->c[1]->c[0]->f = rt->c[1];
    rt->c[1]->pushup();
    rt->pushup();
}
void insert(int pos, info *a, int n)//insert before pos, [1,n]
{
    pos -= shift;

```

```

    assert(1 <= pos && pos <= rt->siz - 1);
    split(pos, pos - 1);
    rt->c[1]->c[0] = build(a, n);
    rt->c[1]->c[0]->f = rt->c[1];
    rt->c[1]->pushup();
    rt->pushup();
}
void erase(int pos)
{
    pos -= shift;
    assert(1 <= pos && pos <= rt->siz - 2);
    split(pos, pos);
    delete rt->c[1]->c[0];
    rt->c[1]->c[0] = 0;
    rt->c[1]->pushup();
    rt->pushup();
}
void erase(int l, int r)
{
    l -= shift; r -= shift + 1;
    if (l - 1 == r) return;
    assert(1 <= l && l <= r && r <= rt->siz - 2);
    split(l, r);
    rt->c[1]->c[0]->clear();
    rt->c[1]->c[0] = 0;
    rt->c[1]->pushup();
    rt->pushup();
}
void modify(int pos, info x)
{
    pos -= shift;
    assert(1 <= pos && pos <= rt->siz - 2);
    findnth(pos + 1, 0);
    rt->v = x; rt->pushup();
}
void modify(int l, int r, tag w)
{
    l -= shift; r -= shift + 1;
    if (l - 1 == r) return;
    assert(1 <= l && l <= r && r <= rt->siz - 2);
    split(l, r);
    node *x = rt->c[1]->c[0];
    *x += w;
    rt->c[1]->pushup();
    rt->pushup();
}
info ask(int l, int r)
{
    l -= shift; r -= shift + 1;
    assert(1 <= l && l <= r && r <= rt->siz - 2);
    split(l, r);
    return rt->c[1]->c[0]->s;
}
~splay() { rt->clear(); }
#undef _rev
};
struct Q

```

```

{
    bool rev;
    Q() :rev(0) { }
    Q(bool c) :rev(c) { }
    void operator+=(const Q &o)
    {
        rev ^= o.rev;
    }
};
struct P
{
    ll s;
    void operator+=(const Q &o) const
    { }
    P operator+(const P &o) const { return{s + o.s}; }
};

```

2.16 fhq-treap

洛谷模板：普通平衡树。

$O((n+q)\log n)$, $O(n)$ 。

```

const int N = 1.1e6 + 2;
int c[N][2], v[N], w[N], s[N];
int n, i, x, y, ds, val, kth, p, q, z, rt, la, m, ans;
void pushup(const int x)
{
    s[x] = s[c[x][0]] + s[c[x][1]] + 1;
}
void split_val(int now, int &x, int &y) //调用外部val, 相等归入y
{
    if (!now) return x = y = 0, void();
    if (val <= v[now]) split_val(c[y = now][0], x, c[now][0]);
    else split_val(c[x = now][1], c[now][1], y);
    pushup(now);
}
void split_kth(int now, int &x, int &y) //调用外部kth, 左子树大小为 kth
{
    if (!now) return x = y = 0, void();
    if (kth <= s[c[now][0]]) split_kth(c[y = now][0], x, c[now][0]);
    else kth -= s[c[now][0]] + 1, split_kth(c[x = now][1], c[now][1], y);
    pushup(now);
}
int merge(int x, int y) //小根ver.
{
    if (!(x && y)) return x | y;
    if (w[x] < w[y]) { c[x][1] = merge(c[x][1], y); pushup(x); return x; }
    else { c[y][0] = merge(x, c[y][0]); pushup(y); return y; }
}
int main()
{
    cin >> n >> m; srand(998244353);
    for (i = 1; i <= n; i++)
    {
        cin >> x;
        val = v[++ds] = x;
    }
}

```



```

    w[ds] = rand();
    s[ds] = 1;
    split_val(rt, p, q);
    rt = merge(merge(p, ds), q);
}
while (m--)
{
    cin >> y >> x;
    x ^= la;
    if (y == 4) // 找到第 x 小的
    {
        kth = x; split_kth(rt, p, q); x = p;
        while (c[x][1]) x = c[x][1];
        ans ^= (la = v[x]); rt = merge(p, q);
        continue;
    }
    val = x; // 注意这一步
    if (y == 1) // 插入 x
    {
        v[++ds] = x; w[ds] = rand(); s[ds] = 1;
        split_val(rt, p, q); rt = merge(merge(p, ds), q);
        continue;
    }
    if (y == 2) // 删除一个 x
    {
        split_val(rt, p, q);
        z = q;
        if (q)
        {
            i = q;
            while (c[i][0]) i = c[i][0];
            if (v[i] == x)
            {
                kth = 1;
                split_kth(q, i, z);
            }
        }
        rt = merge(p, z); continue;
    }
    if (y == 3) // 询问 x 的排名 (比 x 小的数字个数 + 1)
    {
        split_val(rt, p, q); ans ^= (la = s[p] + 1);
        rt = merge(p, q); continue;
    }
    if (y == 5) // 询问比 x 小的最大值
    {
        split_val(rt, p, q); x = p;
        while (c[x][1]) x = c[x][1]; ans ^= (la = v[x]);
        rt = merge(p, q); continue;
    }
    ++val; split_val(rt, p, q); x = q; // 询问比 x 大的最小值
    while (c[x][0]) x = c[x][0];
    ans ^= (la = v[x]); rt = merge(p, q);
}
cout << ans << endl;
}

```

2.17 笛卡尔树的线性建树

$p[1, 2, \dots, n]$ 是原序列, c 表示子结点。

笛卡尔树满足堆性质 (权值小于等于子结点权值), 并且中序遍历是原序列。

$O(n)$, $O(n)$ 。

```
int c[N][2], p[N], st[N];
int main()
{
    int i, n, tp = 0;
    cin >> n;
    for (i = 1; i <= n; i++)
    {
        cin >> p[i]; st[tp + 1] = 0;
        while ((tp) && (p[st[tp]] > p[i])) --tp;
        c[c[st[tp]][1] = i][0] = st[tp + 1]; st[++tp] = i;
    }
}
```

2.18 扫描线

求矩形并的面积和周长 (包括内周长)

$O((n + q) \log n)$, $O(n + q)$ 。

```
using T = ll;
vector<T> fun(vector<tuple<T, T, T, T>> &a)
{
    vector<T> x;
    for (auto [x1, y1, x2, y2] : a)
    {
        x.push_back(x1);
        x.push_back(x2);
    }
    sort(all(x)); x.resize(unique(all(x)) - x.begin());
    for (auto &[x1, y1, x2, y2] : a)
    {
        x1 = lower_bound(all(x), x1) - x.begin();
        x2 = lower_bound(all(x), x2) - x.begin();
    }
    return x;
}
struct sgt
{
    int n, z, y, d;
    vector<T> cnt, &p;
    vector<int> mn, lz;
    void build(int x, int l, int r)
    {
        cnt[x] = p[min(r, n - 1)] - p[l];
        if (l + 1 == r) return;
        int c = x * 2, m = l + r >> 1;
        build(c, l, m); build(c + 1, m, r);
    }
    sgt(vector<T> &p) : n(p.size()), p(p), cnt(n * 4), mn(n * 4), lz(n * 4) { build(1, 0, n); }
    void dfs(int x, int l, int r)
    {
        if (z <= l && r <= y)
```

```

    {
        mn[x] += d;
        lz[x] += d;
        return;
    }
    int c = x * 2, m = l + r >> 1;
    if (lz[x])
    {
        lz[c] += lz[x]; lz[c + 1] += lz[x];
        mn[c] += lz[x]; mn[c + 1] += lz[x];
        lz[x] = 0;
    }
    if (z < m) dfs(c, l, m);
    if (m < y) dfs(c + 1, m, r);
    mn[x] = min(mn[c], mn[c + 1]);
    cnt[x] = cnt[c] * (mn[x] == mn[c]) + cnt[c + 1] * (mn[x] == mn[c + 1]);
}
void modify(int l, int r, int dt)
{
    z = l;
    y = r;
    d = dt;
    dfs(1, 0, n);
}
};
T area(vector<tuple<T, T, T, T>> a)//[x1,y1,x2,y2], x1<y1, x2<y2
{
    int n = a.size(), i;
    auto X = fun(a);
    vector<tuple<T, int, T, T>> b(n * 2);
    for (i = 0; i < n; i++)
    {
        auto [x1, y1, x2, y2] = a[i];
        b[i] = {y1, -1, x1, x2};
        b[i + n] = {y2, 1, x1, x2};
    }
    sort(all(b), greater<>());
    sgt s(X);
    T lst = 0, ans = 0;
    for (auto [y, d, l, r] : b)
    {
        ans += (lst - y) * (X.back() - X[0] - s.cnt[1]);
        s.modify(l, r, d);
        lst = y;
    }
    return ans;
}
T perimeter_x(vector<tuple<T, T, T, T>> a)
{
    int n = a.size(), i;
    auto X = fun(a);
    vector<tuple<T, int, T, T>> b(n * 2);
    for (i = 0; i < n; i++)
    {
        auto [x1, y1, x2, y2] = a[i];
        b[i] = {y1, -1, x1, x2};
        b[i + n] = {y2, 1, x1, x2};
    }

```

```

    }
    sort(all(b), greater<>());
    sgt s(X);
    T lst = s.cnt[1], ans = 0;
    for (auto [y, d, l, r] : b)
    {
        s.modify(l, r, d);
        T cur = s.cnt[1];
        ans += abs(lst - cur);
        lst = cur;
    }
    return ans;
}
T perimeter(vector<tuple<T, T, T, T>> a)//[x1,y1,x2,y2], x1<y1, x2<y2
{
    T ansx = perimeter_x(a);
    for (auto &[x1, y1, x2, y2] : a)
    {
        swap(x1, y1);
        swap(x2, y2);
    }
    T ansy = perimeter_x(a);
    return ansx + ansy;
}

```

2.19 Segmenttree Beats!

核心是 P (tag) 和 Q (info) 的维护。线段树部分是套的模板，并非全都有用。

1. l, r, k : 对于所有的 $i \in [l, r]$, 将 A_i 加上 k (k 可以为负数)。
2. l, r, v : 对于所有的 $i \in [l, r]$, 将 A_i 变成 $\min(A_i, v)$ 。
3. l, r : 求 $\sum_{i=l}^r A_i$ 。
4. l, r : 对于所有的 $i \in [l, r]$, 求 A_i 的最大值。
5. l, r : 对于所有的 $i \in [l, r]$, 求 B_i 的最大值。

其中 B_i 是 A_i 的历史最大值。

```

struct P
{
    ll tg, L, R;
    P(ll a = 0, ll b = -inf, ll c = inf) :tg(a), L(b), R(c) { }
    void operator+=(P o)
    {
        o.L -= tg; o.R -= tg; tg += o.tg;
        if (L >= o.R) L = R = o.R;
        else if (R <= o.L) L = R = o.L;
        else cmax(L, o.L), cmin(R, o.R);
    }
};
struct Q
{
    ll mx0, cmx, mx1, mn0, cmn, mn1, cnt, sum;

```

```

Q() :mx0(-inf), cmx(0), mx1(-inf), mn0(inf), cmn(0), mn1(inf), cnt(0), sum(0) { }
Q(ll x) :mx0(x), cmx(1), mx1(-inf), mn0(x), cmn(1), mn1(inf), cnt(1), sum(x) { }
bool operator+=(const P &o)
{
    if (o.L == o.R)
    {
        ll c = cnt;
        *this = Q(o.L + o.tg);
        cnt = cmx = cmn = c;
        sum = cnt * (o.L + o.tg);
        return 1;
    }
    if (o.L >= mn1 || o.R <= mx1) return 0;
    if (mx0 == mn0)
    {
        mn0 = min(o.R, max(mx0, o.L));
        sum += cnt * (mn0 - mx0);
        mx0 = mn0;
    }
    else
    {
        if (o.L > mn0)
        {
            sum += (o.L - mn0) * cmn;
            mn0 = o.L;
            cmx(mx1, o.L);
        }
        if (o.R < mx0)
        {
            sum += (o.R - mx0) * cmx;
            mx0 = o.R;
            cmn(mn1, o.R);
        }
    }
    if (o.tg)
    {
        sum += o.tg * cnt;
        mx0 += o.tg;
        mx1 += o.tg;
        mn0 += o.tg;
        mn1 += o.tg;
    }
    return 1;
}
};

Q operator+(const Q &a, const Q &b)
{
    Q res;
    res.sum = a.sum + b.sum;
    res.cnt = a.cnt + b.cnt;
    res.mx0 = max(a.mx0, b.mx0);
    res.mx1 = max(a.mx1, b.mx1);
    if (res.mx0 == a.mx0) res.cmx += a.cmx; else cmx(res.mx1, a.mx0);
    if (res.mx0 == b.mx0) res.cmx += b.cmx; else cmx(res.mx1, b.mx0);

    res.mn0 = min(a.mn0, b.mn0);
    res.mn1 = min(a.mn1, b.mn1);
}

```

```

    if (res.mn0 == a.mn0) res.cmn += a.cmn; else cmin(res.mn1, a.mn0);
    if (res.mn0 == b.mn0) res.cmn += b.cmn; else cmin(res.mn1, b.mn0);

    return res;
}
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cout << fixed << setprecision(15);
    int n, q, i;
    cin >> n >> q;
    vector<ll> a(n);
    cin >> a;
    sgt<Q, P> s(a.data(), 0, n - 1);
    while (q--)
    {
        int op, l, r;
        cin >> op >> l >> r;
        --r;
        if (op == 3)
        {
            ll res = s.ask(l, r).sum;
            cout << res << '\n';
        }
        else
        {
            ll b;
            cin >> b;
            if (op == 0) s.modify(l, r, {0, -inf, b});
            else if (op == 1) s.modify(l, r, {0, b});
            else s.modify(l, r, {b});
        }
    }
}

```

2.20 k -d 树（二进制分组）

均摊 $O(\log^2 n)$ 插入, $O(\sqrt{n})$ 矩形查询。

```

#define tmp1 template<class T>
using ll = long long;
tmp1 struct P
{
    ll x, y;
    T v;
};
tmp1 struct Q
{
    ll x[2], y[2];
    bool t;
    T s;
    Q() { }
    Q(const P<T> &a)
    {
        x[0] = x[1] = a.x;
        y[0] = y[1] = a.y;
        s = a.v;
    }
}

```

```

    }
};

template<bool> cmp0(const P<T> &a, const P<T> &b) { return a.x < b.x; }
template<bool> cmp1(const P<T> &a, const P<T> &b) { return a.y < b.y; }
template<struct kdt
{
    vector<P<T>> c;
    vector<Q<T>> a;
    ll m, u, d, l, r;
    T ans;
    bool fir;
    void build(int x, P<T> *b, int n)
    {
        if (x == 1)
        {
            a.resize(m = n << 1);
            a[x].t = 0;
            c.resize(n);
            for (int i = 0; i < n; i++) c[i] = b[i];
        }
        if (n == 1)
        {
            a[x] = Q<T>(b[0]);
            return;
        }
        int mid = n >> 1, c = x << 1;
        nth_element(b, b + mid, b + n, a[x].t ? cmp1<T> : cmp0<T>);
        a[c].t = a[c | 1].t = a[x].t ^ 1;
        build(c, b, mid);
        build(c | 1, b + mid, n - mid);
        a[x].s = a[c].s + a[c | 1].s;
        a[x].x[0] = min(a[c].x[0], a[c | 1].x[0]);
        a[x].x[1] = max(a[c].x[1], a[c | 1].x[1]);
        a[x].y[0] = min(a[c].y[0], a[c | 1].y[0]);
        a[x].y[1] = max(a[c].y[1], a[c | 1].y[1]);
    }
    void find(int x)
    {
        if (x >= m || a[x].x[1] < u || a[x].x[0] > d || a[x].y[1] < l || a[x].y[0] > r) return;
        if (u <= a[x].x[0] && a[x].x[1] <= d && l <= a[x].y[0] && a[x].y[1] <= r)
        {
            ans = fir ? a[x].s : ans + a[x].s;
            fir = 0;
            return;
        }
        find(x << 1); find(x << 1 | 1);
    }
}
pair<bool, T> find(ll x1, ll y1, ll x2, ll y2)
{
    fir = 1;
    ans = { };
    u = x1; d = x2;
    l = y1; r = y2;
    find(1);
    return {!fir, ans};
}
};

```

```

const int N = 2e5 + 2, M = 18;
templ struct KDT
{
    kdt<T> s[M];
    P<T> a[N];
    int n, m, i;
    KDT() { n = 0; }
    KDT(int N, ll *x, ll *y, T *w)//[0,n)
    {
        n = N;
        int i, j;
        for (i = 0; i < n; i++) a[i] = {x[i], y[i], w[i]};
        for (i = j = 0; n >> i; i++) if (n >> i & 1) s[i].build(1, a + j, 1 << i), j += 1 << i;
    }
    void insert(ll x, ll y, T w)//插入 (x,y) 的一个数 w
    {
        a[0] = {x, y, w}; m = 1;
        for (i = 0; n & 1 << i; i++) for (auto u : s[i].c) a[m++] = u;
        s[i].build(1, a, m);
        ++n;
    }
    pair<bool, T> ask(ll x, ll y, ll xx, ll yy)//查询 [x,xx]*[y,yy] 的和
    {
        T ans;
        bool fir = 1;
        for (i = 0; 1 << i <= n; i++) if (1 << i & n)
        {
            auto [_ , tmp] = s[i].find(x, y, xx, yy);
            if (!_ ) continue;
            ans = fir ? tmp : ans + tmp;
            fir = 0;
        }
        return {!fir, ans};
    }
};
int x[N], y[N], w[N];
int main()
{
    ios::sync_with_stdio(0); cin.tie(0); cout.tie(0);
    int n, q, i;
    cin >> n >> q;
    for (i = 0; i < n; i++) cin >> x[i] >> y[i] >> w[i];
    KDT<ll> s(n, x, y, w);
    while (q--)
    {
        int op, x, y, w;
        cin >> op >> x >> y >> w;
        if (op == 0) s.insert(x, y, w); else
        {
            cin >> op;
            cout << s.ask(x, y, w - 1, op - 1) << '\n';
        }
    }
    return 0;
}

```


2.21 双端队列全局查询

对一个支持结合律的信息 T ，维护 deque 内信息的和。总复杂度线性。

```
template<class T> struct dq
{
    vector<T> l, sl, r, sr;
    void push_front(const T &o)
    {
        sl.push_back(sl.size() ? o + sl.back() : o);
        l.push_back(o);
    }
    void push_back(const T &o)
    {
        sr.push_back(sr.size() ? sr.back() + o : o);
        r.push_back(o);
    }
    void pop_front()
    {
        if (l.size()) sl.pop_back(), l.pop_back();
        else
        {
            assert(r.size());
            int n = r.size(), m, i;
            if (m = n - 1 >> 1)
            {
                l.resize(m); sl.resize(m);
                for (i = 1; i <= m; i++) l[m - i] = r[i];
                sl[0] = l[0];
                for (i = 1; i < m; i++) sl[i] = l[i] + sl[i - 1];
            }
            for (i = m + 1; i < n; i++) r[i - (m + 1)] = r[i];
            m = n - (m + 1);
            r.resize(m); sr.resize(m);
            if (m)
            {
                sr[0] = r[0];
                for (i = 1; i < m; i++) sr[i] = sr[i - 1] + r[i];
            }
        }
    }
    void pop_back()
    {
        if (r.size()) sr.pop_back(), r.pop_back();
        else
        {
            assert(l.size());
            int n = l.size(), m, i;
            if (m = n - 1 >> 1)
            {
                r.resize(m); sr.resize(m);
                for (i = 1; i <= m; i++) r[m - i] = l[i];
                sr[0] = r[0];
                for (i = 1; i < m; i++) sr[i] = sr[i - 1] + r[i];
            }
            for (i = m + 1; i < n; i++) l[i - (m + 1)] = l[i];
            m = n - (m + 1);
            l.resize(m); sl.resize(m);
        }
    }
}
```

```

        if (m)
        {
            sl[0] = l[0];
            for (i = 1; i < m; i++) sl[i] = l[i] + sl[i - 1];
        }
    }
}
template<class TT> TT ask(TT r)
{
    if (sl.size()) r = r + sl.back();
    if (sr.size()) r = r + sr.back();
    return r;
}
T ask()
{
    assert(sl.size() || sr.size());
    if (sl.size() && sr.size()) return sl.back() + sr.back();
    return sl.size() ? sl.back() : sr.back();
}
};

```

2.22 静态矩形加矩形和

```

const ull p = 998244353;
struct Q
{
    int n, m;
    ull w;
    int typ;
    bool operator<(const Q &o) const
    {
        if (n != o.n) return n < o.n;
        return typ < o.typ;
    }
};
template<class T> struct tork
{
    vector<T> a;
    int n;
    tork(const vector<T> &b) :a(all(b))
    {
        sort(all(a));
        a.resize(unique(all(a)) - a.begin());
        n = a.size();
    }
    tork(const T *first, const T *last) :a(first, last)
    {
        sort(all(a));
        a.resize(unique(all(a)) - a.begin());
        n = a.size();
    }
    void get(T &x) { x = lower_bound(all(a), x) - a.begin() + 1; }
    T operator[](const int &x) { return a[x]; }
};
struct bit
{

```

```

vector<ull> a;
int n;
bit() { }
bit(int nn) : n(nn), a(nn + 1) { }
template<class T> bit(int nn, T *b) : n(nn), a(nn + 1)
{
    for (int i = 1; i <= n; i++) a[i] = b[i];
    for (int i = 1; i <= n; i++) if (i + (i & -i) <= n) a[i + (i & -i)] += a[i];
}
void add(int x, ull y)
{
    // cerr<<"add "<<x<<" by "<<y<<endl;
    assert(1 <= x && x <= n);
    if ((a[x] += y) >= p) a[x] -= p;
    while ((x += x & -x) <= n) if ((a[x] += y) >= p) a[x] -= p;
}
ull sum(int x)
{
    // cerr<<"sum "<<x;
    assert(0 <= x && x <= n);
    ull r = a[x];
    while (x ^= x & -x) r += a[x];
    // cerr<<"= "<<r<<endl;
    return r % p;
}
ull sum(int x, int y)
{
    return (sum(y) + p - sum(x - 1)) % p;
}
};
struct matrix
{
    int l, d, r, u;
    ull w;
};
vector<ull> rec_add_rec_sum(const vector<matrix> &op, const vector<matrix> &query)
{
    vector<Q> a[4];
    int n = op.size(), m = query.size(), i;
    for (auto &v : a) v.reserve(n + m << 2);
    for (auto [l, d, r, u, w] : op) // [l,r)*[d,u) += w
    {
        a[0].push_back({l, d, w * l % p * d % p, -1});
        a[1].push_back({l, d, w * l % p, -1});
        a[2].push_back({l, d, w * d % p, -1});
        a[3].push_back({l, d, w, -1});
        w = (p - w) % p;
        a[0].push_back({l, u, w * l % p * u % p, -1});
        a[1].push_back({l, u, w * l % p, -1});
        a[2].push_back({l, u, w * u % p, -1});
        a[3].push_back({l, u, w, -1});
        a[0].push_back({r, d, w * r % p * d % p, -1});
        a[1].push_back({r, d, w * r % p, -1});
        a[2].push_back({r, d, w * d % p, -1});
        a[3].push_back({r, d, w, -1});
        w = (p - w) % p;
        a[0].push_back({r, u, w * r % p * u % p, -1});
    }
}

```

```

        a[1].push_back({r, u, w * r % p, -1});
        a[2].push_back({r, u, w * u % p, -1});
        a[3].push_back({r, u, w, -1});
    }
    i = 0;
    for (auto [l, d, r, u, w] : query) // ask sum of [l,r]*[d,u)
    {
        a[0].push_back({l, d, 1, i});
        a[1].push_back({l, d, (p * 2 - d) % p, i});
        a[2].push_back({l, d, (p * 2 - 1) % p, i});
        a[3].push_back({l, d, (ull)l * d % p, i});
        a[0].push_back({l, u, p - 1, i});
        a[1].push_back({l, u, u % p, i});
        a[2].push_back({l, u, 1 % p, i});
        a[3].push_back({l, u, (p * 2 - 1) * u % p, i});
        a[0].push_back({r, u, 1, i});
        a[1].push_back({r, u, (p * 2 - u) % p, i});
        a[2].push_back({r, u, (p * 2 - r) % p, i});
        a[3].push_back({r, u, (ull)u * r % p, i});
        a[0].push_back({r, d, p - 1, i});
        a[1].push_back({r, d, d % p, i});
        a[2].push_back({r, d, r % p, i});
        a[3].push_back({r, d, (p * 2 - d) * r % p, i});
        ++i;
    }
    assert(a[0].size() == n + m << 2);
    vector<ull> ans(m);
    auto cal = [&](vector<Q> a) {
        int n = a.size(), i;
        vector<int> b(n);
        for (i = 0; i < n; i++) b[i] = (a[i].m == a[i].typ >= 0), a[i].n == a[i].typ >= 0;
        sort(all(a));
        tork t(b);
        for (i = 0; i < n; i++) t.get(a[i].m);
        int m = t.a.size();
        bit s(m);
        for (auto [n, m, w, typ] : a) if (typ >= 0) ans[typ] = (ans[typ] + s.sum(m) * w) % p;
    else s.add(m, w);
    };
    for (auto &v : a) cal(v);
    return ans;
}

```

2.23 线段树分裂

```

namespace sgt
{
#define ask_kth
    int L = 0, R = 1e9;
    void set_bound(int l, int r) { L = l; R = r; }
    typedef ll info;
    const info E = 0; // 找不到会返回 E
    const int N = 8e6 + 5;
#define lc(x) (a[x].lc)
#define rc(x) (a[x].rc)
#define s(x) (a[x].s)

```

```

struct node
{
    int lc, rc;
    info s;
};
node a[N];
vector<int> id;
int ids = 0, pos, z, y;
bool fir;
info tmp;
int npt()
{
    int x;
    if (id.size()) x = id.back(), id.pop_back();
    else x = ++ids;
    lc(x) = rc(x) = 0;
    return x;
}
void pushup(int &x)
{
    if (lc(x) && rc(x)) s(x) = s(lc(x)) + s(rc(x));
    else if (lc(x)) s(x) = s(lc(x));
    else if (rc(x)) s(x) = s(rc(x));
    else id.push_back(x), x = 0;
}
void insert(int &x, int l, int r)
{
    if (l == r)
    {
        if (!x) x = npt(), s(x) = tmp;
        else s(x) = s(x) + tmp;
        return;
    }
    if (!x) x = npt();
    int mid = l + r >> 1;
    if (pos <= mid)
    {
        insert(lc(x), l, mid);
        if (rc(x)) s(x) = s(lc(x)) + s(rc(x)); else s(x) = s(lc(x));
    }
    else
    {
        insert(rc(x), mid + 1, r);
        if (lc(x)) s(x) = s(lc(x)) + s(rc(x)); else s(x) = s(rc(x));
    }
}
void modify(int &x, int l, int r)
{
    if (!x) x = npt();
    if (l == r)
    {
        s(x) = tmp;
        return;
    }
    int mid = l + r >> 1;
    if (pos <= mid)
    {

```

```

        modify(lc(x), l, mid);
        if (rc(x)) s(x) = s(lc(x)) + s(rc(x)); else s(x) = s(lc(x));
    }
    else
    {
        modify(rc(x), mid + 1, r);
        if (lc(x)) s(x) = s(lc(x)) + s(rc(x)); else s(x) = s(rc(x));
    }
}

int merge(int x1, int x2, int l, int r)
{
    if (!(x1 && x2)) return x1 | x2;
    if (l == r) { s(x1) = s(x1) + s(x2); return x1; }
    int mid = l + r >> 1;
    lc(x1) = merge(lc(x1), lc(x2), l, mid);
    rc(x1) = merge(rc(x1), rc(x2), mid + 1, r);
    pushup(x1);
    return x1;
}

void ask(int x, int l, int r)
{
    if (!x) return;
    if (z <= l && r <= y)
    {
        if (fir) tmp = s(x), fir = 0; else tmp = tmp + s(x);
        return;
    }
    int mid = l + r >> 1;
    if (z <= mid) ask(lc(x), l, mid);
    if (y > mid) ask(rc(x), mid + 1, r);
}

void split(int &x1, int &x2, int l, int r)
{
    assert(!x1);
    if (!x2) return;
    if (z <= l && r <= y) { x1 = x2; x2 = 0; return; }
    x1 = npt();
    int mid = l + r >> 1;
    if (z <= mid) split(lc(x1), lc(x2), l, mid);
    if (y > mid) split(rc(x1), rc(x2), mid + 1, r);
    pushup(x1); pushup(x2);
}

info *b;
void build(int &x, int l, int r)
{
    x = npt();
    if (l == r) { s(x) = b[l]; return; }
    int mid = l + r >> 1;
    build(lc(x), l, mid); build(rc(x), mid + 1, r);
    s(x) = s(lc(x)) + s(rc(x));
}

struct set
{
    int rt;
    set() :rt(0) { }
    set(info *a) :rt(0) { b = a; build(rt, L, R); }
    void modify(int p, const info &o) { pos = p; tmp = o; sgt::modify(rt, L, R); }
}

```

```

void insert(int p, const info &o) { pos = p; tmp = o; sgt::insert(rt, L, R); }
void join(set &o) { rt = merge(rt, o.rt, L, R); o.rt = 0; }
info ask(int l, int r)
{
    z = l; y = r; fir = 1;
    sgt::ask(rt, L, R);
    return fir ? E : tmp;
}
set split(int l, int r)
{
    z = l; y = r; set p;
    sgt::split(p.rt, rt, L, R);
    return p;
}
#ifdef ask_kth
int kth(info k)
{
    int x = rt, l = L, r = R, mid;
    if (k > s(x)) return -1;
    s(0) = 0;
    while (l < r)
    {
        mid = l + r >> 1;
        if (s(lc(x)) >= k) x = lc(x), r = mid;
        else k -= s(lc(x)), x = rc(x), l = mid + 1;
    }
    return l;
}
#endif
};
#undef lc
#undef rc
#undef s
}
typedef sgt::set tree;

```

2.24 手写 bitset

```

struct Bitset
{
    using ull = unsigned long long;
#define all(x) (x).begin(), (x).end()
    const static ull B = -1llu;
    int n;
    vector<ull> a;
    Bitset() { }
    Bitset(int n) : n(n), a(n + 63 >> 6) { assert(n); }
    bool test(int x) const { assert(x >= 0 && x < n); return a[x >> 6] >> (x & 63) & 1; }
    bool operator[](int x) const { return test(x); }
    void set(int x, bool y) { assert(x >= 0 && x < n); a[x >> 6] = (a[x >> 6] & (B ^ 1llu << (x & 63))) | ((ull)y << (x & 63)); }
    void set(int x) { assert(x >= 0 && x < n); a[x >> 6] |= 1llu << (x & 63); }
    void set() { memset(a.data(), 0xff, a.size() * sizeof a[0]); if (n & 63) a.back() &= (1llu << (n & 63)) - 1; }
    void reset(int x) { assert(x >= 0 && x < n); a[x >> 6] &= ~(1llu << (x & 63)); }
    void reset() { memset(a.data(), 0, a.size() * sizeof a[0]); }

```

```

int count() const
{
    int r = 0;
    for (ull x : a) r += __builtin_popcountll(x);
    return r;
}
int count(int l, int r) const//[l,r)
{
    if (l == r) return 0;
    if (l >> 6 == r >> 6) return __builtin_popcountll(a[l >> 6] >> (l & 63) & (1llu << r - l)
- 1);
    int ans = 0;
    ans += __builtin_popcountll(a[l >> 6] >> (l & 63));
    ++(l >>= 6);
    if (r & 63) ans += __builtin_popcountll(a[r >> 6] & (1llu << (r & 63)) - 1);
    r >>= 6;
    while (l < r) ans += __builtin_popcountll(a[l++]);
    return ans;
}
Bitset &operator|=(const Bitset &o)
{
    assert(n == o.n);
    for (int i = 0; i < a.size(); i++) a[i] |= o.a[i];
    return *this;
}
Bitset operator|(Bitset o) { o |= *this; return o; }
Bitset &operator&=(const Bitset &o)
{
    assert(n == o.n);
    for (int i = 0; i < a.size(); i++) a[i] &= o.a[i];
    return *this;
}
Bitset operator&(Bitset o) { o &= *this; return o; }
Bitset &operator^=(const Bitset &o)
{
    assert(n == o.n);
    for (int i = 0; i < a.size(); i++) a[i] ^= o.a[i];
    return *this;
}
Bitset operator^(Bitset o) { o ^= *this; return o; }
Bitset operator~() const
{
    auto r = *this;
    for (ull &x : r.a) x = ~x;
    if (n & 63) r.a.back() &= (1ull << (n & 63)) - 1;
    return r;
}
Bitset &operator<<=(int x)
{
    if (x >= n) return reset(), *this;
    assert(x >= 0);
    int y = x >> 6;
    x &= 63;
    if (x == 0)
    {
        for (int i = (int)a.size() - 1; i >= y; i--) a[i] = a[i - y];
        if (n & 63) a.back() &= (1llu << (n & 63)) - 1;
    }
}

```



```

        memset(a.data(), 0, y * sizeof a[0]);
        return *this;
    }
    for (int i = (int)a.size() - 1; i > y; i--) a[i] = a[i - y] << x | a[i - y - 1] >> 64 - x
;
    a[y] = a[0] << x;
    memset(a.data(), 0, y * sizeof a[0]);
    // fill_n(a.begin(),y,0);
    if (n & 63) a.back() &= (1llu << (n & 63)) - 1;
    return *this;
}
Bitset operator<<(int x)
{
    auto r = *this;
    r <<= x;
    return r;
}
Bitset &operator>>=(int x)
{
    if (x >= n) return reset(), *this;
    assert(x >= 0);
    int y = x >> 6, R = (int)a.size() - y - 1;
    x &= 63;
    if (x == 0)
    {
        for (int i = 0; i <= R; i++) a[i] = a[i + y];
        memset(a.data() + R + 1, 0, y * sizeof a[0]);
        return *this;
    }
    for (int i = 0; i < R; i++) a[i] = a[i + y] >> x | a[i + y + 1] << 64 - x;
    a[R] = a.back() >> x;
    memset(a.data() + R + 1, 0, y * sizeof a[0]);
    return *this;
}
Bitset operator>>(int x)
{
    auto r = *this;
    r >>= x;
    return r;
}
void range_set(int l, int r)//[l,r) to 1
{
    if (l == r) return;
    if (l >> 6 == r >> 6)
    {
        a[l >> 6] |= (1llu << r - 1) - 1 << (l & 63);
        return;
    }
    if (l & 63)
    {
        a[l >> 6] |= ~((1llu << (l & 63)) - 1);//[l&63,64)
        l += 64;
    }
    if (r & 63) a[r >> 6] |= (1llu << (r & 63)) - 1;
    l >>= 6; r >>= 6;
    memset(a.data() + 1, 0xff, (r - 1) * sizeof a[0]);
}

```

```

void range_reset(int l, int r)//[l,r) to 0
{
    if (l == r) return;
    if (l >> 6 == r >> 6)
    {
        a[l >> 6] &= ~((1llu << r - 1) - 1 << (l & 63));
        return;
    }
    if (l & 63)
    {
        a[l >> 6] &= (1llu << (l & 63)) - 1;
        l += 64;
    }
    if (r & 63) a[r >> 6] &= ~((1llu << (r & 63)) - 1);
    l >>= 6; r >>= 6;
    memset(a.data() + l, 0, (r - l) * sizeof a[0]);
}

void range_set(int l, int r, bool x)//[l,r)
{
    if (x) range_set(l, r);
    else range_reset(l, r);
}

int size() const { return n; }
int _Find_first() const
{
    for (int i = 0; i < a.size(); i++) if (a[i]) return i * 64 + __lg(a[i] & -a[i]);
    return n;
}

int _Find_next(int x) const
{
    assert(x >= 0 && x < n);
    ++x;
    if (x == n) return n;
    int y = x & 63; x >>= 6;
    if (a[x] >> y) return x * 64 + __lg(a[x] >> y & -(a[x] >> y)) + y;
    ++x;
    while (x < a.size() && !a[x]) ++x;
    return x == a.size() ? n : x * 64 + __lg(a[x] & -a[x]);
}

int _Find_last() const
{
    for (int i = a.size() - 1; i >= 0; i--) if (a[i]) return i * 64 + __lg(a[i]);
    return -1;
}

int _Find_prev(int x) const
{
    assert(x >= 0 && x < n);
    --x;
    if (x == -1) return -1;
    int y = x & 63; x >>= 6;
    if (y < 63)
    {
        if (a[x] & (1llu << y + 1) - 1) return x * 64 + __lg(a[x] & (1llu << y + 1) - 1);
        --x;
    }
    while (x >= 0 && !a[x]) --x;
    return x == -1 ? -1 : x * 64 + __lg(a[x]);
}

```

```

    }
    string to_string() const
    {
        int n = size(), i;
        string s(n, '0');
        for (i = 0; i < n; i++) s[n - i - 1] += test(i);
        return s;
    }
};

istream &operator>>(istream &cin, Bitset &o)
{
    string s;
    cin >> s;
    int n = s.size(), i;
    o.reset();
    assert(n <= o.size());
    for (i = 0; i < n; i++) o.set(i, s[n - i - 1] - '0');
    return cin;
}

ostream &operator<<(ostream &cout, const Bitset &o) { return cout << o.to_string(); }

```

2.25 区间众数

```

template<class T> struct mode//[0,n)
{
    int n, ksz, m;
    vector<T> b;
    vector<vector<int>> pos, f;
    vector<int> a, blk, id, l;
    mode(const vector<T> &c) :n(c.size()), ksz(max<int>(1, sqrt(n))), m((n + ksz - 1) / ksz), b(c),
    pos(n), f(m, vector<int>(m)), a(n), blk(n), id(n), l(m + 1)
    {
        int i, j, k;
        sort(all(b)); b.resize(unique(all(b)) - b.begin());
        for (i = 0; i < n; i++)
        {
            a[i] = lower_bound(all(b), c[i]) - b.begin();
            id[i] = pos[a[i]].size();
            pos[a[i]].push_back(i);
        }
        for (i = 0; i < n; i++) blk[i] = i / ksz;
        for (i = 0; i <= m; i++) l[i] = min(i * ksz, n);
        vector<int> cnt(b.size());
        for (i = 0; i < m; i++)
        {
            fill(all(cnt), 0);
            pair<int, int> cur = {0, 0};
            for (j = i; j < m; j++)
            {
                for (k = l[j]; k < l[j + 1]; k++) cmax(cur, pair{++cnt[a[k]], a[k]});
                f[i][j] = cur.second;
            }
        }
    }
};

pair<T, int> ask(int L, int R)//返回最大众数

```

```

{
    assert(0 <= L && L < R && R <= n);
    int val = blk[L] == blk[R - 1] ? 0 : f[blk[L] + 1][blk[R - 1] - 1], i;
    int cnt = lower_bound(all(pos[val]), R) - lower_bound(all(pos[val]), L);
    for (i = min(R, 1[blk[L] + 1]) - 1; i >= L; i--)
    {
        auto &v = pos[a[i]];
        while (id[i] + cnt < v.size() && v[id[i] + cnt] < R) ++cnt, val = a[i];
        if (a[i] > val && id[i] + cnt - 1 < v.size() && v[id[i] + cnt - 1] < R) val = a[i];
    }
    for (i = max(L, 1[blk[R - 1]]); i < R; i++)
    {
        auto &v = pos[a[i]];
        while (id[i] >= cnt && v[id[i] - cnt] >= L) ++cnt, val = a[i];
        if (a[i] > val && id[i] >= cnt - 1 && v[id[i] - cnt + 1] >= L) val = a[i];
    }
    return {b[val], cnt};
}
};

```

2.26 表达式树

传入表达式，输出表达式树。

输入的第二个参数是全体括号以外的运算符，每个运算符要记录字符优先级和是否右结合。优先级数字越大，越优先计算，且优先级必须为正整数。

输出的第一个参数是子结点数组，第二个参数是每个结点对应的字符，第三个参数是根。结点编号从 1 开始。

输出的表达式树满足每个结点对应一个字符。若包含数字串，则视为相邻数码之间加一个并号，表示“数码链接”这个运算符。你不需要，也不应该手动加入这个并号。

如果表达式非法，将返回根为 0。不允许一元运算符（负号），不允许省略乘号，不允许出现字母（除非字母是运算符）。

如果需要在支持字母作为数字，修改所有包含 `isdigit` 的部分。

由于存在“数码链接”，在 `dfs` 树的时候最好记录一下子树大小，便于链接时计算（你不能在链接时直接看右子树的数字大小，因为有可能有前导 0）。

```

struct Q
{
    char ch;
    int prec;
    bool right;
};

tuple<vector<array<int, 2>>, vector<char>, int> parse_expr(string s, vector<Q> op) {
    static int idx[128];
    int maxp = 0, pos = 0, n, err = 0, i;
    {
        string t;
        for (char c : s)
        {
            if (t.size() && isdigit(t.back()) && isdigit(c)) t += '#';
            t += c;
        }
        swap(s, t);
        n = s.size();
    }
}

```

```

for (i = 0; i < op.size(); ++i)
{
    idx[op[i].ch] = i + 1;
    cmax(maxp, op[i].prec);
}
op.push_back({'#', ++maxp, 0});
idx['#'] = op.size();
vector<array<int, 2>> c(1);
vector<char> ch(1);
auto node = [&](char x) {
    c.push_back({0, 0});
    ch.push_back(x);
    return c.size() - 1;
};
function<int(int)> parse = [&](int lv) -> int {
    int u;
    if (lv > maxp)
    {
        if (pos < n && s[pos] == '(')
        {
            pos++;
            u = parse(1);
            if (err != (pos >= n || s[pos++] != ')')) return 0;
            return u;
        }
        else if (pos < n && isdigit(s[pos])) return u = node(s[pos++]);
        else return err = 1, 0;
    }
    else
    {
        u = parse(lv + 1);
        while (!err && pos < n)
        {
            char ch = s[pos];
            int i = idx[ch] - 1;
            if (i >= 0 && op[i].prec == lv)
            {
                ++pos;
                int v = node(ch), w = parse(lv + !op[i].right);
                c[v] = {u, w};
                u = v;
            }
            else break;
        }
        return u;
    }
};
int root = parse(0);
for (auto [ch, _, __] : op) idx[ch] = 0;
if (err || pos != n) return {{ }, { }, 0};
return {c, ch, root};
}
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cout << fixed << setprecision(15);
    string s;

```

```

getline(cin, s);
vector<Q> op = {
    {'|', 1, 0},
    {'&', 2, 0},
};
auto [c, ch, root] = parse_expr(s, op);
assert(root);
function<array<int, 3>(int)> dfs = [&](int u)->array<int, 3> {
    if (isdigit(ch[u])) return {ch[u] - '0', 0, 0};
    auto [l, r1, r2] = dfs(c[u][0]);
    if (ch[u] == '|')
    {
        if (!l) return {1, r1, r2 + 1};
        auto [r, r3, r4] = dfs(c[u][1]);
        return {r, r1 + r3, r2 + r4};
    }
    else
    {
        if (!l) return {0, r1 + 1, r2};
        auto [r, r3, r4] = dfs(c[u][1]);
        return {r, r1 + r3, r2 + r4};
    }
};
auto [r0, r1, r2] = dfs(root);
cout << r0 << endl << r1 << ' ' << r2 << endl;
}

```

2.27 区间排序区间复合

时空 $O((n + m) \log V)$, 其中 V 是排序关键字的值域, 要求排序关键字唯一。实际效率较低, 10^5 要 1.2s。

构造函数传入排序关键字与信息, 以及排序关键字的值域范围。

与其他板子不同的是, 所有区间都是左闭右开的, 下标从 0 开始。

```

template<class T, class info> struct range_sort
{
    enum type { inc, dec };
    int n;
    T pl, pr;
    vector<int> root, lc, rc, sz;
    vector<info> s, rs;
    struct treap
    {
        int n, rt;
        vector<ui> pr;
        vector<int> sz, num, lc, rc, lz, rev;
        vector<info> v, rv, s, rs;
        void reverse(int x)
        {
            lz[x] ^= 1;
            rev[x] ^= 1;
            swap(lc[x], rc[x]);
            swap(s[x], rs[x]);
            swap(v[x], rv[x]);
        }
        void pushdown(int x)
    }
};

```

```

{
    if (lz[x])
    {
        if (lc[x]) reverse(lc[x]);
        if (rc[x]) reverse(rc[x]);
        lz[x] = 0;
    }
}

void pushup(int x)
{
    sz[x] = sz[lc[x]] + sz[rc[x]] + num[x];
    s[x] = v[x];
    rs[x] = rv[x];
    if (lc[x])
    {
        s[x] = s[lc[x]] + s[x];
        rs[x] = rs[x] + rs[lc[x]];
    }
    if (rc[x])
    {
        s[x] = s[x] + s[rc[x]];
        rs[x] = rs[rc[x]] + rs[x];
    }
}

int kth;
void split_kth(int u, int &x, int &y)
{
    if (!u) return x = y = 0, void();
    pushdown(u);
    if (kth < sz[lc[u]]) split_kth(lc[y = u], x, lc[u]);
    else kth -= sz[lc[u]] + num[u], split_kth(rc[x = u], rc[u], y);
    pushup(u);
}

void split_lst(int &x, int &y)
{
    pushdown(x);
    if (rc[x])
    {
        split_lst(rc[x], y);
        pushup(x);
    }
    else
    {
        y = x;
        x = lc[x];
        lc[y] = 0;
        pushup(y);
    }
}

int merge(int x, int y)
{
    if (!x || !y) return x + y;
    if (pr[x] < pr[y])
    {
        pushdown(x);
        rc[x] = merge(rc[x], y);
        pushup(x);
    }
}

```

```

        return x;
    }
    pushdown(y);
    lc[y] = merge(x, lc[y]);
    pushup(y);
    return y;
}
treap(int n) : rt(0), pr(n), sz(n), num(n), lc(n), rc(n), lz(n), v(n), rv(n), s(n), rs(n),
rev(n)
{
    static mt19937 rnd(chrono::steady_clock::now().time_since_epoch().count());
    generate(all(pr), rnd);
}
void init(const vector<int> &index, const vector<info> &a)
{
    n = index.size() - 1;
    for (int i = 1; i <= n; i++)
    {
        s[i] = rs[i] = v[i] = a[index[i]];
        num[i] = sz[i] = 1;
        rt = merge(rt, i);
    }
}
};
treap t;
int np()
{
    lc.push_back(0);
    rc.push_back(0);
    sz.push_back(0);
    s.push_back({ });
    rs.push_back({ });
    return lc.size() - 1;
}
void pushup(int x)
{
    if (!x) return;
    sz[x] = sz[lc[x]] + sz[rc[x]];
    if (lc[x] && rc[x])
    {
        s[x] = s[lc[x]] + s[rc[x]];
        rs[x] = rs[rc[x]] + rs[lc[x]];
    }
    else if (lc[x]) s[x] = s[lc[x]], rs[x] = rs[lc[x]];
    else if (rc[x]) s[x] = s[rc[x]], rs[x] = rs[rc[x]];
}
void insert(int x, T l, T r, T p, const info &v)
{
    if (l + 1 == r)
    {
        s[x] = rs[x] = v;
        sz[x] = 1;
        return;
    }
    T mid = midpoint(l, r);
    if (p < mid)
    {

```



```

        if (!lc[x]) lc[x] = np();
        insert(lc[x], l, mid, p, v);
    }
    else
    {
        if (!rc[x]) rc[x] = np();
        insert(rc[x], mid, r, p, v);
    }
    pushup(x);
}

range_sort(vector<pair<T, info>> a, T pl, T _pr)
: n(a.size()), pl(pl), pr(_pr - pl), root(n + 1), t(n + 3) {
    np();
    for (int i = 0; i < n; i++)
    {
        a[i].first -= pl;
        root[i + 1] = np();
        insert(root[i + 1], 0, pr, a[i].first, a[i].second);
    }
    t.init(root, s);
}

int merge(int x, int y, T l, T r)
{
    if (!x || !y) return x + y;
    T mid = midpoint(l, r);
    lc[x] = merge(lc[x], lc[y], l, mid);
    rc[x] = merge(rc[x], rc[y], mid, r);
    pushup(x);
    return x;
}

pair<int, int> split(int x, int k, T l, T r)
{
    if (x == 0) return {0, 0};
    if (l + 1 == r) return {0, x};
    T mid = midpoint(l, r);
    if (k < sz[lc[x]])
    {
        auto [u, v] = split(lc[x], k, l, mid);
        lc[x] = v;
        pushup(x);
        if (!sz[x]) x = 0;
        if (!u) return {0, x};
        int y = np();
        lc[y] = u; pushup(y);
        return {y, x};
    }
    auto [u, v] = split(rc[x], k - sz[lc[x]], mid, r);
    rc[x] = u;
    pushup(x);
    if (!sz[x]) x = 0;
    if (!v) return {x, 0};
    int y = np();
    rc[y] = v; pushup(y);
    return {x, y};
}

void set_treap(int i, int u, bool swp)
{

```

```

    root[i] = u;
    if (swp)
    {
        t.s[i] = t.v[i] = rs[u];
        t.rs[i] = t.rv[i] = s[u];
    }
    else
    {
        t.s[i] = t.v[i] = s[u];
        t.rs[i] = t.rv[i] = rs[u];
    }
    t.sz[i] = t.num[i] = sz[u];
    t.lc[i] = t.rc[i] = 0;
    t.rev[i] = swp;
}
pair<int, int> find(int i)
{
    if (i == t.sz[t.rt]) return {t.rt, 0};
    t.kth = i;
    int x, y, z, r1, r2;
    t.split_kth(t.rt, x, z);
    t.split_lst(x, y);
    int k = i - t.sz[x], tot = t.num[y];
    if (t.rev[y] && k)
    {
        k = t.num[y] - k;
        tie(r1, r2) = split(root[y], k, 0, pr);
        if (r1) set_treap(i, r2, 1);
        set_treap(i + 1, r1, 1);
        x = t.merge(x, r2 ? i : 0);
        z = t.merge(i + 1, z);
        return {x, z};
    }
    else
    {
        tie(r1, r2) = split(root[y], k, 0, pr);
        if (r1) set_treap(i, r1, t.rev[y]);
        set_treap(i + 1, r2, t.rev[y]);
        x = t.merge(x, r1 ? i : 0);
        z = t.merge(i + 1, z);
        return {x, z};
    }
}
tuple<int, int, int> split_range(int l, int r)
{
    auto [_ , z] = find(r);
    t.rt = _;
    auto [x, y] = find(l);
    return {x, y, z};
}
void modify(int i, const pair<T, info> &rhs)
{
    assert(rhs.first >= pl && rhs.first < pl + pr);
    auto [x, y, z] = split_range(i, i + 1);
    root[y] = np();
    insert(root[y], 0, pr, rhs.first - pl, rhs.second);
    set_treap(y, root[y], 0);
}

```

```

        t.rt = t.merge(t.merge(x, y), z);
    }
    info ask(int l, int r)
    {
        auto [x, y, z] = split_range(l, r);
        info res = t.s[y];
        t.rt = t.merge(t.merge(x, y), z);
        return res;
    }
    void merge_dfs(int x, int y)
    {
        if (!y) return;
        if (x != y) root[x] = merge(root[x], root[y], 0, pr), root[y] = 0;
        merge_dfs(x, t.lc[y]);
        merge_dfs(x, t.rc[y]);
    }
    void sort(int l, int r, type op)
    {
        auto [x, y, z] = split_range(l, r);
        merge_dfs(y, y);
        set_treap(y, root[y], op == dec);
        t.rt = t.merge(t.merge(x, y), z);
    }
};
const ull p = 998244353;
struct info
{
    ull k, b;
    info operator+(const info &rhs) const { return {(k * rhs.k) % p, (rhs.b + rhs.k * b) % p}; }
    bool operator<(const info &rhs) const { return 0; }
    ull operator()(ull x) const { return (k * x + b) % p; }
};
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cout << fixed << setprecision(15);
    int n, m, i;
    cin >> n >> m;
    vector<pair<int, info>> a(n);
    for (auto &[x, y] : a)
    {
        auto &[k, b] = y;
        cin >> x >> k >> b;
    }
    range_sort s(a, 0, (int)1e9 + 1);
    while (m--)
    {
        int op;
        cin >> op;
        if (op == 0)
        {
            int i, p;
            ull k, b;
            cin >> i >> p >> k >> b;
            a[i] = {p, {k, b}};
            s.modify(i, {p, {k, b}});
        }
    }
}

```

```
    else
    {
        int l, r;
        cin >> l >> r;
        if (op == 1)
        {
            ull x;
            cin >> x;
            cout << s.ask(l, r)(x) << '\n';
        }
        else s.sort(l, r, op == 2 ? s.inc : s.dec);
    }
}
```

3 数学

3.1 矩阵类（较新）

```

using ull = unsigned long long;
const ull p = 998244353;
ull ksm(ull x, ull y)
{
    ull r = 1;
    while (y)
    {
        if (y & 1) r = r * x % p;
        x = x * x % p; y >>= 1;
    }
    return r;
}

struct matrix;
matrix E(int n);
struct matrix : vector<vector<ull>>
{
    explicit matrix(int n = 0, int m = 0) : vector(n, vector<ull>(m)) { }
    pair<int, int> sz() const { if (size()) return {size(), back().size()}; return {0, 0}; }
    matrix &operator+=(const matrix &b)
    {
        assert(sz() == b.sz());
        auto [n, m] = sz();
        for (int i = 0; i < n; i++) for (int j = 0; j < m; j++) ((*this)[i][j] += b[i][j]) %= p;
        return *this;
    }
    matrix &operator-=(const matrix &b)
    {
        assert(sz() == b.sz());
        auto [n, m] = sz();
        for (int i = 0; i < n; i++) for (int j = 0; j < m; j++) ((*this)[i][j] += p - b[i][j]) %=
        p;
        return *this;
    }
    matrix operator*(const matrix &b) const
    {
        auto [n, m] = sz();
        auto [_, q] = b.sz();
        assert(m == _);
        int i, j, k;
        matrix c(n, q);
        for (k = 0; k < m; k++)
        {
            for (i = 0; i < n; i++) for (j = 0; j < q; j++) c[i][j] += (*this)[i][k] * b[k][j];
            if ((k & 15) == 15) for (auto &v : c) for (ull &x : v) x %= p;
        }
        for (auto &v : c) for (ull &x : v) x %= p;
        static_assert(-1llu / p / p > 17);
        return c;
    }
    matrix operator+(const matrix &b) const { auto a = *this; return a += b; }
    matrix operator-(const matrix &b) const { auto a = *this; return a -= b; }
    matrix &operator*=(const matrix &b) { return *this = *this * b; }
    matrix &operator*=(ull k) { for (auto &v : *this) for (ull &x : v) x = x * k % p; return *

```

```

this; }
matrix operator*(ull k) const { auto a = *this; return a *= k; }
matrix transpose() const
{
    auto [n, m] = sz();
    matrix res(m, n);
    for (int i = 0; i < n; i++) for (int j = 0; j < m; j++) res[j][i] = (*this)[i][j];
    return res;
}
int rank() const
{
    auto [n, m] = sz();
    vector<vector<ull>> a = n <= m ? *this : transpose();
    if (n > m) ::swap(n, m);
    int i, j, k, l, r = 0;
    for (i = 0, j = 0; i < n && j < m; j++)
    {
        for (k = i; k < n; k++) if (a[k][j]) break;
        if (k == n) continue;
        ::swap(a[i], a[k]);
        ull iv = ksm(a[i][j], p - 2);
        for (k = j; k < m; k++) a[i][k] = a[i][k] * iv % p;
        for (k = i + 1; k < n; k++) for (l = j + 1; l < m; l++) a[k][l] = (a[k][l] + (p - a[k][j]) * a[i][l]) % p;
        ++i; ++r;
    }
    return r;
}
vector<ull> poly() const// | kE - A |
{
    auto [n, m] = sz();
    vector<vector<ull>> a = *this;
    assert(n == m);
    int i, j, k;
    for (i = 1; i < n; i++)
    {
        for (j = i; j < n && !a[j][i - 1]; j++);
        if (j == n) continue;
        if (j > i)
        {
            ::swap(a[i], a[j]);
            for (k = 0; k < n; k++) ::swap(a[k][j], a[k][i]);
        }
        ull r = a[i][i - 1];
        for (j = 0; j < n; j++) a[j][i] = a[j][i] * r % p;
        r = ksm(r, p - 2);
        for (j = i - 1; j < n; j++) a[i][j] = a[i][j] * r % p;
        for (j = i + 1; j < n; j++)
        {
            r = a[j][i - 1];
            for (k = 0; k < n; k++) a[k][i] = (a[k][i] + a[k][j] * r) % p;
            r = p - r;
            for (k = i - 1; k < n; k++) a[j][k] = (a[j][k] + a[i][k] * r) % p;
        }
    }
    vector g(n + 1, vector<ull>(n + 1));
    g[0][0] = 1;

```

```

for (i = 0; i < n; i++)
{
    ull r = p - 1, rr;
    for (j = i; j >= 0; j--)//第 j 行选第 n 列
    {
        rr = r * a[j][i] % p;
        for (k = 0; k <= j; k++) g[i + 1][k] = (g[i + 1][k] + rr * g[j][k]) % p;
        if (j) r = r * a[j][j - 1] % p;
    }
    for (k = 1; k <= i + 1; k++) (g[i + 1][k] += g[i][k - 1]) %= p;
}
auto f = g[n];
//if (n & 1) for (i = 0; i <= n; i++) if (f[i]) f[i] = p - f[i];
return f;
}
ull det() const
{
    auto [n, m] = sz();
    vector<vector<ull>> a = *this;
    assert(n == m);
    int i, j, k;
    ull r = 1;
    for (i = 0; i < n; i++)
    {
        for (j = i; j < n; j++) if (a[j][i]) break;
        if (j == n) return 0;
        if (i != j) r = p - r, ::swap(a[i], a[j]);
        (r *= a[i][i]) %= p;
        ull iv = ksm(a[i][i], p - 2);
        for (j = i; j < n; j++) a[i][j] = a[i][j] * iv % p;
        for (j = i + 1; j < n; j++) for (k = i + 1; k < n; k++) a[j][k] = (a[j][k] + (p - a[i][k]) * a[j][i]) % p;
    }
    return r % p;
}
tuple<int, vector<ull>, vector<vector<ull>>> gauss(const vector<ull> &b) const//Ax=b, rank of
base, one sol, base
{
    auto [n, m] = sz();
    if (b.size() != n) return {-1, { }, { }};
    vector<vector<ull>> a = *this;
    int i, j, k, R = m;
    for (i = 0; i < n; i++) a[i].push_back(b[i]);
    vector<int> fix(m, -1);
    for (i = k = 0; i < m; i++)
    {
        for (j = k; j < n; j++) if (a[j][i]) break;
        if (j == n) continue;
        fix[i] = k; --R;
        ::swap(a[k], a[j]);
        auto &u = a[k];
        ull x = ksm(u[i], p - 2);
        for (j = i; j <= m; j++) u[j] = u[j] * x % p;
        for (auto &v : a) if (v.data() != u.data())
        {
            x = p - v[i];
            for (j = i; j <= m; j++) v[j] = (v[j] + x * u[j]) % p;
        }
    }
}

```

```

    }
    ++k;
}
for (i = k; i < n; i++) if (a[i][m]) return {-1, { }, { }};
vector<ull> r(m);
vector<vector<ull>> c;
for (i = 0; i < m; i++) if (fix[i] != -1) r[i] = a[fix[i]][m];
for (i = 0; i < m; i++) if (fix[i] == -1)
{
    vector<ull> r(m);
    r[i] = 1;
    for (j = 0; j < m; j++) if (fix[j] != -1) r[j] = (p - a[fix[j]][i]) % p;
    c.push_back(r);
}
return {R, r, c};
}
optional<matrix> inverse() const
{
    auto [n, m] = sz();
    assert(n == m);
    vector<int> ih(n, -1), jh(n, -1);
    matrix a = *this;
    int i, j, k;
    for (k = 0; k < n; k++)
    {
        for (i = k; i < n; i++) if (ih[k] == -1) for (j = k; j < n; j++) if (a[i][j])
        {
            ih[k] = i;
            jh[k] = j;
            break;
        }
        if (ih[k] == -1) return { };
        ::swap(a[k], a[ih[k]]);
        for (i = 0; i < n; i++) ::swap(a[i][k], a[i][jh[k]]);
        if (!a[k][k]) return { };
        a[k][k] = ksm(a[k][k], p - 2);
        for (i = 0; i < n; i++) if (i != k) (a[k][i] *= a[k][k]) %= p;
        for (i = 0; i < n; i++) if (i != k) for (j = 0; j < n; j++) if (j != k)
            (a[i][j] += (p - a[i][k]) * a[k][j]) %= p;
        for (i = 0; i < n; i++) if (i != k) (a[i][k] *= p - a[k][k]) %= p;
    }
    for (k = n - 1; k >= 0; k--)
    {
        ::swap(a[k], a[jh[k]]);
        for (i = 0; i < n; i++) ::swap(a[i][k], a[i][ih[k]]);
    }
    return a;
}
matrix adjugate() const
{
    auto [n, m] = sz();
    assert(n == m);
    int R = rank();
    if (n == 1) return E(1);
    if (R == n) return *inverse() * det();
    if (R == n - 1)
    {

```



```

    int i, j, k, l;
    auto [_ , x, dx] = gauss(vector<ull>(n));
    auto [__, y, dy] = transpose().gauss(vector<ull>(n));
    if (count(all(x), 0) == n) x = dx[0];
    if (count(all(y), 0) == n) y = dy[0];
    for (k = 0; k < n; k++) if (x[k]) break;
    for (l = 0; l < n; l++) if (y[l]) break;
    assert(k < n && l < n);
    matrix res(n, n), c(n - 1, n - 1);
    for (i = 0; i < n; i++) if (i != l) for (j = 0; j < n; j++) if (j != k) c[i - (i > l)
] [j - (j > k)] = (*this)[i][j];
    for (i = 0; i < n; i++) for (j = 0; j < n; j++) res[i][j] = x[i] * y[j] % p;
    ull t = c.det() * ksm((k + l & 1) ? p - res[k][l] : res[k][l], p - 2) % p;
    assert(res[k][l]);
    assert(c.det());
    assert(t);
    return res * t;
}
return matrix(n, n);
};

istream &operator>>(istream &cin, matrix &r) { for (auto &v : r) for (ull &x : v) cin >> x;
return cin; }

ostream &operator<<(ostream &cout, const matrix &r) { auto [n, m] = r.sz(); for (int i = 0; i < n
; i++) for (int j = 0; j < m; j++) cout << r[i][j] << "\n" [j + 1 == m]; return cout; }

matrix E(int n) { matrix r(n, n); for (int i = 0; i < n; i++) r[i][i] = 1; return r; }

matrix pow(matrix a, long long k)
{
    assert(k >= 0);
    auto [n, m] = a.sz();
    assert(n == m);
    matrix r = k & 1 ? a : E(n);
    k >>= 1;
    while (k)
    {
        a *= a;
        if (k & 1) r *= a;
        k >>= 1;
    }
    return r;
}

matrix pow2(matrix a, long long k)
{
    vector<ull> f = a.poly();
    int n = f.size() - 1, i, j;
    if (!n) return matrix();
    if (n == 1) return E(1) * ksm(a[0][0], k);
    assert(f[n] == 1);
    vector<ull> r(n), x(n), t(n * 2);
    r[0] = x[1] = 1;
    for (ull &x : f) x = (p - x) % p;
    reverse(all(f));
    fill(all(t), 0);
    if (k & 1)
    {
        for (i = 0; i < n; i++) for (j = 0; j < n; j++) t[i + j] = (t[i + j] + r[i] * x[j]) % p;
        for (i = n * 2 - 2; i >= n; i--) for (j = 1; j <= n; j++) t[i - j] = (t[i - j] + f[j] * t

```

```

[i]) % p;
    for (i = 0; i < n; i++) r[i] = t[i];
}
k >>= 1;
while (k)
{
    fill(all(t), 0);
    for (i = 0; i < n; i++) for (j = 0; j < n; j++) t[i + j] = (t[i + j] + x[i] * x[j]) % p;
    for (i = n * 2 - 2; i >= n; i--) for (j = 1; j <= n; j++) t[i - j] = (t[i - j] + f[j] * t
[i]) % p;
    for (i = 0; i < n; i++) x[i] = t[i];
    if (k & 1)
    {
        fill(all(t), 0);
        for (i = 0; i < n; i++) for (j = 0; j < n; j++) t[i + j] = (t[i + j] + r[i] * x[j]) %
p;
        for (i = n * 2 - 2; i >= n; i--) for (j = 1; j <= n; j++) t[i - j] = (t[i - j] + f[j]
* t[i]) % p;
        for (i = 0; i < n; i++) r[i] = t[i];
    }
    k >>= 1;
}
matrix res(n, n);
int b = ceil(sqrt(n));
vector<matrix> s(b + 1);
s[0] = E(n); s[1] = a;
for (i = 2; i <= b; i++) s[i] = s[i - 1] * a;
for (i = b - 1; i >= 0; i--)
{
    res *= s[b];
    for (j = min(n, (i + 1) * b) - 1; j >= i * b; j--) res += s[j - i * b] * r[j];
}
return res;
}

```

3.2 在线 $O(1)$ 逆元

预处理复杂度为 $O(p^{\frac{2}{3}})$ 。

```

namespace online_inv
{
    using ui = unsigned;
    using ull = unsigned long long;
    const ull p = 1e9 + 7, n = 1010, m = n * n, N = m + 2;
    static_assert(n * n * n > p);
    int l[N], r[N];
    ull y[N];
    bool s[N];
    ull _inv[N * 2], i, j, k;
    void init_inv()
    {
        _inv[1] = 1;
        for (i = 2; i < m * 2; i++)
        {
            j = p / i;
            _inv[i] = (p - j) * _inv[p - i * j] % p;
        }
    }
}

```

```

    s[0] = y[0] = 1;
    for (i = 1; i < n; i++) for (j = i; j < n; j++) if (!s[k = i * m / j])
    {
        y[k] = j;
        s[k] = 1;
    }
    l[0] = 1;
    for (i = 1; i <= m; i++) l[i] = s[i] ? y[i] : l[i - 1];
    r[m] = 1;
    for (i = m - 1; ~i; i--) r[i] = s[i] ? y[i] : r[i + 1];
    for (i = 0; i <= m; i++) y[i] = min(l[i], r[i]);
}
inline ull inv(const ull &x)
{
    assert(x && x < p);
    if (x < m * 2) return _inv[x];
    k = x * m / p;
    j = y[k] * x % p;
    return (j < m * 2 ? _inv[j] : p - _inv[p - j]) * y[k] % p;
}
bool _ = (init_inv(), 0);
}
using online_inv::inv, online_inv::p;

```

3.3 Strassen 矩阵乘法

没用，不如卡常。 $O(n^{\log_2 7})$ 。

```

#include "bits/stdc++.h"
using namespace std;
using ui = unsigned;
using ull = unsigned long long;
const ui p = 998244353;
const ull fh = 1ull << 31;
struct Q
{
    ui **a;
    int n;
    Q() { n = 0; }
    void clear()
    {
        for (int i = 0; i < n; i++) delete[] a[i];
        if (n) delete[] a; n = 0;
    }
    Q(int nn)//不能传入不是 2 的幂的数!
    {
        n = nn;
        assert(n == (n & -n));
        a = new ui * [n];
        for (int i = 0; i < n; i++) a[i] = new ui[n], memset(a[i], 0, n * sizeof a[0][0]);
    }
    const Q &operator=(const Q &b)
    {
        if (this == &b) return *this;
        clear(); n = b.n;
        a = new ui * [n];
        for (int i = 0; i < n; i++) a[i] = new ui[n], memcpy(a[i], b.a[i], n * sizeof a[0][0]);
    }
}

```

```

    return *this;
}
~Q() { clear(); }
Q operator+(const Q &b)
{
    Q c(n);
    for (int i = 0; i < n; i++) for (int j = 0; j < n; j++) if ((c.a[i][j] = a[i][j] + b.a[i][j]) >= p) c.a[i][j] -= p;
    return c;
}
Q operator-(const Q &b)
{
    Q c(n);
    for (int i = 0; i < n; i++) for (int j = 0; j < n; j++) if ((c.a[i][j] = a[i][j] - b.a[i][j]) & fh) c.a[i][j] += p;
    return c;
}
Q operator*(Q &b)
{
    Q c(n);
    if (n <= 128)
    {
        for (int i = 0; i < n; i++) for (int k = 0; k < n; k++) for (int j = 0; j < n; j++) c.a[i][j] = (c.a[i][j] + (ull)a[i][k] * b.a[k][j]) % p;
        return c;
    }
    Q A[2][2], B[2][2], s[10], p[5];
    n >>= 1;
    int i, j, k, l;
    for (i = 0; i < 2; i++) for (j = 0; j < 2; j++)
    {
        A[i][j] = Q(n);
        for (k = 0; k < n; k++) memcpy(A[i][j].a[k], a[k + i * n] + j * n, n * sizeof a[0][0]);
        B[i][j] = Q(n);
        for (k = 0; k < n; k++) memcpy(B[i][j].a[k], b.a[k + i * n] + j * n, n * sizeof a[0][0]);
    }
    s[0] = B[0][1] - B[1][1];
    s[1] = A[0][0] + A[0][1];
    s[2] = A[1][0] + A[1][1];
    s[3] = B[1][0] - B[0][0];
    s[4] = A[0][0] + A[1][1];
    s[5] = B[0][0] + B[1][1];
    s[6] = A[0][1] - A[1][1];
    s[7] = B[1][0] + B[1][1];
    s[8] = A[0][0] - A[1][0];
    s[9] = B[0][0] + B[0][1];
    p[0] = A[0][0] * s[0];
    p[1] = s[1] * B[1][1];
    p[2] = s[2] * B[0][0];
    p[3] = A[1][1] * s[3];
    p[4] = s[4] * s[5];
    A[0][0] = p[4] + p[3] - p[1] + s[6] * s[7];
    A[0][1] = p[0] + p[1];
    A[1][0] = p[2] + p[3];
    A[1][1] = p[4] + p[0] - p[2] - s[8] * s[9];
}

```

```

        for (i = 0; i < 2; i++) for (j = 0; j < 2; j++) for (k = 0; k < n; k++) memcpy(c.a[k + i
* n] + j * n, A[i][j].a[k], n * sizeof a[0][0]);
        n <<= 1;
        return c;
    }
};
int main()
{
    int i, j, n, m, k;
    ios::sync_with_stdio(0); cin.tie(0);
    cin >> n >> m >> k;
    auto f = [](int x) {
        if (x <= 1) return 1;
        return 1 << (32 - __builtin_clz(x - 1));
    };
    int N = max({f(n), f(m), f(k)});
    Q a(N), b(N);
    for (i = 0; i < n; i++) for (j = 0; j < m; j++) cin >> a.a[i][j];
    for (i = 0; i < m; i++) for (j = 0; j < k; j++) cin >> b.a[i][j];
    a = a * b;
    for (i = 0; i < n; i++) for (j = 0; j < k; j++) cout << a.a[i][j] << "\n" [j + 1 == k];
}

```

3.4 扩展欧拉定理

求 $a \uparrow\uparrow b \bmod c$ 。前面的 Prime 命名空间只是求 φ 用的。
 特别注意，这里的 i64 不是无符号。

```

namespace Prime
{
    using ui = unsigned;
    const int N = 1e6 + 2;
    const auto M = (N - 111) * (N - 1);
    ui pr[N], mn[N], phi[N], cnt;
    int mu[N];
    void init_prime()
    {
        ui i, j, k;
        phi[1] = mu[1] = 1;
        for (i = 2; i < N; i++)
        {
            if (!mn[i])
            {
                pr[cnt++] = i;
                phi[i] = i - 1; mu[i] = -1;
                mn[i] = i;
            }
            for (j = 0; (k = i * pr[j]) < N; j++)
            {
                mn[k] = pr[j];
                if (i % pr[j] == 0)
                {
                    phi[k] = phi[i] * pr[j];
                    break;
                }
                phi[k] = phi[i] * (pr[j] - 1);
            }
        }
    }
}

```

```

        mu[k] = -mu[i];
    }
}
//for (i=2;i<N;i++) if (mu[i]<0) mu[i]+=p;
}
template<typename T> T getphi(T x)
{
    assert(M >= x);
    T r = x;
    for (ui i = 0; i < cnt && (T)pr[i] * pr[i] <= x && x >= N; i++) if (x % pr[i] == 0)
    {
        ui y = pr[i], tmp;
        x /= y;
        while (x == (tmp = x / y) * y) x = tmp;
        r = r / y * (y - 1);
    }
    if (x >= N) return r / x * (x - 1);
    while (x > 1)
    {
        ui y = mn[x], tmp;
        x /= y;
        while (x == (tmp = x / y) * y) x = tmp;
        r = r / y * (y - 1);
    }
    return r;
}
template<typename T> vector<pair<T, ui>> getw(T x)
{
    assert(M >= x);
    vector<pair<T, ui>> r;
    for (ui i = 0; i < cnt && (T)pr[i] * pr[i] <= x && x >= N; i++) if (x % pr[i] == 0)
    {
        ui y = pr[i], z = 1, tmp;
        x /= y;
        while (x == (tmp = x / y) * y) x = tmp, ++z;
        r.push_back({y, z});
    }
    if (x >= N)
    {
        r.push_back({x, 1});
        return r;
    }
    while (x > 1)
    {
        ui y = mn[x], z = 1, tmp;
        x /= y;
        while (x == (tmp = x / y) * y) x = tmp, ++z;
        r.push_back({y, z});
    }
    return r;
}
int _ = (init_prime(), 0);
}
using Prime::pr, Prime::phi, Prime::getw;
using Prime::mu, Prime::getphi;
ll ksm(ll x, ll y, ll p)
{

```

```

x = (x - p) % p + p;
ll r = 1;
while (y)
{
    if (y & 1) r = (r * x - p) % p + p;
    x = (x * x - p) % p + p;
    y >>= 1;
}
return r;
}
struct Q
{
    vector<ll> p;
    Q(ll mod) :p{mod}
    {
        while (p.back() > 1) p.push_back(getphi(p.back()));
    }
    ll operator()(ll a, ll b)
    {
        assert(b);
        if (!a) return (b + 1 & 1) % p[0];
        ll r = 1, i = min<ll>(b, p.size());
        while ((--i) >= 0) r = ksm(a, r, p[i]);
        return r % p[0];
    }
    ll operator()(vector<ll> a)
    {
        assert(a.size());
        ll r = 1, i = 0, j;
        while (i < a.size() && i < p.size() && a[i]) ++i;
        if (i < a.size() && i < p.size())
        {
            j = i;
            while (j < a.size() && !a[j]) ++j;
            a[i] = j - i - 1 & 1;
            ++i;
        }
        while ((--i) >= 0) r = ksm(a[i], r, p[i]);
        return r % p[0];
    }
};

```

3.5 exgcd

$O(\log p)$, $O(\log p)$ 。

递归版：

```

int exgcd(int a, int b, int c)//ax+by=c,return x
{
    if (a == 0) return c / b;
    return (c - (ll)b * exgcd(b % a, a, c)) / a % b;
}

```

递推版：

```

pair<ll, ll> exgcd(ll a, ll b, ll c)//ax+by=c, {-1,-1} 无解, b=0 返回 {c/a,0}, 否则返回最小非负 x
{

```

```

assert(a || b);
ll d = gcd(a, b);
if (c % d) return {-1, -1};
if (!b) return {c / a, 0};
if (a < 0) a = -a, b = -b, c = -c;
ll x = 1, x1 = 0, p = a, q = b, k;
b = abs(b);
while (b)
{
    k = a / b;
    x -= k * x1; a -= k * b;
    swap(x, x1);
    swap(a, b);
}
b = abs(q / d);
x = (ll)((c / d) % b) * (x % b) % b;
if (x < 0) x += b;
return {x, (ll)((c - (ll)p * x) / q)};
}
ll fun(ll a, ll b, ll p) // ax=b(mod p)
{
    return exgcd(a, -p, b).first % p;
}

```

3.6 exCRT

CRT: 设 $M = \prod_{i=1}^n m_i$, $t_i \times \frac{M}{m_i} \equiv 1 \pmod{m_i}$, 则 $x \equiv \sum_{i=1}^n a_i t_i \frac{M}{m_i}$ 。

以下为 exCRT, 与 CRT 无关。实现了一个类 Q, 表示一条方程, 支持合并。

```

namespace CRT
{
    pair<ll, ll> exgcd(ll a, ll b, ll c)
    {
        assert(a || b);
        ll d = gcd(a, b);
        if (c % d) return {-1, -1};
        if (!b) return {c / a, 0};
        if (a < 0) a = -a, b = -b, c = -c;
        ll x = 1, x1 = 0, p = a, q = b, k;
        b = abs(b);
        while (b)
        {
            k = a / b;
            x -= k * x1; a -= k * b;
            swap(x, x1);
            swap(a, b);
        }
        b = abs(q / d);
        x = x * (c / d) % b;
        if (x < 0) x += b;
        return {x, (c - p * x) / q};
    }
    struct Q
    {
        ll p, r; // 0 <= r < p
        Q operator+(const Q &o) const

```



```

    {
        if (p == 0 || o.p == 0) return {0, 0};
        auto [x, y] = exgcd(p, -o.p, r - o.r);
        if (x == -1 && y == -1) return {0, 0};
        ll q = lcm(p, o.p);
        return {q, ((r - x * p) % q + q) % q};
    }
};
}
using CRT::Q;

```

3.7 exBSGS

$O(\sqrt{n})$ 。哈希表 ht 可以用 map 代替。

```

namespace BSGS
{
    template<int N, class X, class Y, ui p = (int)1e6 + 7> struct ht//个数, 定义域, 值域
    {
        Y a[N], def;
        X v[N];
        int fir[p + 2], nxt[N], st[p + 2];//和模数相适应
        int tp, ds;//自定义模数
        ht(Y def = 0) :def(def) { memset(fir, 0, sizeof fir); tp = ds = 0; }
        Y &operator[](X x)//位置, 值
        {
            ui y = x % p;
            for (int i = fir[y]; i; i = nxt[i])
                if (v[i] == x)
                    return a[i];
            v[++ds] = x; a[ds] = def;
            if (!fir[y]) st[++tp] = y;
            nxt[ds] = fir[y]; fir[y] = ds;
            return a[ds];
        }
        Y find(X x)
        {
            ui y = x % p;
            int i;
            for (i = fir[y]; i; i = nxt[i]) if (v[i] == x) return a[i];
            return def;
        }
        void clear()
        {
            ++tp;
            while (--tp) fir[st[tp]] = 0;
            ds = 0;
        }
    };
    const int N = 1e6;
    ht<N, ui, ui> s;
    ull a, B, p, ia;
    ull ksm(ull x, ull y, ull p)
    {
        ull r = 1;
        while (y)
        {

```

```

        if (y & 1) (r *= x) %= p;
        (x *= x) %= p; y >>= 1;
    }
    return r;
}
void init(ui _a, ui _p, ui _B = 0)
{
    s.clear();
    B = _B; p = _p; a = _a;
    if (!B) B = sqrt(p) + 2;
    assert(B < N);
    ull x = 1; ia = 0;
    for (int i = 0; i < B; i++)
    {
        ui &y = s[x];
        if (y) return;
        y = i + 1;
        (x *= a) %= p;
    }
    ia = ksm(x, p - 2, p);
}
int calc(ull b)
{
    if (!a) return 1 - min((int)b, 2);
    for (ui i = 0; i * B < p; i++)
    {
        ui x = s.find(b);
        if (x) return i * B + x - 1;
        (b *= ia) %= p;
    }
    return -1;
}
int bsgs(ui a, ui b, ui p)
{
    s.clear();
    a %= p; b %= p;
    if (!a) return 1 - min((int)b, 2); // 含 -1
    ui i, k, x, y;
    ull j;
    x = sqrt(p) + 2;
    assert(x < N);
    for (i = 0, j = 1; i < x; i++, (j *= a) %= p)
    {
        if (j == b) return i;
        s[j * b % p] = i + 1;
    }
    k = j;
    for (i = 1; i <= x; i++, (j *= k) %= p)
        if (y = s.find(j))
            return (ull)i * x - y + 1;
    return -1;
}
bool isprime(ui p)
{
    if (p <= 1) return 0;
    for (ui i = 2; i * i <= p; i++) if (p % i == 0) return 0;
    return 1;
}

```

```

}
int exgcd(int a, int b)
{
    if (a == 1) return 1;
    return (1 - (ll)b * exgcd(b % a, a)) / a; //not ull
}
int exbsgs(ui a, ui b, ui p) //a^x = b (mod p)
{
    // if (isprime(p)) return bsgs(a, b, p);
    a %= p; b %= p;
    ui i, k, x;
    int cnt = 0;
    ull j, y = __lg(p);
    for (i = 0, j = 1 % p; i <= y; i++, (j *= a) %= p)
        if (j == b)
            return i;
    y = 1;
    while (1)
    {
        if ((x = gcd(a, p)) == 1) break;
        if (b % x) return -1; //no sol
        ++cnt;
        p /= x; b /= x;
        (y *= a / x) %= p;
    }
    a %= p;
    b = (ull)b * ((int)p + exgcd(y, p)) % p;
    int r = bsgs(a, b, p);
    return r == -1 ? -1 : r + cnt;
}
}
using BSGS::bsgs, BSGS::exbsgs, BSGS::init, BSGS::calc;

```

3.8 exLucas

求组合数。

```

struct binom
{
    using pa = pair<ui, ui>;
    ull p;
    vector<pa> a;
    vector<vector<ui>> b, ib, pw, pb;
    vector<ui> ph, xs;
    ull ksm(ull x, ll y, ull p)
    {
        ull r = 1;
        while (y)
        {
            if (y & 1) r = r * x % p;
            x = x * x % p;
            y >>= 1;
        }
        return r;
    }
    ull f(ll n, ll m, ll nm, int i)
    {

```

```

    auto [qi, pi] = a[i];
    ull r = 1;
    ll c1 = 0, c2 = 0;
    while (n)
    {
        r = r * b[i][n % pi] % pi * ib[i][m % pi] % pi * ib[i][nm % pi] % pi;
        c2 += n / pi - m / pi - nm / pi;
        n /= qi, m /= qi, nm /= qi;
        c1 += n - m - nm;
    }
    return 1llu * pw[i][min<int>(c1, pw[i].size() - 1)] * pb[i][c2 % ph[i]] % pi * r % pi;
}
ull operator()(ll n, ll m)
{
    if (m < 0 || n < m) return 0;
    ull r = 0;
    for (int i = 0; i < a.size(); i++) r = (r + xs[i] * f(n, m, n - m, i)) % p;
    return r;
}
binom(ull p) :p(p)
{
    int i, j;
    ull x = p, y, z;
    for (i = 2; i * i <= x; i++) if (x % i == 0)
    {
        z = x; x /= i;
        while (1)
        {
            y = x / i;
            if (i * y == x) x = y; else break;
        }
        a.push_back({i, z / x});
    }
    if (x > 1) a.push_back({x, x});
    int n = a.size();
    b = ib = pw = pb = vector<vector<ui>>(n);
    ph = xs = vector<ui>(n);
    for (i = 0; i < n; i++)
    {
        auto [qi, pi] = a[i];
        ph[i] = pi / qi * (qi - 1);
        xs[i] = ksm(p / pi, ph[i] - 1, p) * (p / pi) % p;
    }
    for (i = 0; i < n; i++)
    {
        auto [qi, pi] = a[i];
        b[i] = ib[i] = vector<ui>(pi, 1);
        for (j = 1; j < pi; j++) b[i][j] = 1llu * b[i][j - 1] * (j % qi == 0 ? 1 : j) % pi;
        ib[i][pi - 1] = ksm(b[i][pi - 1], ph[i] - 1, pi);
        for (j = pi - 1; j; j--) ib[i][j - 1] = 1llu * ib[i][j] * (j % qi == 0 ? 1 : j) % pi;
        pw[i] = {1};
        while (pw[i].back()) pw[i].push_back(1llu * pw[i].back() * qi % pi);
        pb[i].resize(ph[i], 1);
        for (j = 1; j < ph[i]; j++) pb[i][j] = 1llu * pb[i][j - 1] * b[i][pi - 1] % pi;
    }
}
};

```

```

int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cout << fixed << setprecision(15);
    int T, p; cin >> T >> p;
    binom s(p);
    while (T--)
    {
        ll n, m;
        cin >> n >> m;
        cout << s(n, m) << '\n';
    }
}

```

3.9 杜教筛

求 $\varphi(n)$ 的前缀和。

核心：构造 g 满足 $h(n) = \sum_{d|n} f(d)g(\frac{n}{d})$ 容易计算，

则有 $\sum_{i=1}^n h(i) = \sum_{i=1}^n g(i) \sum_{j=1}^{\lfloor n/i \rfloor} f(j)$,

故 $g(1) \sum_{j=1}^n f(j) = \sum_{i=1}^n h(i) - \sum_{i=2}^n g(i) \sum_{j=1}^{\lfloor n/i \rfloor} f(j)$,

则 f 前缀和可以递归求解。

```

namespace du_seive
{
    using ui = unsigned int;
    using ull = unsigned long long;
    unordered_map<ull, ui> mp;
    const int N = 1e7 + 2;
    const ui p = 998244353;
    ui pr[N], phi[N];
    ui cnt;
    void init()
    {
        cnt = 0; phi[1] = 1;
        int i, j;
        for (i = 2; i < N; i++)
        {
            if (!phi[i])
            {
                pr[cnt++] = i;
                phi[i] = i - 1;
            }
            for (j = 0; i * pr[j] < N; j++)
            {
                if (i % pr[j] == 0)
                {
                    phi[i * pr[j]] = phi[i] * pr[j];
                    break;
                }
                phi[i * pr[j]] = phi[i] * (pr[j] - 1);
            }
            if ((phi[i] += phi[i - 1]) >= p) phi[i] -= p;
        }
    }
}

```

```

    }
}
ui get_phi_sum(ull n)
{
    if (n < N) return phi[n];
    if (mp.count(n)) return mp[n];
    ui sum = 0;
    for (ull i = 2, j, k; i <= n; i = j + 1)
    {
        j = n / (k = n / i);
        sum = (sum + (ull)get_phi_sum(k) * (j - i + 1)) % p;
    }
    ui nn = n % p;
    sum = (nn * (nn + 1llu) / 2 + p - sum) % p;
    mp[n] = sum;
    return sum;
}
int _ = (init(), 0);
}
using du_seive::get_phi_sum;

```

3.10 $\mu^2(n)$ 前缀和

10^{18} , 0.46s。

$$\mu^2(n) = \sum_{d^2|n} \mu(d)$$

```

const int N = 5e7 + 5;
int pr[N / 8], cnt, mu[N];
bool ed[N];
void init()
{
    ui i, j, k;
    mu[1] = 1;
    for (i = 2; i < N; i++)
    {
        if (!ed[i]) pr[++cnt] = i, mu[i] = -1;
        for (j = 1; pr[j] * i < N; j++)
        {
            ed[pr[j] * i] = 1;
            if (i % pr[j] == 0) break;
            mu[pr[j] * i] = -mu[i];
        }
        mu[i] += mu[i - 1];
    }
}
ll sum_mu(ll n)
{
    if (n < N) return mu[n];
    ll r = 1, i, j, k;
    for (i = 2; i <= n; i = j + 1)
    {
        j = n / (k = n / i);
        r -= sum_mu(k) * (j - i + 1);
    }
    return r;
}

```

```

11 sum_mu2(11 n)
{
    11 r = 0, i, j, k, l, s = 0, t;
    for (i = 1; i * i <= n; i = j + 1)
    {
        k = n / (i * i);
        j = sqrtl(n / k);
        t = sum_mu(j);
        r += k * (t - s);
        s = t;
    }
    return r;
}
int main()
{
    11 n;
    init();
    cin >> n;
    cout << sum_mu2(n) << endl;
}

```

3.11 线性规划

用法：构造函数指明目标函数系数，add 函数增加限制。额外的限制是 $x_i \geq 0$ 。

```

using db = long double; // __float128
struct linear
{
    static const int N = 45; // n+m
    db r[N][N];
    int col[N], row[N];
    const db eps = 1e-10, inf = 1e9; // 1e-17
    int n, m;
    template<class T> linear(const vector<T> &a) // target: maximize \sum a(i-1)xi
    {
        memset(r, 0, sizeof r);
        memset(col, 0, sizeof col);
        memset(row, 0, sizeof row);
        n = a.size(); m = 0;
        for (int i = 1; i <= n; i++) r[0][i] = -a[i - 1];
    }
    template<class T> void add(const vector<T> &a, db b) // limit: \sum a(i-1)xi <= b
    {
        assert(a.size() == n);
        ++m;
        for (int i = 1; i <= n; i++) r[m][i] = -a[i - 1];
        r[m][0] = b;
    }
    void pivot(int k, int t)
    {
        swap(row[k + n], row[t]);
        db rkt = -r[k][t];
        int i, j;
        for (i = 0; i <= n; i++) r[k][i] /= rkt;
        r[k][t] = -1 / rkt;
        for (i = 0; i <= m; i++) if (i != k)
        {

```

```

        db rit = r[i][t];
        if (rit >= -eps && rit <= eps) continue;
        for (j = 0; j <= n; j++) if (j != t) r[i][j] += rit * r[k][j];
        r[i][t] = r[k][t] * rit;
    }
}

bool init()
{
    int i;
    for (i = 1; i <= n + m; i++) row[i] = i;
    while (1)
    {
        int q = 1;
        auto b_min = r[1][0];
        for (i = 2; i <= m; i++) if (r[i][0] < b_min) b_min = r[i][0], q = i;
        if (b_min + eps >= 0) return 1;
        int p = 0;
        for (i = 1; i <= n; i++) if (r[q][i] > eps && (!p || row[i] > row[p])) p = i;
        if (!p) break;
        pivot(q, p);
    }
    return 0;
}

bool simplex()
{
    while (1)
    {
        int t = 1, k = 0, i;
        for (i = 2; i <= n; i++) if (r[0][i] < r[0][t]) t = i;
        if (r[0][t] >= -eps) return 1;
        db ratio_min = inf;
        for (i = 1; i <= m; i++) if (r[i][t] < -eps)
        {
            db ratio = -r[i][0] / r[i][t];
            if (!k || ratio < ratio_min || ratio <= ratio_min + eps && row[i] > row[k])
            {
                ratio_min = ratio;
                k = i;
            }
        }
        if (!k) break;
        pivot(k, t);
    }
    return 0;
}

void solve(int type)
{
    if (!init())
    {
        cout << "Infeasible\n";
        return;
    }
    if (!simplex())
    {
        cout << "Unbounded\n";
        return;
    }
}

```



```

    cout << (long double)(-r[0][0]) << '\n';
    if (type)
    {
        int i;
        memset(col + 1, 0, n * sizeof col[0]);
        for (i = n + 1; i <= n + m; i++) col[row[i]] = i;
        for (i = 1; i <= n; i++) cout << (long double)(col[i] ? r[col[i] - n][0] : 0) << "\n";
    }
    }
};

```

3.12 线性插值 (k 次幂和)

$O(m)$, $O(m)$ 。

```

ull interpolation(vector<ull> a, ull n)
{
    int m = a.size(), i;
    vector<ull> ans(2);
    n %= p;
    if (n < m) return a[n];
    ull k = ifac[m - 1];
    for (i = m - 1; i >= 0; i--)
    {
        (a[i] *= k) %= p;
        (k *= n - i) %= p;
    }
    k = 1;
    for (i = 0; i < m; i++)
    {
        (ans[(m ^ i) & 1] += a[i] * k) %= p;
        k = k * inv[i + 1] % p * (n - i) % p * (m - i - 1) % p;
    }
    return (ans[1] + p - ans[0]) % p;
}

ull sum_of_kth_power(ull n, ull k)
{
    if (n == 0) return 0;
    ull m = min(n + 1, k + 2);
    int i;
    vector<ull> s(m);
    vector<int> pr, ed(m); pr.reserve(m / 4);
    s[1] = 1;
    for (i = 2; i < m; i++)
    {
        if (!ed[i]) s[i] = ksm(i, k), pr.push_back(i);
        for (int j : pr) if (i * j < m)
        {
            s[i * j] = s[i] * s[j] % p;
            if (i % j == 0) break;
        }
        else break;
    }
    for (i = 1; i < m; i++) (s[i] += s[i - 1]) %= p;
    return interpolation(s, n);
}

```

3.13 幂乘等比

$\sum_{i=0}^{n-1} x^i i^d$ 以及 $\sum_{i=0}^{\infty} x^i i^d$
 $O(d)$ 。

```
vector<ull> cal_ik(int n, int k)
{
    int i, j, x, c = 0;
    vector<ull> f(n + 1);
    f[0] = k == 0;
    if (n >= 1) f[1] = 1;
    if (n <= 1) return f;
    vector<int> pr((n / log(n)) * (1 + 1.2762 / log(n)) + 2);
    vector<char> ed(n + 1);
    for (i = 2; i <= n; i++)
    {
        if (!ed[i]) pr[c++] = i, f[i] = ksm(i, k);
        for (j = 0; (x = i * pr[j]) <= n; j++)
        {
            ed[x] = 1;
            f[x] = f[i] * f[pr[j]] % p;
            if (i % pr[j] == 0) break;
        }
    }
    return f;
}

ull sum_n(ull n, ull x, int d)
{
    x %= p;
    if (x == 0) return n > 0 && d == 0;
    vector<ull> id = cal_ik(d + 1, d), y(d + 2);
    ull xp = 1, ix = ksm(x, p - 2);
    int i;
    for (i = 0; i <= d; i++)
    {
        y[i + 1] = (y[i] + xp * id[i]) % p;
        (xp *= x) %= p;
    }
    if (n <= d + 1) return y[n];
    if (x == 1) return interpolation(y, n);
    array<ull, 2> s{0, 0};
    for (i = 0; i <= d + 1; i++)
    {
        (s[d + i & 1] += C(d + 1, i) * xp % p * y[i] % p) %= p;
        (xp *= ix) %= p;
    }
    ull q = (s[1] + p - s[0]) * ksm(ksm((1 + p - x) % p, d + 1), p - 2) % p, ixp = 1;
    y.pop_back();
    for (ull &e : y)
    {
        e = (q + p - e) * ixp % p;
        (ixp *= ix) %= p;
    }
}
```

```

    return (q + (p - ksm(x, n)) * interpolation(y, n)) % p;
}
ull sum_inf(ull x, int d)
{
    if (x == 0) return d == 0;
    assert(x != 1 && x < p);
    if (d == 0) return ksm((1 + p - x) % p, p - 2);
    ull f, xm1 = ksm(x - 1, p - 2), y = x * xm1 % p;
    f = ksm(y - 1, p - 2) * (ksm(y, d + 1) - 1) % p;
    vector<ull> g = cal_ik(d, d);
    xm1 = p - xm1;
    ull r = 0, k = ksm(y, p - 2), a = ksm(y, d) * (x - 1) % p;
    for (int i = 0; i <= d; i++)
    {
        r = (r + xm1 * f % p * g[i]) % p;
        (xm1 *= p - y) %= p;
        f = (f * (1 + p - x) + C(d + 1, d - i) * a) % p;
        (a *= k) %= p;
    }
    return r;
}

```

3.14 原根

你应该传入的是模数 m 和你提供的 `getw` 函数，用于分解质因数。

以下以我的 pollard rho 模板为例。如果在现场赛且范围不大，手动求出更为方便。

定理：只有 $2, 4, p^k, 2p^k$ 有原根。因此可以优化到只分解一次，但没必要。

```

namespace get_root
{
    using ull = unsigned long long;
    using u128 = __uint128_t;
    ull ksm(ull x, ull y, ull p)
    {
        ull r = 1;
        while (y)
        {
            if (y & 1) r = (u128)r * x % p;
            x = (u128)x * x % p; y >>= 1;
        }
        return r;
    }
    template<class T> ll getrt(ull m, T getw)
    {
        assert(m);
        if (m <= 4) return (int)m - 1;
        ull phi = m;
        auto w = getw(m);
        if (w.size() >= 3 || m % 4 == 0 || w.size() == 2 && w[0] != 2) return -1;
        for (ull x : w) phi = phi / x * (x - 1);
        w = getw(phi);
        for (ull &x : w) x = phi / x;
        for (ull i = 2; i < m; i++) if (gcd(i, m) == 1)
        {
            for (ull x : w) if (ksm(i, x, m) == 1) goto no;
            return i;
        }
    }
}

```

```

        no::;
    }
    return -1;
}
}
using get_root::getrt;
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cout << fixed << setprecision(15);
    ull p;
    cin >> p;
    cout << getrt(p, [&](ull m) {
        auto ww = pr::getw(m);
        vector<ull> w;
        for (auto [p, k] : ww) w.push_back(p);
        return w;
    }) << '\n';
}

```

3.15 筛全部原根

```

#include "bits/stdc++.h"
using namespace std;
using ll = long long;
const int N = 1e6 + 2;
int ss[N], mn[N], fmn[N], phi[N];
int t, n, gs, i, d;
bool ed[N], av[N], yg[N], hv[N];
double inv[N];
void getfac(int x, int *a, int &n)
{
    int y = x, z;
    if (1 ^ x & 1)
    {
        a[n = 1] = 2; x >>= 1; while (1 ^ x & 1) x >>= 1;
    }
    while (x > 1)
    {
        x = 1e-9 + (x * inv[a[++n] = z = mn[x]]);
        while (x % z == 0) x = 1e-9 + x * inv[z];
    }
    for (i = 1; i <= n; i++) av[a[i]] = 0, a[i] = 1e-9 + (y * inv[a[i]]);
}
int ksm(int x, int y, int p)
{
    int r = 1;
    while (y)
    {
        if (y & 1) r = (ll)r * x % p;
        x = (ll)x * x % p; y >>= 1;
    }
    return r;
}
bool ck(int x, int *a, int n, int p)
{

```

```

    for (int i = 1; i <= n; i++) if (ksm(x, a[i], p) == 1) return 0;
    return 1;
}
void getrt(int x, int d)
{
    if (!hv[x]) return puts("0\n"), void();
    static int a[30];
    int n = 0, y, i, g = 0, c = d; y = phi[x];
    fill(av + 1, av + y + 1, 1);
    getfac(y, a, n);
    for (i = 1; i < x; i++) if (__gcd(i, x) == 1 && ck(i, a, n, x)) break;
    yg[g = i] = 1; //g就是最小原根
    int j = (ll)g * g % x;
    for (i = 2; i < y; i++, j = (ll)j * g % x) yg[j] = av[i] = av[mn[i]] & av[fmn[i]];
    printf("%d\n", phi[y]);
    for (i = 1; i < x; i++) if (yg[i])
    {
        yg[i] = 0;
        if (--c == 0) printf("%d┐", i), c = d;
    }puts("");
}
void init()
{
    int i, j, k, n = N - 1;
    mn[1] = phi[1] = 1;
    for (i = 1; i <= n; i++) inv[i] = 1.0 / i;
    for (i = 2; i <= n; i++)
    {
        if (!ed[i]) phi[mn[i] = ss[++gs] = i] = i - 1, hv[i] = 1;
        for (j = 1; j <= gs && (k = ss[j] * i) <= n; j++)
        {
            ed[k] = 1; mn[k] = ss[j];
            if (i % ss[j] == 0) { phi[k] = phi[i] * ss[j]; hv[k] = hv[i]; break; }
            phi[k] = phi[i] * (ss[j] - 1);
        }
    }
    for (i = n; i; i--) fmn[i] = 1e-9 + (i * inv[mn[i]]), hv[i] |= (1 ^ i & 1) && hv[i >> 1];
    for (i = 8; i <= n; i <= 1) hv[i] = 0;
}
int main()
{
    init();
    scanf("%d", &t);
    while (t--)
    {
        scanf("%d%d", &n, &d);
        getrt(n, d);
    }
}

```

3.16 高斯消元（浮点数）

$O(n^3)$, $O(n^2)$ 。

```

namespace Gauss
{
    typedef double db;

```

```

const db eps = 1e-8;
template<class T> pair<vector<db>, int> solve(const vector<vector<T>> &A)//和为 0。返回秩，负
数无解
{
    assert(A.size());
    int n = A.size(), m = A[0].size() - 1, i, j, k, l, r, fg = 1;
    db a[n][m + 1], b;
    vector<int> w;
    for (i = 0; i < n; i++) for (j = 0; j <= m; j++) a[i][j] = A[i][j];
    for (i = l = r = 0; i < n && l < m; i++, l++)
    {
        k = i;
        for (j = i + 1; j < n; j++) if (fabs(a[j][l]) > fabs(a[k][l])) k = j;
        if (fabs(a[k][l]) < eps) { --i; continue; }
        if (i != k) for (j = l; j <= m; j++) swap(a[i][j], a[k][j]);
        w.push_back(l);
        b = 1 / a[i][l]; ++r; a[i][l] = 1;
        for (j = l + 1; j <= m; j++) a[i][j] *= b;
        for (j = 0; j < n; j++) if (i != j)
        {
            b = a[j][l]; a[j][l] = 0;
            for (k = l + 1; k <= m; k++) a[j][k] -= b * a[i][k];
        }
    }
    vector<db> X(m);
    for (j = 0; j < w.size(); j++) X[w[j]] = -a[j][m];
    for (j = 0; j < n && ~fg; j++)
    {
        b = a[j][m];
        for (k = 0; k < m; k++) b += X[k] * a[j][k];
        if (fabs(b) > eps) fg = -1;
    }
    return {X, r * fg};
}
//?

```

3.17 行列式求值（任意模数）

$O(n^3)$, $O(n^2)$ 。

原理：辗转相除。注意这个 $\log p$ 并不在 n^3 上。

```

#include "bits/stdc++.h"
using namespace std;
using ll = long long;
const int N = 502, p = 998244353;
int cal(int a[][N], int n)
{
    int i, j, k, r = 1, fh = 0, l;
    for (i = 1; i <= n; i++)
    {
        k = i;
        for (j = i + 1; j <= n; j++) if (a[j][i]) { k = j; break; }
        if (a[k][i] == 0) return 0;
        if (i != k) { swap(a[k], a[i]); fh ^= 1; }
        for (j = i + 1; j <= n; j++)

```

```

    {
        if (a[j][i] > a[i][i]) swap(a[j], a[i]), fh ^= 1;
        while (a[j][i])
        {
            l = a[i][i] / a[j][i];
            for (k = i; k <= n; k++) a[i][k] = (a[i][k] + (ll)(p - l) * a[j][k]) % p;
            swap(a[j], a[i]); fh ^= 1;
        }
    }
    r = (ll)r * a[i][i] % p;
}
if (fh) return (p - r) % p;
return r;
}
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    int n, i, j;
    static int a[N][N];
    cin >> n;
    for (i = 1; i <= n; i++) for (j = 1; j <= n; j++) cin >> a[i][j];
    cout << cal(a, n) << endl;
}

```

3.18 BM/稀疏矩阵系列

BM: 给定 $\{a\}$, 求最短的 $\{r\}$ 满足 $\sum_{j=0}^{m-1} a_{i-j-1}r_j = a_i$.

safe 宏用于验证结果正确性, 可不定义。实现了稀疏矩阵的行列式和求解方程组。

```

vector<ui> bm(const vector<ui> &a)
{
    vector<ui> r, lst;
    int n = a.size(), m = 0, q = 0, i, j, k = -1;
    ui D = 0;
    for (i = 0; i < n; i++)
    {
        ui cur = 0;
        for (j = 0; j < m; j++) cur = (cur + (ull)a[i - j - 1] * r[j]) % p;
        cur = (a[i] + p - cur) % p;
        if (!cur) continue;
        if (k == -1)
        {
            k = i;
            D = cur;
            r.resize(m = i + 1);
            continue;
        }
        auto v = r;
        ui x = (ull)cur * ksm(D, p - 2) % p;
        if (m < q + i - k) r.resize(m = q + i - k);
        (r[i - k - 1] += x) %= p;
        ui *b = r.data() + i - k;
        x = (p - x) % p;
        for (j = 0; j < q; j++) b[j] = (b[j] + (ull)x * lst[j]) % p;
        if (v.size() + k < lst.size() + i)

```

```

        {
            lst = v;
            q = v.size();
            k = i;
            D = cur;
        }
    }
    return r;
}

#define safe
struct Q
{
    int x, y;
    ui w;
};

mt19937_64 rnd(9980);
vector<ui> minpoly(int n, const vector<Q> &a)//[0,n),max:1
{
    for (auto [x, y, w] : a) assert(min(x, y) >= 0 && max(x, y) < n);
    vector<ui> u(n), v(n), b(n * 2 + 1), tmp(n);
    int i;
    for (ui &x : u) x = rnd() % p;
    for (ui &x : v) x = rnd() % p;
    assert(*min_element(all(u)) && *min_element(all(v)));
    for (ui &r : b)
    {
        for (i = 0; i < n; i++) r = (r + (ull)u[i] * v[i]) % p;
        fill(all(tmp), 0);
        for (auto [x, y, w] : a) tmp[x] = (tmp[x] + (ull)w * v[y]) % p;
        swap(v, tmp);
    }
    auto r = bm(b);
#ifdef safe
    for (ui &x : u) x = rnd() % p;
    for (ui &x : v) x = rnd() % p;
    for (ui &r : b)
    {
        for (i = 0; i < n; i++) r = (r + (ull)u[i] * v[i]) % p;
        fill(all(tmp), 0);
        for (auto [x, y, w] : a) tmp[x] = (tmp[x] + (ull)w * v[y]) % p;
        swap(v, tmp);
    }
    auto rr = bm(b);
    assert(r == rr);
#endif
    reverse(all(r));
    for (ui &x : r) if (x) x = p - x;
    r.push_back(1);
    return r;
}

ui det(int n, vector<Q> a)//[0,m)
{
    vector<ui> b(n);
    for (ui &x : b) x = rnd() % p;
    assert(*min_element(all(b)));
    for (auto &[x, y, w] : a) w = (ull)w * b[x] % p;
    ui r = minpoly(n, a)[0], tmp = 1;

```



```

    for (ui x : b) tmp = (ull)tmp * x % p;
    r = (ull)r * ksm(tmp, p - 2) % p;
#ifdef safe
    for (ui &x : b) x = rnd() % p;
    assert(*min_element(all(b)));
    for (auto &[x, y, w] : a) w = (ull)w * b[x] % p;
    ui rr = minpoly(n, a)[0], tmpp = 1;
    for (ui x : b) tmpp = (ull)tmpp * x % p;
    rr = (ull)rr * ksm(tmpp, p - 2) % p * ksm(tmp, p - 2) % p;
    assert(r == rr);
#endif
    return n & 1 ? (p - r) % p : r;
}
vector<ui> gauss(const vector<Q> &a, vector<ui> v)
{
    int n = v.size(), i, j;
    for (auto [x, y, w] : a) assert(0 <= x && x < n && 0 <= y && y < n);
    vector<ui> u(n), b(2 * n + 1), tmp(n), tv = v;
    for (ui &x : u) x = rnd() % p;
    assert(*min_element(all(u)));
    for (ui &r : b)
    {
        for (i = 0; i < n; i++) r = (r + (ull)u[i] * v[i]) % p;
        fill(all(tmp), 0);
        for (auto [x, y, w] : a) tmp[x] = (tmp[x] + (ull)w * v[y]) % p;
        swap(v, tmp);
    }
    auto f = bm(b);
    f.insert(f.begin(), p - 1);
    int m = (int)f.size() - 2;
    v = tv; fill(all(u), 0);
    ui x;
    for (i = 0; i <= m; i++)
    {
        x = f[m - i];
        for (j = 0; j < n; j++) u[j] = (u[j] + (ull)v[j] * x) % p;
        fill(all(tmp), 0);
        for (auto [x, y, w] : a) tmp[x] = (tmp[x] + (ull)w * v[y]) % p;
        swap(v, tmp);
    }
    x = ksm((p - f.back()) % p, p - 2);
    for (ui &y : u) y = (ull)y * x % p;
#ifdef safe
    for (auto [x, y, w] : a) tv[x] = (tv[x] + (ull)(p - w) * u[y]) % p;
    assert(!*min_element(all(tv)));
#endif
    return u;
}

```

3.19 Min_25 筛

$f(p^k) = p^k(p^k - 1)$, 求 $\sum_{i=1}^n f(i)$ 。这个的原理我了解的不多, 因此没有更多注释。

```

const int N = 1e5 + 2, p = 1e9 + 7, i6 = 166666668;
ll fs[N << 1], m;
int ss[N], ys[N << 1], s[N], f[N << 1], g[N << 1], ls[N << 1], cs[N << 1];

```

```

int gs, n, i, j, k, cnt, ct, ans, sq;
bool ed[N];
int S(ll n, int x)
{
    int r, i, j, l;
    ll k;
    if (ss[x] >= n) return 0;
    if (n > sq) r = g[ys[m / n]]; else r = g[n];
    if ((r = r - s[x]) < 0) r += p;
    for (i = x + 1; (ll)ss[i] * ss[i] <= n; i++) for (j = 1, k = ss[i]; k <= n; j++, k *= ss[i])
    {
        l = (k - 1) % p;
        r = (r + (ll)l * (l + 1) % p * ((j != 1) + S(n / k, i))) % p;
    }
    return r;
}
int main()
{
    n = 1e5;
    for (i = 2; i <= n; i++)
    {
        if (!ed[i]) ss[++gs] = i;
        for (j = 1; (j <= gs) && (i * ss[j] <= n); j++)
        {
            ed[i * ss[j]] = 1;
            if (i % ss[j] == 0) break;
        }
    }
    ss[gs + 1] = 1e6;
    s[1] = ss[1] * ss[1];
    for (i = 2; i <= gs; i++) s[i] = (s[i - 1] + (ll)ss[i] * ss[i]) % p; //s 是多项式在素数位置的前缀和
    memcpy(cs, s, sizeof(s));
    ll i, j, k, x, z; scanf("%lld", &m);
    sq = n = sqrt(m); while ((ll)(n + 1) * (n + 1) <= m) ++n;
    cnt = n - 1;
    for (i = n; i <= m; i = j + 1) { j = m / (m / i); ++cnt; } ct = cnt++;
    for (i = 1; i <= m; i = j + 1)
    {
        j = m / (k = m / i);
        if (k <= n) g[fs[k] = k] = (k * (k + 1) * (k << 1 | 1) / 6 - 1) % p; //这里是多项式前缀和 (不含1)
        else
        {
            z = k % p; //一样
            g[ys[j] = --cnt] = (z * (z + 1) % p * (z << 1 | 1) % p + p - 6) * i6 % p; fs[cnt] = k;
        }
    }
    cnt = ct;
    for (j = 1; (j <= gs) && (z = (ll)ss[j] * ss[j]); j++) for (i = cnt; z <= fs[i]; i--)
    {
        x = fs[i] / ss[j]; if (x > n) x = ys[m / x];
        g[i] = (g[i] + (ll)(p - ss[j]) * ss[j] % p * (g[x] - s[j - 1] + p)) % p; //另一处需要修改的
    }
    memcpy(ls, g, sizeof(g));
    s[1] = ss[1];
    for (i = 2; i <= gs; i++) s[i] = s[i - 1] + ss[i];
}

```

```

cnt = n - 1;
for (i = n; i <= m; i = j + 1) { j = m / (m / i); ++cnt; } ct = cnt++;
for (i = 1; i <= m; i = j + 1)
{
    j = m / (k = m / i);
    if (k <= n) g[fs[k] = k] = ((k * (k + 1) >> 1) - 1) % p;
    else
    {
        z = k % p;
        g[ys[j] = --cnt] = (z * (z + 1) - 2 >> 1) % p; fs[cnt] = k;
    }
}
cnt = ct;
for (j = 1; (j <= gs) && (z = (ll)ss[j] * ss[j]); j++) for (i = cnt; z <= fs[i]; i--)
{
    x = fs[i] / ss[j]; if (x > n) x = ys[m / x];
    g[i] = (g[i] + (ll)(p - ss[j]) * (g[x] - s[j - 1] + p)) % p;
}
for (i = 1; i <= cnt; i++) if ((g[i] = ls[i] - g[i]) < 0) g[i] += p;
for (i = 1; i <= gs; i++) if ((s[i] = cs[i] - s[i]) < 0) s[i] += p;
ans = S(m, 0) + 1; if (ans == p) ans = 0; printf("%d", ans);
}

```

这是一个常数较小的版本，实现的是质数个数。需要注意评测机 double 性能。

```

#include "bits/stdc++.h"
using namespace std;
using ll = long long;
const int N = 3.2e5 + 2;
ll s[N];
int ss[N], ys[N], gs = 0;
bool ed[N];
ll cal(ll m)
{
    static ll g[N << 1], fs[N << 1];
    ll i, j, k, x;
    int n;
    int p, q, cnt;
    n = round(sqrt(m));
    q = lower_bound(ss + 1, ss + gs + 1, n) - ss;
    memset(g, 0, sizeof(g)); memset(ys, 0, sizeof(ys)); cnt = n - 1;
    for (i = n; i <= m; i = j + 1) { j = m / (m / i); ++cnt; } int ct = cnt++;
    for (i = 1; i <= m; i = j + 1)
    {
        j = m / (k = m / i);
        if (k <= n) g[fs[k] = k] = k - 1; else { g[ys[j] = --cnt] = k - 1; fs[cnt] = k; }
    } cnt = ct;
    for (j = 1; j <= q; j++) for (i = cnt; (ll)ss[j] * ss[j] <= fs[i]; i--)
    {
        x = fs[i] / ss[j]; if (x > n) x = ys[m / x];
        g[i] -= g[x] - j + 1;
    }
    return g[cnt]; // 这里 g[cnt-i+1] 表示的是 [1, m/i] 的答案
}
int main()
{
    int n, i, j, t;
    n = 3.2e5;
}

```

```

for (i = 2; i <= n; i++)
{
    if (!ed[i]) ss[++gs] = i;
    for (j = 1; (j <= gs) && (i * ss[j] <= n); j++)
    {
        ed[i * ss[j]] = 1;
        if (i % ss[j] == 0) break;
    }
}
s[1] = ss[1];
for (i = 2; i <= gs; i++) s[i] = s[i - 1] + ss[i];
t = 1;
ll m;
while (t--) cin >> m, cout << cal(m) << '\n';
}

```

3.20 光速乘

$O(n2^n)$, $O(2^n)$ 。

```

ll mul(ll x, ll y)
{
    x = x * y - (ll)((ldb)x / p * y + 1e-8) * p;
    if (x < 0) return x + p; return x;
}

```

3.21 二次剩余

当前实现是 Cipolla, 要求模数为素数 ($P = 2$ 已特判)。

```

namespace cipolla
{
    typedef unsigned int ui;
    typedef unsigned long long ull;
    ui p, w;
    struct Q
    {
        ull x, y;
        Q operator*(const Q &o) const { return {(x * o.x + y * o.y % p * w) % p, (x * o.y + y * o.x) % p}; }
    };
    ui ksm(ull x, ui y)
    {
        ull r = 1;
        while (y)
        {
            if (y & 1) r = r * x % p;
            x = x * x % p; y >>= 1;
        }
        return r;
    }
    Q ksm(Q x, ui y)
    {
        Q r = {1, 0};
        while (y)
        {

```

```

        if (y & 1) r = r * x;
        x = x * x; y >>= 1;
    }
    return r;
}
ui mosqrt(ui x, ui P) //P 为素数, 0<=x<P
{
    if (x == 0 || P == 2) return x;
    p = P;
    if (ksm(x, p - 1 >> 1) != 1) return -1;
    ui y;
    mt19937 rnd(chrono::steady_clock::now().time_since_epoch().count());
    do y = rnd() % p, w = ((ull)y * y + p - x) % p; while (ksm(w, p - 1 >> 1) <= 1); //not for
    p=2
    y = ksm({y, 1}, p + 1 >> 1).x;
    if (y * 2 > p) y = p - y; //两解取小
    return y;
}
}
using cipolla::mosqrt;

```

3.22 k 次剩余

两个版本都按素数模数的乘法群使用；合数模数需要分解后 CRT。

```

namespace get_root
{
    bool ied = 0;
    const int N = 1e5 + 5;
    vector<ui> pr;
    bool ed[N];
    void init()
    {
        pr.reserve(N);
        for (ui i = 2; i < N; i++)
        {
            if (!ed[i]) pr.push_back(i);
            for (ui x : pr)
            {
                if (i * x >= N) break;
                ed[i * x] = 1;
                if (i % x == 0) break;
            }
        }
    }
    ui ksm(ui x, ui y, ui p)
    {
        ui r = 1;
        while (y)
        {
            if (y & 1) r = (ull)r * x % p;
            x = (ull)x * x % p; y >>= 1;
        }
        return r;
    }
    vector<ui> getw(ui n)

```

```

{
    vector<ui> w;
    for (ui x : pr)
    {
        if (x * x > n) break;
        if (n % x == 0)
        {
            w.push_back(x);
            n /= x;
            for (ui i = n / x; n == x * i; i = n / x) n /= x;
        }
    }
    if (n > 1) w.push_back(n);
    return w;
}

int getrt(ui n)
{
    if (n <= 2) return n - 1;
    if (!ed[4]) init();
    auto w = getw(n);
    ui ph = n;
    for (ui x : w) ph = ph / x * (x - 1);
    w = getw(ph);
    for (ui &x : w) x = ph / x;
    for (ui i = 2; i < n; i++) if (gcd(i, n) == 1)
    {
        for (ui x : w) if (ksm(i, x, n) == 1) goto no;
        return i;
    }
    no::;
}

return -1;
}

namespace BSGS
{
    using ui = unsigned;
    using ull = unsigned long long;
    template<int N, class T, class TT> struct ht//个数, 定义域, 值域
    {
        const static int p = 1e6 + 7, M = p + 2;
        TT a[N];
        T v[N];
        int fir[p + 2], nxt[N], st[p + 2];//和模数相适应
        int tp, ds;//自定义模数
        ht() { memset(fir, 0, sizeof fir); tp = ds = 0; }
        void mdf(T x, TT z)//位置, 值
        {
            ui y = x % p;
            for (int i = fir[y]; i; i = nxt[i]) if (v[i] == x) return a[i] = z, void();//若不可能
            //重复不需要 for
            v[++ds] = x; a[ds] = z;
            if (!fir[y]) st[++tp] = y;
            nxt[ds] = fir[y]; fir[y] = ds;
        }
        TT find(T x)
        {
            ui y = x % p;

```

```

    int i;
    for (i = fir[y]; i; i = nxt[i]) if (v[i] == x) return a[i];
    return 0; //返回值和是否判断依据要求决定
}
void clear()
{
    ++tp;
    while (--tp) fir[st[tp]] = 0;
    ds = 0;
}
};
const int N = 5e4;
ht<N, ui, ui> s;
int exgcd(int a, int b)
{
    if (a == 1) return 1;
    return (1 - (ll)b * exgcd(b % a, a)) / a; //not ull
}
int bsgs(ui a, ui b, ui p)
{
    s.clear();
    a %= p; b %= p;
    if (!a) return 1 - min((int)b, 2); //含 -1
    ui i, j, k, x, y;
    x = sqrt(p) + 2;
    for (i = 0, j = 1; i < x; i++, j = (ull)j * a % p)
    {
        if (j == b) return i;
        s.mdf((ull)j * b % p, i + 1);
    }
    k = j;
    for (i = 1; i <= x; i++, j = (ull)j * k % p) if (y = s.find(j)) return (ull)i * x - y +
1;
    return -1;
}
bool isprime(ui p)
{
    if (p <= 1) return 0;
    for (ui i = 2; i * i <= p; i++) if (p % i == 0) return 0;
    return 1;
}
int exbsgs(ui a, ui b, ui p) //a^x=b(mod p)
{
    //if (isprime(p)) return bsgs(a,b,p);
    a %= p; b %= p;
    ui i, j, k, x, y = __lg(p), cnt = 0;
    for (i = 0, j = 1 % p; i <= y; i++, j = (ull)j * a % p) if (j == b) return i;
    y = 1;
    while (1)
    {
        if ((x = gcd(a, p)) == 1) break;
        if (b % x) return -1; //no sol
        ++cnt;
        p /= x; b /= x;
        y = (ull)y * (a / x) % p;
    }
    a %= p;

```

```

        b = (ull)b * (p + exgcd(y, p)) % p;
        int r = bsgs(a, b, p);
        return r == -1 ? -1 : r + cnt;
    }
}
pair<ll, ll> exgcd(ll a, ll b, ll c) // ax+by=c, {-1,-1} 无解, b=0 返回 {c/a,0}, 否则返回最小非负 x
{
    assert(a || b);
    if (!b) return {c / a, 0};
    if (a < 0) a = -a, b = -b, c = -c;
    ll d = gcd(a, b);
    if (c % d) return {-1, -1};
    ll x = 1, x1 = 0, p = a, q = b, k;
    b = abs(b);
    while (b)
    {
        k = a / b;
        x -= k * x1; a -= k * b;
        swap(x, x1);
        swap(a, b);
    }
    b = abs(q / d);
    x = x * (c / d) % b;
    if (x < 0) x += b;
    return {x, (c - p * x) / q};
}
ll fun(ll a, ll b, ll p) // ax=b(mod p)
{
    return exgcd(-p, a, b).second % p;
}
using get_root::getrt;
using BSGS::bsgs, BSGS::exbsgs;
int nth_root(ui k, ui y, ui p) // x^k=y(mod p)
{
    if (k == 0) return y == 1 ? 0 : -1;
    if (y == 0) return 0;
    int g = getrt(p);
    if (g == -1) return -1;
    ll z = bsgs(g, y, p);
    if (z == -1) return -1;
    ll x = fun(k, z, p - 1);
    if (x == -1) return -1;
    return get_root::ksm(g, x, p);
}

```

网上的超快版本

```

#define popcount __builtin_popcount
using namespace std;
typedef long long ll;
//using ll=__int128_t;
typedef pair<ll, int> P;
ll gcd(ll a, ll b) {
    if (b == 0) return a;
    return gcd(b, a % b);
}
ll powmod(ll a, ll k, ll mod) {
    ll ap = a, ans = 1;

```



```

while (k) {
    if (k & 1) {
        ans = (__int128)ans * ap % mod;
    }
    ap = (__int128)ap * ap % mod;
    k >>= 1;
}
return ans;
}
ll inv(ll a, ll m) {
    ll b = m, x = 1, y = 0;
    while (b > 0) {
        ll t = a / b;
        swap(a -= t * b, b);
        swap(x -= t * y, y);
    }
    return (x % m + m) % m;
}
vector<P> fac(ll x) {
    vector<P> ret;
    for (ll i = 2; i * i <= x; i++) {
        if (x % i == 0) {
            int e = 0;
            while (x % i == 0) {
                x /= i;
                e++;
            }
            ret.push_back({i, e});
        }
    }
    if (x > 1) ret.push_back({x, 1});
    return ret;
}
//mt19937_64 mt(334);
mt19937 mt(334);
ll solve1(ll p, ll q, int e, ll a) {
    int s = 0;
    ll r = p - 1, qs = 1, qp = 1;
    while (r % q == 0) {
        r /= q;
        qs *= q;
        s++;
    }
    for (int i = 0; i < e; i++) qp *= q;
    ll d = qp - inv(r % qp, qp);
    ll t = ((__int128)d * r + 1) / qp;
    ll at = powmod(a, t, p), inva = inv(a, p);
    if (e >= s) {
        if (powmod(at, qp, p) != a) return -1;
        else return at;
    }
    //uniform_int_distribution<long long> rnd(1, p-1);
    uniform_int_distribution<long long> rnd(1, p - 1);
    ll rv;
    while (1) {
        rv = powmod(rnd(mt), r, p);
        if (powmod(rv, qs / q, p) != 1) break;
    }
}

```

```

}
int i = 0;
ll qi = 1, sq = 1;
while (sq * sq < q) sq++;
auto ml = [&](ll x, ll y) { return (ll)((__int128)x * y % (p - 1)); };
while (i < s - e) {
    ll qq = qs / qp / qi / q;
    vector<P> v(sq);
    ll rvi = powmod(rv, ml(ml(qp, qq), p - 2), p), rvp = powmod(rv, ml(ml(sq, qp), qq), p);
    ll x = powmod((__int128)powmod(at, qp, p) * inva % p, ml(qq, p - 2), p), y = 1;
    for (int j = 0; j < sq; j++) {
        v[j] = P(x, j);
        x = (__int128)x * rvi % p;
    }
    sort(v.begin(), v.end());
    ll z = -1;
    for (int j = 0; j < sq; j++) {
        int l = lower_bound(v.begin(), v.end(), P(y, 0)) - v.begin();
        if (l < v.size() && v[l].first == y) {
            z = v[l].second + j * sq;
            break;
        }
    }
    y = (__int128)y * rvp % p;
}
if (z == -1) return -1;
at = (__int128)at * powmod(rv, z, p) % p;
i++;
qi *= q;
rv = powmod(rv, q, p);
}
return at;
}
ll solve0(ll p, ll q, ll r, ll a) {
    ll d = q - inv(r % q, q);
    ll t = ((__int128)d * r + 1) / q;
    ll at = powmod(a, t, p), inva = inv(a, p);
    if (powmod(at, q, p) != a) return -1;
    else return at;
}
ll solve(ll p, ll k, ll a)//p k y
{
    if (k == 0)
    {
        if (a == 1) return 1;
        return -1;
    }
    if (a == 0) return 0;
    if (p == 2 || a == 1) return 1;
    ll a1 = a;
    ll g = gcd(p - 1, k);
    ll c = inv(k / g % ((p - 1) / g), (p - 1) / g);
    a = powmod(a, c, p);
    if (g == 1) {
        if (powmod(a, k, p) == a1) return a;
        else return -1;
    }
    ll g1 = gcd(g, (p - 1) / g), g2 = g;

```

```

vector<P> f1 = fac(g1), f;
for (auto r : f1) {
    ll q = r.first;
    int e = 0;
    while (g2 % q == 0) {
        g2 /= q;
        e++;
    }
    f.push_back({q, e});
}
ll ret = 1, gp = 1;
if (g2 > 1) {
    ll x = solve0(p, g2, (p - 1) / g2, a);
    if (x == -1) return -1;
    ret = x, gp *= g2;
}
for (auto r : f) {
    ll qp = 1;
    for (int i = 0; i < r.second; i++) qp *= r.first;
    ll x = solve1(p, r.first, r.second, a);
    if (x == -1) return -1;
    if (gp == 1) {
        ret = x, gp *= qp;
        continue;
    }
    ll s = inv(gp % qp, qp), t = (1 - (__int128)gp * s) / qp;
    if (t >= 0) ret = powmod(ret, t, p);
    else ret = powmod(ret, p - 1 + t % (p - 1), p);
    if (s >= 0) x = powmod(x, s, p);
    else x = powmod(x, p - 1 + s % (p - 1), p);
    ret = (__int128)ret * x % p;
    gp *= qp;
}
if (powmod(ret, k, p) != a1) return -1;
return ret;
}

```

3.23 FWT/子集卷积

$O(n2^n)$, $O(2^n)$ 。注意全都是无符号的。

```

void fwt_and(vector<ull> &A) //本质: 母集和
{
    ull n = A.size(), *a = A.data(), i, j, k, l, *f, *g;
    for (i = 1; i < n; i = 1)
    {
        l = i * 2;
        for (j = 0; j < n; j += l)
        {
            f = a + j; g = a + j + i;
            for (k = 0; k < i; k++) f[k] += g[k];
        }
        if (l == n || i == 1 << 10) for (ull &x : A) x %= p;
    }
}

void ifwt_and(vector<ull> &A)
{

```

```

ull n = A.size(), *a = A.data(), i, j, k, l, *f, *g;
for (i = 1; i < n; i = l)
{
    l = i * 2;
    for (j = 0; j < n; j += l)
    {
        f = a + j; g = a + j + i;
        for (k = 0; k < i; k++) f[k] += p * i - g[k];
    }
    if (l == n || i == 1 << 10) for (ull &x : A) x %= p;
}
}

void fwt_or(vector<ull> &A)//本质: 子集和
{
    ull n = A.size(), *a = A.data(), i, j, k, l, *f, *g;
    for (i = 1; i < n; i = l)
    {
        l = i * 2;
        for (j = 0; j < n; j += l)
        {
            f = a + j; g = a + j + i;
            for (k = 0; k < i; k++) g[k] += f[k];
        }
        if (l == n || i == 1 << 10) for (ull &x : A) x %= p;
    }
}

void ifwt_or(vector<ull> &A)
{
    ull n = A.size(), *a = A.data(), i, j, k, l, *f, *g;
    for (i = 1; i < n; i = l)
    {
        l = i * 2;
        for (j = 0; j < n; j += l)
        {
            f = a + j; g = a + j + i;
            for (k = 0; k < i; k++) g[k] += p * i - f[k];
        }
        if (l == n || i == 1 << 10) for (ull &x : A) x %= p;
    }
}

void fwt_xor(vector<ull> &A)
{
    ull n = A.size(), *a = A.data(), i, j, k, l, *f, *g;
    for (i = 1; i < n; i = l)
    {
        l = i * 2;
        for (j = 0; j < n; j += l)
        {
            f = a + j; g = a + j + i;
            for (k = 0; k < i; k++)
            {
                if ((f[k] += g[k]) >= p) f[k] -= p;
                g[k] = (f[k] + 2 * (p - g[k])) % p;
            }
        }
    }
}
}

```

```

void ifwt_xor(vector<ull> &A)
{
    ull n = A.size(), *a = A.data(), i, j, k, l, *f, *g, x = p + 1 >> 1, y = 1;
    for (i = 1; i < n; i = l)
    {
        l = i * 2;
        for (j = 0; j < n; j += l)
        {
            f = a + j; g = a + j + i;
            for (k = 0; k < i; k++)
            {
                if ((f[k] += g[k]) >= p) f[k] -= p;
                g[k] = (f[k] + 2 * (p - g[k])) % p;
            }
        }
        y = y * x % p;
    }
    for (i = 0; i < n; i++) a[i] = a[i] * y % p;
}

vector<ull> fst(const vector<ull> &s, const vector<ull> &t)
{
    int n = s.size(), m = __builtin_ctz(n), i, j, k;
    vector<ull> a[m + 1], b[m + 1], c[m + 1], r(n);
    for (i = 0; i <= m; i++) a[i].resize(n), b[i].resize(n), c[i].resize(n);
    for (i = 0; i < n; i++)
    {
        k = __builtin_popcount(i);
        a[k][i] = s[i];
        b[k][i] = t[i];
    }
    for (i = 0; i < m; i++) fwt_or(a[i]), fwt_or(b[i]); //如果魔改, 上限需改为 m
    for (i = 0; i <= m; i++) for (j = 0; j <= i; j++) for (k = 0; k < n; k++) c[i][k] = (c[i][k]
+ (ull)a[j][k] * b[i - j][k]) % p;
    for (i = 1; i <= m; i++) ifwt_or(c[i]); //如果魔改, 下限需改为 0
    for (i = 0; i < n; i++) r[i] = c[__builtin_popcount(i)][i];
    return r;
}

```

3.24 NTT

一种较快的 NTT（尤其是对于卷积以外的用途），但不推荐在不熟悉的情况下直接使用。一般的卷积可以参照字符串部分通配符的字符串匹配，其余的用途可以参照其他板子。

如果确实需要卡常，建议先抄写需要的函数，并递归地找到需要补的内容。

注意事项：所有 ull 为无符号。始终保证数组大小为 2^n ，不应当使用 `resize` 而应该使用取模来调整长度。三种卷积对应的运算符见注释。

需要特别小心其长度的变化，注意不要越界。如果修改模数，`dft` 和 `hf_dft` 处有一个参数也要修改。

常见函数如下（带 new 的基本上都是较快但较长的）：

卷积 `operator*`，循环卷积 `operator&`，差卷积 `operator^`，求逆 `operator~`（包含一个较短版，被注释了），分治 `cdq`，对数 `ln`，指数 `exp`, `exp_cdq`, `exp_new`，开方 `sqrt`, `sqrt_new`，幂函数 `pow(Q, ull)`, `pow(Q, string)`, `pow2(Q, ull)`, `pow(Q, ull, Q)`，整除与取模 `div`, `mod`, `div_mod`，线性递推 `recurrent`, `recurrent_new`, `recurrent_interval`，连乘 `prod`, `prod_new`，多点求值 `evaluation`, `evaluation_new`，阶乘 `factorial`，快速插值 `interpolation`，复合（逆）

comp, comp_inv, 多项式平移 shift, 区间点值平移 shift, Z 变换 czt, 贝尔数 ($[n]$ 划分等价类方案数) Bell, 斯特林数 S1_row, S1_column, S2_row, S2_column, signed_S1_row, 伯努利数 Bernoulli, 划分数 Partition, 最大公因式 gcd, 求根 root, 模多项式意义的逆 inverse, 转牛顿级数 poly_to_newton, poly_to_facpow, newton_to_poly, facpow_to_poly, 下降幂乘法 mul_facpow。

```
#include <optional>
namespace NTT
{
    using ull = unsigned long long;
    const ull g = 3, p = 998244353;
    const int N = 1 << 21; // 务必修改
    ull inv[N + 1], fac[N + 1], ifac[N + 1]; // 非必要
    int _ = []() { // 非必要
        ifac[0] = fac[0] = fac[1] = ifac[1] = inv[1] = 1;
        for (int i = 2, j; i <= N; i++)
        {
            j = p / i;
            inv[i] = (p - j) * inv[p - i * j] % p;
            fac[i] = fac[i - 1] * i % p;
            ifac[i] = ifac[i - 1] * inv[i] % p;
        }
        return 0;
    }();
    ull w[N];
    int r[N];
    ull ksm(ull x, ull y)
    {
        if (y == p - 2 && x < N) return inv[x];
        ull r = 1;
        while (y)
        {
            if (y & 1) r = r * x % p;
            x = x * x % p;
            y >>= 1;
        }
        return r;
    }
    void init(int n)
    {
        static int pr = 0, pw = 0;
        if (pr == n) return;
        int b = __lg(n) - 1, i, j, k;
        for (i = 1; i < n; i++) r[i] = r[i >> 1] >> 1 | (i & 1) << b;
        if (pw < n)
        {
            for (j = 1; j < n; j = k)
            {
                k = j * 2;
                ull wn = ksm(g, (p - 1) / k);
                w[j] = 1;
                for (i = j + 1; i < k; i++) w[i] = w[i - 1] * wn % p;
            }
            pw = n;
        }
        pr = n;
    }
}
```

```

int cal(int x) { return 1 << __lg(max(x, 1) * 2 - 1); }
struct Q :vector<ull>
{
    bool flag;
    Q &operator%=(int n) { assert((n & -n) == n); resize(n); return *this; }
    Q operator%(int n) const
    {
        assert((n & -n) == n);
        if (size() <= n)
        {
            auto f = *this;
            return f %= n;
        }
        return Q(vector(begin(), begin() + n));
    }
    int deg() const
    {
        int n = size() - 1;
        while (n >= 0 && begin()[n] == 0) --n;
        return n;
    }
    explicit Q(int x = 1, bool f = 0) :flag(f), vector<ull>(cal(x)) { }//小心: {}会调用这条而非下一条
    Q(const vector<ull> &o, bool f = 0) :Q(o.size(), f) { copy(all(o), begin()); }
    Q(const initializer_list<ull> &o, bool f = 0) :Q(vector(o), f) { }
    ull fx(ull x)
    {
        ull r = 0;
        for (auto it = rbegin(); it != rend(); ++it) r = (r * x + *it) % p;
        return r;
    }
    void dft()
    {
        int n = size(), i, j, k;
        ull y, *f, *g, *wn, *a = data();
        init(n);
        for (i = 1; i < n; i++) if (i < r[i]) ::swap(a[i], a[r[i]]);
        for (k = 1; k < n; k *= 2)
        {
            wn = w + k;
            for (i = 0; i < n; i += k * 2)
            {
                {
                    g = (f = a + i) + k;
                    for (j = 0; j < k; j++)
                    {
                        y = g[j] * wn[j] % p;
                        g[j] = f[j] + y - p;
                        f[j] += y;
                    }
                }
                //此处要求 14*p*p<=2^64。如果调整模数，需要修改 12。
                if (__lg(n / k) % 12 == 1) for (i = 0; i < n; i++) a[i] %= p;
            }
        }
        if (flag)
        {
            y = ksm(n, p - 2);
            for (i = 0; i < n; i++) a[i] = a[i] * y % p;
            reverse(a + 1, a + n);
        }
    }
};

```

```

    }
    flag ^= 1;
}
void hf_dft()
{
    assert(size() >= 2 && flag);
    int n = size() / 2, i, j, k;
    ull x, y, *f, *g, *wn, *a = data();
    init(n);
    for (i = 1; i < n; i++) if (i < r[i]) ::swap(a[i], a[r[i]]);
    for (k = 1; k < n; k *= 2)
    {
        wn = w + k;
        for (i = 0; i < n; i += k * 2)
        {
            g = (f = a + i) + k;
            for (j = 0; j < k; j++)
            {
                y = g[j] * wn[j] % p;
                g[j] = f[j] + p - y;
                f[j] += y;
            }
        }
        if (__lg(n / k) % 12 == 1) for (i = 0; i < n; i++) a[i] %= p;
    }
    if (flag)
    {
        x = ksm(n, p - 2);
        for (i = 0; i < n; i++) a[i] = a[i] * x % p;
        reverse(a + 1, a + n);
    }
    flag ^= 1;
}
Q operator<<(int m) const
{
    int n = deg(), i;
    Q r(n + m + 1);
    for (i = 0; i <= n; i++) r[i + m] = at(i);
    return r;
}
Q operator>>(int m) const
{
    int n = deg(), i;
    if (n < m) return Q();
    Q r(n + 1 - m);
    for (i = m; i <= n; i++) r[i - m] = at(i);
    return r;
}
};
Q shrink(Q f) { return f %= cal(f.deg() + 1); }
ostream &operator<<(ostream &cout, const Q &o)
{
    int n = o.deg();
    if (n < 0) return cout << "[0]";
    cout << "[" << o[n];
    for (int i = n - 1; i >= 0; i--) cout << ", " << o[i];
    return cout << "]";
}

```



```

}
Q der(const Q &f)
{
    ull n = f.size(), i;
    Q r(n);
    for (i = 1; i < n; i++) r[i - 1] = f[i] * i % p;
    return r;
}
Q integral(const Q &f)
{
    ull n = f.size(), i;
    Q r(n);
    for (i = 1; i < n; i++) r[i] = f[i - 1] * inv[i] % p;
    return r;
}
Q &operator+=(Q &f, ull x) { (f[0] += x) %= p; return f; }
Q operator+(Q f, ull x) { return f += x; }
Q &operator-=(Q &f, ull x) { (f[0] += p - x) %= p; return f; }
Q operator-(Q f, ull x) { return f -= x; }
Q &operator*=(Q &f, ull x) { for (ull &y : f) (y *= x) %= p; return f; }
Q operator*(Q f, ull x) { return f *= x; }
Q &operator+=(Q &f, const Q &g)
{
    f %= max(f.size(), g.size());
    for (int i = 0; i < g.size(); i++) f[i] = (f[i] + g[i]) % p;
    return f;
}
Q operator+(Q f, const Q &g) { return f += g; }
Q &operator-=(Q &f, const Q &g)
{
    f %= max(f.size(), g.size());
    for (int i = 0; i < g.size(); i++) f[i] = (f[i] + p - g[i]) % p;
    return f;
}
Q operator-(Q f, const Q &g) { return f -= g; }
Q &operator*=(Q &f, Q g) //卷积
{
    if (f.flag | g.flag)
    {
        int n = f.size(), i;
        assert(n == g.size());
        if (!f.flag) f.dft();
        if (!g.flag) g.dft();
        for (i = 0; i < n; i++) (f[i] *= g[i]) %= p;
        f.dft();
    }
    else
    {
        int n = cal(f.size() + g.size() - 1), i, j;
        int m1 = f.deg(), m2 = g.deg();
        if ((ull)m1 * m2 > (ull)n * __lg(n) * 8)
        {
            (f %= n).dft(); (g %= n).dft();
            for (i = 0; i < n; i++) (f[i] *= g[i]) %= p;
            f.dft();
        }
        else

```

```

        {
            vector<ull> r(max(0, m1 + m2 + 1));
            for (i = 0; i <= m1; i++) for (j = 0; j <= m2; j++) (r[i + j] += f[i] * g[j]) %=
p;

            f = Q(n);
            copy(all(r), f.begin());
        }
    }
    return f;
}
Q operator*(Q f, const Q &g) { return f *= g; }
Q &operator*=(Q &f, Q g) //循环卷积
{
    assert(f.size() == g.size());
    int n = f.size(), i;
    if (!f.flag) f.dft();
    if (!g.flag) g.dft();
    for (i = 0; i < n; i++) (f[i] *= g[i]) %= p;
    f.dft();
    return f;
}
Q operator&(Q f, const Q &g) { return f &= g; }
Q &operator^=(Q &f, Q g) //差卷积
{
    int n = f.size();
    g %= n;
    reverse(all(g));
    f *= g;
    rotate(f.begin(), n - 1 + all(f));
    return f %= n;
}
Q operator^(Q f, const Q &g) { return f ^= g; }
Q sqr(Q f)
{
    assert(!f.flag);
    int n = f.size() * 2, i;
    (f %= n).dft();
    for (i = 0; i < n; i++) f[i] = f[i] * f[i] % p;
    f.dft();
    return f;
}
/*Q operator~(const Q &f)
{
    Q r;
    r[0]=ksm(f[0],p-2);
    for (int i=1; i<=f.size(); i*=2) r=-((f%i)*r-2)*r%i;
    return r;
} //trivial, 5e5 750ms*/
Q operator~(const Q &f)
{
    Q q, r, g;
    int n = f.size(), i, j, k;
    r[0] = ksm(f[0], p - 2);
    for (j = 2; j <= n; j *= 2)
    {
        k = j / 2;
        g = (r %= j) % k;
    }
}

```

```

        r.dft();
        q = f % j * r;
        fill_n(q.begin(), k, 0);
        r *= q;
        copy(all(g), r.begin());
        for (i = k; i < j; i++) r[i] = (p - r[i]) % p;
    }
    return r;
} //5e5 200ms, inv(1 6 3 4 9)=(1 998244347 33 998244169 1020)
Q &operator/=(Q &f, const Q &g) { int n = f.size(); return (f *= ~g) %= n; }
Q operator/(Q f, const Q &g) { return f /= g; }
void cdq(Q &f, Q &g, int l, int r) //g_0=1, i*g_i=g_{i-j}*f_j, use for cdq
{
    static vector<Q> cd;
    int i, m = l + r >> 1, n = r - l, nn = n >> 1;
    if (r - l == f.size())
    {
        g = Q(n);
        cd.clear();
        for (i = 2; i <= n; i *= 2)
        {
            cd.emplace_back(i);
            Q &h = cd.back();
            h %= i;
            copy_n(f.begin(), i, h.begin());
            h.dft();
        }
    }
    if (l + 1 == r)
    {
        g[l] = 1 ? g[l] * inv[l] % p : 1;
        return;
    }
    cdq(f, g, l, m);
    Q h(n);
    copy_n(g.begin() + l, nn, h.begin());
    h *= cd[_lg(n) - 1];
    for (i = m; i < r; i++) (g[i] += h[i - l]) %= p;
    cdq(f, g, m, r);
}
Q exp_cdq(Q f)
{
    Q g;
    int n = f.size(), i;
    for (i = 1; i < n; i++) f[i] = f[i] * i % p;
    cdq(f, g, 0, n);
    return g;
} //5e5 455ms
Q ln(const Q &f) { return integral(der(f) / f); }
//5e5 330ms, ln(1 2 3 4 5)=(0 2 1 665496236 499122177)
Q exp(Q f)
{
    Q r; r[0] = 1;
    for (int i = 1; i <= f.size(); i *= 2) (r *= f % i - ln(r % i) + 1) %= i;
    return r;
} //5e5 700ms, exp(0 4 2 3 5)=(1 4 10 665496257 665496281)
Q exp_new(Q b)

```

```

{
    Q h, f, r, u, v, bj;
    int n = b.size(), i, j, k;
    r[0] = h[0] = 1;
    for (j = 2; j <= n; j *= 2)
    {
        f = bj = der(b % j); k = j / 2; fill(k + all(bj), 0);
        h.dft(); u = der(r) & h;
        v = (r & h) % j - 1 & bj;
        for (i = 0; i < k; i++) f[i + k] = (p * p + u[i] - v[i] - f[i] - f[i + k]) % p, f[i]
= 0;
        f[k - 1] = (f[j - 1] + v[k - 1]) % p;
        u = (r %= j) & integral(f);
        for (i = k; i < j; i++) r[i] = (p - u[i]) % p;
        if (j < n) h = ~r;
    }
    return r;
} //5e5 420ms
optional<ull> mosqrt(ull x)
{
    static mt19937 rnd(chrono::steady_clock::now().time_since_epoch().count());
    static ull W;
    struct P
    {
        ull x, y;
        P operator*(const P &a) const
        {
            return {(x * a.x + y * a.y % p * W) % p, (x * a.y + y * a.x) % p};
        }
    };
    if (x == 0) return {0};
    if (ksm(x, p - 1 >> 1) != 1) return { };
    ull y;
    do y = rnd() % p; while (ksm(W = (y * y % p + p - x) % p, p - 1 >> 1) <= 1); //not for p=2
    y = [&](P x, ull y) {
        P r{1, 0};
        while (y)
        {
            if (y & 1) r = r * x;
            x = x * x; y >>= 1;
        }
        return r.x;
    }({y, 1}, p + 1 >> 1);
    return {y * 2 < p ? y : p - y};
}
optional<Q> sqrt(Q f)
{
    const static ull i2 = p + 1 >> 1;
    Q r;
    int n = f.size(), i, l;

    for (i = 0; i < n; i++) if (f[i]) break;
    if (i == n) return f;
    if (i & 1) return { };
    l = i / 2;
    copy(i + all(f), f.begin());
    fill(n - i + all(f), 0);

```

```

    auto rt = mosqrt(f[0]);
    if (rt) r[0] = rt.value(); else return { };
    for (i = 2; i <= n; i *= 2) r = (sqr(r) + f % i) / (r % i) % i * i2;

    copy_backward(all(r) - 1, r.end());
    fill_n(r.begin(), 1, 0);

    return {r};
} //5e5 530ms, sqrt(0 0 4 2 3)=(0 2 499122177 311951361 171573248)
optional<Q> sqrt_new(Q f)
{
    const static ull i2 = p + 1 >> 1;
    Q q, r;
    int n = f.size(), i, j, k, l;

    for (i = 0; i < n; i++) if (f[i]) break;
    if (i == n) return f;
    if (i & 1) return { };
    l = i / 2;
    copy(i + all(f), f.begin());
    fill(n - i + all(f), 0);

    auto rt = mosqrt(f[0]);
    if (rt) r[0] = rt.value(); else return { };
    for (j = 2; j <= n; j *= 2)
    {
        k = j / 2; (q = r).dft(); (q &= q) %= j;
        for (i = k; i < j; i++) q[i] = (q[i - k] + p * 2 - f[i] - f[i - k]) * i2 % p, q[i - k]
    ] = 0;
        q &= ~r % j; r %= j;
        for (i = k; i < j; i++) r[i] = (p - q[i]) % p;
    }

    copy_backward(all(r) - 1, r.end());
    fill_n(r.begin(), 1, 0);

    return {r};
} //5e5 280ms
Q pow(Q b, ull m) //不应传入超过 int 内容
{
    assert(m <= 1llu << 32);
    int n = b.size(), i, j = n, k;
    for (i = 0; i < n; i++) if (b[i]) { j = i; break; }
    if (j == n) return b[0] = !m, b;
    if (j * m >= n) return Q(n);
    copy(j + all(b), b.begin());
    fill(n - j + all(b), 0);
    k = b[0]; j *= m;
    b = exp_new(ln(b * ksm(k, p - 2)) * m) * ksm(k, m);
    copy_backward(all(b) - j, b.end());
    fill_n(b.begin(), j, 0);
    return b;
}
Q pow(Q b, string s)
{
    int n = b.size(), i, j = n, k;

```

```

    for (i = 0; i < n; i++) if (b[i]) { j = i; break; }
    if (j == n) return b[0] = s == "0", b;
    if (j && (s.size() > 8 || j * stoll(s) >= n)) return Q(n);
    ull m0 = 0, m1 = 0;
    for (auto c : s) m0 = (m0 * 10 + c - '0') % p, m1 = (m1 * 10 + c - '0') % (p - 1);
    copy(j + all(b), b.begin());
    fill(n - j + all(b), 0);
    k = b[0]; j *= m0;
    b = exp_new(ln(b * ksm(k, p - 2)) * m0) * ksm(k, m1);
    copy_backward(all(b) - j, b.end());
    fill_n(b.begin(), j, 0);
    return b;
} //5e5 1e18 700ms
Q pow2(Q b, ull m)
{
    int n = b.size();
    Q r(n); r[0] = 1;
    while (m)
    {
        if (m & 1) (r *= b) %= n;
        if (m >>= 1) b = sqr(b) % n;
    }
    return r;
} //5e5 1e18 7425ms
Q div(Q f, Q g)
{
    int n = 0, m = 0, i;
    for (i = f.size() - 1; i >= 0; i--) if (f[i]) { n = i + 1; break; }
    for (i = g.size() - 1; i >= 0; i--) if (g[i]) { m = i + 1; break; }
    assert(m);
    if (n < m) return Q(1);
    reverse(f.begin(), f.begin() + n);
    reverse(g.begin(), g.begin() + m);
    n = n - m + 1; m = cal(n);
    f = (f % m) / (g % m) % m;
    fill(n + all(f), 0);
    reverse(f.begin(), f.begin() + n);
    return f;
}
Q mod(const Q &a, const Q &b)
{
    if (a.deg() < b.deg()) return shrink(a);
    Q r = (a - b * div(a, b));
    return shrink(r %= min(r.size(), b.size()));
}
Q pow(Q x, ull y, Q f)
{
    Q r(1);
    r[0] = 1;
    while (y)
    {
        if (y & 1) r = mod(r * x, f);
        if (y >>= 1) x = mod(sqr(x), f);
    }
    return r;
}
pair<Q, Q> div_mod(Q a, Q b)

```

```

{
    if (b.size() <= 16)
    {
        int n = a.deg(), m = b.deg(), i, j;
        if (m == 0)
        {
            assert(b[0]);
            a *= ksm(b[0], p - 2);
            return {a, Q()};
        }
        assert(m > 0);
        Q q(n - m + 1);
        ull k = b[m], ik = ksm(k, p - 2);
        for (i = 0; i < m; i++) (b[i] *= ik) %= p;
        for (i = n - m; i >= 0; i--)
        {
            q[i] = a[i + m];
            for (j = 0; j <= m; j++) (a[i + j] += (p - q[i]) * b[j]) %= p;
            (q[i] *= ik) %= p;
        }
        return {q, a % min<ull>(a.size(), cal(m))};
    }
    Q q = div(a, b);
    (a -= (b * q));
    return {q, a %= min(a.size(), b.size())};
}
//5e5 430ms (1 2 3 4)=(916755018 427819009)*(5 6 7)+(407446676 346329673)
// Q cdq_inv(const Q &f) { return ~(f-1)*(p-1); } //g_0=1,g_i=g_{i-j}*f_j ?
ull recurrent(const vector<ull> &f, const vector<ull> &a, ull m)//常系数齐次线性递推, find a_m
,a_n=a_{n-i}*f_i,f_1...k,a_0...k-1
{
    if (m < a.size()) return a[m];
    assert(f.size() == a.size() + 1 && f[0] == 0);
    int k = a.size(), n = cal(k + 1) * 2, i;
    ull ans = 0;
    Q h(n), g(2);
    for (i = 1; i <= k; i++) h[k - i] = (p - f[i]) % p;
    h[k] = g[1] = 1;
    Q r = pow(g, m, h);
    k = min(k, (int)r.size());
    for (i = 0; i < k; i++) ans = (ans + a[i] * r[i]) % p;
    return ans;
}
//1e5 1e18 8500ms
ull recurrent_new(const vector<ull> &f, const vector<ull> &a, ull m)//常系数齐次线性递推, find
a_m,a_n=a_{n-i}*f_i,f_1...k,a_0...k-1
{
    const static ull i2 = p + 1 >> 1;
    if (m < a.size()) return a[m];
    assert(f.size() == a.size() + 1 && f[0] == 0);
    int k = a.size(), n = cal(k + 1), i;
    Q g(n * 2), h(n * 2);
    for (h[0] = i = 1; i <= k; i++) h[i] = (p - f[i]) % p;
    copy(all(a), g.begin());
    g &= h; fill(k++ + all(g), 0);
    vector<ull> res(n);
    while (m)
    {

```

```

    if (m & 1)
    {
        ull x = p - g[0];
        for (i = 1; i < k; i += 2) res[i >> 1] = x * h[i] % p;
        copy_n(g.begin() + 1, k - 1, g.begin());
        g[k - 1] = 0;
    }
    g.dft(); h.dft();
    ull *a = g.data(), *b = h.data(), *c = a + n, *d = b + n;
    for (i = 0; i < n; i++) g[i] = (a[i] * d[i] + b[i] * c[i]) % p * i2 % p;
    for (i = 0; i < n; i++) h[i] = h[i] * h[i ^ n] % p;
    g.hf_dft(); h.hf_dft();
    fill(k + all(g), 0);
    if (m & 1) for (i = 0; i < k; i++) (g[i] += res[i]) %= p;
    fill(k + all(h), 0);
    m >>= 1;
}
assert(h[0] == 1);
return g[0];
} //1e5 1e18 1000ms
vector<ull> recurrent_interval(const vector<ull> &f, const vector<ull> &a, ull L, ull R) //常
系数齐次线性递推, find a_[L,R], a_n=a_{n-i}*f_i, f_1...k, a_0...k-1
{
    assert(f.size() == a.size() + 1 && f[0] == 0);
    int k = a.size(), n = cal(k + 1) * 2, i, len = R - L;
    ull ans = 0, m = L;
    Q h(n), g(2), r;
    for (i = 1; i <= k; i++) h[k - i] = (p - f[i]) % p;
    h[k] = g[1] = r[0] = 1;
    while (m)
    {
        if (m & 1) r = mod(r * g, h);
        if (m >>= 1) g = mod(sqr(g), h);
    }
    Q F(f), A(a);
    F[0] = p - 1;
    A *= F;
    A %= cal(k);
    fill(k + all(A), 0);
    n = cal(len + k);
    F %= n;
    A *= ~F;
    r %= cal(k);
    reverse(r.begin(), r.begin() + k);
    r *= A;
    r.erase(r.begin(), r.begin() + k - 1);
    r.resize(len);
    return r;
} //1e5 1e18 5e5 10000ms
Q prod(const vector<Q> &a)
{
    if (!a.size()) return {1};
    function<Q(int, int)> dfs = [&](int l, int r) {
        if (r - l == 1) return a[l];
        int m = l + r >> 1;
        return shrink(dfs(l, m) * dfs(m, r));
    };
};

```



```

    return dfs(0, a.size());
}
Q prod_new(const vector<Q> &a)
{
    if (!a.size()) return {1};
    struct cmp
    {
        bool operator()(const Q &f, const Q &g) const { return f.size() > g.size(); }
    };
    priority_queue<Q, vector<Q>, cmp> q(all(a));
    while (q.size() > 1)
    {
        auto f = q.top(); q.pop();
        f = shrink(f * q.top()); q.pop();
        q.push(f);
    }
    return q.top();
}
vector<ull> evaluation(const Q &f, const vector<ull> &X)
{
    int m = X.size(), n = f.size() - 1, i, j;
    vector<Q> pro(m * 4 + 4);
    while (n > 1 && !f[n]) --n;
    vector<ull> y(m);
    function<void(int, int, int)> build = [&](int x, int l, int r) {
        if (l + 1 == r)
        {
            pro[x] = Q(vector{(p - X[l]) % p, 1llu});
            return;
        }
        int mid = l + r >> 1, c = x * 2;
        build(c, l, mid); build(c + 1, mid, r);
        pro[x] = shrink(pro[c] * pro[c + 1]);
    };
    function<void(int, int, int, Q, int)> dfs = [&](int x, int l, int r, Q f, int d) {
        const static int limit = 256;
        if (d >= r - 1) f = shrink(mod(f, pro[x]));
        if (r - l < limit)
        {
            for (int i = l; i < r; i++) y[i] = f.fx(X[i]);
            return;
        }
        int mid = l + r >> 1, c = x * 2;
        dfs(c, l, mid, f, d);
        dfs(c + 1, mid, r, f, d);
    };
    build(1, 0, m);
    dfs(1, 0, m, f, n);
    return y;
}
//131072 880ms
vector<ull> evaluation_new(Q f, const vector<ull> &X)//多项式多点求值
{
    int m = X.size(), i, j;
    vector<ull> y(m);
    if (X.size() <= 10)
    {
        for (i = 0; i < m; i++) y[i] = f.fx(X[i]);
    }

```

```

    return y;
}
int n = f.size();
while (n > 1 && !f[n - 1]) --n;
f.resize(cal(n));
vector<Q> pro(m * 4 + 4);
function<void(int, int, int)> build = [&](int x, int l, int r) {
    if (l == r)
    {
        pro[x] = Q(vector{1llu, (p - X[l]) % p});
        return;
    }
    int m = l + r >> 1, c = x * 2;
    build(c, l, m); build(c + 1, m + 1, r);
    pro[x] = shrink(pro[c] * pro[c + 1]);
};
function<void(int, int, int, Q)> dfs = [&](int x, int l, int r, Q f) {
    const static int limit = 30;
    if (r - l + 1 <= limit)
    {
        int m = r - l + 1, m1, m2, mid = l + r >> 1, i, j, k;
        static ull g[limit + 2], g1[limit + 2], g2[limit + 2];
        m1 = m2 = r - l;
        copy_n(f.data(), m, g1);
        copy_n(g1, m, g2);
        for (i = mid + 1; i <= r; i++, --m1) for (k = 0; k < m1; k++) g1[k] = (g1[k] + g1
[k + 1] * (p - X[i])) % p;
        for (i = l; i <= mid; i++, --m2) for (k = 0; k < m2; k++) g2[k] = (g2[k] + g2[k +
1] * (p - X[i])) % p;
        for (i = l; i <= mid; i++)
        {
            copy_n(g1, (m = m1) + 1, g);
            for (j = l; j <= mid; j++) if (i != j)
            {
                for (k = 0; k < m; k++) g[k] = (g[k] + g[k + 1] * (p - X[j])) % p;
                --m;
            }
            y[i] = g[0];
        }
        for (i = mid + 1; i <= r; i++)
        {
            copy_n(g2, (m = m2) + 1, g);
            for (j = mid + 1; j <= r; j++) if (i != j)
            {
                for (k = 0; k < m; k++) g[k] = (g[k] + g[k + 1] * (p - X[j])) % p;
                --m;
            }
            y[i] = g[0];
        }
        return;
    }
    int mid = l + r >> 1, c = x * 2, n = f.size();
    f.dft();
    for (auto [x, len] : {pair{c, r - mid}, {c + 1, mid - l + 1}})
    {
        pro[x] %= n;
        reverse(all(pro[x])); pro[x] &= f;
    }
}

```

```

        rotate(all(pro[x]) - 1, pro[x].end());
        pro[x] %= cal(len);
        fill(len + all(pro[x]), 0);
    }
    dfs(c, l, mid, pro[c + 1]);
    dfs(c + 1, mid + 1, r, pro[c]);
};
build(1, 0, m - 1);
pro[1] %= f.size();
(f ^= ~pro[1]) %= cal(m);
fill(min(m, n) + all(f), 0);
dfs(1, 0, m - 1, f);
return y;
} //131072 460ms
ull factorial(ull n)
{
    if (n >= p) return 0;
    if (n <= 1) return 1 % p;
    ull B = ::sqrt(n), i;
    vector F(B, Q({0, 1}));
    for (i = 0; i < B; i++) F[i][0] = i + 1;
    auto f = prod(F);
    vector<ull> x(B);
    for (i = 0; i < B; i++) x[i] = i * B;
    ull r = 1;
    auto y = evaluation(f, x);
    for (i = 0; i < B; i++) r = r * y[i] % p;
    for (i = B * B + 1; i <= n; i++) r = r * i % p;
    return r;
} //998244352 170ms
vector<ull> getinvs(vector<ull> a)
{
    int n = a.size(), i;
    if (n <= 2)
    {
        for (i = 0; i < n; i++) a[i] = ksm(a[i], p - 2);
        return a;
    }
    vector<ull> l(n), r(n);
    l[0] = a[0]; r[n - 1] = a[n - 1];
    for (i = 1; i < n; i++) l[i] = l[i - 1] * a[i] % p;
    for (i = n - 2; i; i--) r[i] = r[i + 1] * a[i] % p;
    ull x = ksm(l[n - 1], p - 2);
    a[0] = x * r[1] % p; a[n - 1] = x * l[n - 2] % p;
    for (i = 1; i < n - 1; i++) a[i] = x * l[i - 1] % p * r[i + 1] % p;
    return a;
}
Q interpolation(const vector<ull> &X, const vector<ull> &y) //多项式快速插值
{
    assert(X.size() == y.size());
    int n = X.size(), i, j;
    if (n <= 1) return Q(y);
    if (1)
    {
        auto vv = X; sort(all(vv));
        assert(unique(all(vv)) - vv.begin() == n);
    }
}

```

```

vector<Q> sum(4 * n + 4), pro(4 * n + 4);
function<void(int, int, int)> build = [&](int x, int l, int r) {
    if (l == r)
    {
        sum[x] = Q(vector{(p - X[l]) % p, 1llu});
        return;
    }
    int mid = l + r >> 1, c = x * 2;
    build(c, l, mid); build(c + 1, mid + 1, r);
    sum[x] = shrink(sum[c] * sum[c + 1]);
};
build(1, 0, n - 1);
auto v = evaluation_new(sum[1] = der(sum[1]), X);
assert(v.size() == n);
auto Y = getinvs(v);
for (i = 0; i < n; i++) Y[i] = Y[i] * y[i] % p;
function<void(int, int, int)> dfs = [&](int x, int l, int r) {
    if (l == r)
    {
        pro[x][0] = Y[l];
        return;
    }
    int c = x * 2, mid = l + r >> 1;
    dfs(c, l, mid); dfs(c + 1, mid + 1, r);
    pro[x] = shrink((pro[c] * sum[c + 1]) + (pro[c + 1] * sum[c]));
};
dfs(1, 0, n - 1);
return pro[1] %= cal(n);
} //131072 1150ms
Q comp(const Q &f, Q g) //多项式复合  $f(g(x)) = [x^i]f(x)g(x)^i$ 
{
    int n = f.size(), l = ceil(::sqrt(n)), i, j;
    assert(n >= g.size()); //返回  $n-1$  次多项式
    vector<Q> a(l + 1), b(l);
    a[0] %= n; a[0][0] = 1; a[1] = g;
    g %= n * 2;
    Q u = g, v(n);
    g.dft();
    for (i = 2; i <= l; i++) a[i] = ((u &= g) %= n), u %= n * 2;
    for (i = 2; i < l; i++)
    {
        u.dft(); b[i - 1] = u;
        u &= b[1]; fill(n + all(u), 0);
    }
    u.dft(); b[l - 1] = u;
    for (i = 0; i < l; i++)
    {
        fill(all(v), 0);
        for (j = 0; j < l; j++) if (i * l + j < n) v += a[j] * f[i * l + j];
        if (i == 0) u = v; else u += ((v %= n * 2) &= b[i]) %= n;
    }
    return u;
} //n^2+n*sqrt n*log n, 8000 350ms
Q comp_inv(Q f) //多项式复合逆  $g(f(x))=x$ , 求  $g$ ,  $[x^n]g = ([x^{n-1}](x/f)^n)/n$ , 要求常数 0 一次非 0
{
    assert(!f[0] && f[1]);
    int n = f.size(), l = ceil(::sqrt(n)), i, j, k, m; //l>=2

```

```

rotate(f.begin(), 1 + all(f));
f = ~f;
vector<Q> a(1 + 1), b(1);
Q u, v;
u = a[1] = f;
u %= n * 2; (v = u).dft();
for (i = 2; i <= 1; i++)
{
    u &= v;
    fill(n + all(u), 0);
    a[i] = u;
}
b[0] %= n; b[0][0] = 1; b[1] = u; (v = u).dft();
for (i = 2; i < 1; i++)
{
    u &= v;
    fill(n + all(u), 0);
    b[i] = u;
}
u %= n; u[0] = 0;
for (i = 0; i < 1; i++) for (j = 1; j <= 1; j++) if (i * 1 + j < n)
{
    m = i * 1 + j - 1;
    ull r = 0, *f = b[i].data(), *g = a[j].data();
    for (k = 0; k <= m; k++) r = (r + f[k] * g[m - k]) % p;
    u[m + 1] = r * inv[m + 1] % p;
}
return u;
} //8000 200ms
Q shift(Q f, ull c) //get f(x+c), c in [0, p)
{
    int n = f.size(), i, j;
    Q g(n);
    for (i = 0; i < n; i++) (f[i] *= fac[i]) %= p;
    g[0] = 1;
    for (i = 1; i < n; i++) g[i] = g[i - 1] * c % p;
    for (i = 0; i < n; i++) (g[i] *= ifac[i]) %= p;
    f ^= g;
    for (i = 0; i < n; i++) (f[i] *= ifac[i]) %= p;
    return f;
} //5e5 200ms (1 2 3 4 5) 3 -> (547 668 309 64 5)
vector<ull> shift(vector<ull> y, ull c, ull m) // [0, n) 点值 -> [c, c+m) 点值
{
    assert(y.size());
    if (y.size() == 1) return vector(m, y[0]);
    vector<ull> r, res;
    r.reserve(m);
    int n = y.size(), i, j, mm = m;
    while (c < n && m) r.push_back(y[c++]), --m;
    if (c + m > p)
    {
        res = shift(y, 0, c + m - p);
        m = p - c;
    }
    if (!m) { r.insert(r.end(), all(res)); return r; }
    int len = cal(m + n - 1), l = m + n - 1;
    for (i = n & 1; i < n; i += 2) y[i] = (p - y[i]) % p;

```

```

for (i = 0; i < n; i++) y[i] = y[i] * ifac[i] % p * ifac[n - 1 - i] % p;
y.resize(len);
Q f, g;
vector<ull> v(m + n - 1);
c -= n - 1;
for (i = 0; i < l; i++) v[i] = (c + i) % p;
f = Q(y); g = Q(getinvs(v)) % len;
f *= g;
vector<ull> u(m);
for (i = n - 1; i < l; i++) u[i - (n - 1)] = f[i];
v.resize(m);
for (i = 0; i < m; i++) v[i] = c + i;
v = getinvs(v); c += n;
ull tmp = 1;
for (i = c - n; i < c; i++) tmp = tmp * i % p;
for (i = 0; i < m; i++) (u[i] *= tmp) %= p, tmp = tmp * (c + i) % p * v[i] % p;
r.insert(r.end(), all(u));
r.insert(r.end(), all(res));
assert(r.size() == mm);
return r;
} //5e5 430ms, (1 4 9 16) 3 5 -> (16 25 36 49 64)
vector<ull> czt(Q f, ull c, int m) //求  $f(c^{-[0,m]})$ 。核心  $ij=C(i+j,2)-C(i,2)-C(j,2)$ 
{
    const static ull B = 1e5;
    static ull a[B + 2], b[B + 2];
    int i, n = f.size();
    if (m == 0) return { };
    if (c == 0)
    {
        vector<ull> r(m, f.fx(0));
        r[0] = f.fx(1);
        return r;
    }
    if (min(n, m) <= 16)
    {
        vector<ull> r(m);
        ull j;
        for (i = 0, j = 1; i < m; i++) r[i] = f.fx(j), j = j * c % p;
        return r;
    }
    ull l = cal(m += n - 1);
    ui cur = min<ui>(B, max({m, 4}));
    auto mic = [&](ull x) { return a[x % cur] * b[x / cur] % p; };
    Q g(1);
    assert(B * B > p);
    a[0] = b[0] = g[0] = g[1] = 1;
    for (i = 1; i <= cur; i++) a[i] = a[i - 1] * c % p;
    for (i = 1; i <= cur; i++) b[i] = b[i - 1] * a[cur] % p;
    for (i = 2; i < m; i++) g[i] = mic(i * (i - 1llu) / 2 % (p - 1));
    vector<ull> ig(m, 1);
    for (i = 2; i < m; i++) ig[i] = ig[i - 1] * g[i] % p;
    ull iv = ksm(ig[m - 1], p - 2);
    for (i = m - 1; i >= 2; i--)
    {
        ig[i] = ig[i - 1] * iv % p;
        if (i < n)
            (f[i] *= ig[i]) %= p;
    }
}

```

```

        (iv *= g[i]) %= p;
    }
    reverse(all(f)); (f %= 1) &= g;
    vector<ull> r(f.begin() + n - 1, f.begin() + m); m -= n - 1;
    for (i = 2; i < m; i++) (r[i] *= ig[i]) %= p;

    return r;
} //luogu 1e6 500ms

vector<ull> dft(const vector<ull> &a)
{
    int n = a.size();
    if (n <= 1) return a;
    assert((p - 1) % n == 0);
    if ((n & -n) == n)
    {
        Q q(a, 0);
        q.dft();
        return vector<ull>(q.begin(), q.begin() + n);
    }
    ull wn = ksm(g, (p - 1) / n);
    return czt(Q(a), wn, n);
}

vector<ull> idft(const vector<ull> &a)
{
    int n = a.size();
    if (n <= 1) return a;
    assert((p - 1) % n == 0);

    ull inv_wn = ksm(g, p - 1 - (p - 1) / n);
    vector<ull> res = czt(Q(a), inv_wn, n);

    ull inv_n = ksm(n, p - 2);
    for (ull &x : res) x = x * inv_n % p;
    return res;
}

vector<ull> Bell(int n) //B(0...n)
{
    ++n;
    Q f(n);
    int i;
    for (i = 1; i < n; i++) f[i] = ifac[i];
    f = exp_new(f);
    for (i = 2; i < n; i++) f[i] = f[i] * fac[i] % p;
    return vector<ull>(f.begin(), f.begin() + n);
}

vector<ull> S1_row(int n, int m) //S1(n,0...m), O(nlogn), unsigned
{
    int cm = cal(++m);
    if (n == 0)
    {
        vector<ull> r(m);
        r[0] = 1;
        return r;
    }
    function<Q(int)> dfs = [&](int n) {

```

```

    if (n == 1)
    {
        Q f(2);
        f[1] = 1;
        return f;
    }
    Q f = dfs(n / 2);
    f *= shift(f, n / 2);
    if (n & 1)
    {
        f %= cal(n + 1);
        for (int i = n; i; i--) f[i] = f[i - 1];
        // for (int i=1; i<=n; i++) f[i]=f[i-1];
        --n;
        for (int i = 0; i <= n; i++) f[i] = (f[i] + f[i + 1] * n) % p;
    }
    if (f.size() > cm) f %= cm;
    return f;
};
Q f = dfs(n);
if (f.size() < cm) f %= cm;
return vector<ull>(f.begin(), f.begin() + m);
}
vector<ull> S1_column(int n, int m)//S1(0...n,m),0(nlogn)
{
    if (m == 0)
    {
        vector<ull> r(n + 1);
        r[0] = 1;
        return r;
    }
    Q f(n + 1);
    int i;
    for (i = 1; i <= n; i++) f[i] = inv[i];
    f = pow(f, m);
    for (i = m; i <= n; i++) f[i] = f[i] * fac[i] % p * ifac[m] % p;
    return vector<ull>(f.begin(), f.begin() + n + 1);
}
vector<ull> S2_row(int n, int m)//S2(n,0...m),0(mlogm)
{
    if (m == 0) return {n == 0}; // by codex
    int tm = ++m, i, j, cnt = 0;
    if (n == 0)
    {
        vector<ull> r(m);
        r[0] = 1;
        return r;
    }
    m = min(m, n + 1);
    vector<ull> pr(m), pw(m);
    pw[1] = 1;
    for (i = 2; i < m; i++)
    {
        if (!pw[i]) pr[cnt++] = i, pw[i] = ksm(i, n);
        for (j = 0; i * pr[j] < m; j++)
        {
            pw[i * pr[j]] = pw[i] * pw[pr[j]] % p;

```



```

        if (i % pr[j] == 0) break;
    }
}
Q f(m), g(m);
for (i = 0; i < m; i += 2) f[i] = ifac[i];
for (i = 1; i < m; i += 2) f[i] = p - ifac[i];
// for (i=1; i<m; i++) g[i]=pw[i]*ifac[i]%p;
for (i = 1; i < m; i++) g[i] = ksm(i, n) * ifac[i] % p;
f *= g;
vector<ull> r(f.begin(), f.begin() + m);
r.resize(tm);
return r;
} //5e5 150ms
vector<ull> S2_column(int n, int m) //S2(0...n,m), O(nlogn)
{
    if (m == 0)
    {
        vector<ull> r(n + 1);
        r[0] = 1;
        return r;
    }
    Q f(n + 1);
    int i;
    for (i = 1; i <= n; i++) f[i] = ifac[i];
    f = pow(f, m);
    for (i = m; i <= n; i++) f[i] = f[i] * fac[i] % p * ifac[m] % p;
    return vector<ull>(f.begin(), f.begin() + n + 1);
} //5e5 640ms
vector<ull> signed_S1_row(int n, int m)
{
    auto v = S1_row(n, m);
    for (int i = 1 ^ n & 1; i <= m; i += 2) v[i] = (p - v[i]) % p;
    return v;
} //5e5 190ms
vector<ull> Bernoulli(int n) //B(0...n)
{
    int i;
    Q f(++n);
    for (i = 0; i < n; i++) f[i] = ifac[i + 1];
    f = ~f;
    for (i = 0; i < n; i++) f[i] = f[i] * fac[i] % p;
    return vector<ull>(f.begin(), f.begin() + n);
} //5e5 180ms
vector<ull> Partition(int n) //P(0...n), 拆分数
{
    Q f(++n);
    int i, l = 0, r = 0;
    while (--l) if (3 * l * l - l >= n * 2) break;
    while (++r) if (3 * r * r - r >= n * 2) break;
    ++l;
    for (i = l + abs(l) % 2; i < r; i += 2) f[3 * i * i - i >> 1] = 1;
    for (i = l + abs(l + 1) % 2; i < r; i += 2) f[3 * i * i - i >> 1] = p - 1;
    f = ~f;
    return vector<ull>(f.begin(), f.begin() + n);
} //5e5 150ms
struct reg
{

```

```

Q a00, a01, a10, a11;
reg operator*(const reg &o) const
{
    return {
        shrink(a00 * o.a00 + a01 * o.a10),
        shrink(a00 * o.a01 + a01 * o.a11),
        shrink(a10 * o.a00 + a11 * o.a10),
        shrink(a10 * o.a01 + a11 * o.a11)};
}
pair<Q, Q> operator*(const pair<Q, Q> &o) const
{
    const auto &[b0, b1] = o;
    return {shrink(a00 * b0 + a01 * b1), shrink(a10 * b0 + a11 * b1)};
}
} E = {{vector{1llu}}, Q(), Q(), {vector{1llu}}};
ostream &operator<<(ostream &cout, const reg &o)
{
    return cout << "[" << o.a00 << ", " << o.a01 << "]\n"
        << "[" << o.a10 << ", " << o.a11 << "]\n";
}
reg hgcd(Q a, Q b)
{
    int m = a.deg() + 1 >> 1;
    if (m == 0) return E; // by codex
    if (b.deg() < m) return E;
    reg r = hgcd(a >> m, b >> m);
    auto [c, d] = r * pair{a, b};
    if (d.deg() < m) return r;
    auto [q, e] = div_mod(c, d);
    r.a00 -= shrink(q * r.a10);
    r.a01 -= shrink(q * r.a11);
    swap(r.a00, r.a10);
    swap(r.a01, r.a11);
    if (e.deg() < m) return r;
    int k = 2 * m - d.deg();
    auto s = hgcd(d >> k, e >> k);
    return s * r;
}
Q gcd(Q a, Q b)
{
    if (a.deg() < b.deg()) swap(a, b);
    while (b.deg() >= 0)
    {
        a = mod(a, b);
        swap(a, b);
        auto tmp = hgcd(a, b);
        tie(a, b) = tmp * pair{a, b};
    }
    if (a.deg() == -1) return a;
    ull k = ksm(a[a.deg()], p - 2);
    for (int i = 0; i < a.size(); i++) a[i] = a[i] * k % p;
    return a;
}
vector<ull> root(Q f)
{
    Q x(2);
    x[1] = 1;

```

```

x = pow(x, p, f);
if (x.size() < 2) x %= 2;
(x[1] += p - 1) %= p;
f = gcd(f, x);
vector<ull> res;
static mt19937 rnd(chrono::steady_clock::now().time_since_epoch().count());
function<void(Q)> dfs = [&](Q f) {
    int n = f.deg(), i;
    if (n <= 0) return;
    if (n == 1)
    {
        res.push_back((p - f[0]) % p);
        return;
    }
    Q g(n);
    for (i = 0; i < n; i++) g[i] = rnd() % p;
    g = gcd(pow(g, (p - 1) / 2, f) - 1, f);
    dfs(g); dfs(div(f, g));
};
dfs(f);
sort(all(res));
assert(unique(all(res)) == res.end());
return res;
} // 4000 950ms
optional<Q> inverse(Q a, Q m)
{
    Q b = m;
    vector<pair<reg, Q>> buf;
    a = mod(a, b);
    swap(a, b);
    while (b.deg() >= 0)
    {
        auto [q, r] = div_mod(a, b);
        swap(a, r); swap(a, b);
        auto tmp = hgcd(a, b);
        tie(a, b) = tmp * pair{a, b};
        buf.emplace_back(move(tmp), q);
    }
    if (a.deg()) return { };
    reg res = E;
    reverse(all(buf));
    for (const auto &[tmp, q] : buf)
    {
        res = res * tmp;
        res.a00 -= shrink(q * res.a01);
        res.a10 -= shrink(q * res.a11);
        swap(res.a00, res.a01);
        swap(res.a10, res.a11);
    }
    return {res.a01 * ksm(a[0], p - 2)};
} // 5e4 950ms
vector<ull> poly_to_newton(Q f, const vector<ull> &X) // f=sum(r_i*prod_{j<i}(x-x_j))
{
    int n = X.size();
    if (n == 0) return { };
    assert(f.deg() < n);
    vector<ull> res(n);

```

```

vector<Q> sum(4 * n + 4);
function<void(int, int, int)> build = [&](int x, int l, int r) {
    if (l == r)
    {
        if (r != n - 1) sum[x] = Q(vector{(p - X[l]) % p, 1llu});
        return;
    }
    int mid = l + r >> 1, c = x * 2;
    build(c, l, mid); build(c + 1, mid + 1, r);
    sum[x] = shrink(sum[c] * sum[c + 1]);
};
build(1, 0, n - 1);
function<void(Q &, int, int, int)> dfs = [&](Q &f, int x, int l, int r) {
    if (l == r)
    {
        res[l] = f[0];
        return;
    }
    if (l + 1 == r)
    {
        res[r] = f[1];
        res[l] = (f[0] + f[1] * X[l]) % p;
        return;
    }
    int c = x * 2, m = l + r >> 1;
    {
        Q g;
        tie(g, f) = div_mod(f, sum[c]);
        dfs(g, c | 1, m + 1, r);
    }
    dfs(f, c, l, m);
};
dfs(f, 1, 0, n - 1);
return res;
} //131072 559 ms
vector<ull> poly_to_facpow(const Q &f, int n)
{
    vector<ull> x(n);
    iota(all(x), 0);
    return poly_to_newton(f, x);
} //luogu 1e5 368 ms
Q newton_to_poly(const vector<ull> &g, const vector<ull> &X)
{
    int n = g.size();
    if (n == 0) return Q();
    assert(n == X.size());
    vector<Q> sum(4 * n + 4);
    Q f(n);
    function<void(int, int, int)> build = [&](int x, int l, int r) {
        if (l == r)
        {
            if (r != n - 1) sum[x] = Q(vector{(p - X[l]) % p, 1llu});
            return;
        }
        int mid = l + r >> 1, c = x * 2;
        build(c, l, mid); build(c + 1, mid + 1, r);
        sum[x] = shrink(sum[c] * sum[c + 1]);
    };
    build(1, 0, n - 1);
    return sum[1];
}

```

```

};
build(1, 0, n - 1);
function<Q(int, int, int)> dfs = [&](int x, int l, int r) {
    if (l == r) return Q(vector<ull>{g[l]});
    if (l + 1 == r) return Q(vector<ull>{g[l] + g[r] * (p - X[l]) % p, g[r]});
    int c = x * 2, m = l + r >> 1;
    return (sum[c] *= dfs(c + 1, m + 1, r)) += dfs(c, l, m);
};
return dfs(1, 0, n - 1);
}
Q facpow_to_poly(const vector<ull> &g)
{
    vector<ull> x(g);
    iota(all(x), 0);
    return newton_to_poly(g, x);
} //luogu 2e5 449 ms
vector<ull> mul_facpow(const vector<ull> &a, const vector<ull> &b)
{
    int m = a.size() + b.size() - 1, n = cal(m * 2), i;
    Q f(a), g(b), h(n);
    f %= n, g %= n;
    for (i = 0; i < m; i++) h[i] = ifac[i];
    h.dft();
    f *= h, g *= h;
    for (i = 0; i < m; i++) (f[i] *= g[i] * fac[i] % p) %= p;
    fill(m + all(f), 0);
    rotate(h.begin(), n / 2 + all(h));
    f *= h;
    vector<ull> r(m);
    for (i = 0; i < m; i++) r[i] = f[i];
    return r;
} //luogu 1e5 131 ms
}
using NTT::p;
using poly = NTT::Q;

```

3.25 MTT

如果长度较长，可以考虑将 p_3 替换为 $5 \times 2^{25} + 1$ 。

```

namespace MTT
{
    template<ull p> constexpr ull ksm(ull x, ull y = p - 2)
    {
        ull r = 1;
        while (y)
        {
            if (y & 1) r = r * x % p;
            x = x * x % p;
            y >>= 1;
        }
        return r;
    }
    int cal(int x) { return 1 << __lg(max(x, 1) * 2 - 1); }
    const int N = 1 << 22;
    const ull p = 1e9 + 7, g = 3,
        p1 = 7 << 26 | 1, p2 = 119 << 23 | 1, p3 = 479 << 21 | 1, //三模，原根都是 3，非常好

```

```

    inv_p1 = ksm<p2>(p1), inv_p12 = ksm<p3>(p1 * p2 % p3), _p12 = p1 * p2 % p; //三模, 1 关于 2
    逆, 1*2 关于 3 逆, 1*2 mod 3
int r[N];
struct P
{
    ull v1, v2, v3;
    P operator+(const P &o) const { return {v1 + o.v1, v2 + o.v2, v3 + o.v3}; }
    P operator-(const P &o) const { return {v1 + p1 - o.v1, v2 + p2 - o.v2, v3 + p3 - o.v3}; }
}
P operator*(const P &o) const { return {v1 * o.v1, v2 * o.v2, v3 * o.v3}; }
void operator+=(const P &o) { v1 += o.v1, v2 += o.v2, v3 += o.v3; }
void operator-=(const P &o) { v1 += p1 - o.v1, v2 += p2 - o.v2, v3 += p3 - o.v3; }
void operator*=(const P &o) { v1 *= o.v1, v2 *= o.v2, v3 *= o.v3; }
void mod() { v1 %= p1, v2 %= p2, v3 %= p3; }
};
P w[N];
void init(int n)
{
    static int pr = 0, pw = 0;
    if (pr == n) return;
    int b = __lg(n) - 1, i, j, k;
    for (i = 1; i < n; i++) r[i] = r[i >> 1] >> 1 | (i & 1) << b;
    if (pw < n)
    {
        for (j = 1; j < n; j = k)
        {
            k = j * 2;
            P wn = {ksm<p1>(g, (p1 - 1) / k), ksm<p2>(g, (p2 - 1) / k), ksm<p3>(g, (p3 - 1) /
k)}};
            w[j] = {1, 1, 1};
            for (i = j + 1; i < k; i++) w[i] = w[i - 1] * wn, w[i].mod();
        }
        pw = n;
    }
    pr = n;
}
void dft(vector<P> &a, int o = 0)
{
    int n = a.size(), i, j, k;
    P *f, *g, *wn, *b = a.data(), x, y;
    init(n);
    for (i = 1; i < n; i++) if (i < r[i]) swap(a[i], a[r[i]]);
    for (k = 1; k < n; k *= 2)
    {
        wn = w + k;
        for (i = 0; i < n; i += k * 2)
        {
            f = b + i; g = b + i + k;
            for (j = 0; j < k; j++)
            {
                y = g[j] * wn[j];
                y.mod();
                g[j] = f[j] - y;
                f[j] += y;
            }
        }
    }
    if (k * 2 == n || k == 1 << 14) for (P &x : a) x.mod();
}

```

```

    }
    if (o)
    {
        x = {ksm<p1>(n), ksm<p2>(n), ksm<p3>(n)};
        for (P &y : a) y *= x, y.mod();
        reverse(1 + all(a));
    }
}
struct Q :vector<ull>
{
    Q(int x = 1) :vector(x) { }
    Q &operator%=(int n) { resize(n); return *this; }
};
Q &operator*=(Q &f, const Q &g)
{
    int n = f.size() + g.size() - 1, m = cal(n), i;
    vector<P> F(m, {0, 0, 0}), G(m, {0, 0, 0});
    for (i = 0; i < f.size(); i++) F[i] = {f[i] % p1, f[i] % p2, f[i] % p3};
    for (i = 0; i < g.size(); i++) G[i] = {g[i] % p1, g[i] % p2, g[i] % p3};
    dft(F); dft(G);
    for (i = 0; i < m; i++) F[i] *= G[i], F[i].mod();
    dft(F, 1);
    f %= n;
    ull x;
    for (i = 0; i < n; i++)
    {
        auto [r1, r2, r3] = F[i];
        x = (r2 + p2 - r1) * inv_p1 % p2 * p1 + r1;
        f[i] = ((x + p3 - r3) % p3 * (p3 - inv_p12) % p3 * _p12 + x) % p;
    }
    return f;
} //5e5 440ms
Q operator*(Q f, const Q &g) { return f *= g; }
}
using MTT::p;
using poly = MTT::Q;

```

3.26 FFT

```

namespace FFT
{
#define all(x) (x).begin(), (x).end()
    typedef double db;
    const int N = 1 << 21;
    const db pi = 3.14159265358979323846;
    struct comp
    {
        db x, y;
        comp operator+(const comp &o) const { return {x + o.x, y + o.y}; }
        comp operator-(const comp &o) const { return {x - o.x, y - o.y}; }
        comp operator*(const comp &o) const { return {x * o.x - y * o.y, o.x * y + x * o.y}; }
        comp operator*(const db &o) const { return {x * o, y * o}; }
        void operator*=(const comp &o) { *this = {x * o.x - y * o.y, o.x * y + x * o.y}; }
        void operator*=(const db &o) { x *= o; y *= o; }
        void operator/=(const db &o) { x /= o; y /= o; }
    };
}

```

```

long long dtol(const double &x) { return fabs(round(x)); }
const comp I{0, -1};
ostream &operator<<(ostream &cout, const comp &o) { cout << o.x; if (o.y >= 0) cout << '+';
return cout << o.y << 'i'; }
int r[N];
char c;
comp Wn[N];
void init(int n)
{
    static int preone = -1;
    if (n == preone) return;
    preone = n;
    int b, i;
    b = __builtin_ctz(n) - 1;
    for (i = 1; i < n; i++) r[i] = r[i >> 1] >> 1 | (i & 1) << b;
    for (i = 0; i < n; i++) Wn[i] = {cos(pi * i / n), sin(pi * i / n)};
}
int cal(int x) { return 1u << 32 - __builtin_clz(max(x, 2) - 1); }
struct Q
{
    vector<comp> a;
    int deg;
    comp *pt() { return a.data(); }
    Q(int n = 0)
    {
        deg = n;
        a.resize(cal(n));
    }
    void dft(int xs = 0)//1,0
    {
        int i, j, k, l, n = a.size(), d;
        comp w, wn, b, c, *f = pt(), *g, *a = f;
        init(n);
        if (xs) reverse(a + 1, a + n);//spe
        for (i = 0; i < n; i++) if (i < r[i]) swap(a[i], a[r[i]]);
        for (i = 1, d = 0; i < n; i = 1, d++)
        {
            //wn={cos(pi/i),(xs?-1:1)*sin(pi/i)};
            l = i << 1;
            for (j = 0; j < n; j += l)
            {
                //w={1,0};
                f = a + j; g = f + i;
                for (k = 0; k < i; k++)
                {
                    w = Wn[k * (n >> d)];
                    b = f[k]; c = g[k] * w;
                    f[k] = b + c;
                    g[k] = b - c;
                    //w*=wn;
                }
            }
        }
        if (xs) for (i = 0; i < n; i++) a[i] /= n;
    }
    void operator|=(Q o)
    {

```



```

        int n = deg + o.deg - 1, m = cal(n), i;
        a.resize(m); o.a.resize(m);
        dft(); o.dft();
        for (i = 0; i < m; i++) a[i] *= o.a[i];
        dft(1);
        for (i = n; i < m; i++) a[i] = { };
        deg = n;
    }
    Q operator|(Q o) const { o |= *this; return o; }
};
Q mul(Q a, const Q &b)//三次变两次, 仅实数, 注意精度
{
    int n = a.deg + b.deg - 1, m = cal(n), i;
    a.a.resize(m);
    for (i = 0; i < b.deg; i++) a.a[i] = {a.a[i].x, b.a[i].x};
    a.dft();
    for (i = 0; i < m; i++) a.a[i] *= a.a[i];
    a.dft(1);
    for (i = 0; i < n; i++) a.a[i] = {a.a[i].y * .5};
    for (i = n; i < m; i++) a.a[i] = { };
    a.deg = n;
    return a;
}
void ddt(Q &a, Q &b)//double dft, 仅实数, 注意精度
{
    comp x, y;
    int n = a.a.size(), i;
    assert(n == b.a.size());
    for (i = 0; i < n; i++) a.a[i] = {a.a[i].x, b.a[i].x};
    a.dft();
    for (i = 0; i < n; i++) b.a[i] = {a.a[i].x, -a.a[i].y};
    reverse(b.pt() + 1, b.pt() + n);
    for (i = 0; i < n; i++)
    {
        x = a.a[i]; y = b.a[i];
        a.a[i] = (x + y) * .5;
        b.a[i] = (y - x) * .5 * I;
    }
}
}
using FFT::dtol;

```

3.27 约数个数和

$$O(\sqrt[3]{n} \log n)。$$

```

#include"bits/stdc++.h"
using ll = long long;
using lll = __int128;
using namespace std;

void myw(lll x) {
    if (!x) return;
    myw(x / 10); printf("%d", (int)(x % 10));
}

struct vec {

```

```

    ll x, y;
    vec(ll x0 = 0, ll y0 = 0) { x = x0, y = y0; }
    vec operator +(const vec b) { return vec(x + b.x, y + b.y); }
};

ll N;
vec stk[1000005]; int len;
vec P;
vec L, R;

bool ninR(vec a) { return N < (lll)a.x * a.y; }
bool steep(ll x, vec a) { return (lll)N * a.x <= (lll)x * x * a.y; }

lll Solve() {
    len = 0;
    ll cbr = cbrt(N), sqr = sqrt(N);
    P.x = N / sqr, P.y = sqr + 1;
    lll ans = 0;
    stk[++len] = vec(1, 0); stk[++len] = vec(1, 1);
    while (1) {
        L = stk[len--];
        while (ninR(vec(P.x + L.x, P.y - L.y)))
            ans += (lll)P.x * L.y + (lll)(L.y + 1) * (L.x - 1) / 2,
            P.x += L.x, P.y -= L.y;
        if (P.y <= cbr) break;
        R = stk[len];
        while (!ninR(vec(P.x + R.x, P.y - R.y))) L = R, R = stk[--len];
        while (1) {
            vec mid = L + R;
            if (ninR(vec(P.x + mid.x, P.y - mid.y))) R = stk[++len] = mid;
            else if (steep(P.x + mid.x, R)) break;
            else L = mid;
        }
    }
    for (int i = 1; i < P.y; i++) ans += N / i;
    return ans * 2 - sqr * sqr;
}

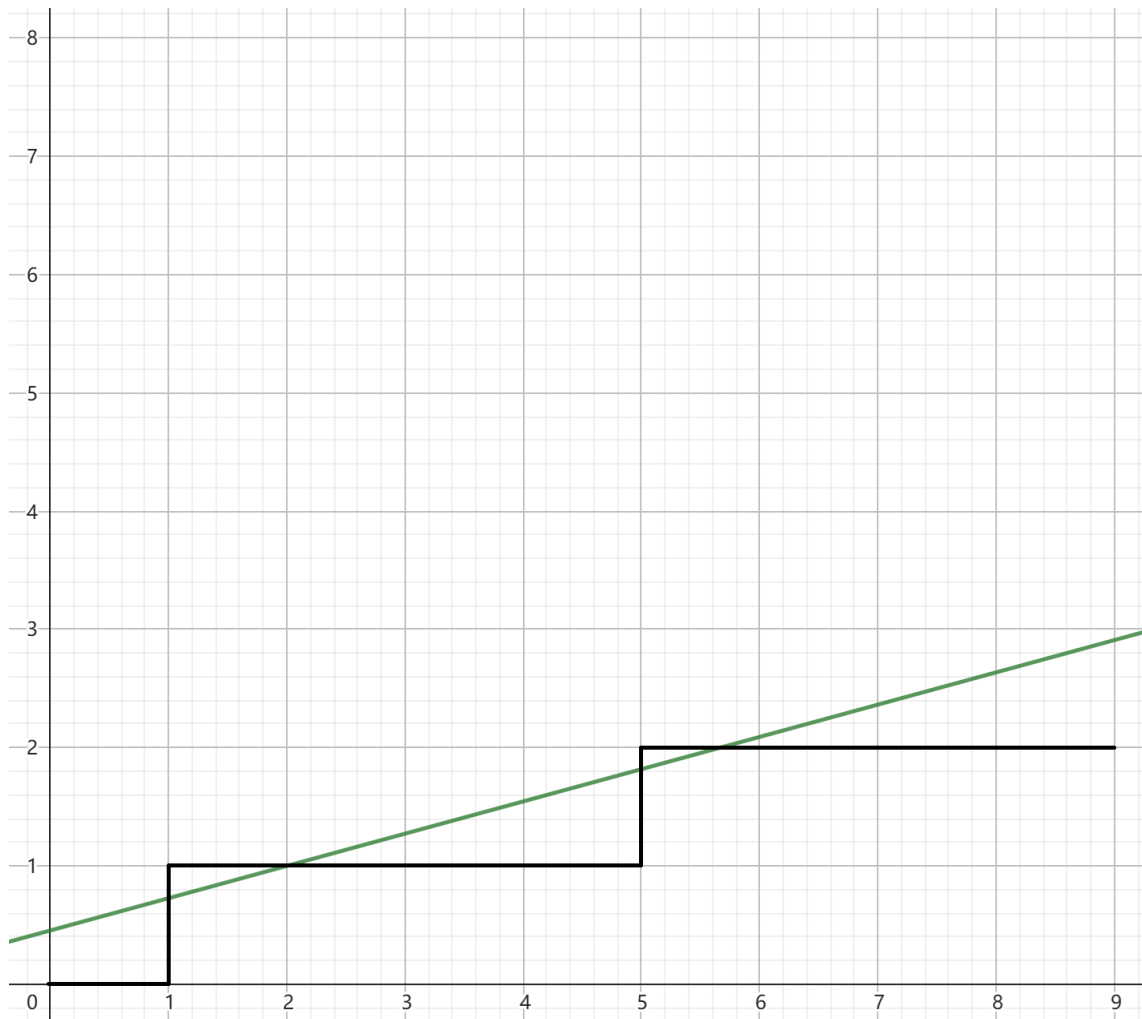
int T;

int main() {
    scanf("%d", &T);
    while (T--) {
        scanf("%lld", &N);
        myw(Solve()); printf("\n");
    }
}

```

3.28 万能欧几里得/min of mod of linear

题意/接口: $\sum_{x=0}^n \lfloor \frac{ax+b}{m} \rfloor \quad (0 \leq a, b < m)$ 。



原理：考虑紧贴着斜线的折线的答案。每个 `nd` 表示的是一段折线，你需要实现 `operator+` 来计算出拼接两个折线之后的答案。注意，需要实现默认构造。

你需要传入的 `dx` 和 `dy` 表示向上和向右的折线的答案（也就是边界）。图中对应的折线为 `RURRRRURRRRR`。

以对 x 求和为例，通常的处理手段是在 `dx` 时计算 $x = 1$ 的答案，在 `dy` 时更新辅助数组。

注意，如果 $x = 0$ 处有答案，你需要手动计算它，并保证传入的 $b < m$ 。

如果这段向上的竖线对后面有影响，你可以在一开始先加一个 `ksm(dy, ...)`，但你仍然可能需要手动计算 $x = 0$ 。

如果答案要取模，特别注意 y 有可能比模数大！不卡时间请使用 `int128`。

```
struct nd
{
    ll x, y, sy;
    nd operator+(const nd &o) const
    {
        return {x + o.x, y + o.y, sy + o.sy + y * o.x};
    }
};
nd ksm(nd a, ll k)
{
    nd res{ };
    while (k)
    {
        if (k & 1) res = res + a;
        a = a + a; k >>= 1;
    }
    return res;
}
```

```

}
nd sol(int a, int b, int m, int n, nd dx, nd dy)//[0,n] (ax+b)/m 0<=b<m
{
    if (!n) return { };
    if (a >= m) return sol(a % m, b, m, n, ksm(dy, a / m) + dx, dy);
    ll c = ((ll)n * a + b) / m;
    if (!c) return ksm(dx, n);
    ll cnt = n - ((ll)m * c - b - 1) / a;
    return ksm(dx, (m - b - 1) / a) + dy + sol(m, (m - b - 1) % a, a, c - 1, dy, dx) + ksm(dx,
cnt);
}
ll sum_of_floor_of_linear(int a, int b, int m, int n)//[0,n] sum((ax+b)/m)
{
    nd dx = {1, 0, 0}, dy = {0, 1, 0};
    int nb = (b % m + m) % m;
    return sol(a, nb, m, n, dx, dy).sy + (ll)(b - nb) / m * (n + 1);
}
int min_of_mod_of_linear(int a, int b, int p, int n)//[0,n] min((ax+b) mod p)
{
    ll s = sum_of_floor_of_linear(a, b, p, n);
    int l = 0, r = p - 1, mid;
    while (l < r)
    {
        mid = (l + r + 1) / 2;
        if (sum_of_floor_of_linear(a, b - mid, p, n) >= s) l = mid;
        else r = mid - 1;
    }
    return l;
}

```

3.29 高斯整数类

圆上整点的基础。

```

ll roundiv(ll x, ll y)
{
    return x >= 0 ? (x + y / 2) / y : (x - y / 2) / y;
}
struct Q
{
    ll x, y;
    Q operator~() const { return {x, -y}; }
    ll len2() const { return x * x + y * y; }
    Q operator+(const Q &o) const { return {x + o.x, y + o.y}; }
    Q operator-(const Q &o) const { return {x - o.x, y - o.y}; }
    Q operator*(const Q &o) const { return {x * o.x - y * o.y, x * o.y + y * o.x}; }
    Q operator/(const Q &o) const
    {
        Q t = *this * ~o;
        ll l = o.len2();
        return {roundiv(t.x, l), roundiv(t.y, l)};
    }
    Q operator%(const Q &o) const { return *this - *this / o * o; }
};
Q gcd(Q a, Q b)
{
    if (a.len2() > b.len2()) swap(a, b);

```

```

while (a.len2())
{
    b = b % a;
    swap(a, b);
}
return b;
}

```

3.30 Miller Rabin/Pollard Rho

1s: 200 组 10^{18} 。

如果你只需要做 `int` 以内的分解，你可以改为

```

using ll = int;
using lll = long long;

```

```

namespace pr
{
    using ll = long long;
    using lll = __int128;
    using pa = pair<ll, int>;
    ll ksm(ll x, ll y, const ll p)
    {
        ll r = 1;
        while (y)
        {
            if (y & 1) r = (lll)r * x % p;
            x = (lll)x * x % p; y >>= 1;
        }
        return r;
    }
}

namespace miller
{
    const int p[7] = {2, 3, 5, 7, 11, 61, 24251};
    ll s, t;
    bool test(ll n, int p)
    {
        if (p >= n) return 1;
        ll r = ksm(p, t, n), w;
        for (int j = 0; j < s && r != 1; j++)
        {
            w = (lll)r * r % n;
            if (w == 1 && r != n - 1) return 0;
            r = w;
        }
        return r == 1;
    }
    bool prime(ll n)
    {
        if (n < 2 || n == 46'856'248'255'981) return 0;
        for (int i = 0; i < 7; ++i) if (n % p[i] == 0) return n == p[i];
        s = __builtin_ctz(n - 1); t = n - 1 >> s;
        for (int i = 0; i < 7; ++i) if (!test(n, p[i])) return 0;
        return 1;
    }
}

```

```

using miller::prime;
mt19937_64 rnd(chrono::steady_clock::now().time_since_epoch().count());
namespace rho
{
    void nxt(ll &x, ll &y, ll &p) { x = ((lll)x * x + y) % p; }
    ll find(ll n, ll C)
    {
        ll l, r, d, p = 1;
        l = rnd() % (n - 2) + 2, r = 1;
        nxt(r, C, n);
        int cnt = 0;
        while (l ^ r)
        {
            p = (lll)p * llabs(l - r) % n;
            if (!p) return gcd(n, llabs(l - r));
            ++cnt;
            if (cnt == 127)
            {
                cnt = 0;
                d = gcd(llabs(l - r), n);
                if (d > 1) return d;
            }
            nxt(l, C, n); nxt(r, C, n); nxt(r, C, n);
        }
        return gcd(n, p);
    }
    vector<pa> w;
    vector<ll> d;
    void dfs(ll n, int cnt)
    {
        if (n == 1) return;
        if (prime(n)) return w.emplace_back(n, cnt), void();
        ll p = n, C = rnd() % (n - 1) + 1;
        while (p == 1 || p == n) p = find(n, C++);
        int r = 1; n /= p;
        while (n % p == 0) n /= p, ++r;
        dfs(p, r * cnt); dfs(n, cnt);
    }
    vector<pa> getw(ll n)
    {
        w = vector<pa>(0); dfs(n, 1);
        if (n == 1) return w;
        sort(w.begin(), w.end());
        int i, j;
        for (i = 1, j = 0; i < w.size(); i++) if (w[i].first == w[j].first) w[j].second += w[i].second; else w[++j] = w[i];
        w.resize(j + 1);
        return w;
    }
    void dfss(int x, ll n)
    {
        if (x == w.size()) return d.push_back(n), void();
        dfss(x + 1, n);
        for (int i = 1; i <= w[x].second; i++) dfss(x + 1, n * w[x].first);
    }
    vector<ll> getd(ll n)
    {

```

```
        getw(n); d = vector<ll>(0); dfss(0, 1);
        sort(d.begin(), d.end());
        return d;
    }
}
using rho::getw, rho::getd;
using miller::prime;
ull rand_prime(ull l, ull r)//[l, r]
{
    assert(l <= r);
    static mt19937_64 rnd(234);
    ull p = 0;
    while (!prime(p)) p = rnd() % (r - l + 1) + l;
    return p;
}
}
using pr::getw, pr::getd, pr::prime, pr::rand_prime;
```

4 字符串

4.1 字典树 (trie 树)

```
struct trie
{
    const static int N = 1e6 + 2, M = 52;
    int c[N][M], sz[N]; //sz 维护有多少个以当前字符串为前缀的字符串。
    int cnt = 1;
    void insert(string s)
    {
        int u = 1;
        ++sz[u];
        for (char ch : s)
        {
            assert(ch >= 0 && ch < M);
            int &v = c[u][ch];
            if (!v) v = ++cnt;
            u = v;
            ++sz[u];
        }
        //此时 u 是字符串结束位置。你可以在此存储结点信息。
    }
    int match(string s) //返回字符串结束位置。可能为 0。
    {
        int u = 1;
        for (char ch : s)
        {
            assert(ch >= 0 && ch < M);
            u = c[u][ch];
            if (!u) return 0;
        }
        return u;
    }
    void clear()
    {
        memset(c, 0, (cnt + 1) * sizeof c[0]);
        memset(sz, 0, (cnt + 1) * sizeof sz[0]);
        cnt = 1;
    }
} t;
```

4.2 AC 自动机

注意 AC 自动机与 trie 不同的地方在于，根必须是 0。

题意：给你一个文本串 S 和 n 个模式串 $T_1 \sim T_n$ ，请你分别求出每个模式串 T_i 在 S 中出现的次数。

```
struct AC
{
    const static int N = 3e6 + 2, M = 26;
    int c[N][M], sz[N], pos[N], f[N], app[N]; //sz 维护有多少个以当前字符串为前缀的字符串。
    int cnt = 0, id = 0;
    vector<int> q;
    void insert(string s)
    {
```



```

    int u = 0;
    ++sz[u];
    for (char ch : s)
    {
        assert(ch >= 0 && ch < M);
        int &v = c[u][ch];
        if (!v) v = ++cnt;
        u = v;
        ++sz[u];
    }
    pos[id++] = u;
    //此时 u 是字符串结束位置。你可以在此存储结点信息。
}
vector<int> match(string s)//返回答案。复杂度 O(结点数)
{
    int u = 0, i;
    for (char ch : s)
    {
        assert(ch >= 0 && ch < M);
        u = c[u][ch];
        ++app[u];
    }
    for (int u : q) app[f[u]] += app[u];
    vector<int> r(id);
    for (i = 0; i < id; i++) r[i] = app[pos[i]];
    memset(app, 0, (cnt + 1) * sizeof app[0]);
    return r;
}
void clear()
{
    memset(c, 0, (cnt + 1) * sizeof c[0]);
    memset(f, 0, (cnt + 1) * sizeof f[0]);
    memset(sz, 0, (cnt + 1) * sizeof sz[0]);
    cnt = id = 0;
}
void build()
{
    q.clear();
    int ql = 0;
    for (int i = 0; i < M; i++) if (c[0][i]) q.push_back(c[0][i]);
    while (ql < q.size())
    {
        int u = q[ql++];
        for (int i = 0; i < M; i++) if (c[u][i])
        {
            q.push_back(c[u][i]);
            f[c[u][i]] = c[f[u]][i];
        }
        else c[u][i] = c[f[u]][i];
    }
    reverse(all(q));
}
} s;
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    int n, i;

```

```

cin >> n;
while (n--)
{
    string t;
    cin >> t;
    for (char &c : t) c -= 'a';
    s.insert(t);
}
s.build();
string t;
cin >> t;
for (char &c : t) c -= 'a';
auto res = s.match(t);
for (int x : res) cout << x << '\n';
}

```

4.3 hash

在调试时，可以把 `base` 设置为 10 的幂方便输出。可能建议把第一个模数也设置为 1，但未测试是否有奇怪的问题。但要注意，此时不应当使用接近 10 的幂次的模数。

$O(n)$, $O(n)$ 。

建议抄 `int128` 版本。目前来看，`int128` 版本的既好写又快。

`__int128` 版本：

```

namespace sh
{
    using ull = unsigned long long;
    using lll = __uint128_t;
    const int N = 1e6 + 5;
    const ull p = (ull)1e18 - 11, b = 137;
    ull m[N];
    int i = []() {
        m[0] = 1;
        for (int i = 1; i < N; i++) m[i] = (lll)m[i - 1] * b % p;
        return 0;
    }();
    struct str
    {
        int n;
        vector<ull> a;
        template<class T> str(const vector<T> &s) : n(s.size()), a(n + 1)
        {
            for (i = 0; i < n; i++) a[i + 1] = ((lll)a[i] * b + s[i]) % p;
        }
        template<class T> str(const basic_string<T> &s) : n(s.size()), a(n + 1) // 直接去掉模板换成
string 也可以
        {
            for (i = 0; i < n; i++) a[i + 1] = ((lll)a[i] * b + s[i]) % p;
        }
        ull getv(int l, int r) // [l, r)
        {
            return (a[r] + (lll)(p - a[l]) * m[r - l]) % p;
        }
        int lcp(int i, int j)
        {

```

```

        if (i == j) return n - i;
        int l = 0, r = n - max(i, j), mid;
        while (l < r)
        {
            mid = (l + r + 1) >> 1;
            if (getv(i, i + mid) == getv(j, j + mid)) l = mid;
            else r = mid - 1;
        }
        return l;
    }
};
}
using sh::str;

```

双模数版本:

特别注意这里 m 数组预处理的不是幂次, 而是幂次的相反数。

其返回值是两个 32 位数拼接而成的, 要改动比较麻烦。

```

namespace sh
{
    using ui = unsigned int;
    using ull = unsigned long long;
    const int N = 1e6 + 5;
    const ui p1 = 2'034'452'107, p2 = 2'013'074'419;
    struct pa
    {
        ui v1, v2;
        pa(ui v = 0) : v1(v), v2(v) { }
        pa(ui v1, ui v2) : v1(v1), v2(v2) { }
        pa operator*(const pa &o) const { return pa(1llu * v1 * o.v1 % p1, 1llu * v2 * o.v2 % p2)
    }; }
    pa fma(const pa &a, const pa &b, const pa &c) { return pa((1llu * a.v1 * b.v1 + c.v1) % p1,
    (1llu * a.v2 * b.v2 + c.v2) % p2); }
    const pa b = {137, 149}, inv = {1'603'801'661, 1'024'053'074};
    pa m[N];
    int i = []() {
        m[0] = {p1 - 1, p2 - 1};
        for (int i = 1; i < N; i++) m[i] = m[i - 1] * b;
        return 0;
    }();
    struct str
    {
        int n;
        vector<pa> a;
        template<class T> str(const vector<T> &s) : n(s.size()), a(n + 1)
        {
            for (i = 0; i < n; i++) a[i + 1] = fma(a[i], b, s[i]);
        }
        template<class T> str(const basic_string<T> &s) : n(s.size()), a(n + 1) // 直接去掉模板换成
string 也可以
        {
            for (i = 0; i < n; i++) a[i + 1] = fma(a[i], b, s[i]);
        }
        ull getv(int l, int r) // [l, r)
        {
            auto [x, y] = fma(a[l], m[r - l], a[r]);
            return (ull)x << 32 | y;
        }
    };
}

```

```

    }
    int lcp(int i, int j)
    {
        if (i == j) return n - i;
        int l = 0, r = n - max(i, j), mid;
        while (l < r)
        {
            mid = (l + r + 1) >> 1;
            if (getv(i, i + mid) == getv(j, j + mid)) l = mid;
            else r = mid - 1;
        }
        return l;
    }
};
using sh::str;

```

4.4 KMP

$O(n)$, $O(n)$ 。

```

template<class T> struct str//[0,n)
{
    int n;
    vector<int> nxt;//长度 - 1, 就是 father
    T s;
    str(const T &s) : n(s.size()), nxt(n, -1), s(s)
    {
        int i, j = -1;
        for (i = 1; i < n; i++)
        {
            while (j != -1 && s[i] != s[j + 1]) j = nxt[j];
            nxt[i] = j += s[i] == s[j + 1];
        }
    }
    vector<int> match(const T &t)//find s(str) in t (start pos)
    {
        int m = t.size(), i, j = -1;
        vector<int> r;
        for (i = 0; i < m; i++)
        {
            while (j != -1 && t[i] != s[j + 1]) j = nxt[j];
            if ((j += t[i] == s[j + 1]) == n - 1) j = nxt[j], r.push_back(i - n + 1);
        }
        return r;
    }
};

```

4.5 manacher

$O(n)$, $O(n)$ 。

ex[i] 表示以 $i/2$ 为回文中心的回文串长度。

如果 t 可能包含 $\$$ #, 你需要修改字符。

```
vector<int> manacher(const string &t)
```

```

{
    string S = "$#";
    int n = t.size(), i, r = 1, m = 0;
    for (i = 0; i < n; i++) S += t[i], S += '#';
    S += '#';
    char *s = S.data() + 2;
    n = n * 2 - 1;
    vector<int> ex(n);
    ex[0] = 2;
    for (i = 1; i < n; i++)
    {
        ex[i] = i < r ? min(ex[m * 2 - i], r - i + 1) : 1;
        while (s[i + ex[i]] == s[i - ex[i]]) ++ex[i];
        if (i + ex[i] - 1 > r) r = i + ex[m = i] - 1;
    }
    for (int &x : ex) --x;
    return ex;
}

pair<vector<int>, vector<pair<int, int>>> distinct_palindrome(const string &t)
// [1,r)
{
    string S = "$#";
    int n = t.size(), i, r = 1, m = 0;
    for (i = 0; i < n; i++) (S += t[i]) += '#';
    S += '#';
    char *s = S.data() + 2;
    n = n * 2 - 1;
    vector<int> ex(n);
    vector<pair<int, int>> res = {{0, 1}};
    ex[0] = 2;
    for (i = 1; i < n; i++)
    {
        if (i < r) ex[i] = min(ex[m * 2 - i], r - i + 1);
        else
        {
            ex[i] = 1;
            if (!(i & 1)) res.push_back({i / 2, i / 2 + 1});
        }
        while (s[i + ex[i]] == s[i - ex[i]])
        {
            if (s[i - ex[i]] != '#')
                res.push_back({i - ex[i] >> 1, i + ex[i] + 2 >> 1});
            ++ex[i];
        }
        if (i + ex[i] - 1 > r) r = i + ex[m = i] - 1;
    }
    for (int &x : ex) --x;
    return {ex, res};
}

```

4.6 SA

$O((n + \sum) \log n)$, $O(n + \sum)$ 。

功能：查询两个后缀的 lcp。单次询问复杂度 $O(1)$ 。

下标从 0 开始。

```

struct SA
{
    int n;
    vector<vector<int>> st;
    vector<int> sa, rk, h;
    int lcp(int x, int y)
    {
        if (x == y) return n - x;
        x = rk[x]; y = rk[y];
        if (x > y) swap(x, y);
        ++x;
        int z = __lg(y - x + 1);
        return min(st[z][x], st[z][y - (1 << z) + 1]);
    }
    SA(vector<int> a) : n(a.size()), st(__lg(n) + 1, vector<int>(n + 1)), sa(n), h(n)
    {
        int i, j, m, cnt;
        m = *min_element(all(a));
        for (int &x : a) x -= m;
        m = *max_element(all(a)) + 1;
        vector<int> id(n), s(max(n, m));
        rk = a;
        for (int i : rk) ++s[i];
        partial_sum(all(s), s.begin());
        for (i = n - 1; i >= 0; i--) sa[--s[rk[i]]] = i;
        for (j = 1; j <= n; j <= 1)
        {
            fill_n(s.begin(), m, 0);
            cnt = j;
            iota(all(id) - (n - j), n - j);
            for (int i : sa) if (i >= j) id[cnt++] = i - j;
            for (int i : rk) ++s[i];
            partial_sum(all(s), s.begin());
            for (i = n - 1; i >= 0; i--) sa[--s[rk[id[i]]]] = id[i];
            id[sa[0]] = cnt = 0;
            for (i = 1; i < n; i++)
                if (sa[i] + j < n && sa[i - 1] + j < n && rk[sa[i]] == rk[sa[i - 1]] && rk[sa[i]
+ j] == rk[sa[i - 1] + j])
                    id[sa[i]] = cnt;
                else
                    id[sa[i]] = ++cnt;
            swap(rk, id);
            if ((m = cnt + 1) == n) break;
        }
        j = 0;
        for (i = 0; i < n; i++) if (rk[i])
        {
            cnt = sa[rk[i] - 1];
            while (i + j < n && cnt + j < n && a[i + j] == a[cnt + j]) ++j;
            h[rk[i]] = j;
            if (j) --j;
        }
        st[0] = h;
        for (j = 0; j < __lg(n); j++)
            for (i = 0, m = n - (1 << j + 1); i <= m; i++)
                st[j + 1][i] = min(st[j][i], st[j][i + (1 << j)]);
    }
}

```

```

template<class T> SA(const T &s) :SA([&]() {
    vector<int> a; a.reserve(s.size());
    for (auto x : s) a.push_back(x);
    return a;
}()) { }
};

```

4.7 SAM

$O(n\Sigma)$, $O(2n\Sigma)$ 。

```

template<int M> struct sam//M: 字符集大小
{
    vector<array<int, M>> c;
    vector<int> len, fa, ep;
    int np, cd;
    sam() :c(2), len(2), fa(2), ep(2), np(1), cd(0) { }
    void insert(int ch)
    {
        int p = np, q, nq;
        np = c.size();
        len.push_back(++cd);
        fa.push_back(0);
        c.push_back({ });
        ep.push_back(cd);
        while (p && !c[p][ch]) c[p][ch] = np, p = fa[p];
        if (!p)
        {
            fa[np] = 1;
            return;
        }
        q = c[p][ch];
        if (len[q] == len[p] + 1)
        {
            fa[np] = q;
            return;
        }
        nq = c.size();
        len.push_back(len[p] + 1);
        c.push_back(c[q]);
        fa.push_back(fa[q]);
        ep.push_back(ep[q]);
        fa[np] = fa[q] = nq;
        c[p][ch] = nq;
        while (c[p = fa[p]][ch] == q) c[p][ch] = nq;
    }
    vector<int> match(const string &s)//返回每个前缀最长匹配长度
    {
        vector<int> r;
        r.reserve(s.size());
        int p = 1, nl = 0;
        for (auto ch : s)
        {
            if (c[p][ch]) ++nl, p = c[p][ch];
            else
            {
                while (p && c[p][ch] == 0) p = fa[p];
            }
        }
    }
};

```

```

        if (p == 0) p = 1, nl = 0; else nl = len[p] + 1, p = c[p][ch];
    }
    r.push_back(nl);
}
return r;
}
array<int, 3> max_match(const string &s)//返回长度, 结尾(开)
{
    array<int, 3> r{0, 0, 0};
    int p = 1, nl = 0, i = 0;
    for (auto ch : s)
    {
        if (c[p][ch]) ++nl, p = c[p][ch];
        else
        {
            while (p && c[p][ch] == 0) p = fa[p];
            if (p == 0) p = 1, nl = 0; else nl = len[p] + 1, p = c[p][ch];
        }
        cmax(r, array{nl, ep[p], i + 1});
        ++i;
    }
    if (r[0] == 0) return { };
    return r;
}
};

```

4.8 ukkonen 后缀树

$O(n)$, $O(2n \sum)$ 。

```

void dfs(int x, int lf)
{
    if (!fir[x])
    {
        siz[x][1] = 1;
        return;
    }
    int i, j;
    for (i = fir[x]; i; i = nxt[i])
    {
        j = c[x][lj[i]];
        if ((f[j] <= m) && (t[j] >= m)) ++siz[x][0];
        dfs(zd[j], t[j] - f[j] + 1);
        siz[x][0] += siz[zd[j]][0];
        siz[x][1] += siz[zd[j]][1];
        if ((t[j] == n) && (f[j] <= m)) --siz[x][1];
    }
    ans += (ll)siz[x][0] * siz[x][1] * lf;
}

void add(int a, int b, int cc, int d)
{
    zd[++bbs] = b;
    t[bbs] = d;
    c[a][s[f[bbs] = cc]] = bbs;
}

void add(int x, int y)
{
}

```



```

    lj[++bs] = y;
    nxt[bs] = fir[x];
    fir[x] = bs;
}
s[++m] = 26;
fa[1] = point = ds = 1;
for (i = 1; i <= m; i++)
{
    ad = 0; ++remain;
    while (remain)
    {
        if (r == 0) edge = i;
        if ((j = c[point][s[edge]]) == 0)
        {
            fa[++ds] = 1;
            fa[ad] = point;
            add(ad = point, ds, edge, m);
            add(point, s[edge]);
        }
        else
        {
            if ((t[j] != m) && (t[j] - f[j] + 1 <= r))
            {
                r -= t[j] - f[j] + 1;
                edge += t[j] - f[j] + 1;
                point = zd[j];
                continue;
            }
            if (s[f[j] + r] == s[i]) { ++r; fa[ad] = point; break; }
            fa[fa[ad] = ++ds] = 1;
            add(ad = ds, zd[j], f[j] + r, t[j]);
            add(ds, s[i]); add(ds, s[f[j] + r]); fa[++ds] = 1;
            add(ds - 1, ds, i, m);
            zd[j] = ds - 1; t[j] = f[j] + r - 1;
        }
        --remain;
        if ((r) && (point == 1))
        {
            --r; edge = i - remain + 1;
        }
        else point = fa[point];
    }
}
for (i = 1; i <= ds; i++) for (j = fir[i]; j; j = nxt[j]) { len[j] = t[c[i][lj[j]]] - f[c[i][lj[j]]] + 1; lj[j] = zd[c[i][lj[j]]]; }

```

4.9 ukkonen 后缀树（重构）

```

struct suffixtree
{
    const static int M = 27;
    struct P
    {
        int v, w;
    };
    struct Q

```

```

{
    int f, t, v; //t=0: n
};
vector<Q> edges;
vector<vector<P>> e;
vector<array<int, M>> c;
vector<int> s, fa, dep, siz;
int n, point, ds, remain, r, edge;
bool bd;
suffixtree() :c(2), fa({0, 1}), edges(1), e(2)
{
    n = remain = r = edge = bd = 0;
    point = ds = 1;
}
suffixtree(const string &s) :c(2), fa({0, 1}), edges(1), e(2)
{
    n = remain = r = edge = bd = 0;
    point = ds = 1;
    reserve(s.size());
    for (auto c : s) insert(c - 'a');
    insert(26);
}
void reserve(int len)
{
    ++len;
    s.reserve(len);
    len = len * 2 + 2;
    c.reserve(len);
    fa.reserve(len);
    e.reserve(len);
}
inline void add(int a, int b, int cc, int d)
{
    assert(edges.size());
    c[a][s[cc]] = edges.size();
    edges.push_back({cc, d, b});
}
void insert(int ch) //[0, |S|)
{
    assert(ds == fa.size() - 1 && ds == c.size() - 1 && n == s.size() && ds == e.size() - 1);
    assert(ch >= 0 && ch < M);
    s.push_back(ch);
    int ad = 0;
    ++remain;
    while (remain)
    {
        if (!r) edge = n;
        if (int m = c[point][s[edge]]; !m)
        {
            assert(!m);
            fa.push_back(1); c.push_back({ }); e.push_back({ });
            fa[ad] = point;
            add(ad = point, ++ds, edge, -1);
            e[point].push_back({s[edge]});
            //add(point, s[edge]);
        }
        else

```

```

{
    assert(m);
    auto [f, t, v] = edges[m];
    if (t >= 0 && t - f + 1 <= r)
    {
        assert(t != n);
        r -= t - f + 1;
        edge += t - f + 1;
        point = v;
        continue;
    }
    assert(f + r <= n);
    if (s[f + r] == s[n])
    {
        ++r;
        fa[ad] = point;
        break;
    }
    fa.push_back(1); c.push_back({ }); e.push_back({ });
    fa.push_back(1); c.push_back({ }); e.push_back({ });
    fa[ad] = ++ds;
    add(ad = ds, v, f + r, t);
    e[ds].push_back({s[n]});
    e[ds].push_back({s[f + r]});
    //add(ds, s[n]); add(ds, s[f + r]);
    ++ds; add(ds - 1, ds, n, -1);
    edges[m] = {f, f + r - 1, ds - 1};
}
--remain;
if (r && point == 1)
{
    --r;
    edge = n - remain + 1;
}
else point = fa[point];
}
++n;
}
void build_edge()
{
    bd = 1;

    //其余信息
    dep.resize(ds + 1);
    siz.resize(ds + 1);

    int i, j;
    for (i = 1; i <= ds; i++) for (auto &[v, w] : e[i])
    {
        j = c[i][v];
        v = edges[j].v;
        w = (edges[j].t >= 0 ? edges[j].t : n - 1) - edges[j].f + 1;
    }
}
void out()
{
    int i;

```

```

    for (i = 1; i <= ds; i++) for (int j : c[i]) if (j)
    {
        auto [f, t, v] = edges[j];
        if (t == -1) t = n - 1;
        cerr << i << ' ' << v << ' ';
        //cerr<<i<<" -> "<<v<<": ";
        for (int k = f; k <= t; k++) cerr << char('a' + s[k]);
        cerr << endl;
    }
}
ll ans;
void dfs(int u)
{
    assert(bd);
    ++ans;
    for (auto [v, w] : e[u])
    {
        //dep[v]=dep[u]+w;
        dfs(v);
        ans += w - 1;
    }
}
ll fun()
{
    ans = 0;
    build_edge();
    dfs(1);
    return ans - n;
}
};

```

4.10 Z 函数

表示每个后缀和母串的 lcp。

```

vector<int> Z(const string &s)
{
    int n = s.size(), i, l, r;
    vector<int> z(n);
    z[0] = n;
    for (i = 1, l = r = 0; i < n; i++)
    {
        if (i <= r && z[i - l] < r - i + 1) z[i] = z[i - l];
        else
        {
            z[i] = max(0, r - i + 1);
            while (i + z[i] < n && s[i + z[i]] == s[z[i]]) ++z[i];
        }
        if (i + z[i] - 1 > r) l = i, r = i + z[i] - 1;
    }
    return z;
}

```

4.11 最小表示法

找到一个串的循环同构串中字典序最小的那个，将这个串直接变过去。常见应用：环哈希（基环树哈希）。

如果只需要找到起点下标，在 `rotate` 前返回 $\min\{i, j\}$ 即可。

$O(n)$, $O(1)$ 。

```
template<class T> void min_order(vector<T> &a)
{
    int n = a.size(), i, j, k;
    a.resize(n * 2);
    for (i = 0; i < n; i++) a[i + n] = a[i];
    i = k = 0; j = 1;
    while (i < n && j < n && k < n)
    {
        T x = a[i + k], y = a[j + k];
        if (x == y) ++k; else
        {
            (x > y ? i : j) += k + 1;
            j += (i == j);
            k = 0;
        }
    }
    a.resize(n);
    // [min(i, j), n) + [0, min(i, j))
    rotate(a.begin(), min(i, j) + all(a));
}
```

4.12 带通配符的字符串匹配

原理：匹配等价于 $\sum (f_i - g_i)^2 = 0$ 。带通配符等价于 $\sum f_i g_i (f_i - g_i)^2 = 0$ ，展开即可。

这里也是较为推荐的 NTT 版本，直接实现任意长度的多项式相乘，便于一般情况的运用。不需要提前做任何 init。

```
namespace NTT
{
    const int N = 1 << 22;
    const ui p = 998244353, g = 3;
    inline ui ksm(ui x, ui y)
    {
        ui ans = 1;
        while (y)
        {
            if (y & 1) ans = 1llu * ans * x % p;
            y >>= 1; x = 1llu * x * x % p;
        }
        return ans;
    }
    ui r[N], w[N];
    void ntt(vector<ui> &a)
    {
        int n = a.size(), i, j, k;
        for (i = 0; i < n; i++) if (i < r[i]) swap(a[i], a[r[i]]);
        for (k = 1; k < n; k <= 1)
        {
            for (i = 0; i < n; i += k << 1)
```

```

    {
        for (j = 0; j < k; j++)
        {
            ui x = a[i + j], y = 1llu * a[i + j + k] * w[j + k] % p;
            a[i + j] = (x + y) % p; a[i + j + k] = (x + p - y) % p;
        }
    }
}

vector<ui> mul(vector<ui> a, vector<ui> b)
{
    if (a.size() == 0 || b.size() == 0) return { };
    int m = a.size() + b.size() - 1;
    int n = 1 << __lg(m * 2 - 1);
    int i, j, base = __lg(n) - 1;
    ui inv = ksm(n, p - 2);
    for (i = 1; i < n; i++) r[i] = r[i >> 1] >> 1 | (i & 1) << base;
    for (j = 1; j < n; j <= 1)
    {
        ui wn = ksm(3, (p - 1) / (j << 1));
        w[j] = 1;
        for (i = 1; i < j; i++) w[j + i] = 1llu * w[j + i - 1] * wn % p;
    }
    a.resize(n); b.resize(n);
    ntt(a); ntt(b);
    for (i = 0; i < n; i++) a[i] = 1llu * a[i] * b[i] % p;
    ntt(a); reverse(1 + all(a)); a.resize(n = m);
    for (i = 0; i < n; i++) a[i] = 1llu * a[i] * inv % p;
    return a;
}

vector<int> match(const string &s, const string &t)
{
    using NTT::p, NTT::mul;
    static mt19937 rnd(chrono::steady_clock::now().time_since_epoch().count());
    static array<ui, 256> c;
    static bool initd = 0;
    if (!initd)
    {
        initd = 1;
        for (ui &x : c) x = rnd() % NTT::p;
        c['*'] = 0; //通配符
    }
    int n = s.size(), m = t.size(), i, j;
    if (n < m) return { };
    vector<int> ans;
    vector<ui> f(n), ff(n), fff(n), g(m), gg(m), ggg(m);
    for (i = 0; i < n; i++)
    {
        f[i] = c[s[i]];
        ff[i] = 1llu * f[i] * f[i] % p;
        fff[i] = 1llu * ff[i] * f[i] % p;
    }
    for (i = 0; i < m; i++)
    {
        g[i] = c[t[m - i - 1]];
        gg[i] = 1llu * g[i] * g[i] % p;
    }
}

```

```

        ggg[i] = 1llu * gg[i] * g[i] % p;
    }
    auto fffg = mul(fff, g), ffgg = mul(ff, gg), fgfg = mul(f, ggg);
    for (i = 0; i <= n - m; i++) if ((fffg[m - 1 + i] + fgfg[m - 1 + i] + 2 * (NTT::p - ffgg[m - 1 + i])) % NTT::p == 0) ans.push_back(i);
    return ans;
}

```

快一些的版本，手动拆开了多项式乘法。

```

const int N = 1 << 22;
const ui p = 998244353, g = 3;
inline ui ksm(ui x, ui y)
{
    ui ans = 1;
    while (y)
    {
        if (y & 1) ans = 1llu * ans * x % p;
        y >>= 1; x = 1llu * x * x % p;
    }
    return ans;
}
ui r[N], w[N];
void ntt(vector<ui> &a)
{
    int n = a.size(), i, j, k;
    for (k = 1; k < n; k <= 1)
    {
        for (i = 0; i < n; i += k << 1)
        {
            for (j = 0; j < k; j++)
            {
                ui x = a[i + j], y = 1llu * a[i + j + k] * w[j + k] % p;
                a[i + j] = (x + y) % p; a[i + j + k] = (x - y) % p;
            }
        }
    }
}
vector<int> match(string s, string t, char ch = '*')// 单次程序运行中中通配符不可修改
{
    static mt19937 rnd(chrono::steady_clock::now().time_since_epoch().count());
    static array<ui, 256> c;
    static bool initd = 0;
    if (!initd)
    {
        initd = 1;
        for (ui &x : c) x = rnd() % p;
        // for (int i=0; i<256; i++) c[i]=i-96;
        c[ch] = 0;//通配符
    }
    int n = s.size(), m = t.size(), i, j;
    if (n < m) return { };
    vector<int> ans;
    int N = 1 << __lg(n * 2 - 1), base = __lg(N) - 1;
    vector<ui> f(N), ff(N), fff(N), g(N), gg(N), ggg(N);
    reverse(all(t));
    s.resize(N, ch), t.resize(N, ch);
    for (i = 0; i < N; i++)

```

```

{
    r[i] = r[i >> 1] >> 1 | (i & 1) << base;
    if (i < r[i])
    {
        swap(s[i], s[r[i]]);
        swap(t[i], t[r[i]]);
    }
}
for (j = 1; j < N; j <= 1)
{
    ui wn = ksm(3, (p - 1) / (j << 1));
    w[j] = 1;
    for (i = 1; i < j; i++) w[j + i] = 1llu * w[j + i - 1] * wn % p;
}
for (i = 0; i < N; i++)
{
    f[i] = c[s[i]];
    ff[i] = 1llu * f[i] * f[i] % p;
    fff[i] = 1llu * ff[i] * f[i] % p;
    g[i] = c[t[i]];
    gg[i] = 1llu * g[i] * g[i] % p;
    ggg[i] = 1llu * gg[i] * g[i] % p;
}
ntt(f); ntt(ff); ntt(fff); ntt(g); ntt(gg); ntt(ggg);
for (i = 0; i < N; i++) f[i] = (1llu * fff[i] * g[i] + 1llu * f[i] * ggg[i] + 2llu * (p - ff[
i]) * gg[i]) % p;
for (i = 0; i < N; i++) if (i < r[i]) swap(f[i], f[r[i]]);
ntt(f); reverse(1 + all(f));
for (i = 0; i <= n - m; i++) if (f[m + i - 1] == 0) ans.push_back(i);
return ans;
}

```


5 图论

5.1 最小生成树相关

5.1.1 切比雪夫距离最小生成树

原理：先转曼哈顿距离，再用曼哈顿的板子。

$O(n \log n)$, $O(n)$ 。

```
const int N = 3e5 + 2, M = N << 2;
struct P
{
    int u, v;
    ll w;
    P(int a = 0, int b = 0, ll c = 0) : u(a), v(b), w(c) { }
    bool operator<(const P &o) const { return w < o.w; }
};
struct Q
{
    ll x, y;
    int id;
    Q(ll a = 0, ll b = 0, int c = 0) : x(a), y(b), id(c) { }
    bool operator<(const Q &o) const { return x != o.x ? x > o.x : y > o.y; }
};
ll ans;
P lb[M];
Q a[N], b[N];
int f[N];
ll c[N];
int n, m, i, x, y;
struct bit
{
    ll a[N];
    int pos[N], n;
    void init(int &nn)
    {
        memset(a + 1, 0x7f, (n = nn) * sizeof a[0]);
        memset(pos + 1, 0, n * sizeof pos[0]);
    }
    void mdf(int x, const ll y, const int z)
    {
        if (a[x] > y) a[x] = y, pos[x] = z;
        while (x -= x & -x) if (a[x] > y) a[x] = y, pos[x] = z;
    }
    int sum(int x)
    {
        ll r = a[x];
        int rr = pos[x];
        while ((x += x & -x) <= n) if (a[x] < r) r = a[x], rr = pos[x];
        return rr;
    }
};
bit s;
void cal()
{
    int i, x, y;
    s.init(n);
```

```

memcpy(b + 1, a + 1, sizeof(Q) * n);
sort(a + 1, a + n + 1);
for (i = 1; i <= n; i++) c[i] = a[i].y - a[i].x;
sort(c + 1, c + n + 1);
for (i = 1; i <= n; i++)
{
    if (x = s.sum(y = lower_bound(c + 1, c + n + 1, a[i].y - a[i].x) - c))
        lb[+m] = P(a[x].id, a[i].id, a[x].x + a[x].y - a[i].x - a[i].y); //谨防 int 爆
    s.mdf(y, a[i].y + a[i].x, i);
}
memcpy(a + 1, b + 1, sizeof(Q) * n);
}
int getf(int x) { return f[x] == x ? x : f[x] = getf(f[x]); }
int main()
{
    cin >> n;
    for (i = 1; i <= n; i++) {
        cin >> a[f[i] = a[i].id = i].x >> a[i].y;
        swap(a[i].x, a[i].y); a[i] = Q(a[i].x + a[i].y, a[i].x - a[i].y, i);
    }
    cal(); for (i = 1; i <= n; i++) swap(a[i].x, a[i].y);
    cal(); for (i = 1; i <= n; i++) a[i].y = -a[i].y;
    cal(); for (i = 1; i <= n; i++) swap(a[i].x, a[i].y);
    cal(); sort(lb + 1, lb + m + 1);
    for (i = 1; i <= m; i++) if ((x = getf(lb[i].u)) != (y = getf(lb[i].v))) f[x] = y, ans += lb[i].w;
    cout << ans / 2 << endl;
}

```

5.1.2 最小乘积生成树

题意：每条边有两个属性 x_i, y_i ，你需要最小化 $(\sum x_i)(\sum y_i)$ 。

你需要实现的是 sol1，即按照 val 为权值的答案。 val_i 是根据 x_i, y_i 计算的。

```

#include "bits/stdc++.h"
using namespace std;
typedef long long ll;
const int N = 202, M = 10002;
struct P
{
    int x, y;
    P(int a = 0, int b = 0) : x(a), y(b) { }
    bool operator<(const P &o) const { return (ll)x * y < (ll)o.x * o.y || (ll)x * y == (ll)o.x * o.y && x < o.x; }
};
struct Q
{
    int u, v, x, y, val;
    bool operator<(const Q &o) const { return val < o.val; }
};
P ans = P(1e9, 1e9), l, r;
Q a[M];
int f[N];
int n, m, i;
int getf(int x)
{
    if (f[x] == x) return x;
}

```

```

    return f[x] = getf(f[x]);
}
P sol1()
{
    P r = P(0, 0);
    for (i = 1; i <= n; i++) f[i] = i;
    sort(a + 1, a + m + 1);
    for (i = 1; i <= m; i++) if (getf(a[i].u) != getf(a[i].v))
    {
        f[f[a[i].u]] = f[a[i].v];
        r.x += a[i].x, r.y += a[i].y;
    }
    return r;
}
void sol2(P l, P r)
{
    for (i = 1; i <= m; i++) a[i].val = (r.x - l.x) * a[i].y + (l.y - r.y) * a[i].x;
    P np = sol1();
    ans = min(ans, np);
    if ((ll)(r.x - l.x) * (np.y - l.y) - (ll)(r.y - l.y) * (np.x - l.x) >= 0) return;
    sol2(l, np); sol2(np, r);
}
int main()
{
    cin >> n >> m;
    for (i = 1; i <= m; i++) cin >> a[i].u >> a[i].v >> a[i].x >> a[i].y, ++a[i].u, ++a[i].v;
    for (i = 1; i <= m; i++) a[i].val = a[i].x; l = sol1();
    for (i = 1; i <= m; i++) a[i].val = a[i].y; r = sol1();
    ans = min(ans, min(l, r)); sol2(l, r);
    cout << ans.x << '□' << ans.y << endl;
}

```

5.1.3 最小斯坦纳树

题意：让给定点集连通的最小生成树（不要求全图连通）

$O(3^k n + 2^k m \log m)$ 。

分为有方案与无方案两个版本，点标号 $[0, n)$ 。

```

ll steiner(int n, const vector<tuple<int, int, ll>> &eg, vector<int> id)//[0,n)
{
    using pa = pair<ll, int>;
    int m = id.size(), i;
    vector f(1 << m, vector<ll>(n, inf));
    for (i = 0; i < m; i++) f[1 << i][id[i]] = 0;
    vector<vector<pair<int, ll>>> e(n);
    for (auto [u, v, w] : eg) e[u].push_back({v, w}), e[v].push_back({u, w});
    for (int S = 1; S < 1 << m; S++)
    {
        auto &d = f[S];
        priority_queue<pa, vector<pa>, greater<pa>> q;
        for (i = 0; i < n; i++)
        {
            for (int T = S - 1 & S; T; T = T - 1 & S) cmin(d[i], f[T][i] + f[S ^ T][i]);
            if (d[i] != inf) q.push({d[i], i});
        }
        while (q.size())

```

```

    {
        auto [_, u] = q.top(); q.pop();
        if (_ != d[u]) continue;
        for (auto [v, w] : e[u]) if (cmin(d[v], d[u] + w)) q.push({d[v], v});
    }
}
return *min_element(all(f.back()));
}
pair<ll, vector<int>> steiner_construct(int n, const vector<tuple<int, int, ll>> &eg, vector<int>
    id)//[0,n)
{
    using pa = pair<ll, int>;
    int m = id.size(), i;
    vector f(1 << m, vector<ll>(n, inf));
    vector pre(1 << m, vector(n, pair{-1, -1}));
    for (i = 0; i < m; i++) f[1 << i][id[i]] = 0;
    vector<vector<tuple<int, ll, int>>> e(n);
    i = 0;
    for (auto [u, v, w] : eg) e[u].push_back({v, w, i}), e[v].push_back({u, w, i++});
    for (int S = 1; S < 1 << m; S++)
    {
        auto &d = f[S];
        priority_queue<pa, vector<pa>, greater<pa>> q;
        for (i = 0; i < n; i++)
        {
            for (int T = S - 1 & S; T; T = T - 1 & S)
                if (cmin(d[i], f[T][i] + f[S ^ T][i]))
                    pre[S][i] = {-2, T};

            if (d[i] != inf) q.push({d[i], i});
        }
        while (q.size())
        {
            auto [_, u] = q.top(); q.pop();
            if (_ != d[u]) continue;
            for (auto [v, w, id] : e[u]) if (cmin(d[v], d[u] + w))
            {
                q.push({d[v], v});
                pre[S][v] = {u, id};
            }
        }
    }
    vector<int> chosen;
    int S = (1 << m) - 1, u = min_element(all(f[S])) - f[S].begin();
    auto dfs = [&](auto &&dfs, int S, int u) {
        auto [x, y] = pre[S][u];
        while (x >= 0)
        {
            u = x;
            chosen.push_back(y);
            tie(x, y) = pre[S][u];
        }
        if (x == -1) return;
        dfs(dfs, y, u); dfs(dfs, S ^ y, u);
    };
    dfs(dfs, S, u);
    sort(all(chosen));
}

```

```

    return {f[S][u], chosen};
}
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cout << fixed << setprecision(15);
    int n, m, q, i;
    cin >> n >> m;
    vector<tuple<int, int, ll>> eg(m);
    cin >> eg >> q;
    vector<int> id(q);
    cin >> id;
    auto [ans, eid] = steiner_construct(n, eg, id);
    cout << ans << ' ' << eid.size() << '\n' << eid << endl;
}

```

5.2 最短路相关

5.2.1 全源最短路与判负环

使用 floyd 实现全源最短路与判负环。注意边权较大时可能需要考虑 int128.

```

#include "bits/stdc++.h"
using namespace std;
typedef long long ll;
typedef pair<int, int> pa;
typedef tuple<int, int, int> tp;
const int N = 152;
const ll inf = 5e8;
ll dis[N][N];
bool ne[N][N];
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    while (1)
    {
        int n, m, q, i, j, k;
        cin >> n >> m >> q;
        if (tp(n, m, q) == tp(0, 0, 0)) return 0;
        ll oo = inf * inf;
        for (i = 0; i < n; i++) fill_n(dis[i], n, oo), fill_n(ne[i], n, 0);
        for (i = 0; i < n; i++) dis[i][i] = 0;
        while (m--)
        {
            int u, v, w;
            cin >> u >> v >> w;
            dis[u][v] = min(dis[u][v], (ll)w);
        }
        for (k = 0; k < n; k++) for (i = 0; i < n; i++) if (dis[i][k] < oo) for (j = 0; j < n; j
        ++)) if (dis[k][j] < oo)
            dis[i][j] = max(min(dis[i][j], dis[i][k] + dis[k][j]), -inf * 2);
        for (k = 0; k < n; k++) if (dis[k][k] < 0) for (i = 0; i < n; i++) if (dis[i][k] < oo)
        for (j = 0; j < n; j++) if (dis[k][j] < oo) ne[i][j] = 1;
        while (q--)
        {
            int u, v;
            cin >> u >> v;

```

```

        if (dis[u][v] > inf) cout << "Impossible\n"; else if (ne[u][v]) cout << "-Infinity\n"
; else cout << dis[u][v] << '\n';
    }
    cout << '\n';
}
}

```

5.2.2 Dijkstra/SPFA/Johnson

Johnson 不适用于图中存在负环的情况，因为负环不一定是可以经过的。

$O(nm \log m)$, $O(n + m)$ 。

```

vector<ll> spfa(const vector<vector<pair<int, ll>>> &e, int s)
{
    int n = e.size(), i;
    assert(n);
    queue<int> q;
    vector<int> len(n), ed(n);
    vector<ll> dis(n, inf);
    q.push(s); dis[s] = 0;
    while (q.size())
    {
        int u = q.front(); q.pop();
        ed[u] = 0;
        for (auto [v, w] : e[u]) if (cmin(dis[v], dis[u] + w))
        {
            len[v] = len[u] + 1;
            if (len[v] > n) return { };
            if (!ed[v])
            {
                ed[v] = 1;
                q.push(v);
            }
        }
    }
    return dis;
}

vector<ll> spfa(const vector<vector<pair<int, ll>>> &e)
{
    int n = e.size(), i;
    assert(n);
    queue<int> q;
    vector<int> len(n), ed(n, 1);
    vector<ll> dis(n);
    for (i = 0; i < n; i++) q.push(i);
    while (q.size())
    {
        int u = q.front(); q.pop();
        ed[u] = 0;
        for (auto [v, w] : e[u]) if (cmin(dis[v], dis[u] + w))
        {
            len[v] = len[u] + 1;
            if (len[v] > n) return { };
            if (!ed[v])
            {
                ed[v] = 1;
                q.push(v);
            }
        }
    }
}

```

```

    }
    }
}
return dis;
}
vector<ll> dijk(const vector<vector<pair<int, ll>>> &e, int s)
{
    int n = e.size();
    using pa = pair<ll, int>;
    vector<ll> d(n, inf);
    vector<int> ed(n);
    priority_queue<pa, vector<pa>, greater<pa>> q;
    d[s] = 0; q.push({0, s});
    while (q.size())
    {
        int u = q.top().second; q.pop();
        ed[u] = 1;
        for (auto [v, w] : e[u]) if (cmin(d[v], d[u] + w)) q.push({d[v], v});
        while (q.size() && ed[q.top().second]) q.pop();
    }
    return d;
}
vector<vector<ll>> dijk(const vector<vector<pair<int, ll>>> &e)
{
    vector<vector<ll>> r;
    for (int i = 0; i < e.size(); i++) r.push_back(dijk(e, i));
    return r;
}
vector<vector<ll>> john(vector<vector<pair<int, ll>>> e)
{
    int n = e.size(), i, j;
    assert(n);
    auto h = spfa(e);
    if (!h.size()) return { };
    for (i = 0; i < n; i++) for (auto &[v, w] : e[i]) w += h[i] - h[v];
    auto r = dijk(e);
    for (i = 0; i < n; i++) for (j = 0; j < n; j++) if (r[i][j] != inf) r[i][j] -= h[i] - h[j];
    return r;
}

```

5.2.3 无向图最小环

原理：floyd 外层循环本质是计算只经过 $\leq k$ 的点的最短路。因此枚举环上标号最大的，在做这一轮转移之前正好是不经过它的最短路。

$O(n^3)$, $O(n^2)$ 。

```

int f[N][N], jl[N][N];
int n, m, c, ans = inf, i, j, k, x, y, z;
int main()
{
    cin >> n >> m;
    memset(f, 0x3f, sizeof(f));
    memset(jl, 0x3f, sizeof(jl));
    while (m--)
    {
        cin >> x >> y >> z;

```

```

    if (jl[x][y] != jl[0][0]) ans = min(ans, jl[x][y] + z);
    jl[x][y] = jl[y][x] = f[x][y] = f[y][x] = min(f[y][x], z);
}
for (k = 1; k <= n; k++)
{
    for (i = 1; i < k; i++) if (jl[k][i] != jl[0][0]) for (j = 1; j < i; j++)
        if (jl[k][j] != jl[0][0]) ans = min(ans, jl[k][i] + jl[k][j] + f[i][j]);
    for (i = 1; i <= n; i++) if (i != k) for (j = 1; j <= n; j++)
        if ((j != i) && (j != k)) f[i][j] = min(f[i][j], f[i][k] + f[k][j]);
}
if (ans == inf) cout << "No solution.\n"; else cout << ans << endl;
}

```

5.2.4 输出负环

```

#include "bits/stdc++.h"
using namespace std;
const int N = 34;
struct Q
{
    int v, w, c;
    Q() { }
    Q(int x, int y, int z) :v(x), w(y), c(z) { }
};
vector<Q> lj[N];
int dis[N], cnt[N], pt[N], S;
Q pre[N], st[N];
int n, m, ans, tp;
bool ed[N];
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cin >> n >> m;
    while (m--)
    {
        int x, y, z, w;
        cin >> x >> y >> z >> w;
        lj[x].emplace_back(y, w, z);
        lj[y].emplace_back(x, 0, -z);
    }
    for (int i = 1; i <= n; i++) lj[0].emplace_back(i, 1, 0);
    while (1)
    {
        memset(dis, -0x3f, sizeof dis); dis[0] = 0;
        for (int i = 0; i <= n; i++) ed[i] = cnt[i] = 0; S = -1;
        queue<int> q; q.push(0);
        while (!q.empty())
        {
            int u = q.front(); q.pop(); ed[u] = 0;
            for (auto &[v, w, c] : lj[u]) if (w && dis[v] < dis[u] + c)
            {
                dis[v] = dis[u] + c; pre[v] = Q(u, w, c);
                if (!ed[v])
                {
                    if (++cnt[v] > n + 1) { S = v; goto aa; }
                    ed[v] = 1; q.push(v);
                }
            }
        }
    }
}

```



```

    }
}
}
aa:;
if (S == -1) break;
{
    static bool ed[N];
    memset(ed, 0, sizeof ed);
    while (!ed[S]) ed[S] = 1, S = pre[S].v;
}
st[tp = 1] = pre[S]; pt[1] = S;
int x = pre[S].v;
while (x != S)
{
    st[++tp] = pre[x]; pt[tp] = x;
    x = pre[x].v;
    assert(tp <= n + 5);
}
int fl = 1e9;
for (int j = 1; j <= tp; j++) fl = min(fl, st[j].w);
assert(fl);
for (int j = 1; j <= tp; j++)
{
    ans += fl * st[j].c;
    int nn = 0;
    for (auto &[v, w, c] : lj[st[j].v]) if (v == pt[j] && st[j].c == c && st[j].w == w) {
++nn; w -= fl; break; }
    for (auto &[v, w, c] : lj[pt[j]]) if (v == st[j].v && st[j].c + c == 0) { ++nn; w +=
fl; break; } assert(nn == 2);
}
}
cout << ans << endl;
}

```

5.3 二分图与网络流建图

以下约定，若为二分图则 n, m 表示两侧点数，否则仅 n 表示全图点数。

5.3.1 二分图边染色

留坑待填。

结论： $\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$ ，二分图时 $\chi'(G) = \Delta(G)$ 。 $\Delta(G)$ 为图的最大度。

5.3.2 二分图最小点集覆盖

$ans = \text{maxmatch}$ ，方案如下。

输入为左侧点数、右侧点数、边数。

```

#include "bits/stdc++.h"
using namespace std;
const int N = 5e3 + 2;
vector<int> e[N];
int ed[N], lk[N], kl[N], flg[N], now;
bool dfs(int u)
{
    for (int v : e[u]) if (ed[v] != now)

```

```

    {
        ed[v] = now;
        if (!lk[v] || dfs(lk[v])) return lk[v] = u;
    }
    return 0;
}
void dfs2(int u)
{
    for (int v : e[u]) if (!flg[v]) flg[v] = 1, dfs2(lk[v]);
}
int main()
{
    int n, m, k, i, r = 0;
    cin >> n >> m >> k;
    while (k--)
    {
        int u, v;
        cin >> u >> v;
        e[u].push_back(v);
    }
    for (i = 1; i <= n; i++) dfs(now = i);
    for (i = 1; i <= m; i++) if (lk[i]) kl[lk[i]] = i;
    for (i = 1; i <= n; i++) if (!kl[i]) dfs2(i);
    vector<int> A[2];
    for (i = 1; i <= m; i++) if (lk[i])
    {
        if (flg[i]) A[1].push_back(i); else A[0].push_back(lk[i]);
    }
    for (int j = 0; j < 2; j++)
    {
        cout << A[j].size();
        for (int x : A[j]) cout << ' ' << x; cout << '\n';
    }
}

```

5.3.3 二分图最大独立集

$ans = n + m - \text{maxmatch}$, 方案是最小点集覆盖的补集。

5.3.4 二分图最小边覆盖

$ans = n + m - \text{maxmatch}$, 方案是最大匹配加随便一些边（用于覆盖失配点）。无解当且仅当有孤立点，算法会视为单选孤立点（无边）。这个定理对一般图也成立。

5.3.5 有向无环图最小不相交链覆盖

$ans = n - \text{maxmatch}$, 其中二分图建图方法是拆入点和出点（实现时直接跑一次二分图就行，不用额外处理），注意不需要传递闭包。方案如下。

```

#include "bits/stdc++.h"
using namespace std;
const int N = 152;
vector<int> e[N];
int lk[N], kl[N], ed[N], now;
bool dfs(int u)
{

```

```

for (int v : e[u]) if (ed[v] != now)
{
    ed[v] = now;
    if (!lk[v] || dfs(lk[v])) return lk[v] = u;
}
return 0;
}
int main()
{
    int n, m, i;
    ios::sync_with_stdio(0); cin.tie(0);
    cin >> n >> m;
    while (m--)
    {
        int u, v;
        cin >> u >> v;
        e[u].push_back(v);
    }
    int r = 0;
    for (i = 1; i <= n; i++) r += dfs(now = i);
    for (i = 1; i <= n; i++) kl[lk[i]] = i;
    for (i = 1; i <= n; i++) if (ed[i] != -1 && !lk[i])
    {
        vector<int> ans;
        int u = i;
        while (u)
        {
            ed[u] = -1;
            ans.push_back(u);
            u = kl[u];
        }
        for (int j = 0; j < ans.size(); j++) cout << ans[j] << " " [j + 1 == ans.size()];
    }
    cout << n - r << endl;
}

```

5.3.6 有向无环图最大互不可达集

$ans = n - \text{maxmatch}$, 其中二分图建图方法是拆入点和出点 (实现时直接跑一次二分图就行, 不用额外处理), 注意需要传递闭包。方案?

5.3.7 最大权闭合子图

若 $v_i > 0$, $s \rightarrow i$ 流量 v_i ; 若 $v_i < 0$, $i \rightarrow t$ 流量 $-v_i$ 。若原图 $u \rightarrow v$ 可花费 w 代价违抗, 流量 w , 否则 $+\infty$ 。答案为 $\sum_{v_i > 0} v_i - \text{maxflow}$ 。方案?

5.4 匹配与网络流相关代码

5.4.1 二分图最大权匹配

```

namespace KM
{
    const int N = 405; // 点数
    typedef long long ll; // 答案范围
    const ll inf = 1e16;

```

```

int lk[N], kl[N], pre[N], q[N], n, h, t;
ll sl[N], e[N][N], lx[N], ly[N];
bool edx[N], edy[N];
bool ck(int v)
{
    if (edy[v] = 1, kl[v]) return edx[q[++t] = kl[v]] = 1;
    while (v) swap(v, lk[kl[v] = pre[v]]);
    return 0;
}
void bfs(int u)
{
    fill_n(sl + 1, n, inf);
    memset(edx + 1, 0, n * sizeof edx[0]);
    memset(edy + 1, 0, n * sizeof edy[0]);
    q[h = t = 1] = u; edx[u] = 1;
    while (1)
    {
        while (h <= t)
        {
            int u = q[h++], v;
            ll d;
            for (v = 1; v <= n; v++) if (!edy[v] && sl[v] >= (d = lx[u] + ly[v] - e[u][v]))
if (pre[v] = u, d) sl[v] = d; else if (!ck(v)) return;
        }
        int i;
        ll m = inf;
        for (i = 1; i <= n; i++) if (!edy[i]) m = min(m, sl[i]);
        for (i = 1; i <= n; i++)
        {
            if (edx[i]) lx[i] -= m;
            if (edy[i]) ly[i] += m; else sl[i] -= m;
        }
        for (i = 1; i <= n; i++) if (!edy[i] && !sl[i] && !ck(i)) return;
    }
}
template<class TT> ll max_weighted_match(int N, const vector<tuple<int, int, TT>> &edges)//lk
[[1,n]]->[1,n]
{
    int i; n = N;
    memset(lk + 1, 0, n * sizeof lk[0]);
    memset(kl + 1, 0, n * sizeof kl[0]);
    memset(ly + 1, 0, n * sizeof ly[0]);
    for (i = 1; i <= n; i++) fill_n(e[i] + 1, n, 0); // 若不需保证匹配边最多, 置 0 即可, 否则 -
inf/N
    for (auto [u, v, w] : edges) e[u][v] = max(e[u][v], (ll)w);
    for (i = 1; i <= n; i++) lx[i] = *max_element(e[i] + 1, e[i] + n + 1);
    for (i = 1; i <= n; i++) bfs(i);
    ll r = 0;
    for (i = 1; i <= n; i++)
        if (!e[i][lk[i]]) // 若不需保证匹配边最多, 需要去掉这些辅助边
            kl[lk[i]] = 0, lk[i] = 0;
        else
            r += e[i][lk[i]];
    return r;
}
using KM::max_weighted_match, KM::lk, KM::kl, KM::e;

```

5.4.2 一般图最大匹配

```

namespace blossom_tree
{
    const int N = 1005;
    vector<int> e[N];
    int lk[N], rt[N], f[N], dfn[N], typ[N], q[N];
    int id, h, t, n;
    int lca(int u, int v)
    {
        ++id;
        while (1)
        {
            if (u)
            {
                if (dfn[u] == id) return u;
                dfn[u] = id; u = rt[f[lk[u]]];
            }
            swap(u, v);
        }
    }
    void blm(int u, int v, int a)
    {
        while (rt[u] != a)
        {
            f[u] = v;
            v = lk[u];
            if (typ[v] == 1) typ[q[++t] = v] = 0;
            rt[u] = rt[v] = a;
            u = f[v];
        }
    }
    void aug(int u)
    {
        while (u)
        {
            int v = lk[f[u]];
            lk[lk[u] = f[u]] = u;
            u = v;
        }
    }
    void bfs(int root)
    {
        memset(typ + 1, -1, n * sizeof typ[0]);
        iota(rt + 1, rt + n + 1, 1);
        typ[q[h = t = 1] = root] = 0;
        while (h <= t)
        {
            int u = q[h++];
            for (int v : e[u])
            {
                if (typ[v] == -1)
                {
                    typ[v] = 1; f[v] = u;
                    if (!lk[v]) return aug(v);
                    typ[q[++t] = lk[v]] = 0;
                }
            }
        }
    }
}

```

```

    }
    else if (!typ[v] && rt[u] != rt[v])
    {
        int a = lca(rt[u], rt[v]);
        blm(v, u, a); blm(u, v, a);
    }
}
}
}
int max_general_match(int N, vector<pair<int, int>> edges)//[1,n]
{
    n = N; id = 0;
    memset(f + 1, 0, n * sizeof f[0]);
    memset(dfn + 1, 0, n * sizeof dfn[0]);
    memset(lk + 1, 0, n * sizeof lk[0]);
    int i;
    for (i = 1; i <= n; i++) e[i].clear();
    mt19937 rnd(114);
    shuffle(all(edges), rnd);
    for (auto [u, v] : edges)
    {
        e[u].push_back(v), e[v].push_back(u);
        if (!(lk[u] || lk[v])) lk[u] = v, lk[v] = u;
    }
    int r = 0;
    for (i = 1; i <= n; i++) if (!lk[i]) bfs(i);
    for (i = 1; i <= n; i++) r += !!lk[i];
    return r / 2;
}
}
using blossom_tree::max_general_match, blossom_tree::lk;

```

5.4.3 一般图最大权匹配

$n = 400$: UOJ 600ms, Luogu 135ms

```

namespace weighted_blossom_tree
{
#define d(x) (lab[x.u]+lab[x.v]-e[x.u][x.v].w*2)
    const int N = 403 * 2;//两倍点数
    typedef long long ll;//总和大小
    typedef int T;//权值大小
    //均不允许无符号
    const T inf = numeric_limits<int>::max() >> 1;
    struct Q
    {
        int u, v;
        T w;
    } e[N][N];
    T lab[N];
    int n, m = 0, id, h, t, lk[N], sl[N], st[N], f[N], b[N][N], s[N], ed[N], q[N];
    vector<int> p[N];
    void upd(int u, int v) { if (!sl[v] || d(e[u][v]) < d(e[sl[v]][v])) sl[v] = u; }
    void ss(int v)
    {
        sl[v] = 0;
    }
}

```

```

    for (int u = 1; u <= n; u++) if (e[u][v].w > 0 && st[u] != v && !s[st[u]]) upd(u, v);
}
void ins(int u) { if (u <= n) q[++t] = u; else for (int v : p[u]) ins(v); }
void mdf(int u, int w)
{
    st[u] = w;
    if (u > n) for (int v : p[u]) mdf(v, w);
}
int gr(int u, int v)
{
    if ((v = find(all(p[u]), v) - p[u].begin()) & 1)
    {
        reverse(1 + all(p[u]));
        return (int)p[u].size() - v;
    }
    return v;
}
void stm(int u, int v)
{
    lk[u] = e[u][v].v;
    if (u <= n) return;
    Q w = e[u][v];
    int x = b[u][w.u], y = gr(u, x), i;
    for (i = 0; i < y; i++) stm(p[u][i], p[u][i ^ 1]);
    stm(x, v);
    rotate(p[u].begin(), y + all(p[u]));
}
void aug(int u, int v)
{
    int w = st[lk[u]];
    stm(u, v);
    if (!w) return;
    stm(w, st[f[w]]);
    aug(st[f[w]], w);
}
int lca(int u, int v)
{
    for (++id; u | v; swap(u, v))
    {
        if (!u) continue;
        if (ed[u] == id) return u;
        ed[u] = id; //????????v?? 这是原作者的注释, 我也不知道是啥
        if (u = st[lk[u]]) u = st[f[u]];
    }
    return 0;
}
void add(int u, int a, int v)
{
    int x = n + 1, i, j;
    while (x <= m && st[x]) ++x;
    if (x > m) ++m;
    lab[x] = s[x] = st[x] = 0; lk[x] = lk[a];
    p[x].clear(); p[x].push_back(a);
    for (i = u; i != a; i = st[f[j]]) p[x].push_back(i), p[x].push_back(j = st[lk[i]]), ins(j);
    //复制, 改一处
    reverse(1 + all(p[x]));
    for (i = v; i != a; i = st[f[j]]) p[x].push_back(i), p[x].push_back(j = st[lk[i]]), ins(j);
}

```

```

);
    mdf(x, x);
    for (i = 1; i <= m; i++) e[x][i].w = e[i][x].w = 0;
    memset(b[x] + 1, 0, n * sizeof b[0][0]);
    for (int u : p[x])
    {
        for (v = 1; v <= m; v++) if (!e[x][v].w || d(e[u][v]) < d(e[x][v])) e[x][v] = e[u][v]
    }, e[v][x] = e[v][u];
        for (v = 1; v <= n; v++) if (b[u][v]) b[x][v] = u;
    }
    ss(x);
}
void ex(int u) // s[u] == 1
{
    for (int x : p[u]) mdf(x, x);
    int a = b[u][e[u][f[u]].u], r = gr(u, a), i;
    for (i = 0; i < r; i += 2)
    {
        int x = p[u][i], y = p[u][i + 1];
        f[x] = e[y][x].u;
        s[x] = 1; s[y] = 0;
        sl[x] = 0; ss(y);
        ins(y);
    }
    s[a] = 1; f[a] = f[u];
    for (i = r + 1; i < p[u].size(); i++) s[p[u][i]] = -1, ss(p[u][i]);
    st[u] = 0;
}
bool on(const Q &e)
{
    int u = st[e.u], v = st[e.v], a;
    if (s[v] == -1)
    {
        f[v] = e.u; s[v] = 1;
        a = st[lk[v]];
        sl[v] = sl[a] = s[a] = 0;
        ins(a);
    }
    else if (!s[v])
    {
        a = lca(u, v);
        if (!a) return aug(u, v), aug(v, u), 1;
        else add(u, a, v);
    }
    return 0;
}
bool bfs()
{
    memset(s + 1, -1, m * sizeof s[0]);
    memset(sl + 1, 0, m * sizeof sl[0]);
    h = 1; t = 0;
    int i, j;
    for (i = 1; i <= m; i++) if (st[i] == i && !lk[i]) f[i] = s[i] = 0, ins(i);
    if (h > t) return 0;
    while (1)
    {
        while (h <= t)

```



```

    {
        int u = q[h++], v;
        if (s[st[u]] != 1) for (v = 1; v <= n; v++) if (e[u][v].w > 0 && st[u] != st[v])
        {
            if (d(e[u][v])) upd(u, st[v]); else if (on(e[u][v])) return 1;
        }
    }
    T x = inf;
    for (i = n + 1; i <= m; i++) if (st[i] == i && s[i] == 1) x = min(x, lab[i] >> 1);
    for (i = 1; i <= m; i++) if (st[i] == i && sl[i] && s[i] != 1) x = min(x, d(e[sl[i]][i]) >> s[i] + 1);
    for (i = 1; i <= n; i++) if (~s[st[i]]) if ((lab[i] += (s[st[i]] * 2 - 1) * x) <= 0)
return 0;
    for (i = n + 1; i <= m; i++) if (st[i] == i && ~s[st[i]]) lab[i] += (2 - s[st[i]] *
4) * x;
    h = 1; t = 0;
    for (i = 1; i <= m; i++) if (st[i] == i && sl[i] && st[sl[i]] != i && !d(e[sl[i]][i])
&& on(e[sl[i]][i])) return 1;
    for (i = n + 1; i <= m; i++) if (st[i] == i && s[i] == 1 && !lab[i]) ex(i);
}
return 0;
}
template<class TT> ll max_weighted_general_match(int N, const vector<tuple<int, int, TT>> &
edges)//[1,n], 返回权值
{
    memset(ed + 1, 0, m * sizeof ed[0]);
    memset(lk + 1, 0, m * sizeof lk[0]);
    n = m = N; id = 0;
    iota(st + 1, st + n + 1, 1);
    int i, j;
    T wm = 0;
    ll r = 0;
    for (i = 1; i <= n; i++) for (j = 1; j <= n; j++) e[i][j] = {i, j, 0};
    for (auto [u, v, w] : edges) wm = max(wm, e[v][u].w = e[u][v].w = max(e[u][v].w, (T)w));
    for (i = 1; i <= n; i++) p[i].clear();
    for (i = 1; i <= n; i++) for (j = 1; j <= n; j++) b[i][j] = i * (i == j);
    fill_n(lab + 1, n, wm);
    while (bfs());
    for (i = 1; i <= n; i++) if (lk[i]) r += e[i][lk[i]].w;
    return r / 2;
}
#undef d
}
using weighted_blossom_tree::max_weighted_general_match, weighted_blossom_tree::lk;
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    int n, m;
    cin >> n >> m;
    vector<tuple<int, int, long long>> edges(m);
    for (auto &[u, v, w] : edges) cin >> u >> v >> w;
    cout << max_weighted_general_match(n, edges) << '\n';
    for (int i = 1; i <= n; i++) cout << lk[i] << "\n"[i == n];
}

```

5.4.4 网络流

包含多种费用流方案，spfa，dijkstra，以及 capacity scaling 的 spfa。

```
namespace net
{
    const int N = 4e5 + 50; //number of nodes
    namespace flow
    {
        const ll inf = 4e18;
        struct Q
        {
            int v;
            ll w;
            int id;
        };
        vector<Q> e[N];
        vector<Q>::iterator fir[N];
        int fc[N], q[N];
        int n, s, t;
        int bfs()
        {
            for (int i = 0; i < n; i++)
            {
                fir[i] = e[i].begin();
                fc[i] = 0;
            }
            int p1 = 0, p2 = 0, u;
            fc[s] = 1; q[0] = s;
            while (p1 <= p2)
            {
                int u = q[p1++];
                for (auto [v, w, id] : e[u]) if (w && !fc[v])
                {
                    q[++p2] = v;
                    fc[v] = fc[u] + 1;
                }
            }
            return fc[t];
        }
        ll dfs(int u, ll maxf)
        {
            if (u == t) return maxf;
            ll j = 0, k;
            for (auto &it = fir[u]; it != e[u].end(); ++it)
            {
                auto &[v, w, id] = *it;
                if (w && fc[v] == fc[u] + 1 && (k = dfs(v, min(maxf - j, w))))
                {
                    j += k;
                    w -= k;
                    e[v][id].w += k;
                    if (j == maxf) return j;
                }
            }
            fc[u] = 0;
            return j;
        }
    }
}
```

```

pair<ll, vector<ll>> max_flow(int _n, const vector<tuple<int, int, ll>> &edges, int _s,
int _t)//[0,n]
{
    s = _s; t = _t; n = _n + 1;
    for (int i = 0; i < n; i++) e[i].clear();
    for (auto [u, v, w] : edges)
    {
        e[u].push_back({v, w, (int)e[v].size()});
        e[v].push_back({u, 0, (int)e[u].size() - 1});
    }
    ll r = 0;
    while (bfs()) r += dfs(s, inf);
    vector<ll> ans, id(n);
    for (auto [u, v, w] : edges)
    {
        ++id[u];
        ans.push_back(e[v][id[v]++].w);
    }
    return {r, ans};
}
}
using flow::max_flow, flow::fc;
namespace match
{
    int lk[N], kl[N], ed[N];
    vector<int> e[N];
    int max_match(int n, int m, const vector<pair<int, int>> &edges)//lk[[0,n]]->[0,m]
    {
        ++n; ++m;
        int s = n + m, t = n + m + 1, i;
        vector<tuple<int, int, ll>> eg;
        eg.reserve(n + m + edges.size());
        for (i = 0; i < n; i++) eg.push_back({s, i, 1});
        for (i = 0; i < m; i++) eg.push_back({i + n, t, 1});
        for (auto [u, v] : edges) eg.push_back({u, v + n, 1});
        int r = max_flow(t, eg, s, t).first;
        fill_n(lk, n, -1); fill_n(kl, m, -1);
        for (i = 0; i < n; i++) for (auto [v, w, id] : flow::e[i]) if (v < s && !w)
        {
            lk[i] = v - n;
            kl[v - n] = i;
            break;
        }
        return r;
    }
    void dfs(int u)
    {
        for (int v : e[u]) if (!ed[v]) ed[v] = 1, dfs(kl[v]);
    }
    pair<vector<int>, vector<int>> min_cover(int n, int m, const vector<pair<int, int>> &
edges)//[0,n]-[0,m]
    {
        max_match(n++, m++, edges);
        fill_n(ed, m, 0);
        int i;
        for (i = 0; i < n; i++) e[i].clear();
        for (auto [u, v] : edges) e[u].push_back(v);
    }
}

```

```

        for (i = 0; i < n; i++) if (lk[i] == -1) dfs(i);
        vector<int> r[2];
        for (i = 0; i < m; i++) if (kl[i] != -1)
        {
            if (ed[i]) r[1].push_back(i); else r[0].push_back(kl[i]);
        }
        sort(all(r[0]));
        return {r[0], r[1]};
    }
}

using match::max_match, match::min_cover, match::lk, match::kl;
namespace cost_flow
{
    const ll inf = 3e18;
    struct Q
    {
        int v;
        ll w, c;
        int id;
    };
    vector<Q> e[N];
    ll dis[N];
    int pre[N], pid[N], ipd[N];
    bool ed[N];
    int n, s, t;
    pair<ll, ll> spfa()
    {
        queue<int> q;
        fill_n(dis, n, inf);
        memset(ed, 0, n * sizeof ed[0]);
        q.push(s); dis[s] = 0;
        while (q.size())
        {
            int u = q.front(); q.pop(); ed[u] = 0;
            for (auto [v, w, c, id] : e[u]) if (w && cmin(dis[v], dis[u] + c))
            {
                pre[v] = u;
                pid[v] = e[v][id].id;
                ipd[v] = id;
                if (!ed[v]) q.push(v), ed[v] = 1;
            }
        }
        if (dis[t] == inf) return {0, 0};
        ll mw = 9e18;
        for (int i = t; i != s; i = pre[i]) mw = min(mw, e[pre[i]][pid[i]].w);
        for (int i = t; i != s; i = pre[i]) e[pre[i]][pid[i]].w -= mw, e[i][ipd[i]].w += mw;
        return {mw, (lll)mw * dis[t]};
    }
}

tuple<ll, ll, vector<ll>> mcmf_spfa(int _n, const vector<tuple<int, int, ll, ll>> &edges
, int _s, int _t)//[0,n]
{
    s = _s; t = _t; n = _n + 1;
    for (int i = 0; i < n; i++) e[i].clear();
    for (auto [u, v, w, c] : edges)
    {
        e[u].push_back({v, w, c, (int)e[v].size()});
        e[v].push_back({u, 0, -c, (int)e[u].size() - 1});
    }
}

```

```

    }
    ll w = 0, dw;
    lll c = 0, dc;
    do
    {
        tie(dw, dc) = spfa();
        w += dw, c += dc;
    } while (dw);
    vector<ll> ans, id(n);
    for (auto [u, v, w, c] : edges)
    {
        ++id[u];
        ans.push_back(e[v][id[v]++].w);
    }
    return {w, c, ans};
}
pair<ll, ll> spfa_loop()
{
    vector<ll> d(n);
    vector<int> ed(n, 1), cnt(n), pre(n), pid(n);
    queue<int> q;
    int i;
    for (i = 0; i < n; i++) q.push(i);
    while (q.size())
    {
        int u = q.front(); q.pop(); ed[u] = 0;
        for (auto [v, w, c, id] : e[u]) if (w && cmin(d[v], d[u] + c))
        {
            pre[v] = u;
            pid[v] = id;
            if (d[v] < -inf * 2 || (cnt[v] = cnt[u] + 1) > n)
            {
                ll tw = 0, tc = 0;
                for (u = v; ed[u] <= 1; u = pre[u]) ed[u] = 2;
                for (; ed[u] == 2; u = v)
                {
                    v = pre[u];
                    ++e[u][pid[u]].w;
                    --e[v][e[u][pid[u]].id].w;
                    if (e[v][e[u][pid[u]].id].c != -inf) tc += e[v][e[u][pid[u]].id].c;
                    else tw = 1;
                    ed[u] = 3;
                }
                return {tw, tc};
            }
            if (!ed[v]) ed[v] = 1, q.push(v);
        }
    }
    return {0, 0};
}
tuple<ll, lll, vector<ll>> mcmf_spfa_scaling(int _n, vector<tuple<int, int, ll, ll>>
edges, int _s, int _t)//[0,n]
{
    s = _s; t = _t; n = _n + 1;
    for (int i = 0; i < n; i++) e[i].clear();
    ll tot = 1;
    for (auto [u, v, w, c] : edges) tot += w;

```

```

edges.push_back({t, s, tot, -inf}); //最后一项 mcmf:-inf, mcf:0
for (auto [u, v, w, c] : edges)
{
    e[u].push_back({v, 0, c, (int)e[v].size()});
    e[v].push_back({u, 0, -c, (int)e[u].size() - 1});
}
ll w = 0, dw;
lll c = 0, dc;
for (int g = __lg(tot); g >= 0; g--)
{
    vector<int> id(n);
    for (auto [u, v, w, c] : edges)
    {
        (e[u][id[u]++] .w *= 2) += w >> g & 1;
        (e[v][id[v]++] .w *= 2);
    }
    w *= 2, c *= 2;
    do
    {
        tie(dw, dc) = spfa_loop();
        w += dw, c += dc;
    } while (dw || dc < 0);
}
vector<ll> ans, id(n);
edges.pop_back();
e[s].pop_back(), e[t].pop_back();
for (auto [u, v, w, c] : edges)
{
    ++id[u];
    ans.push_back(e[v][id[v]++] .w);
}
return {w, c, ans};
}
tuple<ll, lll, vector<ll>> mcmf_dijk(int _n, const vector<tuple<int, int, ll, ll>> &edges
, int _s, int _t)//[0,n]
{
    s = _s; t = _t; n = _n + 1;
    for (int i = 0; i < n; i++) e[i].clear();
    for (auto [u, v, w, c] : edges)
    {
        e[u].push_back({v, w, c, (int)e[v].size()});
        e[v].push_back({u, 0, -c, (int)e[u].size() - 1});
    }
    static ll h[N];
    auto get_h = [&]() {
        fill_n(h, n, inf);
        memset(ed, 0, n * sizeof ed[0]);
        queue<int> q;
        q.push(s); h[s] = 0;
        while (q.size())
        {
            int u = q.front(); q.pop(); ed[u] = 0;
            for (auto [v, w, c, id] : e[u]) if (w && h[v] > h[u] + c)
            {
                assert(c >= 0);
                h[v] = h[u] + c;
                if (!ed[v]) q.push(v), ed[v] = 1;
            }
        }
    };
    get_h();
    return {h[t], h[s] - h[t], ans};
}

```

```

    }
    }
    return;
};

auto dijkstra = [&]() -> pair<ll, ll> {
    static int fl[N], zl[N];
    int i;
    memset(ed, 0, n * sizeof ed[0]);
    fill_n(dis, n, inf);
    using pa = pair<ll, int>;
    priority_queue<pa, vector<pa>, greater<pa>> q;
    dis[s] = 0; q.push({0, s});
    while (q.size())
    {
        int u = q.top().second;
        q.pop(); ed[u] = 1;
        i = 0;
        for (auto [v, w, c, id] : e[u])
        {
            if (w && cmin(dis[v], dis[u] + c))
            {
                assert(c >= 0);
                fl[v] = id, zl[v] = i, pre[v] = u;
                q.push({dis[v], v});
            }
            ++i;
        }
        while (q.size() && ed[q.top().second]) q.pop();
    }
    if (dis[t] == inf) return {0, 0};
    ll tf = numeric_limits<ll>::max();
    for (i = t; i != s; i = pre[i]) tf = min(tf, e[pre[i]][zl[i]].w);
    for (i = t; i != s; i = pre[i]) e[pre[i]][zl[i]].w -= tf, e[i][fl[i]].w += tf;
    for (int u = 0; u < n; u++) for (auto &[v, w, c, id] : e[u]) c += dis[u] - dis[v];

    return {tf, (ll)tf * (h[t] += dis[t])};
};

get_h();
for (int u = 0; u < n; u++) for (auto &[v, w, c, id] : e[u]) c += h[u] - h[v];
ll w = 0, dw;
lll c = 0, dc;
do
{
    tie(dw, dc) = dijkstra();
    w += dw, c += dc;
} while (dw);
vector<ll> ans, id(n);
for (auto [u, v, w, c] : edges)
{
    ++id[u];
    assert(e[v][id[v]].v == u);
    ans.push_back(e[v][id[v]++].w);
}
return {w, c, ans};
}

}

using cost_flow::mcmf_spfa, cost_flow::mcmf_spfa_scaling, cost_flow::mcmf_dijk;
```

```

namespace bounded_flow
{
    vector<ll> valid_flow(int n, const vector<tuple<int, int, ll, ll>> &edges)
    {
        //返回空 vector 表示无解。最好保证 edges 非空。
        int i, m = edges.size();
        assert(m);
        ++n;
        ll tot = 0;
        static ll cd[N];
        memset(cd, 0, n * sizeof cd[0]);
        for (auto [u, v, l, r] : edges) cd[u] += l, cd[v] -= l;
        vector<tuple<int, int, ll>> eg;
        eg.reserve(n + m);
        for (i = 0; i < n; i++) if (cd[i] > 0) eg.push_back({i, n + 1, cd[i]}), tot += cd[i];
        else if (cd[i] < 0) eg.push_back({n, i, -cd[i]});
        for (auto [u, v, l, r] : edges) eg.push_back({u, v, r - l});
        auto [w, ans] = flow::max_flow(n + 1, eg, n, n + 1);
        if (tot != w) return { };
        ans.erase(all(ans) - m);
        for (i = 0; i < m; i++) ans[i] += get<2>(edges[i]);
        return ans;
    }

    pair<ll, vector<ll>> valid_flow_st(int n, vector<tuple<int, int, ll, ll>> edges, int s,
    int t)
    {
        //返回值 first -1 表示无解
        ll tot = 0;
        for (auto [u, v, l, r] : edges) tot += (u == s) * r;
        edges.push_back({t, s, 0, tot});
        auto ans = valid_flow(n, edges);
        if (!ans.size()) return {-1, { }};
        ans.pop_back();
        assert(flow::e[s].back().v == t);
        assert(flow::e[t].back().v == s);
        return {flow::e[s].back().w, ans};
    }

    pair<ll, vector<ll>> valid_max_flow(int n, const vector<tuple<int, int, ll, ll>> &edges,
    int s, int t)
    {
        //返回值 first -1 表示无解
        auto [r, _] = valid_flow_st(n, edges, s, t);
        if (r == -1) return {-1, { }};
        flow::s = s, flow::t = t;
        flow::e[s].pop_back(), flow::e[t].pop_back();
        while (flow::bfs()) r += flow::dfs(s, flow::inf);
        int m = edges.size(), i;
        vector<ll> ans(m), id(n + 1);
        for (i = 0; i <= n; i++) id[i] = flow::e[i].size();
        for (i = m - 1; i >= 0; i--)
        {
            auto [u, v, l, r] = edges[i];
            --id[u]; ans[i] = flow::e[v][--id[v]].w + l;
        }
        return {r, ans};
    }

    pair<ll, vector<ll>> valid_min_flow(int n, const vector<tuple<int, int, ll, ll>> &edges,
    int s, int t)
    {
        auto [r, _] = valid_flow_st(n, edges, s, t);
    }
}

```



```

    if (r == -1) return {-1, { }};
    flow::s = t; flow::t = s;
    flow::e[s].pop_back(); flow::e[t].pop_back();
    ll d;
    while (r && flow::bfs() && (d = flow::dfs(t, r))) r -= d;
    int m = edges.size(), i;
    vector<ll> ans(m), id(n + 1);
    for (i = 0; i <= n; i++) id[i] = flow::e[i].size();
    for (i = m - 1; i >= 0; i--)
    {
        auto [u, v, l, r] = edges[i];
        --id[u]; ans[i] = flow::e[v][--id[v]].w + 1;
    }
    return {r, ans};
}

}

using bounded_flow::valid_flow, bounded_flow::valid_flow_st, bounded_flow::valid_max_flow,
bounded_flow::valid_min_flow;
namespace bounded_cost_flow
{
    tuple<ll, ll, vector<ll>> valid_mcf(int n, const vector<tuple<int, int, ll, ll, ll>> &
edges, int s, int t)
    {
        // [u,v,l,r,c], mincost flow
        ++n;
        int ss = n, tt = n + 1, m = edges.size(), i;
        static ll cd[N];
        memset(cd, 0, n * sizeof cd[0]);
        for (auto [u, v, l, r, c] : edges) cd[u] += 1, cd[v] -= 1;
        vector<tuple<int, int, ll, ll>> e; e.reserve(n + m + 1);
        ll t1 = 0, t2 = 0;
        for (auto [u, v, l, r, c] : edges) e.push_back({u, v, r - 1, c});
        for (i = 0; i < n; i++) if (cd[i] > 0) e.push_back({i, tt, cd[i], 0}), t2 += cd[i];
        else if (cd[i] < 0) e.push_back({ss, i, -cd[i], 0});
        for (auto [u, v, w, c] : e) t1 += (u == s) * w;
        e.push_back({t, s, t1, 0});
        auto [tw, tc, _] = mcmf_spfa_scaling(tt, e, ss, tt); // checked spfa/dijk
        if (tw != t2) return {-1, -1, { }};
        tw = cost_flow::e[s].back().w;
        for (auto [u, v, l, r, c] : edges) tc += (lll)l * c;
        vector<ll> ans(m), id(n);
        for (i = 0; i < m; i++)
        {
            auto [u, v, l, r, c] = edges[i];
            assert(cost_flow::e[v][id[v]].v == u);
            ++id[u]; ans[i] = cost_flow::e[v][id[v]++].w + 1;
        }
        return {tw, tc, ans};
    }

    tuple<ll, ll, vector<ll>> valid_mcmf(int n, const vector<tuple<int, int, ll, ll, ll>> &
edges, int s, int t)
    {
        // [u,v,l,r,c], mincost max_flow
        auto [tw, tc, _] = valid_mcf(n, edges, s, t);
        if (tw == -1) return {-1, -1, { }};
        cost_flow::e[s].pop_back();
        cost_flow::e[t].pop_back();
        cost_flow::s = s; cost_flow::t = t;
        ll dw;

```

```

    lll dc;
    do
    {
        tie(dw, dc) = cost_flow::spfa();
        tw += dw; tc += dc;
    } while (dw);
    int m = edges.size(), i;
    vector<ll> ans(m), id(n + 1);
    for (i = 0; i < m; i++)
    {
        auto [u, v, l, r, c] = edges[i];
        ++id[u]; ans[i] = cost_flow::e[v][id[v]++].w + l;
    }
    return {tw, tc, ans};
}

tuple<ll, lll, vector<ll>> valid_mcmf_scaling(int n, const vector<tuple<int, int, ll, ll,
ll>> &edges, int s, int t)
{
    //[u,v,l,r,c], mincost max_flow
    using cost_flow::e, cost_flow::spfa_loop;
    auto [tw, tc, _] = valid_mcf(n, edges, s, t);
    if (tw == -1) return {-1, -1, { }};
    e[s].pop_back();
    e[t].pop_back();
    cost_flow::s = s; cost_flow::t = t;
    n = cost_flow::n;
    int m = edges.size(), i, j;
    vector<ll> cur;
    ll tot = 1;
    for (i = 0; i < n; i++) for (auto [v, w, c, id] : cost_flow::e[i]) tot += w;
    for (i = 0; i < n; i++)
    {
        for (auto &[v, w, c, id] : cost_flow::e[i])
            cur.push_back(w), w = 0;
        if (i == t) cur.push_back(tot);
        if (i == s) cur.push_back(0);
    }
    e[t].push_back({s, 0, -cost_flow::inf, (int)e[s].size()});
    e[s].push_back({t, 0, cost_flow::inf, (int)e[t].size() - 1});
    ll w = 0, dw;
    lll c = 0, dc;
    for (int g = __lg(*max_element(all(cur))); g >= 0; g--)
    {
        for (i = j = 0; i < n; i++) for (auto &[v, w, c, id] : e[i]) w = w * 2 + (cur[j
++]) >> g & 1);
        w *= 2, c *= 2;
        do
        {
            tie(dw, dc) = spfa_loop();
            w += dw, c += dc;
        } while (dw || dc < 0);
    }
    vector<ll> ans(m), id(n + 1);
    for (i = 0; i < m; i++)
    {
        auto [u, v, l, r, c] = edges[i];
        ++id[u]; ans[i] = cost_flow::e[v][id[v]++].w + l;
    }
}

```

```

        return {tw + w, tc + c, ans};
    }
}
using bounded_cost_flow::valid_mcf, bounded_cost_flow::valid_mcmf, bounded_cost_flow::
valid_mcmf_scaling;
namespace ne_cost_flow
{
    tuple<ll, ll, vector<ll>> ne_mcmf(int n, const vector<tuple<int, int, ll, ll>> &edges,
int s, int t)
    {
        vector<tuple<int, int, ll, ll, ll>> e;
        for (auto [u, v, w, c] : edges) if (c >= 0) e.push_back({u, v, 0, w, c}); else
        {
            e.push_back({u, v, w, w, c});
            e.push_back({v, u, 0, w, -c});
        }
        auto [tw, tc, res] = valid_mcmf_scaling(n, e, s, t);
        int m = edges.size(), i, j;
        vector<ll> ans(m);
        for (i = j = 0; i < m; i++, j++)
        {
            auto [u, v, w, c] = edges[i];
            if (c >= 0) ans[i] = res[j];
            else ans[i] = w - res[++j];
        }
        assert(j == e.size());
        return {tw, tc, ans};
    }
    tuple<ll, ll, vector<ll>> ne_valid_mcf(int n, const vector<tuple<int, int, ll, ll, ll>>
&edges, int s, int t)
    {
        vector<tuple<int, int, ll, ll, ll>> e;
        for (auto [u, v, l, r, c] : edges) if (c >= 0) e.push_back({u, v, l, r, c}); else
        {
            e.push_back({u, v, r, r, c});
            e.push_back({v, u, 0, r - l, -c});
        }
        auto [tw, tc, res] = valid_mcf(n, e, s, t);
        if (tw == -1) return {-1, -1, { }};
        int m = edges.size(), i, j;
        vector<ll> ans(m);
        for (i = j = 0; i < m; i++, j++)
        {
            auto [u, v, l, r, c] = edges[i];
            if (c >= 0) ans[i] = res[j];
            else ans[i] = r - res[++j];
        }
        assert(j == e.size());
        return {tw, tc, ans};
    }
}
using ne_cost_flow::ne_mcmf, ne_cost_flow::ne_valid_mcf;
}

```

5.4.5 假带花树

一种错误的一般图最大匹配算法，但较难卡掉。推荐在时间不足时作为乱搞使用。

```
mt19937 rnd(3214);
vector<int> lj[N];
int lk[N], ed[N];
int n, m, cnt, i, t, x, y, ans, la;
bool dfs(int x)
{
    ed[x] = cnt; int v;
    shuffle(lj[x].begin(), lj[x].end(), rnd);
    for (auto u : lj[x]) if (ed[v = lk[u]] != cnt)
    {
        lk[v] = 0, lk[u] = x, lk[x] = u;
        if (!v || dfs(v)) return 1;
        lk[v] = u, lk[u] = v, lk[x] = 0;
    }
    return 0;
}
int main()
{
    srand(time(0)); la = -1;
    cin >> n >> m;
    while (m--) cin >> x >> y, lj[x].push_back(y), lj[y].push_back(x);
    while (la != ans)
    {
        memset(ed + 1, 0, n << 2); la = ans;
        for (i = 1; i <= n; i++) if (!lk[i]) ans += dfs(cnt = i);
    }
    cout << ans << '\n';
    for (i = 1; i <= n; i++) cout << lk[i] << "□\n"[i == n];
}
```

5.4.6 Stoer-Wagner 全局最小割

无向图 G 的最小割为：一个去掉后可以使 G 变成两个连通分量，且边权和最小的边集的边权和。

$O(n^3)$ 。可优化到 $O(nm \log n)$ 。

```
#include "bits/stdc++.h"
using namespace std;
namespace StoerWagner
{
    const int N = 602; // 点数
    typedef int T; // 边权和
    T e[N][N], w[N];
    int ed[N], p[N], f[N]; // f 仅输出方案用
    int getf(int u) { return f[u] == u ? u : f[u] = getf(f[u]); }
    template<class TT> pair<T, vector<int>> mincut(int n, const vector<tuple<int, int, TT>> &
edges) // [1, n], 返回某一集合
    {
        vector<int> ans; ans.reserve(n);
        int i, j, m;
        T r;
        r = numeric_limits<T>::max();
        for (i = 1; i <= n; i++) memset(e[i] + 1, 0, n * sizeof e[0][0]);
    }
```

```

    for (auto [u, v, w] : edges) e[u][v] += w, e[v][u] += w;
    fill_n(ed + 1, n, 0);
    iota(f + 1, f + n + 1, 1);
    for (m = n; m > 1; m--)
    {
        fill_n(w + 1, n, 0);
        for (i = 1; i <= n; i++) ed[i] &= 2;
        for (i = 1; i <= m; i++)
        {
            int x = 0;
            for (j = 1; j <= n; j++) if (!ed[j]) break; x = j;
            for (j++; j <= n; j++) if (!ed[j] * w[j] > w[x]) x = j;
            ed[p[i] = x] = 1;
            for (j = 1; j <= n; j++) w[j] += !ed[j] * e[x][j];
        }
        int s = p[m - 1], t = p[m];
        if (r > w[t])
        {
            r = w[t]; ans.clear();
            for (i = 1; i <= n; i++) if (getf(i) == getf(t)) ans.push_back(i);
        }
        for (i = 1; i <= n; i++) e[i][s] = e[s][i] += e[t][i];
        ed[t] = 2;
        f[getf(s)] = getf(t);
    }
    return {r, ans};
}

int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    int n, m;
    cin >> n >> m;
    vector<tuple<int, int, int>> e(m);
    for (auto &[u, v, w] : e) cin >> u >> v >> w;
    auto [_, v] = StoerWagner::mincut(n, e);
    cout << _ << endl;
    static int ed[602];
    for (int x : v) ed[x] = 1;
    for (auto [u, v, w] : e) _ -= w * (ed[u] ^ ed[v]);
    assert(!_);
}

```

5.4.7 最小割树

结论：两个点之间的最小割等于最小割树上两点间最小边权。

直接返回任意两点最小割。

```

template<class T> vector<vector<T>> min_cut(int n, const vector<tuple<int, int, T>> &edges)//[0,n)
{
    int m = edges.size(), i, s, t, cnt = 0;
    vector<int> fir(n, -1), nxt(m * 2, -1), fc(n), q(n);
    vector<pair<int, T>> e(m * 2);
    vector<tuple<T, int, int>> eg;
    auto add = [&](int u, int v, T w) {
        e[cnt] = {v, w};
    };
}

```

```

    nxt[cnt] = fir[u];
    fir[u] = cnt++;
};
for (auto [u, v, w] : edges) add(u, v, w), add(v, u, w);
auto E = e;
auto bfs = [&]() {
    fill(all(fc), 0);
    int ql = 0, qr = 0, u, i;
    fc[q[0] = s] = 1;
    while (ql <= qr)
    {
        u = q[ql++];
        for (int i = fir[u]; i != -1; i = nxt[i])
            if (auto &[v, w] = e[i]; w && !fc[v]) fc[q[++qr] = v] = fc[u] + 1;
    }
    return fc[t];
};
function<T(int, T)> dfs = [&](int u, T maxf) {
    if (u == t) return maxf;
    T j = 0, k;
    for (int i = fir[u]; i != -1; i = nxt[i])
        if (auto &[v, w] = e[i]; w && fc[v] == fc[u] + 1 && (k = dfs(v, min(maxf - j, w))))
        {
            j += k;
            w -= k;
            e[i ^ 1].second += k;
            if (j == maxf) return j;
        }
    fc[u] = 0;
    return j;
};
function<void(vector<int>>> solve = [&](vector<int> id) {
    static mt19937 rnd(chrono::steady_clock::now().time_since_epoch().count());
    if (id.size() <= 1) return;
    vector<int> u(2);
    sample(all(id), u.begin(), 2, rnd);
    s = u[0], t = u[1], e = E;
    T ans = 0;
    while (bfs()) ans += dfs(s, numeric_limits<T>::max());
    auto it = partition(all(id), [&](int u) { return fc[u]; });
    eg.emplace_back(ans, s, t);
    solve(vector(id.begin(), it));
    solve(vector(it, id.end()));
};
solve(range(0, n));
sort(all(eg), greater<>());
vector<basic_string<int>> ver(n);
vector ans(n, vector<T>(n));
vector<int> f(n);
for (i = 0; i < n; i++) ver[i] = {f[i] = i};
function<int(int)> getf = [&](int u) { return f[u] == u ? u : f[u] = getf(f[u]); };
for (auto [w, u, v] : eg)
{
    u = getf(u);
    v = getf(v);
    for (int w1 : ver[u]) for (int w2 : ver[v]) ans[w1][w2] = ans[w2][w1] = w;
    ver[u] += ver[v];
}

```

```

        f[v] = u;
    }
    return ans;
}

```

5.5 缩点相关

5.5.1 双极分解

无向图，图点双连通时对任意 s, t 存在。

含义：确定一个拓扑序，使得按这个拓扑序定向后，入度为 0 的只有 s ，出度为 0 的只有 t 。

```

vector<int> bipolar_orientation(const vector<pair<int, int>> &edges, int n, int s, int t) //[0,n)
{
    assert(s != t);
    vector e(n, vector<int>());
    for (auto [u, v] : edges)
    {
        e[u].push_back(v);
        e[v].push_back(u);
    }
    int cur = 1, i;
    vector<int> pre(n), low(n), p(n);
    pre[s] = 1;
    vector<int> id;
    bool flg = 0;
    function<void(int)> dfs = [&](int x) {
        pre[x] = ++cur;
        low[x] = x;
        for (int y : e[x])
        {
            flg |= y == s;
            if (pre[y] == 0)
            {
                id.push_back(y);
                dfs(y);
                p[y] = x;
                if (pre[low[y]] < pre[low[x]]) low[x] = low[y];
            }
            else if (pre[y] != 0 && pre[y] < pre[low[x]]) low[x] = y;
        }
    };
    dfs(t);
    if (!flg) return { };
    vector<int> sign(n, -1);
    vector<int> l(n), r(n);
    r[s] = t;
    l[t] = s;
    for (int v : id)
    {
        if (sign[low[v]] == -1)
        {
            l[v] = l[p[v]];
            r[l[v]] = v;
            l[p[v]] = v;
            r[v] = p[v];
            sign[p[v]] = 1;
        }
    }
}

```

```

    }
    else
    {
        r[v] = r[p[v]];
        l[r[v]] = v;
        r[p[v]] = v;
        l[v] = p[v];
        sign[p[v]] = -1;
    }
}
vector<int> a(n);
int x;
for (i = 0, x = s; i < n; x = r[x], i++) a[i] = x;
vector<int> ia(n, -1), rd(n), cd(n);
for (i = 0; i < n; i++) ia[a[i]] = i;
if (count(all(ia), -1)) return { };
for (auto [u, v] : edges)
{
    if (ia[u] > ia[v]) swap(u, v);
    ++cd[u]; ++rd[v];
}
for (i = 0; i < n; i++) if (i != s && i != t && (!cd[i] || !rd[i])) return { };
return a;
}

```

5.5.2 点双

一些结论：

判定一个图里是否有（点不重复）偶环：看其所有点双，若存在点数为偶数的或边数多于点数的点双，则存在偶环。

（无自环时）点双的边不交，边双的点不交。点双内的总点数 $O(n)$ ，总边数为 m ，边双内的总点数为 n ，总边数不超过 m 。

关于板子本身：

所有标号从 0 开始。

构造函数传入邻接表和边数，其中 `pair` 的 `second` 是边的标号，正反向编号相同，允许不连续但不能大于等于 m 。

不能处理有自环的情况。你可以直接在图中删除自环后传入。可以处理有重边的情况。

`bcc_node`：每个点双包含的点（已验证）；`bcc_edge`：每个点双包含的边；`bcc_n`：新图点数；`ct`：是否割点（已验证）；`blk`：边所属点双标号。

```

struct node_bcc
{
    int n, id, tp, bcc_n;
    vector<vector<pair<int, int>>> e;
    vector<vector<int>> bcc_node, bcc_edge;
    vector<int> dfn, low, st, ed, blk, ct;
    node_bcc(const vector<vector<pair<int, int>>> &e, int m) :
        n(e.size()), id(0), tp(0), bcc_n(0), e(e), dfn(n, -1), low(n, -1), st(m), ed(m), blk(m),
        ct(n)
    {
        for (int i = 0; i < n; i++) for (auto [v, w] : e[i])
        {
            assert(v >= 0 && v < n);
            assert(w >= 0 && w < m);
        }
    }
}

```



```

    for (int i = 0; i < n; i++) if (dfn[i] == -1) dfs(i, 1);
    bcc_node.resize(bcc_n);
    for (int i = 0; i < n; i++) for (auto [v, w] : e[i]) bcc_node[blk[w]].push_back(i);
    vector<int> flg(n);
    for (auto &v : bcc_node)
    {
        vector<int> t;
        for (int x : v) if (!exchange(flg[x], 1)) t.push_back(x);
        swap(t, v);
        for (int x : v) flg[x] = 0;
    }
    for (int i = 0; i < n; i++) if (e[i].size() == 0)
    {
        bcc_node.push_back({i});
        bcc_edge.push_back({ });
        ++bcc_n;
    }
}

void dfs(int u, bool rt)
{
    dfn[u] = low[u] = id++;
    int cnt = 0;
    for (auto [v, w] : e[u]) if (!ed[w])
    {
        st[tp++] = w;
        ed[w] = 1;
        if (dfn[v] == -1)
        {
            dfs(v, 0);
            ++cnt;
            cmin(low[u], low[v]);
            if (dfn[u] <= low[v])
            {
                ct[u] = cnt > rt;
                bcc_edge.push_back({ });
                do
                {
                    bcc_edge[bcc_n].push_back(st[--tp]);
                    blk[st[tp]] = bcc_n;
                } while (st[tp] != w);
                ++bcc_n;
            }
        }
        else cmin(low[u], dfn[v]);
    }
}

};

int main()
{
    int n, m, i;
    cin >> n >> m;
    vector<vector<pair<int, int>>> e(n);
    for (i = 0; i < m; i++)
    {
        int u, v;
        cin >> u >> v;
        e[u].push_back({v, i});
    }
}

```

```

        e[v].push_back({u, i});
    }
    node_bcc bcc(e, m);
}

```

5.5.3 边双

所有标号从 0 开始。

构造函数传入邻接表和边数，其中 pair 的 second 是边的标号，正反向编号相同，允许不连续但不能大于等于 m 。

可以处理有重边和自环的情况。

bcc_node: 每个边双包含的点（已验证）；bcc_edge: 每个边双包含的边；bcc_n: 新图点数；cur_e: 新图边表；ct: 是否割边；blk: 点所属边双标号。

```

struct edge_bcc
{
    int n, id, tp, bcc_n;
    vector<vector<pair<int, int>>> e, cur_e;
    vector<vector<int>> bcc_node, bcc_edge;
    vector<int> dfn, low, st, blk, ct;
    edge_bcc(const vector<vector<pair<int, int>>> &e, int m) :
        n(e.size()), id(0), tp(0), bcc_n(0), e(e), dfn(n, -1), low(n, -1), st(n), blk(n), ct(m)
    {
        for (int i = 0; i < n; i++) for (auto [v, w] : e[i])
        {
            assert(v >= 0 && v < n);
            assert(w >= 0 && w < m);
        }
        for (int i = 0; i < n; i++) if (dfn[i] == -1) dfs(i, -1);
        cur_e.resize(bcc_n);
        for (int i = 0; i < n; i++) for (auto [v, w] : e[i]) if (ct[w]) cur_e[blk[i]].push_back({
            blk[v], w});
        else bcc_edge[blk[i]].push_back(w);
        vector<int> flg(m);
        for (auto &v : bcc_edge)
        {
            vector<int> t;
            for (int x : v) if (!exchange(flg[x], 1)) t.push_back(x);
            swap(t, v);
        }
    }
    void dfs(int u, int fw)
    {
        dfn[u] = low[u] = id++;
        st[tp++] = u;
        for (auto [v, w] : e[u]) if (w != fw)
        {
            if (dfn[v] == -1)
            {
                dfs(v, w);
                cmin(low[u], low[v]);
                ct[w] = (dfn[u] < low[v]);
            }
            else cmin(low[u], dfn[v]);
        }
        if (dfn[u] == low[u])

```

```

        {
            bcc_node.push_back({ });
            bcc_edge.push_back({ });
            do
            {
                bcc_node[bcc_n].push_back(st[--tp]);
                blk[st[tp]] = bcc_n;
            } while (st[tp] != u);
            ++bcc_n;
        }
    }
};

int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    int n, m, i;
    cin >> n >> m;
    vector<vector<pair<int, int>>> e(n);
    for (i = 0; i < m; i++)
    {
        int u, v;
        cin >> u >> v;
        --u, --v;
        e[u].push_back({v, i});
        e[v].push_back({u, i});
    }
    edge_bcc s(e, m);
    cout << s.bcc_n << '\n';
    for (auto &v : s.bcc_node)
    {
        for (int &x : v) ++x;
        cout << v.size() << '□' << v << '\n';
    }
}

```

5.5.4 Tarjan 强连通分量

所有标号从 0 开始。

构造函数传入邻接表，其中 pair 的 second 是边的标号。

与点边双不同的是，你可以任意传入 second，代码中不会使用它，仅用于新图区分边。

可以处理有重边和自环的情况。

cc_node: 每个 SCC 包含的点（已验证）；cc_n: 新图点数；cur_e: 新图边表；blk: 点所属 SCC 标号。

```

struct scc
{
    int n, id, tp, cc_n;
    vector<vector<pair<int, int>>> e, cur_e;
    vector<vector<int>> cc_node;
    vector<int> dfn, low, st, ed, blk;
    void dfs(int u)
    {
        dfn[u] = low[u] = id++;
        st[tp++] = u; ed[u] = 1;
        for (auto [v, w] : e[u]) if (dfn[v] == -1)
        {

```

```

        dfs(v);
        cmin(low[u], low[v]);
    }
    else if (ed[v]) cmin(low[u], dfn[v]);
    if (low[u] == dfn[u])
    {
        cc_node.push_back({ });
        do
        {
            int v = st[--tp];
            ed[v] = 0;
            blk[v] = cc_n;
            cc_node[cc_n].push_back(v);
        } while (st[tp] != u);
        cc_n++;
    }
}
scc(const vector<vector<pair<int, int>>> &e) :n(e.size()), id(0), tp(0), cc_n(0),
    e(e), cur_e(n), dfn(n, -1), low(dfn), st(n), ed(n), blk(n)
{
    for (int i = 0; i < n; i++) for (auto [v, w] : e[i]) assert(v >= 0 && v < n);
    for (int i = 0; i < n; i++) if (dfn[i] == -1) dfs(i);
    reverse(all(cc_node));
    for (int &x : blk) x = cc_n - x - 1;
    for (int u = 0; u < n; u++)
        for (auto [v, w] : e[u])
            if (blk[u] != blk[v])
                cur_e[blk[u]].push_back({blk[v], w});
    cur_e.resize(cc_n);
}
};
vector<vector<int>> unique(vector<vector<pair<int, int>>> &e)
{
    int n = e.size(), i;
    vector<vector<int>> g(n);
    for (i = 0; i < n; i++)
    {
        for (auto [v, w] : e[i]) g[i].push_back(v);
        sort(all(g[i]));
        g[i].resize(unique(all(g[i])) - g[i].begin());
    }
    return g;
}
}

```

5.5.5 动态强连通分量

给出一个加边序列，solve 会返回每个时间进入强连通分量的边。点标号范围是 $[0, n)$

```

struct union_set
{
    vector<int> f;
    int n;
    union_set() { }
    union_set(int nn) :n(nn), f(nn + 1)
    {
        iota(all(f), 0);
    }
}

```

```

}
int getf(int u) { return f[u] == u ? u : f[u] = getf(f[u]); }
bool merge(int u, int v)
{
    u = getf(u); v = getf(v);
    if (u == v) return 0;
    f[u] = v;
    return 1;
}
bool connected(int u, int v) { return getf(u) == getf(v); }
};
struct edge
{
    int u, v, t;
};
vector<vector<edge>> solve(int n, const auto &eg)//[0,n)
{
    int m = eg.size(), tp = -1, id = 0, fs = 0;
    vector<vector<edge>> res(m);
    vector e(n, vector<int>());
    vector<int> dfn(n, -1), low(n, -1), st(n), ed(n), blk(n), node;
    union_set s(n - 1);
    function<void(int)> dfs = [&](int u) {
        dfn[u] = low[u] = id++;
        ed[st[++tp] = u] = 1;
        for (int v : e[u]) if (dfn[v] != -1)
        {
            if (ed[v]) cmin(low[u], dfn[v]);
        }
        else dfs(v), cmin(low[u], low[v]);
        if (dfn[u] == low[u])
        {
            do
            {
                ed[st[tp]] = 0;
                blk[st[tp]] = fs;
            } while (st[tp--] != u);
            ++fs;
        }
    };
    auto ztef = [&](auto ztef, int l, int r, const vector<edge> &q) {
        if (eg.size() == 0) return;
        if (l + 1 == r)
        {
            if (l < m)
            {
                res[l].insert(res[l].end(), all(q));
                for (auto [u, v, t] : q) s.merge(u, v);
            }
            return;
        }
        int m = (l + r) / 2;
        node.clear();
        for (auto [u, v, t] : q) if (t < m)
        {
            u = s.getf(u);
            v = s.getf(v);

```

```

        e[u].push_back(v);
        node.push_back(u);
        node.push_back(v);
    }
    else break;
    for (int u : node) if (dfn[u] == -1) dfs(u);
    vector<vector<edge>> g(2);
    for (auto [u, v, t] : q) g[t < m && blk[s.f[u]] == blk[s.f[v]]].push_back({u, v, t});
    for (int u : node)
    {
        e[u].clear();
        dfn[u] = low[u] = -1;
    }
    id = fs = 0;
    ztef(ztef, 1, m, g[1]);
    ztef(ztef, m, r, g[0]);
};
ztef(ztef, 0, m + 1, eg);
return res;
}
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cout << fixed << setprecision(15);
    int n, m, i, j;
    cin >> n >> m;
    vector<ull> x(n);
    cin >> x;
    vector<edge> edges(m);
    for (i = 0; i < m; i++)
    {
        auto &[u, v, t] = edges[i];
        cin >> u >> v;
        t = i;
    }
    auto event = solve(n, edges);
    union_set s(n - 1);
    ull ans = 0;
    for (auto e : event)
    {
        for (auto [u, v, t] : e)
        {
            u = s.getf(u);
            v = s.getf(v);
            if (u == v) continue;
            s.f[v] = u;
            (ans += x[u] * x[v]) %= p;
            (x[u] += x[v]) %= p;
        }
        cout << ans << '\n';
    }
}

```

5.5.6 圆方树

题意：求仙人掌上两点最短路。

$O(n + m)$, $O(n + m)$ 。

```
#include "bits/stdc++.h"
using namespace std;
#if !defined(ONLINE_JUDGE)&&defined(LOCAL)
#include "my_header/debug.h"
#else
#define dbg(...); 1;
#endif
typedef unsigned int ui;
typedef long long ll;
#define all(x) (x).begin(), (x).end()
const int N = 3e4 + 2, M = 3e4 + 2; //M 包括方点
struct P
{
    int v, w, id;
    P(int a, int b, int c) : v(a), w(b), id(c) { }
};
struct Q
{
    int v, w;
    Q(int a, int b) : v(a), w(b) { }
};
vector<P> e[N];
vector<Q> fe[M];
int dfn[M], low[N], st[N], len[M], top[M], siz[M], hc[M], dep[M], f[M], rb[N];
bool ed[M]; //ed, dfn, loop, sum, fe, hc, tp, id, cnt, dep[1] 需初始化 (注意倍率), ed 大小为边数
int tp, id, cnt, n;
void dfs1(int u)
{
    dfn[u] = low[u] = ++id;
    st[++tp] = u;
    for (auto [v, w, id] : e[u]) if (!ed[id])
    {
        if (dfn[v]) low[u] = min(low[u], dfn[v]), rb[v] = w; else
        {
            ed[id] = 1;
            dfs1(v);
            if (dfn[u] > low[v]) low[u] = min(low[u], low[v]), rb[v] = w; else
            {
                int ntp = tp;
                while (st[ntp] != v) --ntp;
                if (ntp == tp) //圆圆边
                {
                    --tp;
                    fe[u].emplace_back(v, w);
                    f[v] = u;
                    continue;
                }
                ++cnt; f[cnt] = u;
                for (int i = ntp; i <= tp; i++) f[st[i]] = cnt;
                len[st[ntp]] = w;
                for (int i = ntp + 1; i <= tp; i++) len[st[i]] = len[st[i - 1]] + rb[st[i]];
                len[cnt] = len[st[tp]] + rb[u];
                fe[u].emplace_back(cnt, 0);
                for (int i = ntp; i <= tp; i++) fe[cnt].emplace_back(st[i], min(len[st[i]], len[
                    cnt] - len[st[i]]));
                tp = ntp - 1;
            }
        }
    }
}
```

```

    }
    }
}
void dfs2(int u)
{
    siz[u] = 1;
    for (auto [v, w] : fe[u])
    {
        dep[v] = dep[u] + w;
        dfs2(v);
        siz[u] += siz[v];
        if (siz[v] > siz[hc[u]]) hc[u] = v;
    }
}
void dfs3(int u)
{
    dfn[u] = ++id;
    if (hc[u])
    {
        top[hc[u]] = top[u];
        dfs3(hc[u]);
        for (auto [v, w] : fe[u]) if (v != hc[u]) dfs3(top[v] = v);
    }
}
int lca(int u, int v)
{
    while (top[u] != top[v]) if (dfn[top[u]] > dfn[top[v]]) u = f[top[u]]; else v = f[top[v]]; //
    注意不能用 dep
    return dfn[u] < dfn[v] ? u : v;
}
int find(int u, int v) // u 是根
{
    if (dfn[hc[u]] + siz[hc[u]] > dfn[v]) return hc[u];
    while (f[top[v]] != u) v = f[top[v]];
    return top[v];
}
int dis(int u, int v)
{
    int o = lca(u, v), r = dep[u] + dep[v];
    if (o <= n) return r - (dep[o] << 1);
    u = find(o, u); v = find(o, v);
    if (len[u] > len[v]) swap(u, v);
    return r + min(len[v] - len[u], len[o] - (len[v] - len[u])) - dep[u] - dep[v];
}
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    int m, q, i;
    cin >> n >> m >> q; cnt = n;
    for (i = 1; i <= m; i++)
    {
        int u, v, w;
        cin >> u >> v >> w;
        e[u].emplace_back(v, w, i);
        e[v].emplace_back(u, w, i);
    }
}

```



```

mt19937 rnd(time(0));
for (i = 1; i <= n; i++) shuffle(all(e[i]), rnd);
dfs1(1); id = 0;
dfs2(1);
dfs3(top[1] = 1);
while (q--)
{
    int u, v;
    cin >> u >> v;
    cout << dis(u, v) << '\n';
}
}

```

5.5.7 广义圆方树

建议使用点双来做这个。你只需要对每个点双建一个虚点，向点双内所有原点连边，就得到了广义圆方树。

```

void dfs(int u)
{
    dfn[u] = low[u] = ++id;
    st[++tp] = u;
    for (int v : e[u]) if (dfn[v]) low[u] = min(low[u], dfn[v]); else
    {
        dfs(v);
        low[u] = min(low[u], low[v]);
        if (dfn[u] <= low[v])
        {
            vector cur = {u};
            do
            {
                cur.push_back(st[tp]);
            } while (st[tp--] != v);
            ans.push_back(cur);
        }
    }
}
}

```

5.5.8 2-sat

支持添加一个条件 $\text{add}(u, x, v, y)$ ，表示 $a_u = x \Rightarrow a_v = y$ 。支持设定一个变量的值。
 $O(n + m)$, $O(n + m)$ 。

```

struct sat
{
    vector<vector<int>> e;
    vector<int> dfn, low, st, f, ed;
    int fs, tp, id, n;
    sat(int n) : n(n), e(n * 2), dfn(n * 2, -1), low(n * 2), st(n * 2), f(n * 2, -1), ed(n * 2),
    fs(0), tp(-1), id(0) { }
    void dfs(int u)
    {
        dfn[u] = low[u] = id++;
        ed[u] = 1; st[++tp] = u;
        for (int v : e[u]) if (dfn[v] != -1)
        {

```

```

        if (ed[v]) low[u] = min(low[u], dfn[v]);
    }
    else dfs(v), low[u] = min(low[u], low[v]);
    if (dfn[u] == low[u])
    {
        do
        {
            f[st[tp]] = fs;
            ed[st[tp]] = 0;
        } while (st[tp--] != u);
        ++fs;
    }
}
void add(int u, bool x, int v, bool y)
{
    assert(u >= 0 && u < n && v >= 0 && v < n);
    e[u + x * n].push_back(v + y * n);
    e[v + (y ^ 1) * n].push_back(u + (x ^ 1) * n);
}
void set(int u, bool x)
{
    assert(u >= 0 && u < n);
    e[u + (x ^ 1) * n].push_back(u + x * n);
}
vector<int> getans()
{
    int i;
    for (i = 0; i < n * 2; i++) if (dfn[i] == -1) dfs(i);
    vector<int> r(n);
    for (i = 0; i < n; i++)
    {
        if (f[i] == f[i + n]) return { };
        r[i] = f[i] > f[i + n];
    }
    return r;
}
};

```

5.5.9 Kosaraju 强连通分量 (bitset 优化)

实用意义不大。

$O(\frac{n^2}{w})$, $O(\frac{n^2}{w})$ 。

```

void dfs1(int x)
{
    int i; ed[x] = 0;
    for (i = (lj[x] & ed)._Find_first(); i <= n; i = (lj[x] & ed)._Find_next(i)) dfs1(i);
    sx[--tp] = x;
}
void dfs2(int x)
{
    int i; ed[x] = 0; tv[f[x] = f[0]] += v[x];
    for (i = (fj[x] & ed)._Find_first(); i <= n; i = (fj[x] & ed)._Find_next(i)) dfs2(i);
}
int main()
{
    cin >> n >> m;

```

```

tp = n + 1;
for (i = 1; i <= n; i++) { ed[i] = 1; cin >> v[i]; }
for (i = 1; i <= m; i++)
{
    cin >> x >> y;
    lj[x][y] = 1; fj[y][x] = 1; lb[i][0] = x; lb[i][1] = y;
}
for (i = 1; i <= n; i++) if (ed[i]) dfs1(i);
ed.set();
for (i = 1; i <= n; i++) if (ed[sx[i]]) { ++f[0]; dfs2(sx[i]); }
for (i = 1; i <= m; i++) if (f[lb[i][0]] != f[lb[i][1]])
{
    flj[f[lb[i][0]]].push_back(f[lb[i][1]]); ++rd[f[lb[i][1]]];
}
for (i = 1; i <= f[0]; i++) if (!rd[i]) dl[++wei] = i;
while (tou <= wei)
{
    x = dl[tou++]; g[x] += tv[x];
    for (i = 0; i < flj[x].size(); i++)
    {
        g[flj[x][i]] = max(g[flj[x][i]], g[x]);
        if (--rd[flj[x][i]] == 0) dl[++wei] = flj[x][i];
    }
}
for (i = 1; i <= f[0]; i++) ans = max(ans, g[i]); printf("%d", ans);
}

```

5.6 树上问题

5.6.1 换根 dp

$[0, n)$ 。

第一版要求 $\text{merge}(x, y, W\{\})$ 表示同一父结点坐标系下合并；若合并语义更复杂，使用合并版。
zero 版：

```

template <class T, class W, class F>
struct solver
{
    int n;
    vector<T> f, g;
    vector<int> dep;
    solver(const vector<tuple<int, int, W>> &eg, vector<T> a, const F &merge, int rt = 0, T zero = T{ })
        : n(a.size()), f(a), g(n), dep(n)
    {
        assert(eg.size() == n - 1);
        vector<vector<pair<int, W>>> e(n);
        vector<W> lf(n);
        for (auto [u, v, w] : eg)
        {
            e[u].push_back({v, w});
            e[v].push_back({u, w});
        }
        {
            auto dfs = [&](auto &&dfs, int u) -> void {
                for (auto [v, w] : e[u])

```

```

        {
            erase(e[v], pair{u, w});
            dep[v] = dep[u] + 1;
            lf[v] = w;
            dfs(dfs, v);
            merge(f[u], f[v], w);
        }
    };
    dfs(dfs, rt);
}
{
    auto dfs = [&](auto &&dfs, int u)->void {
        T s = a[u];
        if (u != rt) merge(s, g[u], lf[u]);
        for (auto [v, w] : e[u]) g[v] = s, merge(s, f[v], w);
        s = zero;
        reverse(all(e[u]));
        for (auto [v, w] : e[u]) merge(g[v], s, W{ }), dfs(dfs, v), merge(s, f[v], w);
    };
    dfs(dfs, rt);
}
}
};

```

合并版:

```

template <class T, class W, class F1, class F2>
struct solver
{
    int n;
    vector<T> f, g;
    vector<int> dep;
    solver(const vector<tuple<int, int, W>> &eg, vector<T> a, const F1 &merge, const F2 &merge_ex, int rt = 0)
        :n(a.size()), f(a), g(n), dep(n)
    {
        assert(eg.size() == n - 1);
        vector<vector<pair<int, W>>> e(n);
        vector<W> lf(n);
        for (auto [u, v, w] : eg)
        {
            e[u].push_back({v, w});
            e[v].push_back({u, w});
        }
        {
            auto dfs = [&](auto &&dfs, int u)->void {
                for (auto [v, w] : e[u])
                {
                    erase(e[v], pair{u, w});
                    dep[v] = dep[u] + 1;
                    lf[v] = w;
                    dfs(dfs, v);
                    merge(f[u], f[v], w);
                }
            };
            dfs(dfs, rt);
        }
    }
};

```

```

        auto dfs = [&](auto &&dfs, int u)->void {
            T s = a[u];
            if (u != rt) merge(s, g[u], lf[u]);
            for (auto [v, w] : e[u]) g[v] = s, merge(s, f[v], w);
            s = a[u];
            reverse(all(e[u]));
            for (auto [v, w] : e[u]) merge_ex(g[v], s, w, a[u]), dfs(dfs, v), merge(s, f[v],
w);
        };
        dfs(dfs, rt);
    }
}
};

```

5.6.2 轻重链剖分/DFS 序 LCA

首先 `init(n)`，然后正常存边 $([1, n])$ ，然后 `fun(root)`。
`get_path` 会返回这条路径上的 `dfn` 区间。

```

namespace HLD
{
    const int N = 5e5 + 2;
    vector<int> e[N];
    int dfn[N], nfd[N], dep[N], f[N], siz[N], hc[N], top[N];
    int id, n;
    void dfs1(int u)
    {
        siz[u] = 1;
        for (int v : e[u]) if (v != f[u])
        {
            dep[v] = dep[f[v] = u] + 1;
            dfs1(v);
            siz[u] += siz[v];
            if (siz[v] > siz[hc[u]]) hc[u] = v;
        }
    }
    void dfs2(int u)
    {
        dfn[u] = ++id;
        nfd[id] = u;
        if (hc[u])
        {
            top[hc[u]] = top[u];
            dfs2(hc[u]);
            for (int v : e[u]) if (v != hc[u] && v != f[u]) dfs2(top[v] = v);
        }
    }
    int lca(int u, int v)
    {
        while (top[u] != top[v])
        {
            if (dep[top[u]] < dep[top[v]]) swap(u, v);
            u = f[top[u]];
        }
        if (dep[u] > dep[v]) swap(u, v);
        return u;
    }
}

```

```

int dis(int u, int v)
{
    return dep[u] + dep[v] - (dep[lca(u, v)] << 1);
}
void init(int _n)
{
    n = _n;
    for (int i = 1; i <= n; i++)
    {
        e[i].clear();
        f[i] = hc[i] = 0;
    }
    id = 0;
}
void fun(int root)
{
    dep[root] = 1; dfs1(root); dfs2(top[root] = root);
}
vector<pair<int, int>> get_path(int u, int v) //u->v, 注意可能出现 [r>l] (表示反过来走)
{
    //cerr<<"path from "<<u<<" to "<<v<<": ";
    vector<pair<int, int>> v1, v2;
    while (top[u] != top[v])
    {
        if (dep[top[u]] > dep[top[v]]) v1.push_back({dfn[u], dfn[top[u]]}), u = f[top[u]];
        else v2.push_back({dfn[top[v]], dfn[v]}), v = f[top[v]];
    }
    v1.reserve(v1.size() + v2.size() + 1);
    v1.push_back({dfn[u], dfn[v]});
    reverse(v2.begin(), v2.end());
    for (auto v : v2) v1.push_back(v);
    //for (auto [x,y]:v1) cerr<<"["<<x<<','<<y<<"] ";cerr<<endl;
    return v1;
}
}
using HLD::e, HLD::dfn, HLD::nfd, HLD::dep, HLD::f, HLD::siz, HLD::get_path;
using HLD::init; //5e5
namespace LCA
{
    using HLD::N, HLD::n;
    int st[__lg(N) + 1][N];
    int cmp(const int &x, const int &y) { return dep[x] < dep[y] ? x : y; }
    void fun(int rt)
    {
        HLD::fun(rt);
        assert(f[rt] == 0);
        for (int i = 1; i <= n; i++) st[0][dfn[i] - 1] = f[i];
        for (int j = 0; j < __lg(n); j++)
            for (int i = 1, k = n - (1 << j + 1); i <= k; i++)
                st[j + 1][i] = cmp(st[j][i], st[j][i + (1 << j)]);
    }
    int lca(int u, int v)
    {
        if (u == v) return u;
        u = dfn[u], v = dfn[v];
        if (u > v) swap(u, v);
        int g = __lg(v - u);
    }
}

```

```

        return cmp(st[g][u], st[g][v - (1 << g)]);
    }
    int dis(int u, int v)
    {
        return dep[u] + dep[v] - (dep[lca(u, v)] << 1);
    }
}
using LCA::lca, LCA::fun, LCA::dis;

struct chain
{
    int u, v, w;
    chain(int u = 0, int v = 0) : u(u), v(v) { w = lca(u, v); }
    chain(int u, int v, int w) : u(u), v(v), w(w) { }
};
int deeper(int u, int v) { return dep[u] > dep[v] ? u : v; }
bool have(int u, int v) { return dfn[u] <= dfn[v] && dfn[v] < dfn[u] + siz[u]; }
chain operator&(chain a, chain b)
{
    if (!a.u || !b.u) return { };
    if (dep[a.w] > dep[b.w]) swap(a, b);
    auto [u1, v1, w1] = a;
    auto [u2, v2, w2] = b;
    if (have(w1, w2) && (have(w2, u1) || have(w2, v1)))
    {
        if (w1 == w2)
            return {deeper(lca(u1, v1), lca(u1, v2)),
                    deeper(lca(u2, v1), lca(u2, v2))};
        int w = lca(u1, v1);
        if (w == w2) return {w2, lca(u2, v2), w2};
        if (w == w1) return {w2, deeper(lca(u2, v1), lca(u2, v2)), w2};
        return {w2, w, w2};
    }
    return { };
}
}

```

5.6.3 换根树剖

本质是对普通树剖在换根后的子树进行分类讨论。

设预处理的根是 u ，当前根是 v ，那么 w 的子树如下：

1. $w = v$ ， dfn 区间为 $[1, n]$ 。
2. w 在 u, v 之间， dfn 区间为 $[1, n]$ 去掉 w 前往 v 方向的子树。找到这个子树的方法见 `find` 函数。
3. 其余情况， dfn 区间和原来一致。

```

int find(int x, int y) // 找到 y 向 x 的子树
{
    while ((top[x] != top[y]) && (f[top[x]] != y)) x = f[top[x]];
    if (top[x] == top[y]) return hc[y];
    return top[x];
}

```

$O(n + q \log n)$, $O(n)$ 。

5.6.4 毛毛虫剖分

毛毛虫剖分，一种由轻重链剖分（HLD）推广而成的树上结点重标号方法，支持修改 / 查询一只毛毛虫的信息，并且可以对毛毛虫的身体和足分别修改 / 查询不同信息。

严格强于树剖，而且复杂度和树剖一样哦！

一些定义（默认在一棵树上）：

毛毛虫：一条链和与这条链邻接的所有结点构成的集合。虫身（身体）：毛毛虫的链部分。虫足（足）：毛毛虫除虫身的部分。重标号方法首先重剖求出重链。DFS，若现在处理到结点 u ：若 u 还未被标号，则为其标号。若 u 是重链头，遍历这条重链，将邻接这条链的结点依次标号。先递归重儿子，再递归轻儿子。重标号性质对于重链，除链头外的结点标号连续。对于任意结点，其轻儿子标号连续。对于以重链头为根的子树，与这条重链邻接的所有结点标号连续。这样就可以随便维护毛毛虫信息了，顺便还能维护链信息，子树信息等。

时间复杂度同轻重链剖分。

以 SAM 为例，若我们只保留所有的转移边 (u, v) ，满足到达 u 的路径数目大于到达 v 的路径数目一半，且从 v 出发的路径数目大于从 u 出发的路径数目一半，这样剩余的子图显然会形成若干条链，且每个点恰好在一条链上。这样，我们容易证明，从根结点出发的任何一条路径，至多经过 $O(\log n)$ 条不在链上的转移边（也意味着至多经过 $O(\log n)$ 条链）。

以下是一段示例代码，展示了将一条链对应区间取出来的过程

```
vector<int> e[N];
vector<pair<int, int>> seg[N], qu[N];
int ans[Q];
int dfn[N], dep[N], nfd[N], top[N], f[N], sz[N], hc[N], pre[N], fir[N], lst2[N], rt[N];
int
void insert()
void dfs1(int u)
{
    sz[u] = 1;
    for (int v : e[u]) if (v != f[u])
    {
        dep[v] = dep[u] + 1;
        f[v] = u;
        dfs1(v);
        sz[u] += sz[v];
        if (sz[v] > sz[hc[u]]) hc[u] = v;
    }
    if (f[u]) erase(e[u], f[u]);
}
void dfs2(int u)
{
    static int id = 0;
    //dbg(u);
    if (!dfn[u])
    {
        dfn[u] = ++id;
        nfd[id] = u;
    }
    if (top[u] == u)
    {
        vector<int> stk;
        for (int v = u; v; v = hc[v])
        {
            for (int w : e[v]) if (w != hc[v])
            {
```



```

        dfn[w] = ++id;
        nfd[id] = w;
        pre[v] = id;
        cmin(fir[v], id);
        lst2[v] = id;
    }
    stk.push_back(v);
}
for (int i = (int)stk.size() - 2; i >= 0; i--)
{
    cmin(fir[stk[i]], fir[stk[i + 1]]);
    cmax(lst2[stk[i]], lst2[stk[i + 1]]);
}
for (int i = 1; i < stk.size(); i++)
{
    cmax(pre[stk[i]], pre[stk[i - 1]]);
}
}
//dbg(u);
top[hc[u]] = top[u];
if (hc[u]) dfs2(hc[u]);
for (int v : e[u]) if (v != hc[u]) dfs2(top[v] = v);
}
mt19937 rnd(245);
int main()
{
    memset(fir, 0x3f, sizeof fir);
    ios::sync_with_stdio(0); cin.tie(0);
    cout << fixed << setprecision(15);
    int n, m, q, i, j;
    cin >> n >> m >> q;
    for (i = 1; i < n; i++)
    {
        int u, v;
        //cin >> u >> v;
        u = i + 1;
        v = rnd() % i + 1;
        //v = (i + 1) / 2;
        //v = i / 2 + 1;
        //dbg(u, v);
        e[u].push_back(v);
        e[v].push_back(u);
    }
    dfs1(dep[1] = 1);
    //dbg("??");
    dfs2(top[1] = 1);
    //for (i = 1; i <= n; i++) cerr << i << ": " << dfn[i] << endl;
    for (i = 1; i <= m; i++)
    {
        int u, v;
        //cin >> u >> v;
        u = rnd() % n + 1;
        v = rnd() % n + 1;
        int uu = u, vv = v;
        //dbg(uu, vv);
        auto &w = seg[i];
        while (top[u] != top[v])

```

```

{
    if (dep[top[u]] < dep[top[v]]) swap(u, v);
    w.push_back({fir[top[u]], pre[u]});
    //else w.push_back({fir[top[u]], lst2[top[u]]});
    if (hc[u]) w.push_back({dfn[hc[top[u]]], dfn[hc[u]]});
    else if (top[u] != u) w.push_back({dfn[hc[top[u]]], dfn[u]});
    //dbg(u, v, w);
    //[[fir[top[u]],lst[u]]
    u = f[top[u]];
}
if (dep[u] < dep[v]) swap(u, v);
w.push_back({fir[v], pre[u]});
//else if (!hc[u]) w.push_back({fir[v], lst2[v]});
//dbg(v, lst2[v], fir[v]);
if (hc[u]) w.push_back({dfn[hc[v]], dfn[hc[u]]});
else if (u != v) w.push_back({dfn[hc[v]], dfn[u]});
//dbg(w);
w.push_back({dfn[v], dfn[v]});
if (f[v]) w.push_back({dfn[f[v]], dfn[f[v]]});
erase_if(w, [&](const auto &x) {return x.first > x.second; });
//int len = 0;
//for (auto [l, r] : w) len += r - l + 1;
//for (auto [l, r] : w)
//{
//    for (int j = l; j <= r; j++) cerr << nfd[j] << ' '; cerr << " | ";
//}
//cerr << endl;
//int tl = 0;
//set<int> s = {uu, vv};
//while (uu != vv)
//{
//    if (dep[uu] < dep[vv]) swap(uu, vv);
//    s.insert(all(e[uu]));s.insert(f[uu]);uu = f[uu];
//}
//s.insert(all(e[uu]));
//if (f[uu]) s.insert(f[uu]);
////dbg(s);
//assert(len == s.size());
}
for (i = 1; i <= q; i++)
{
    int l, r;
    cin >> l >> r;
    qu[l].push_back({r, i});
}
for (i = m; i; i--)
{
}
for (i = 1; i <= q; i++) cout << ans[i] << '\n';
//cerr << "??\n";
}

```

5.6.5 树上启发式合并, DSU on tree

一种过时的、基于两次 dfs 的写法，在复杂度要求不严时不如直接存储 set。

流程:

1. dfs 轻子树计算答案，并清空全局统计信息。
2. dfs 重子树统计答案和全局信息。
3. dfs 轻子树统计全局信息。

```
void dfs1(int x)
{
    siz[x] = zdep[x] = 1;
    int i;
    for (i = fir[x]; i; i = nxt[i]) if (lj[i] != f[x])
    {
        dep[lj[i]] = dep[f[lj[i]] = x] + 1;
        dfs1(lj[i]);
        siz[x] += siz[lj[i]];
        if (siz[hc[x]] < siz[lj[i]]) hc[x] = lj[i];
        zdep[x] = max(zdep[x], zdep[lj[i]] + 1);
    }
}

void cal(int x)
{
    int i;
    dl[tou = wei = 1] = x;
    while (tou <= wei)
    {
        ++dp[dep[x = dl[tou++]]];
        if ((dp[dep[x]] > dp[zd]) || (dp[dep[x]] == dp[zd]) && (dep[x] < zd)) zd = dep[x];
        for (i = fir[x]; i; i = nxt[i]) if (lj[i] != f[x]) dl[++wei] = lj[i];
    }
}

void dfs2(int x)
{
    if (!hc[x])
    {
        if (++dp[dep[x]] > dp[zd]) zd = dep[x];
        return;
    }
    int i;
    for (i = fir[x]; i; i = nxt[i]) if ((lj[i] != f[x]) && (lj[i] != hc[x]))
    {
        dfs2(lj[i]);
        memset(dp + dep[lj[i]], 0, zdep[lj[i]] << 2);
    }
    dfs2(hc[x]);
    dp[dep[x]] = 1;
    if (dp[zd] <= 1) zd = dep[x];
    for (i = fir[x]; i; i = nxt[i]) if ((lj[i] != f[x]) && (lj[i] != hc[x])) cal(lj[i]);
    ans[x] = zd - dep[x];
}
```

5.6.6 长链剖分 (k 级祖先)

$O(n \log n + q)$, $O(n)$ 。

```

void dfs1(int x)
{
    int i;
    for (i = 1; i <= er[dep[x] - 1]; i++) f[x][i] = f[f[x][i - 1]][i - 1]; md[x] = dep[x];
    for (i = fir[x]; i; i = nxt[i]) { dep[lj[i]] = dep[x] + 1; dfs1(lj[i]); if (md[lj[i]] > md[dc[x]]) dc[x] = lj[i]; }
    if (dc[x]) md[x] = md[dc[x]];
}

void dfs2(int x)
{
    int i;
    if (dc[x])
    {
        top[dc[x]] = top[x];
        dfs2(dc[x]);
        for (i = fir[x]; i; i = nxt[i]) if (lj[i] != dc[x]) dfs2(top[lj[i]] = lj[i]);
    }
    if (x == top[x])
    {
        c = md[x] - dep[x]; y = x; up[x].push_back(x); down[x].push_back(x);
        for (i = 1; (i <= c) && (y = f[y][0]); i++) up[x].push_back(y); y = x;
        for (i = 1; i <= c; i++) down[x].push_back(y = dc[y]);
    }
}

int main()
{
    int n, q, ans = 0, x, y, c, i;
    ll ta = 0;
    cin >> n >> q >> s;
    for (i = 1; i <= n; i++) { cin >> f[i][0]; if (f[i][0] == 0) rt = i; else add(f[i][0], i); }
    for (i = 2; i <= n; i++) er[i] = er[i >> 1] + 1; dep[rt] = 1;
    dfs1(rt); dfs2(top[rt] = rt);
    for (i = 1; i <= q; i++)
    {
        x = (get(s) ^ ans) % n + 1; y = (get(s) ^ ans) % dep[x];
        //此时计算 x 的 y 级祖先。结果在 ans 中。
        if (y == 0) { ans = x; ta ^= (ll)i * ans; continue; }
        c = dep[x] - y; x = top[f[x][er[y]]];
        if (dep[x] > c) ans = up[x][dep[x] - c]; else ans = down[x][c - dep[x]];
        ta ^= (ll)i * ans;
    }
    cout << ta << endl;
}

```

5.6.7 长链剖分 (dp 合并)

一种常见的实现方法是用指针指向同一片数组区域，使得从链头到链尾正好指向连续的一段数组，就不需要计算偏移量了。

$O(n)$, $O(n)$ 。

```

void dfs1(int x)
{
    top[x] = 1;
    for (int i = fir[x]; i; i = nxt[i]) if (!top[lj[i]])
    {
        dfs1(lj[i]);
    }
}

```

```

        if (len[lj[i]] > len[hc[x]]) hc[x] = lj[i];
    }
    len[x] = len[hc[x]] + 1; top[hc[x]] = 0;
}
void dfs2(int x)
{
    *f[x] = 1; gs[x] = 1;
    if (!hc[x]) return;
    ed[x] = 1; f[hc[x]] = f[x] + 1;
    for (int i = fir[x]; i; i = nxt[i]) if (!ed[lj[i]]) dfs2(lj[i]);
    ans[x] = ans[hc[x]] + 1; gs[x] = gs[hc[x]];
    if (gs[x] == 1) ans[x] = 0;
    for (int i = fir[x]; i; i = nxt[i]) if ((!ed[lj[i]]) && (lj[i] != hc[x]))
    {
        int v = lj[i], *p;
        for (int j = 0; j < len[v]; j++)
        {
            *(p = f[x] + j + 1) += *(f[v] + j);
            if (j + 1 == ans[x]) { gs[x] = *p; continue; }
            if ((*p > gs[x]) || (*p == gs[x]) && (j + 1 < ans[x])) { gs[x] = *p; ans[x] = j + 1; }
        }
    }
    gs[x] = *(f[x] + ans[x]);
    ed[x] = 0;
}

```

5.6.8 LCT

$O(n \log n)$, $O(n)$ 。

makeroot 会变根, split 会把 y 变根, findroot 会把根变根, link 会把 x, y 变根 (y 是新的), cut 会把 x, y 变根 (x 是新的), 注意 swap 子节点可能要 pushup。

代码为动态割边割点。

```

#include "bits/stdc++.h"
using namespace std;
template<class info, class tag> struct lct
{
    vector<array<int, 2>> c;
    vector<int> f, rev, lz, st;
    vector<info> s, v;
    vector<tag> tg;
#ifdef Rev
    vector<info> rs;
#endif
    lct(int n) : f(n + 1), c(n + 1), s(n + 1), v(n + 1), tg(n + 1), rev(n + 1), lz(n + 1), st(n + 1)
    {
#ifdef Rev
        , rs(n + 1)
    }
#endif

    bool nroot(int x) const
    {
        return c[f[x]][0] == x || c[f[x]][1] == x;
    }
}

```

```

void pushup(int x)
{
    int lc = c[x][0], rc = c[x][1];
    s[x] = v[x];
#ifdef Rev
    rs[x] = v[x];
#endif
    if (lc)
    {
        s[x] = s[lc] + s[x];
#ifdef Rev
        rs[x] = rs[x] + rs[lc];
#endif
    }
    if (rc)
    {
        s[x] = s[x] + s[rc];
#ifdef Rev
        rs[x] = rs[rc] + rs[x];
#endif
    }
}

void swp(int x)
{
    swap(c[x][0], c[x][1]);
#ifdef Rev
    swap(s[x], rs[x]);
#endif
    rev[x] ^= 1;
}

void add(int x, const tag &o)
{
    s[x] += o; v[x] += o;
#ifdef Rev
    rs[x] += o;
#endif
    if (lz[x]) tg[x] += o; else tg[x] = o, lz[x] = 1;
}

void pushdown(int x)
{
    if (rev[x])
    {
        for (int y : c[x]) if (y) swp(y);
        rev[x] = 0;
    }
    if (lz[x])
    {
        for (int y : c[x]) if (y) add(y, tg[x]);
        lz[x] = 0;
    }
}

void zigzag(int x)
{
    int y = f[x], z = f[y], typ = (c[y][0] == x);
    if (nroot(y)) c[z][c[z][1] == y] = x;
    f[x] = z; f[y] = x;
    if (c[x][typ]) f[c[x][typ]] = y;
}

```

```

    c[y][typ ^ 1] = c[x][typ]; c[x][typ] = y;
    pushup(y);
}
void splay(int x)
{
    int y, tp;
    st[tp = 1] = y = x;
    while (nroot(y)) st[++tp] = y = f[y];
    while (tp) pushdown(st[tp--]);
    for (; nroot(x); zigzag(x)) if (nroot(y = f[x])) zigzag((c[y][0] == x) ^ (c[f[y]][0] == y)
) ? x : f[x]);
    pushup(x);
}
int access(int x)
{
    int y = 0;
    for (; x; x = f[y = x]) splay(x), c[x][1] = y, pushup(x);
    return y;
}
int findroot(int x)//splay 根为树根, splay 维护树根到 x 的链
{
    access(x); splay(x); pushdown(x);
    while (c[x][0]) pushdown(x = c[x][0]);
    splay(x); return x;
}
void split(int x, int y)//x 为树新根, y 为 splay 新根
{
    makeroot(x); access(y); splay(y);
}
void makeroot(int x)//x 为树、splay 新根
{
    access(x); splay(x); swp(x);
}
void modify(int x, const info &o)
{
    makeroot(x); v[x] = o; pushup(x);
}
void modify(int x, int y, const tag &o)
{
    split(x, y); add(y, o);
}
info ask(int x, int y) { split(x, y); return s[y]; }
bool connected(int x, int y)//注意会改变形态
{
    makeroot(x); return findroot(y) == x;
}
void link(int x, int y)//y 为新根
{
    if (!connected(x, y)) makeroot(f[x] = y);
}
void cut(int x, int y)
{
    if (connected(x, y))//可能本不连通
    {
        pushdown(x);
        if (c[x][1] == y && !c[y][0] && !c[y][1])//可能连通但无边
        {

```

```

        c[x][1] = f[y] = 0;
        pushup(x);
    }
}
}
int lca(int x, int y) { access(x); return access(y); }
vector<int> res;
void dfs(int x)
{
    if (!x) return;
    pushdown(x);
    dfs(c[x][0]); res.push_back(x); dfs(c[x][1]);
}
vector<int> get_path(int x, int y)
{
    res.clear(); split(x, y); dfs(y);
    if (res[0] != x) return { };
    return res;
}
};
const int N = 2e5 + 5, M = 4e5 + 5;
struct tag
{
    void operator+=(const tag &o) const { }
};
void operator+=(int &x, const tag &o) { x = 0; }
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    int n, m, i, r = 0;
    cin >> n >> m;
    lct<int, tag> s(n * 2), t(n + m);
    for (i = 1; i <= n; i++) s.modify(i + n, 1), t.modify(i, 1);
    int bs = n, ds = n;
    while (m--)
    {
        int op, u, v;
        cin >> op >> u >> v;
        u ^= r; v ^= r;
        if (op == 1)
        {
            if (s.connected(u, v))
            {
                s.modify(u, v, { });
                auto c = t.get_path(u, v);
                for (i = 1; i < c.size(); i++) t.cut(c[i - 1], c[i]);
                ++ds;
                for (int x : c) t.link(ds, x);
            }
            else
            {
                s.link(++bs, u);
                s.link(bs, v);
                t.link(++ds, u);
                t.link(ds, v);
            }
        }
    }
}

```



```

else
{
    if (!s.connected(u, v))
    {
        cout << "-1\n";
        continue;
    }
    r = op == 2 ? s.ask(u, v) : t.ask(u, v);
    cout << r << '\n';
}
}
}

```

5.6.9 带子树的 LCT

$O(n \log n)$, $O(n)$ 。

你需要实现的是 `info` 类的 `+`, `+=`, `-=`。

1. `info` 维护的是从上往下的一条链以及这条链挂着的所有轻子树的信息。
2. `a+b` 表示两条链合并后的信息，其中 a 接近根。
3. `a+=b` 表示一条链合并了一个轻子树，即 b 是 a 链尾的子结点。
4. `a-=b` 表示一条链移除了一个轻子树，即 b 是 a 链尾的子节点。

如果有边权，将边 (u, v) 当成点并与 u, v 分别连边。

代码对应题意：

对于满足 $0 \leq e \leq N-2$ 的整数 e ，定义 $f_e(x) = b_e x + c_e$ 。

设 $e_0, e_1, \dots, e_k$ 为从顶点 x 到顶点 y 的简单路径上的边，按顺序排列，并定义

$$P(x, y) = f_{e_0}(f_{e_1}(\dots f_{e_k}(a_y) \dots)).$$

按给定顺序处理 Q 个查询。查询有两种类型：

- `0 w x r`：将 a_w 更新为 x ，然后输出

$$\left(\sum_{v=0}^{N-1} P(r, v) \right) \bmod 998244353.$$

- `1 e y z r`：将 (b_e, c_e) 更新为 (y, z) ，然后输出

$$\left(\sum_{v=0}^{N-1} P(r, v) \right) \bmod 998244353.$$

```

template<class info> struct lct
{
    int n;
    vector<info> sum, sum_rev, val, sum_oth;
    vector<int> f, lz;
    vector<array<int, 2>> c;
    lct(int _n, const info &o) : n(_n + 1), sum(n, o), sum_rev(n, o), val(n, o), sum_oth(n, o), f(
n), lz(n), c(n) { }

```

```

bool nroot(int x) const
{
    return c[f[x]][0] == x || c[f[x]][1] == x;
}
void pushup(int x)
{
    sum[x] = val[x];
    sum[x] += sum_oth[x];
    sum_rev[x] = sum[x];
    sum[x] = sum[c[x][0]] + sum[x] + sum[c[x][1]];
    sum_rev[x] = sum_rev[c[x][1]] + sum_rev[x] + sum_rev[c[x][0]];
}
void rev(int x)
{
    if (x)
    {
        swap(c[x][0], c[x][1]);
        swap(sum[x], sum_rev[x]);
        lz[x] ^= 1;
    }
}
void pushdown(int x)
{
    if (lz[x])
    {
        rev(c[x][0]);
        rev(c[x][1]);
        lz[x] = 0;
    }
}
void zigzag(int x)
{
    int y = f[x], z = f[y], typ = (c[y][0] == x);
    if (nroot(y)) c[z][c[z][1] == y] = x;
    f[x] = z; f[y] = x;
    if (c[x][typ]) f[c[x][typ]] = y;
    c[y][typ ^ 1] = c[x][typ]; c[x][typ] = y;
    pushup(y);
}
void splay(int x)
{
    static vector<int> st(n);
    int y, tp;
    st[tp = 1] = y = x;
    while (nroot(y)) st[++tp] = y = f[y];
    while (tp) pushdown(st[tp--]);
    for (; nroot(x); zigzag(x)) if (!nroot(f[x])) continue; else zigzag((c[f[x]][0] == x) ^ (
c[f[f[x]]][0] == f[x]) ? x : f[x]);
    pushup(x);
}
void access(int x)
{
    for (int y = 0; x; x = f[y = x])
    {
        splay(x);
        sum_oth[x] -= sum[y];
        sum_oth[x] += sum[c[x][1]];
    }
}

```

```

        c[x][1] = y; pushup(x);
    }
}
int findroot(int x)
{
    access(x); splay(x); pushdown(x);
    while (c[x][0]) pushdown(x = c[x][0]);
    splay(x);
    return x;
}
void split(int x, int y)
{
    makeroot(x);
    access(y);
    splay(y);
}
void makeroot(int x)
{
    access(x);
    splay(x);
    rev(x);
}
void link(int x, int y)
{
    makeroot(x);
    if (x != findroot(y))//可能已经连通
    {
        makeroot(y); f[x] = y;
        sum_oth[y] += sum[x];
        pushup(y);
    }
}
void cut(int x, int y)
{
    makeroot(x);
    if (x == findroot(y))//可能本不连通
    {
        pushdown(x);
        if (c[x][1] == y && !c[y][0] && !c[y][1])//可能连通但无边
        {
            c[x][1] = f[y] = 0;
            pushup(x);
        }
    }
}
void set(int x, info y)
{
    makeroot(x);
    val[x] = y;
    pushup(x);
}
};
const ull p = 998244353;
struct Q
{
    ull k, b, sum, sz;
    Q operator+(const Q &o) const

```

```

{
    return {k * o.k % p, (b + k * o.b) % p, (sum + k * o.sum + b * o.sz) % p, sz + o.sz};
}
void operator+=(const Q &o)
{
    (sum += k * o.sum + b * o.sz) %= p;
    sz += o.sz;
}
void operator-=(const Q &o)
{
    (sum += p * p * 2 - k * o.sum - b * o.sz) %= p;
    sz -= o.sz;
}
};
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    int n, m, i;
    cin >> n >> m;
    vector<Q> a(n * 2 + 1);
    for (i = 1; i <= n; i++)
    {
        ull x;
        cin >> x;
        a[i] = {1, 0, x, 1};
    }
    vector<pair<int, int>> edges(n);
    for (i = 1; i < n; i++)
    {
        auto &[u, v] = edges[i];
        ull k, b;
        cin >> u >> v >> k >> b;
        ++u, ++v;
        a[i + n] = {k, b, 0, 0};
    }
    lct<Q> s(n * 2 - 1, Q{1, 0, 0, 0});
    for (i = 1; i < n * 2; i++) s.set(i, a[i]);
    for (i = 1; i < n; i++)
    {
        auto [u, v] = edges[i];
        s.link(u, n + i);
        s.link(v, n + i);
    }
    while (m--)
    {
        int op;
        cin >> op;
        if (op == 0)
        {
            int u;
            ull x;
            cin >> u >> x;
            ++u;
            a[u] = {1, 0, x, 1};
            s.set(u, a[u]);
        }
        else

```

```

    {
        int id;
        ull k, b;
        cin >> id >> k >> b;
        ++id;
        a[id + n] = {k, b, 0, 0};
        s.set(id + n, a[id + n]);
    }
    int rt;
    cin >> rt;
    ++rt;
    s.makeroot(rt);
    cout << s.sum[rt].sum << '\n';
}
}

```

5.6.10 动态 dp (全局平衡二叉树)

意义不大。

$O((n + q) \log n)$, $O(n)$ 。

```

#include <stdio.h>
#include <string.h>
#include <algorithm>
#include <fstream>
using namespace std;
const int N = 1e6 + 2, M = 6e7 + 2, INF = -1e9;
struct matrix
{
    int a[2][2];
};
matrix s[N], js;
matrix operator *(matrix x, matrix y)
{
    {
        js.a[0][0] = max(x.a[0][0] + y.a[0][0], x.a[0][1] + y.a[1][0]);
        js.a[0][1] = max(x.a[0][0] + y.a[0][1], x.a[0][1] + y.a[1][1]);
        js.a[1][0] = max(x.a[1][0] + y.a[0][0], x.a[1][1] + y.a[1][0]);
        js.a[1][1] = max(x.a[1][0] + y.a[0][1], x.a[1][1] + y.a[1][1]);
        return js;
    }
}
int st[N], c[N][2], hc[N], lj[N << 1], nxt[N << 1], fir[N], siz[N], v[N], g[N][2], fa[N], f[N],
    val[N];
int n, m, i, j, x, y, z, dtp, stp, tp, fh, bs, rt, aaa, la;
char dr[M + 5], sc[M];
void pushup(int x)
{
    {
        s[x].a[0][0] = s[x].a[0][1] = g[x][0];
        s[x].a[1][0] = g[x][1]; s[x].a[1][1] = INF;
        if (c[x][0]) s[x] = s[c[x][0]] * s[x];
        if (c[x][1]) s[x] = s[x] * s[c[x][1]];
    }
}
void add(int x, int y)
{
    {
        lj[++bs] = y;
        nxt[bs] = fir[x];
        fir[x] = bs;
        lj[++bs] = x;
    }
}

```

```

    nxt[bs] = fir[y];
    fir[y] = bs;
}
bool nroot(int x)
{
    return ((c[f[x]][0] == x) || (c[f[x]][1] == x));
}
void dfs1(int x)
{
    siz[x] = 1;
    int i;
    for (i = fir[x]; i; i = nxt[i]) if (lj[i] != fa[x])
    {
        fa[lj[i]] = x;
        dfs1(lj[i]);
        siz[x] += siz[lj[i]];
        if (siz[hc[x]] < siz[lj[i]]) hc[x] = lj[i];
    }
}
int build(int l, int r)
{
    if (l > r) return 0;
    int i, tot = 0, upn = 0;
    for (i = l; i <= r; i++) tot += val[i]; tot >>= 1;
    for (i = l; i <= r; i++)
    {
        upn += val[i];
        if (upn >= tot)
        {
            f[c[st[i]][0] = build(l, i - 1)] = st[i];
            f[c[st[i]][1] = build(i + 1, r)] = st[i];
            pushup(st[i]);
            ++aaa;
            return st[i];
        }
    }
}
int dfs2(int x)
{
    int i, j;
    for (i = x; i; i = hc[i]) for (j = fir[i]; j; j = nxt[j]) if ((lj[j] != fa[i]) && (lj[j] != hc[i]))
    {
        f[y = dfs2(lj[j])] = i;
        g[i][0] += max(s[y].a[0][0], s[y].a[1][0]);
        g[i][1] += s[y].a[0][0];
    }
    tp = 0;
    for (i = x; i; i = hc[i]) st[++tp] = i;
    for (i = 1; i < tp; i++) val[i] = siz[st[i]] - siz[st[i + 1]];
    val[tp] = siz[st[tp]];
    return build(1, tp);
}
void change(int x, int y)
{
    g[x][1] += y - v[x]; v[x] = y;
    while (f[x])

```

```

{
    if (nroot(x)) pushup(x);
    else
    {
        g[f[x]][0] -= max(s[x].a[0][0], s[x].a[1][0]);
        g[f[x]][1] -= s[x].a[0][0];
        pushup(x);
        g[f[x]][0] += max(s[x].a[0][0], s[x].a[1][0]);
        g[f[x]][1] += s[x].a[0][0];
    }
    x = f[x];
}
pushup(x);
}
int main()
{
    scanf("%d%d", &n, &m);
    fread(dr + 1, 1, min(M, n * 20 + m * 20), stdin);
    for (i = 1; i <= n; i++)
    {
        read(g[i][1]);
        v[i] = g[i][1];
    }
    for (i = 1; i < n; i++)
    {
        read(x); read(y);
        add(x, y);
    }
    dfs1(1);
    rt = dfs2(1); tp = 0;
    while (m--)
    {
        read(x); read(y);
        change(x ^ 1a, y);
        x = 1a = max(s[rt].a[0][0], s[rt].a[1][0]);
        while (x)
        {
            st[++tp] = x % 10;
            x /= 10;
        }
        while (tp) sc[++stp] = st[tp--] | 48;
        sc[++stp] = 10;
    }
    fwrite(sc + 1, 1, stp, stdout);
}

```

5.6.11 全局平衡二叉树（修改版）

$O((n + q) \log n)$, $O(n)$ 。

```

#include "bits/stdc++.h"
using namespace std;
typedef long long ll;
typedef pair<int, int> pa;
const int N = 1e6 + 2, M = 1e6 + 2;
ll ans;
pa w[N];

```

```

int c[N][2], f[N], fa[N], v[N], s[N], lz[N], lj[M], nxt[M], siz[N], hc[N], fir[N], st[N];
int a[N], top[N];
int n, i, x, y, z, bs, tp, rt;
void add()
{
    lj[++bs] = y; nxt[bs] = fir[x]; fir[x] = bs;
    lj[++bs] = x; nxt[bs] = fir[y]; fir[y] = bs;
}
void pushup(int &x)
{
    s[x] = min(v[x], min(s[c[x][0]], s[c[x][1]]));
}
void pushdown(int &x)
{
    if (lz[x] < 0)
    {
        int cc = c[x][0];
        if (cc)
        {
            lz[cc] += lz[x]; s[cc] += lz[x]; v[cc] += lz[x];
        }
        cc = c[x][1];
        if (cc)
        {
            v[cc] += lz[x]; lz[cc] += lz[x]; s[cc] += lz[x];
        }
        lz[x] = 0;
        return;
    }
}
bool nroot(int &x) { return c[f[x]][0] == x || c[f[x]][1] == x; }
bool cmp(pa &o, pa &p) { return o > p; }
void dfs1(int x)
{
    siz[x] = 1;
    for (int i = fir[x]; i; i = nxt[i]) if (lj[i] != fa[x])
    {
        fa[lj[i]] = x; dfs1(lj[i]); siz[x] += siz[lj[i]];
        if (siz[hc[x]] < siz[lj[i]]) hc[x] = lj[i];
    }
}
int build(int l, int r)
{
    if (l > r) return 0;
    if (l == r)
    {
        l = st[l]; s[l] = v[l] = siz[l] >> 1;
        return l;
    }
    int x = lower_bound(a + l, a + r + 1, a[l] + a[r] >> 1) - a, y = st[x];
    c[y][0] = build(l, x - 1);
    c[y][1] = build(x + 1, r);
    v[y] = siz[y] >> 1;
    if (c[y][0]) f[c[y][0]] = y;
    if (c[y][1]) f[c[y][1]] = y;
    pushup(y);
    return y;
}

```



```

void dfs2(int x)
{
    if (!hc[x]) return;
    int i;
    top[hc[x]] = top[x];
    if (top[x] == x)
    {
        st[tp = 1] = x;
        for (i = hc[x]; i; i = hc[i]) st[++tp] = i;
        for (i = 1; i <= tp; i++) a[i] = siz[st[i]] - siz[hc[st[i]]] + a[i - 1];
        f[build(1, tp)] = fa[x];
    }
    dfs2(hc[x]);
    for (i = fir[x]; i; i = nxt[i]) if (lj[i] != fa[x] && lj[i] != hc[x]) dfs2(top[lj[i]]) = lj[i];
}

void mdf(int x)
{
    int y = x;
    st[tp = 1] = x;
    while (y = f[y]) st[++tp] = y; y = x;
    while (tp) pushdown(st[tp--]);
    while (x)
    {
        --v[x]; --lz[c[x][0]]; --v[c[x][0]]; --s[c[x][0]];
        while (c[f[x]][0] == x) x = f[x]; x = f[x];
    }
    pushup(y);
    while (y = f[y]) pushup(y);
}

int ask(int x)
{
    int y = x;
    st[tp = 1] = x;
    while (y = f[y]) st[++tp] = y;
    while (tp) pushdown(st[tp--]);
    int r = v[x];
    while (x)
    {
        r = min(r, min(v[x], s[c[x][0]]));
        while (c[f[x]][0] == x) x = f[x]; x = f[x];
    }
    return r;
}

signed main()
{
    cin >> n; s[0] = 1e9;
    for (i = 1; i <= n; i++) cin >> w[w[i].second = i].first;
    for (i = 1; i < n; i++) cin >> x >> y, add();
    sort(w + 1, w + n + 1, cmp); dfs1(1); dfs2(top[1] = 1); rt = 1; while (f[rt]) rt = f[rt];
    for (i = 1; i <= n && v[rt]; i++) if (ask(w[i].second)) mdf(w[i].second), ans += w[i].first;
    cout << ans << endl;
}

```

5.6.12 虚树

传入点标号列表，返回虚树边表。自动认为 1 是根，标号从 1 开始。

需要注意的是：在清空的时候需要同时考虑点列表和边表，都清空一下。

你需要提供的是：dep,lca,dfn。

$O(n + \sum k \log n)$, $O(n)$ 。

```
vector<pair<int, int>> get_tree(vector<int> a)
{
    vector<pair<int, int>> edges;
    sort(all(a), [&](int u, int v) { return dfn[u] < dfn[v]; });
    vector<int> st(a.size() + 2);
    int tp = 0;
    auto ins = [&](int u) {
        if (tp == 0)
        {
            st[tp = 1] = u;
            return;
        }
        int v = lca(st[tp], u);
        while (tp > 1 && dep[v] < dep[st[tp - 1]])
        {
            edges.emplace_back(st[tp - 1], st[tp]);
            --tp;
        }
        if (dep[v] < dep[st[tp]]) edges.emplace_back(v, st[tp--]);
        if (!tp || st[tp] != v) st[++tp] = v;
        st[++tp] = u;
    };
    if (a[0] != 1) st[tp = 1] = 1; //先行添加根节点
    for (int u : a) ins(u);
    if (tp) while (--tp) edges.emplace_back(st[tp], st[tp + 1]); //回溯
    return edges;
}
```

5.6.13 点分治

点分治板子的参考意义不大。

$O(n \log n)$, $O(n)$ 。

```
int siz[N], dep[N];
int n, ksiz, md, rt, mn;
bool ed[N];
void find(int u)
{
    ed[u] = 1; siz[u] = 1;
    int mx = 0;
    for (int v : e[u]) if (!ed[v])
    {
        find(v);
        siz[u] += siz[v];
        mx = max(mx, siz[v]);
    }
    mx = max(mx, ksiz - siz[u]);
    if (mn > mx) mn = mx, rt = u;
    ed[u] = 0;
}
```

```

void cal(int u)
{
    md = max(md, dep[u]);
    ed[u] = 1; ++cnt[dep[u]];
    for (int v : e[u]) if (!ed[v])
    {
        dep[v] = dep[u] + 1;
        cal(v);
    }
    ed[u] = 0;
}

void solve(int u)
{
    mn = 1e9;
    find(u);
    ed[rt] = 1;
    vector<int> c;
    for (int v : e[rt]) if (!ed[v])
    {
        c.push_back(v);
        if (siz[v] >= siz[rt]) siz[v] = siz[u] - siz[rt];
    }
    sort(all(c), [&](const int &a, const int &b) {return siz[a] < siz[b]; });
    NTT::Q a(vector<ui>{1});
    NT::Q b(vector<ui>{1});
    for (int v : c)
    {
        md = 0; dep[v] = 1;
        cal(v); ++md;
        vector<ui> d(cnt, cnt + md);
        NTT::Q e(d);
        NT::Q f(d);
        auto g = e & a;
        auto h = f & b;
        for (int i = 0; i < g.a.size(); i++) r1[i] = (r1[i] + g.a[i]) % NTT::p;
        for (int i = 0; i < h.a.size(); i++) r2[i] = (r2[i] + h.a[i]) % NT::p;
        a += e; b += f;
        fill_n(cnt, md, 0);
    }
    for (int v : c)
    {
        ksiz = siz[v];
        solve(v);
    }
}

```

5.6.14 点分树

核心结论：点分树上 lca 出现在原树路径上。

$O(n \log^2 n)$, $O(n \log n)$ 。

```

template<class typC> struct bit
{
    vector<typC> a;
    int n;
    bit() { }
    bit(int nn) :n(nn), a(nn + 1) { }

```

```

template<class T> bit(int nn, T *b) : n(nn), a(nn + 1)
{
    for (int i = 1; i <= n; i++) a[i] = b[i - 1];
    for (int i = 1; i <= n; i++) if (i + (i & -i) <= n) a[i + (i & -i)] += a[i];
}
void add(int x, typC y)
{
    //cerr<<"add "<<x<<" by "<<y<<endl;
    ++x;
    x = clamp(x, 1, n + 1);
    if (x > n) return;
    assert(1 <= x && x <= n);
    a[x] += y;
    while ((x += x & -x) <= n) a[x] += y;
}
typC sum(int x)
{
    //cerr<<"sum "<<x;
    ++x;
    x = clamp(x, 0, n);
    assert(0 <= x && x <= n);
    typC r = a[x];
    while (x ^= x & -x) r += a[x];
    //cerr<<"=" <<r<<endl;
    return r;
}
typC sum(int x, int y)
{
    return sum(y) - sum(x - 1);
}
int lower_bound(typC x)
{
    if (n == 0) return 0;
    int i = __lg(n), j = 0;
    for (; i >= 0; i--) if ((1 << i | j) <= n && a[1 << i | j] < x) j |= 1 << i, x -= a[j];
    return j + 1;
}
};
namespace DFS
{
    typedef long long ll;
    const int N = 1e5 + 5, M = 18;
    ll a[N];
    int st[M][N * 2], lg[N * 2];
    int dep[N], dfn[N], siz[N], f[N], szp[N], szn[N];
    vector<int> e[N], c[N], rg[N];
    bool ed[N];
    int n, ksiz, rt, mn, id;
    int lca(int u, int v)
    {
        u = dfn[u]; v = dfn[v];
        if (u > v) swap(u, v);
        int z = lg[v - u + 1];
        return dep[st[z][u]] < dep[st[z][v - (1 << z) + 1]] ? st[z][u] : st[z][v - (1 << z) + 1];
    }
    int dis(int u, int v)
    {

```

```

        return dep[u] + dep[v] - dep[lca(u, v)] * 2;
    }
    void findroot(int u)
    {
        ed[u] = siz[u] = 1;
        int mx = 0;
        for (int v : e[u]) if (!ed[v])
        {
            findroot(v);
            siz[u] += siz[v];
            mx = max(mx, siz[v]);
        }
        mx = max(mx, ksiz - siz[u]);
        ed[u] = 0;
        if (mn > mx) mn = mx, rt = u;
    }
    int dfs(int u)
    {
        mn = 1e9;
        findroot(u);
        u = rt;
        ed[u] = 1;
        for (int v : e[u]) if (!ed[v] && siz[v] > siz[u]) siz[v] = ksiz - siz[u];
        for (int v : e[u]) if (!ed[v])
        {
            ksiz = siz[v];
            c[u].push_back(dfs(v));
            f[c[u].back()] = u;
        }
        return u;
    }
    void pre_dfs(int u)
    {
        st[0][dfn[u] = ++id] = u;
        ed[u] = 1;
        for (int v : e[u]) if (!ed[v])
        {
            dep[v] = dep[u] + 1;
            pre_dfs(v);
            st[0][++id] = u;
        }
        ed[u] = 0;
    }
    void init(int _n)
    {
        n = _n; id = 0;
        int i;
        for (int i = 1; i <= n; i++)
        {
            e[i].clear();
            c[i].clear(); rg[i].clear();
            a[i] = f[i] = siz[i] = ed[i] = 0;
        }
    }
    void new_dfs(int u)
    {
        siz[u] = 1;
    }

```

```

    for (int v : c[u]) new_dfs(v), siz[u] += siz[v];
    vector<int> &q = rg[u];
    q = {u};
    int ql = 0;
    while (ql < q.size())
    {
        int x = q[ql++];
        for (int v : c[x]) q.push_back(v);
    }
}
void fun()
{
    pre_dfs(1);
    int i, j;
    for (i = 2; i <= id; i++) lg[i] = lg[i >> 1] + 1;
    for (j = 0; j < lg[id]; j++)
    {
        int R = id - (2 << j) + 1;
        for (i = 1; i <= R; i++) st[j + 1][i] = dep[st[j][i]] < dep[st[j][i + (1 << j)]] ? st
[j][i] : st[j][i + (1 << j)];
    }
    ksiz = n;
    rt = dfs(1);
    new_dfs(rt);
}
vector<int> get(int u)
{
    vector<int> st = {u};
    while (u = f[u]) st.push_back(u);
    return st;
}
}
using DFS::init, DFS::fun, DFS::e, DFS::dis, DFS::rg, DFS::get;

```

圆环修改和单点查询:

```

int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cout << fixed << setprecision(15);
    int n, m, i;
    cin >> n >> m;
    vector<int> a(n + 1);
    for (i = 1; i <= n; i++) cin >> a[i];
    DFS::init(n);
    for (i = 1; i < n; i++)
    {
        int u, v;
        cin >> u >> v;
        ++u; ++v;
        e[u].push_back(v);
        e[v].push_back(u);
    }
    DFS::fun();
    vector<bit<ll>> inc(n + 1), dec(n + 1);
    for (i = 1; i <= n; i++)
    {
        int mx = 0;

```

```

    for (int v : rg[i]) cmax(mx, dis(i, v));
    inc[i] = bit<ll>(mx + 1);
    if (i != DFS::rt)
    {
        mx = 0;
        for (int v : rg[i]) cmax(mx, dis(DFS::f[i], v));
        dec[i] = bit<ll>(mx + 1);
    }
}
while (m--)
{
    int op, u;
    cin >> op >> u; ++u;
    if (op == 0)
    {
        int l, r, x;
        cin >> l >> r >> x;
        auto v = get(u);
        int m = v.size();
        for (i = 0; i < m; i++)
        {
            inc[v[i]].add(l - dis(v[i], u), x);
            inc[v[i]].add(r - dis(v[i], u), -x);
        }
        for (i = 0; i + 1 < m; i++)
        {
            dec[v[i]].add(l - dis(v[i + 1], u), x);
            dec[v[i]].add(r - dis(v[i + 1], u), -x);
        }
    }
    else
    {
        ll res = a[u];
        auto v = get(u);
        int m = v.size();
        for (i = 0; i < m; i++) res += inc[v[i]].sum(dis(v[i], u));
        for (i = 0; i + 1 < m; i++) res -= dec[v[i]].sum(dis(v[i + 1], u));
        cout << res << '\n';
    }
}
}

```

单点修改和圆环查询：

```

int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cout << fixed << setprecision(15);
    int n, m, i;
    cin >> n >> m;
    vector<int> a(n + 1);
    for (i = 1; i <= n; i++) cin >> a[i];
    DFS::init(n);
    for (i = 1; i < n; i++)
    {
        int u, v;
        cin >> u >> v;
        ++u; ++v;
    }
}

```

```

        e[u].push_back(v);
        e[v].push_back(u);
    }
    DFS::fun();
    vector<bit<ll>> inc(n + 1), dec(n + 1);
    vector<ll> tmp(n + 1);
    for (i = 1; i <= n; i++)
    {
        int mx = 0;
        for (int v : rg[i])
        {
            int d = dis(i, v);
            cmax(mx, d);
            tmp[d] += a[v];
        }
        inc[i] = bit<ll>(mx + 1, tmp.data());
        fill_n(tmp.begin(), mx + 1, 0);
        if (i != DFS::rt)
        {
            mx = 0;
            for (int v : rg[i])
            {
                int d = dis(DFS::f[i], v);
                cmax(mx, d);
                tmp[d] += a[v];
            }
            dec[i] = bit<ll>(mx + 1, tmp.data());
            fill_n(tmp.begin(), mx + 1, 0);
        }
    }
}
while (m--)
{
    int op, u;
    cin >> op >> u; ++u;
    if (op == 0)
    {
        int x;
        cin >> x;
        auto v = get(u);
        int m = v.size();
        for (i = 0; i < m; i++) inc[v[i]].add(dis(v[i], u), x);
        for (i = 0; i + 1 < m; i++) dec[v[i]].add(dis(v[i + 1], u), x);
    }
    else
    {
        int l, r;
        cin >> l >> r;
        --r;
        ll res = 0;
        auto v = get(u);
        int m = v.size();
        for (i = 0; i < m; i++) res += inc[v[i]].sum(1 - dis(v[i], u), r - dis(v[i], u));
        for (i = 0; i + 1 < m; i++) res -= dec[v[i]].sum(1 - dis(v[i + 1], u), r - dis(v[i + 1], u));
        cout << res << '\n';
    }
}

```


}

5.6.15 (基环) 树哈希

有根树返回每个子树的哈希值，无根树返回树的哈希值（长度至多为 2 的 vector），基环树返回图的哈希值（长度等于环长的 vector）。

```
vector<int> tree_hash(const vector<vector<int>> &e, int root)//[0,n)
{
    int n = e.size();
    static map<vector<int>, int> mp;
    static int id = 0;
    vector<int> h(n), ed(n);
    function<void(int)> dfs = [&](int u) {
        ed[u] = 1;
        vector<int> c;
        for (int v : e[u]) if (!ed[v])
        {
            dfs(v);
            c.push_back(h[v]);
        }
        sort(all(c));
        if (!mp.count(c)) mp[c] = id++;
        h[u] = mp[c];
    };
    dfs(root);
    return h;
}

vector<int> tree_hash(const vector<vector<int>> &e)//[0,n)
{
    int n = e.size();
    if (n == 0) return { };
    vector<int> sz(n), mx(n);
    function<void(int)> dfs = [&](int u) {
        sz[u] = 1;
        for (int v : e[u]) if (!sz[v])
        {
            dfs(v);
            sz[u] += sz[v];
            cmax(mx[u], sz[v]);
        }
        cmax(mx[u], n - sz[u]);
    };
    dfs(0);
    int m = *min_element(all(mx)), i;
    vector<int> rt;
    for (i = 0; i < n; i++) if (mx[i] == m) rt.push_back(i);
    for (int &u : rt) u = tree_hash(e, u)[u];
    sort(all(rt));
    return rt;
}

template<class T> void min_order(vector<T> &a)
{
    int n = a.size(), i, j, k;
    a.resize(n * 2);
    for (i = 0; i < n; i++) a[i + n] = a[i];
    i = k = 0; j = 1;
```

```

while (i < n && j < n && k < n)
{
    T x = a[i + k], y = a[j + k];
    if (x == y) ++k; else
    {
        (x > y ? i : j) += k + 1;
        j += (i == j);
        k = 0;
    }
}
a.resize(n);
//[min(i,j),n)+[0,min(i,j))
rotate(a.begin(), min(i, j) + all(a));
}
vector<int> pseudotree_hash(const vector<vector<int>> &e)//[0,n)
{
    int n = e.size();
    static map<vector<int>, int> mp;
    static int id = 0;
    vector<int> f(n), ed(n), h(n);
    pair lp{-1, -1};
    function<void(int)> dfs = [&](int u) {
        ed[u] = 1;
        for (int v : e[u]) if (!ed[v])
        {
            f[v] = u;
            dfs(v);
        }
        else if (v != f[u]) lp = {u, v};
    };
    dfs(0);
    auto [x, y] = lp;
    vector<int> node = {y};
    do node.push_back(y = f[y]); while (y != x);
    fill(all(ed), 0);
    for (int u : node) ed[u] = 1;
    dfs = [&](int u) {
        ed[u] = 1;
        vector<int> c;
        for (int v : e[u]) if (!ed[v])
        {
            dfs(v);
            c.push_back(h[v]);
        }
        sort(all(c));
        if (!mp.count(c)) mp[c] = id++;
        h[u] = mp[c];
    };
    vector<int> r0;
    for (int u : node)
    {
        dfs(u);
        r0.push_back(h[u]);
    }
    auto r1 = r0;
    reverse(all(r1));
    min_order(r0);
}

```

```
    min_order(r1);  
    return min(r0, r1);  
}
```

5.7 欧拉路相关

5.7.1 构造：字典序最小

```
#include "bits/stdc++.h"  
using namespace std;  
#define all(x) (x).begin(), (x).end()  
const int N = 1e5 + 2;  
vector<int> e[N];  
int rd[N], cd[N];  
vector<int> ans;  
void dfs(int u)  
{  
    while (e[u].size())  
    {  
        int v = e[u].back();  
        e[u].pop_back();  
        dfs(v);  
        ans.push_back(v);  
    }  
}  
int main()  
{  
    ios::sync_with_stdio(0); cin.tie(0);  
    int n, m, i, x = 0, m0;  
    cin >> n >> m; ans.reserve(m + 1); m0 = m;  
    while (m--)  
    {  
        int u, v;  
        cin >> u >> v;  
        e[u].push_back(v);  
        ++cd[u]; ++rd[v];  
    }  
    for (i = 1; i <= n; i++) if (cd[i] != rd[i])  
    {  
        if (abs(cd[i] - rd[i]) > 1) goto no;  
        ++x;  
    }  
    if (x > 2) goto no; x = 0;  
    for (i = 1; i <= n; i++) if (cd[i] > rd[i]) { x = i; break; }  
    if (!x) for (i = 1; i <= n; i++) if (cd[i]) { x = i; break; }  
    if (!x) x = 1;  
    for (i = 1; i <= n; i++) sort(all(e[i])), reverse(all(e[i]));  
    dfs(x); ans.push_back(x); reverse(all(ans));  
    if (ans.size() != m0 + 1) goto no;  
    for (i = 0; i < ans.size(); i++) cout << ans[i] << "\n"[i + 1 == ans.size()];  
    return 0;  
no:cout << "No" << endl;  
}
```

5.7.2 回路/通路构造

$O(n + m)$, $O(n + m)$ 。

```
optional<vector<int>> undirected_euler_cycle(int n, const vector<pair<int, int>> &edges)//[1,n]
    [1,m], 正数表示正向, 负数表示反向
{
    int i = 0;
    vector<int> rd(n + 1), ed(edges.size() + 1), r;
    vector<vector<pair<int, int>>> e(n + 1);
    for (auto [u, v] : edges)
    {
        ++rd[u], ++rd[v];
        e[u].push_back({v, ++i});
        e[v].push_back({u, -i});
    }
    for (i = 1; i <= n; i++) if (rd[i] & 1) return { };
    function<void(int)> dfs = [&](int u) {
        while (e[u].size())
        {
            auto [v, w] = e[u].back();
            e[u].pop_back();
            if (ed[abs(w)]) continue;
            ed[abs(w)] = 1;
            dfs(v);
            r.push_back(w);
        }
    };
    for (i = 1; i <= n; i++) if (rd[i]) { dfs(i); break; }
    reverse(all(r));
    if (r.size() != edges.size()) return { };
    return {r};
}

optional<vector<int>> directed_euler_cycle(int n, const vector<pair<int, int>> &edges)//[1,n]/[1,
m]
{
    int i = 0;
    vector<int> rd(n + 1), cd(n + 1), r;
    vector<vector<pair<int, int>>> e(n + 1);
    for (auto [u, v] : edges)
    {
        ++cd[u], ++rd[v];
        e[u].push_back({v, ++i});
    }
    for (i = 1; i <= n; i++) if (rd[i] != cd[i]) return { };
    function<void(int)> dfs = [&](int u) {
        while (e[u].size())
        {
            auto [v, w] = e[u].back();
            e[u].pop_back();
            dfs(v);
            r.push_back(w);
        }
    };
    for (i = 1; i <= n; i++) if (cd[i]) { dfs(i); break; }
    reverse(all(r));
    if (r.size() != edges.size()) return { };
    return {r};
}
```

```

}
optional<vector<int>> undirected_euler_trail(int n, const vector<pair<int, int>> &edges)//[1,n]
    ][1,m], 正数表示正向, 负数表示反向
{
    int i = 0;
    vector<int> rd(n + 1), ed(edges.size() + 1), r;
    vector<vector<pair<int, int>>> e(n + 1);
    for (auto [u, v] : edges)
    {
        ++rd[u], ++rd[v];
        e[u].push_back({v, ++i});
        e[v].push_back({u, -i});
    }
    int odd = 0;
    for (i = 1; i <= n; i++) odd += rd[i] & 1;
    if (odd > 2) return { };
    function<void(int)> dfs = [&](int u) {
        while (e[u].size())
        {
            auto [v, w] = e[u].back();
            e[u].pop_back();
            if (ed[abs(w)]) continue;
            ed[abs(w)] = 1;
            dfs(v);
            r.push_back(w);
        }
    };
    for (i = 1; i <= n; i++) if (rd[i] & 1) { dfs(i); break; }
    if (i > n)
    {
        for (i = 1; i <= n; i++) if (rd[i]) { dfs(i); break; }
    }
    reverse(all(r));
    if (r.size() != edges.size()) return { };
    return {r};
}
optional<vector<int>> directed_euler_trail(int n, const vector<pair<int, int>> &edges)//[1,n]/[1,
    m]
{
    int i = 0;
    vector<int> rd(n + 1), cd(n + 1), r;
    vector<vector<pair<int, int>>> e(n + 1);
    for (auto [u, v] : edges)
    {
        ++cd[u], ++rd[v];
        e[u].push_back({v, ++i});
    }
    int diff = 0;
    for (i = 1; i <= n; i++)
    {
        if (abs(rd[i] - cd[i]) > 1) return { };
        if (rd[i] != cd[i]) ++diff;
    }
    if (diff > 2) return { };
    function<void(int)> dfs = [&](int u) {
        while (e[u].size())
        {

```

```

        auto [v, w] = e[u].back();
        e[u].pop_back();
        dfs(v);
        r.push_back(w);
    }
};
for (i = 1; i <= n; i++) if (cd[i] > rd[i]) { dfs(i); break; }
if (i > n)
{
    for (i = 1; i <= n; i++) if (cd[i]) { dfs(i); break; }
}
reverse(all(r));
if (r.size() != edges.size()) return { };
return {r};
}
optional<vector<int>> mixed_euler_cycle(int n, const vector<tuple<int, int, bool>> &edges)//true:
    单向, false: 双向
{
    int i = 0, j, m = edges.size();
    vector<int> rd(n + 1), cd(n + 1), r, rev(m + 1);
    vector<vector<pair<int, int>>> e(n + 1);
    for (auto [u, v, d] : edges) ++cd[u], ++rd[v];
    for (i = 1; i <= n; i++) if (cd[i] + rd[i] & 1) return { };
    vector<tuple<int, int, ll>> eg;
    for (auto [u, v, d] : edges) if (!d) eg.push_back({u, v, 1});
    ll sum = 0;
    for (i = 1; i <= n; i++)
    {
        int d = (cd[i] - rd[i]) / 2;
        if (d > 0) eg.push_back({0, i, d}), sum += d;
        else if (d < 0) eg.push_back({i, n + 1, -d});
    }
    auto [w, res] = net::max_flow(n + 1, eg, 0, n + 1);
    if (w != sum) return { };
    vector<pair<int, int>> G(m);
    for (i = j = 0; i < m; i++)
    {
        auto [u, v, d] = edges[i];
        if (d || !res[j++]) G[i] = {u, v};
        else G[i] = {v, u}, rev[i + 1] = 1;
    }
    auto ans = directed_euler_cycle(n, G);
    if (!ans) return ans;
    for (int &x : *ans) if (rev[x]) x = -x;
    return ans;
}

```

5.7.3 回路计数 (BEST 定理) / 生成树计数

$O(n^3)$, $O(n^2)$ 。

以 u 为起点的欧拉回路个数 $sum = T(u) \times \prod_{v=1}^n (out(v) - 1)!$, 其中 $T(u)$ 是以 u 为根的内向树个数 (出度矩阵-邻接矩阵), $out(v)$ 是 v 的出度。若允许循环同构 (如 $1 \rightarrow 2 \rightarrow 1 \rightarrow 3 \rightarrow 1$ 与 $1 \rightarrow 3 \rightarrow 1 \rightarrow 2 \rightarrow 1$), 还需多乘 $out(u)$ 。

```
ull det(vector<vector<ull>> b)
```

```

{
    ull r = 1;
    int n = b.size(), i, j, k;
    for (i = 0; i < n; i++)
    {
        for (j = i; j < n; j++) if (b[j][i]) break;
        if (j == n) return 0;
        swap(b[j], b[i]);
        if (j != i) r = (p - r) % p;
        r = r * b[i][i] % p;
        b[i][i] = ksm(b[i][i], p - 2);
        for (j = n - 1; j >= i; j--) b[i][j] = b[i][j] * b[i][i] % p;
        for (j = i + 1; j < n; j++) for (k = n - 1; k >= i; k--) b[j][k] = (b[j][k] + (p - b[j][i]
    ]) * b[i][k]) % p;
    }
    return r;
}

ull euler_path_count(vector<vector<int>> a, int s, int t)
{
    int n = a.size(), i, j, k;
    ++a[t][s]; s = t;
    vector<int> rd(n), cd(n);
    for (i = 0; i < n; i++) for (j = 0; j < n; j++) cd[i] += a[i][j], rd[j] += a[i][j];
    for (i = 0; i < n; i++) if (cd[i] != rd[i]) return 0;
    vector<int> f(n);
    iota(all(f), 0);
    function<int(int)> getf = [&](int u) { return f[u] == u ? u : f[u] = getf(f[u]); };
    for (i = 0; i < n; i++) for (j = 0; j < n; j++) if (a[i][j]) f[getf(i)] = getf(j);
    ull r = 1;
    vector<int> id;
    for (i = 0; i < n; i++) if (cd[i])
    {
        if (getf(i) != getf(s)) return 0;
        r = r * fac[cd[i] - 1] % p;
        if (i != s) id.push_back(i);
    }
    n = id.size();
    vector b(n, vector<ull>(n));
    for (i = 0; i < n; i++)
    {
        b[i][i] = cd[id[i]] - a[id[i]][id[i]];
        for (j = 0; j < n; j++) if (i != j) b[i][j] = (p - a[id[i]][id[j]]) % p;
    }
    return r * det(b) % p;
}

ull euler_path_count(vector<vector<int>> a)
{
    int n = a.size(), i, j, s = -1, t = -1;
    vector<int> rd(n), cd(n), d(n);
    for (i = 0; i < n; i++) for (j = 0; j < n; j++) cd[i] += a[i][j], rd[j] += a[i][j];
    if (count(all(cd), 0) == n) return 1;
    for (i = 0; i < n; i++) d[i] = cd[i] - rd[i];
    s = max_element(all(d)) - d.begin();
    t = min_element(all(d)) - d.begin();
    ull r = 0;
    if (s == t)
    {

```

```

        for (i = 0; i < n; i++) if (cd[i]) r += euler_path_count(a, i, i);
    }
    else r = euler_path_count(a, s, t);
    return r % p;
}
ull euler_circuit_count(vector<vector<int>> a)
{
    int n = a.size(), i, j;
    for (i = 0; i < n; i++) for (j = 0; j < n; j++) if (a[i][j]) return euler_path_count(a, i, i)
        * ksm(accumulate(all(a[i]), 0llu) % p, p - 2) % p;
    return 1;
}
ull directed_spanning_tree_count(vector<vector<int>> a, int s)
{
    int n = a.size(), i, j;
    vector b(n - 1, vector<ull>(n - 1));
    for (i = 0; i < n; i++) a[i][i] = 0;
    for (i = 0; i < n; i++) if (i != s) for (j = 0; j < n; j++) if (j != s && i != j) b[i - (i >
s)][j - (j > s)] = (p - a[i][j]) % p;
    for (i = 0; i < n; i++) if (i != s) for (j = 0; j < n; j++) (b[i - (i > s)][i - (i > s)] += a
[j][i]) %= p;
    return det(b);
} //外向
ull undirected_spanning_tree_count(vector<vector<int>> a)
{
    int n = a.size(), i, j;
    --n;
    vector b(n, vector<ull>(n));
    for (i = 0; i < n; i++) a[i][i] = 0;
    for (i = 0; i < n; i++) for (j = 0; j < n; j++) if (i != j) b[i][j] = (p - a[i][j]) % p;
    for (i = 0; i < n; i++) b[i][i] = reduce(all(a[i]), 0llu) % p;
    return det(b);
}

```

5.8 三/四元环计数

不能处理有重边和自环的情况。

$O(m\sqrt{m})$, $O(n + m)$ 。

注意四元环数的是边四元环。点四元环需要去掉四点完全图个数 *2, 似乎不太能做?

三元环是可以枚举的, 你可以在 ans 改变时记录三元环 (i, u, v) 。

```

11 triple(const vector<pair<int, int>> &edges) //start from 0
{
    int n = 0, i;
    for (auto [u, v] : edges) n = max({n, u, v});
    ++n;
    vector<int> d(n), id(n), rk(n), cnt(n);
    vector<vector<int>> e(n);
    for (auto [u, v] : edges) ++d[u], ++d[v];
    iota(all(id), 0); sort(all(id), [&](int x, int y) { return d[x] < d[y]; });
    for (i = 0; i < n; i++) rk[id[i]] = i;
    for (auto [u, v] : edges)
    {
        if (rk[u] > rk[v]) swap(u, v);
        e[u].push_back(v);
    }
}

```



```

    ll ans = 0;
    for (i = 0; i < n; i++)
    {
        for (int u : e[i]) cnt[u] = 1;
        for (int u : e[i]) for (int v : e[u]) ans += cnt[v];
        for (int u : e[i]) cnt[u] = 0;
    }
    return ans;
}

ll quadruple(const vector<pair<int, int>> &edges)
{
    int n = 0, i;
    for (auto [u, v] : edges) n = max({n, u, v});
    ++n;
    vector<int> d(n), id(n), rk(n), cnt(n);
    vector<vector<int>> e(n), lk(n);
    for (auto [u, v] : edges) ++d[u], ++d[v];
    iota(all(id), 0); sort(all(id), [&](int x, int y) { return d[x] < d[y]; });
    for (i = 0; i < n; i++) rk[id[i]] = i;
    for (auto [u, v] : edges)
    {
        if (rk[u] > rk[v]) swap(u, v);
        e[u].push_back(v);
        lk[u].push_back(v);
        lk[v].push_back(u);
    }
    ll ans = 0;
    for (i = 0; i < n; i++)
    {
        for (int u : lk[i]) for (int v : e[u]) if (rk[v] > rk[i]) ans += cnt[v]++;
        for (int u : lk[i]) for (int v : e[u]) cnt[v] = 0;
    }
    return ans;
}

map<pair<int, int>, ll> quadruple_of_edge(vector<pair<int, int>> edges)
{
    int n = 0, i;
    for (auto [u, v] : edges) n = max({n, u, v});
    ++n;
    map<pair<int, int>, int> ec;
    for (auto [u, v] : edges)
    {
        if (u > v) swap(u, v);
        ++ec[{u, v}];
    }
    vector<ll> c;
    edges.clear();
    for (auto [_ , cc] : ec) edges.push_back(_), c.push_back(cc);
    vector d(n, 0), id(d), rk(d);
    vector<ll> cnt(n);
    vector<vector<pair<int, int>>> e(n), lk(n);
    for (auto [u, v] : edges) ++d[u], ++d[v];
    iota(all(id), 0); sort(all(id), [&](int x, int y) { return d[x] < d[y]; });
    for (i = 0; i < n; i++) rk[id[i]] = i;
    i = 0;
    for (auto [u, v] : edges)

```

```

{
    if (rk[u] > rk[v]) swap(u, v);
    e[u].push_back({v, i});
    lk[u].push_back({v, i});
    lk[v].push_back({u, i});
    ++i;
}
int m = edges.size();
vector<ll> ans(m);
for (i = 0; i < n; i++)
{
    for (auto [u, w1] : lk[i]) for (auto [v, w2] : e[u]) if (rk[v] > rk[i])
    {
        cnt[v] += c[w1] * c[w2];
    }
    for (auto [u, w1] : lk[i]) for (auto [v, w2] : e[u]) if (rk[v] > rk[i])
    {
        ans[w1] += (cnt[v] - c[w1] * c[w2]) * c[w2];
        ans[w2] += (cnt[v] - c[w1] * c[w2]) * c[w1];
    }
    for (auto [u, w1] : lk[i]) for (auto [v, w2] : e[u]) if (rk[v] > rk[i]) cnt[v] = 0;
}
map<pair<int, int>, ll> mp;
for (i = 0; i < m; i++) mp[edges[i]] = ans[i];
return mp;
}
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cout << fixed << setprecision(15);
    int n, m, i;
    cin >> n >> m;
    vector<pair<int, int>> eg(m);
    cin >> eg;
    auto mp = quadruple_of_edge(eg);
    for (i = 0; i < m; i++)
    {
        auto [u, v] = eg[i];
        if (u > v) swap(u, v);
        cout << mp[{u, v}] << "\n"[i + 1 == m];
    }
}

```

5.9 支配树

u 支配 v 指的是从 S 到 v 的路径必然经过 u 。支配树是保持支配关系不变的树，其中 s 是根， $idom[u]$ 是 u 的父节点。

DAG 版： $O(m \log n)$, $O(n \log n)$ 。

```

int lca(int x, int y)
{
    int i;
    if (dep[x] < dep[y]) swap(x, y);
    for (i = lm[x]; dep[x] != dep[y]; i--) if (dep[f[x][i]] >= dep[y]) x = f[x][i];
    if (x == y) return x;
    for (i = lm[x]; f[x][0] != f[y][0]; i--) if (f[x][i] != f[y][i])

```

```

    {
        x = f[x][i]; y = f[y][i];
    }
    return f[x][0];
}
void dfs(int x)
{
    s[x] = 1;
    int i;
    for (i = sfir[x]; i; i = snxt[i])
    {
        dfs(slj[i]);
        s[x] += s[slj[i]];
    }
}
int main()
{
    dep[0] = -1;
    cin >> n;
    for (i = 1; i <= n; i++)
    {
        cin >> x;
        while (x)
        {
            add(x, i);
            cin >> x;
        }
    }
    dl[tou = wei = 1] = ++n;
    for (i = 1; i < n; i++) if (!rd[i]) add(n, i);
    while (tou <= wei)
    {
        for (i = fir[x = dl[tou++]]; i; i = nxt[i]) if (--rd[lj[i]] == 0) dl[++wei] = lj[i];
        if (i = ffir[x])
        {
            y = flj[i];
            while (i = fnxt[i]) y = lca(y, flj[i]);
            f[x][0] = y;
        }
        else y = 0;
        sadd(y, x);
        f[x][0] = y;
        for (i = 1; i <= 16; i++) if (0 == (f[x][i] = f[f[x][i - 1]][i - 1]))
        {
            lm[x] = i;
            break;
        }
        dep[x] = dep[y] + 1;
    }
    dfs(n);
    for (i = 1; i < n; i++) printf("%d\n", s[i] - 1);
}

```

一般图：标号从 1 开始。

```

vector<int> dom_tree(vector<vector<int>> e, int s)
{
    int n = e.size() - 1, i, id = 0;

```

```

vector<vector<int>> buc(n + 1), ie(n + 1);
vector<int> mn(n + 1), f(n + 1), sdom(n + 1, n + 1), idom(n + 1), dfn(n + 1), nfd(n + 2), pv(
n + 1), ed(n + 1), cf(n + 1);
#define cmp(x,y) (sdom[x] < sdom[y] ? x : y)
auto getf = [&](auto &&getf, int u) ->void {
    if (f[u] == u) return;
    getf(getf, f[u]);
    mn[u] = cmp(mn[u], mn[f[u]]);
    f[u] = f[f[u]];
};
for (i = 1; i <= n; i++) mn[i] = f[i] = i;
{
    auto dfs = [&](auto &&dfs, int u) ->void {
        ed[u] = 1;
        for (int v : e[u]) if (!ed[v]) dfs(dfs, v);
    };
    dfs(dfs, s);
}
for (i = 1; i <= n; i++) if (ed[i]) erase_if(e[i], [&](int v) { return !ed[v]; });
else e[i].clear();
for (i = 1; i <= n; i++) for (int v : e[i]) ie[v].push_back(i);
auto dfs = [&](auto &&dfs, int u) ->void {
    nfd[dfn[u] = ++id] = u;
    for (int v : e[u]) if (!dfn[v]) dfs(dfs, v), cf[v] = u;
};
dfs(dfs, s); dfn[0] = n + 1;
for (i = id; i; i--)
{
    int u = nfd[i], w = 0;
    for (int v : ie[u])
    {
        cmin(sdom[u], dfn[v]);
        if (dfn[v] > dfn[u])
        {
            getf(getf, v);
            w = cmp(w, mn[v]);
        }
    }
    cmin(sdom[u], sdom[w]);
    buc[nfd[sdom[u]]].push_back(u);
    for (int v : buc[u]) getf(getf, v), pv[v] = mn[v];
    for (int v : e[u]) if (cf[v] == u) f[v] = u, mn[v] = cmp(mn[v], mn[u]);
}
for (i = 1; i <= n; i++) idom[nfd[i]] = (sdom[pv[nfd[i]]] == sdom[nfd[i]]) ? nfd[sdom[nfd[i]
]] : idom[pv[nfd[i]]];
idom[s] = s;
return idom;
#undef cmp
}
int main()
{
    int n, m, s;
    cin >> n >> m >> s; ++s;
    vector<vector<int>> e(n + 1);
    for (int i = 1; i <= m; i++)
    {
        int u, v;

```

```

        cin >> u >> v; ++u; ++v;
        e[u].push_back(v);
    }
    auto r = dom_tree(e, s);
    for (int i = 1; i <= n; i++) cout << r[i] - 1 << "\n"[i == n];
}

```

5.10 prufer 与树的互相转化

$O(n)$, $O(n)$ 。

```

vector<int> edges_to_prufer(const vector<pair<int, int>> &eg)//[1,n], 定根为 n
{
    int n = eg.size() + 1, i, j, k;
    vector<int> fir(n + 1), nxt(n * 2 + 1), e(n * 2 + 1), rd(n + 1);
    int cnt = 0;
    for (auto [u, v] : eg)
    {
        e[++cnt] = v; nxt[cnt] = fir[u]; fir[u] = cnt; ++rd[v];
        e[++cnt] = u; nxt[cnt] = fir[v]; fir[v] = cnt; ++rd[u];
    }
    for (i = 1; i <= n; i++) if (rd[i] == 1) break;
    int u = i;
    vector<int> r; r.reserve(n - 2);
    for (j = 1; j < n - 1; j++)
    {
        for (k = fir[u], u = rd[u] = 0; k; k = nxt[k]) if (rd[e[k]])
        {
            r.push_back(e[k]);
            if ((--rd[e[k]] == 1) && (e[k] < i)) u = e[k];
        }
        if (!u) { while (rd[i] != 1) ++i; u = i; }
    }
    return r;
}

vector<pair<int, int>> prufer_to_edges(const vector<int> &p)//[1,n], 定根为 n
{
    int n = p.size(), i, j, k;
    int m = n + 3;
    vector<int> cs(m);
    for (i = 0; i < n; i++) ++cs[p[i]];
    i = 0;
    while (cs[++i]);
    int u = i, v;
    vector<pair<int, int>> r;
    r.reserve(n + 1);
    for (j = 0; j < n; j++)
    {
        cs[u] = 1e9;
        r.push_back({u, v = p[j]});
        if ((--cs[v] == 0) && (v < i)) u = v;
        if (v != u) { while (cs[i] > 0) ++i; u = i; }
    }
    r.push_back({u, n + 2});
    return r;
}

```

5.11 最小密度环

求所有环中边权和除以边数最少的, $O(nm)$ 。更常用的做法是二分 spfa。

```
#include "bits/stdc++.h"
using namespace std;
const int N = 3e3 + 5, M = 1e4 + 5;
const double inf = 1e18;
int u[M], v[M];
double f[N][N], w[M];
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cout << setiosflags(ios::fixed) << setprecision(8);
    int n, m, i, j;
    cin >> n >> m;
    for (i = 1; i <= m; i++)
        cin >> u[i] >> v[i] >> w[i];
    ++n;
    for (i = 1; i <= n; i++)
    {
        fill_n(f[i] + 1, n, inf);
        for (j = 1; j <= m; j++)
            f[i][v[j]] = min(f[i][v[j]], f[i - 1][u[j]] + w[j]);
    }
    double ans = inf;
    for (i = 1; i < n; i++) if (f[n][i] != inf)
    {
        double r = -inf;
        for (j = 1; j < n; j++) r = max(r, (f[n][i] - f[j][i]) / (n - j));
        ans = min(ans, r);
    }
    cout << ans << endl;
}
```

5.12 点染色

结论: $\chi(G) \leq \Delta(G) + 1$, 其中 $\Delta(G)$ 是图的最大度。只有奇圈和完全图取等。
构造方案只能爆搜。

```
vector<int> chromatic_number(int n, const vector<pair<int, int>> &edges)//[0,n)
{
    vector r(n, -1), cur(n, -1);
    vector<vector<int>> e(n);
    int ans = 0, i;
    for (auto [u, v] : edges) e[u].push_back(v), e[v].push_back(u);
    for (i = 0; i < n; i++) ans = max(ans, (int)e[i].size());
    ans += 2;
    vector p(n, vector(ans, 0));
    function<void(int)> dfs = [&](int u) {
        int col = u ? *max_element(cur.begin(), cur.begin() + u) + 1 : 0;
        if (col >= ans) return;
        if (u == n)
        {
            r = cur;
            ans = col;
            return;
        }
    }
```

```

    }
    int i;
    for (int i = 0; i <= col; i++) if (!p[u][i])
    {
        cur[u] = i;
        for (int v : e[u]) ++p[v][i];
        dfs(u + 1);
        for (int v : e[u]) --p[v][i];
    }
};
dfs(0);
return r;
}

```

5.13 最大独立集

爆搜。

```

vector<int> indep_set(int n, const vector<pair<int, int>> &edges)//[0,n)
{
    vector<vector<int>> e(n);
    mt19937 rnd(998);
    vector<int> p(n), q(n), ed(n);
    iota(all(p), 0);
    shuffle(all(p), rnd);
    for (int i = 0; i < n; i++) q[p[i]] = i;
    for (auto [u, v] : edges)
    {
        e[p[u]].push_back(p[v]);
        e[p[v]].push_back(p[u]);
    }
    vector<int> r, cur;
    function<void(int)> dfs = [&](int u) {
        if (cur.size() + n - u <= r.size()) return;
        if (u == n)
        {
            r = cur;
            return;
        }
        if (!ed[u])
        {
            cur.push_back(u);
            for (int v : e[u]) ++ed[v];
            dfs(u + 1);
            for (int v : e[u]) --ed[v];
            cur.pop_back();
        }
        if (ed[u] || e[u].size()) dfs(u + 1);
    }; dfs(0);
    for (int &x : r) x = q[x];
    sort(all(r));
    return r;
}

```

5.14 弦图

单纯点： v 和 v 邻点构成团。

完美消除序列： v_i 在 $\{v_i, v_{i+1}, \dots, v_n\}$ 为单纯点。

$N(v_i) = \{v_j | j > i \wedge (v_i, v_j) \in E\}$, $next(v_i)$ 为 $N(v_i)$ 最靠前的点。

极大团一定是 $\{v\} \cup N(v)$ 。

最大团大小等于色数。

弦图判定：等价于是否存在完美消除序列。首先求出一个完美消除序列，然后判定是否合法。

判定方法：设 $v_{i+1}, \dots, v_n$ 中与 v_i 相邻的依次为 $v'_1, \dots, v'_m$ 。只需判断是否 v'_1 与 $v'_2, \dots, v'_m$ 相邻。

`find_cycle` 若不是弦图，返回一个无弦环（首尾相连但不重复首点）；若是弦图返回空。

LexBFS 算法（我不会写）

每个点有一个字符串 `label`，初始为 0。从 $i = n$ 到 $i = 1$ 确定，选 `label` 字典序最大的 u ，再把 u 邻点的 `label` 后面接一个 i 。

最大势算法：从 v_n 求到 v_1 ，设 $label_i$ 表示 i 与多少个已选点相邻，每次选 $label_i$ 最大的点。

弦图极大团： $\{v | \forall next(w) = v, |N(v)| \geq |N(w)|\}$ 。选出的集合为基本点，按上述极大团构造。

弦图染色：从 v_n 到 v_1 依次选最小可染的色。

最大独立集：从 v_1 到 v_n 能选就选。

最小团覆盖：设最大独立集为 $\{p_m\}$ ，最小团覆盖为 $\{\{p_i\} \cup N(p_i)\}$ 。

区间图：两个区间有边当且仅当交集非空。

区间图是弦图。

代码如下：

```
namespace chordal_graph//下标从 1 开始
{
    const int N = 1e5 + 2;//点数
    bool ed[N];
    vector<int> e[N];
    int n;
    void init(const vector<pair<int, int>> &edges)
    {
        n = 0;
        for (auto [u, v] : edges) n = max({n, u, v});
        for (int i = 1; i <= n; i++) e[i].clear();
        for (auto [u, v] : edges) e[u].push_back(v), e[v].push_back(u);
    }
    vector<int> perfect_seq(const vector<pair<int, int>> &edges)//MCS
    {
        init(edges);
        static int d[N];
        static vector<int> buc[N];
        int i, mx = 0;
        memset(d + 1, 0, n * sizeof d[0]);
        memset(ed + 1, 0, n * sizeof ed[0]);
        for (i = 1; i <= n; i++) buc[i].clear();
        buc[0].resize(n);
        iota(all(buc[0]), 1);
        vector<int> r(n);
        for (i = n - 1; i >= 0; i--)
        {
            int u = 0;
            while (!u)
            {
                while (buc[mx].size() && (ed[buc[mx].back()])) buc[mx].pop_back();
```



```

        else
        {
            ed[u = buc[mx].back()] = 1;
            buc[mx].pop_back();
            goto yes;
        }
        --mx;
    }
yes:;
    r[i] = u;
    for (int v : e[u]) if (!ed[v]) buc[++d[v]].push_back(v), mx = max(mx, d[v]);
}
return r;
}

bool check_perfect_seq(vector<int> a)
{
    static bool ee[N];
    static int pos[N];
    memset(ed + 1, 0, n * sizeof ed[0]);
    memset(ee + 1, 0, n * sizeof ee[0]);
    for (int i = 0; i < n; i++) pos[a[i]] = i;
    for (int u : a)
    {
        int w = 0;
        for (int v : e[u]) if (pos[v] > pos[u] && (!w || pos[v] < pos[w])) w = v;
        if (!w) continue;
        ee[w] = 1;
        for (int v : e[w]) ee[v] = 1;
        for (int v : e[u]) if (pos[v] > pos[u] && !ee[v]) return 0;
        ee[w] = 0;
        for (int v : e[w]) ee[v] = 0;
    }
    return 1;
}

bool check_chordal(const vector<pair<int, int>> &edges) { return check_perfect_seq(
perfect_seq(edges)); }

vector<int> find_cycle(const vector<pair<int, int>> &edges)//若不是弦图，返回一个无弦环。首尾相
连，不重复首点
{
    auto a = perfect_seq(edges);
    static bool ee[N];
    static int pos[N], pre[N];
    memset(ee + 1, 0, n * sizeof ee[0]);
    for (int i = 0; i < n; i++) pos[a[i]] = i;
    for (int u : a)
    {
        int w = 0, vv = 0;
        for (int v : e[u]) if (pos[v] > pos[u] && (!w || pos[v] < pos[w])) w = v;
        if (!w) continue;
        ee[w] = 1;
        for (int v : e[w]) ee[v] = 1;
        for (int v : e[u]) if (pos[v] > pos[u] && !ee[v]) { vv = v; break; }
        ee[w] = 0;
        for (int v : e[w]) ee[v] = 0;
        if (!vv) continue;
        memset(ed + 1, 0, n * sizeof ed[0]);
        memset(pre + 1, 0, n * sizeof pre[0]);
    }
}

```

```

    ed[u] = 1;
    for (int v : e[u]) if (v != w && v != vv) ed[v] = 1;
    queue<int> q;
    q.push(w); pre[w] = -1;
    while (q.size())
    {
        int x = q.front(); q.pop();
        if (x == vv) break;
        for (int y : e[x]) if (!ed[y] && !pre[y]) pre[y] = x, q.push(y);
    }
    if (!pre[vv]) continue;
    vector<int> r;
    for (int x = vv; x != -1; x = pre[x]) r.push_back(x);
    reverse(all(r));
    r.insert(r.begin(), u);
    return r;
}
return { };
}
vector<int> color(int _n, const vector<pair<int, int>> &edges)//返回长度为 _n+1。其中 0 无意义
{
    auto a = perfect_seq(edges);
    reverse(all(a));
    memset(ed + 1, 0, n * sizeof ed[0]);
    vector<int> r(_n + 1);
    for (int u : a)
    {
        for (int v : e[u]) ed[r[v]] = 1;
        int x = 1;
        while (ed[x]) ++x;
        r[u] = x;
        for (int v : e[u]) ed[r[v]] = 0;
    }
    for (int i = n + 1; i <= _n; i++) r[i] = 1;
    return r;
}
vector<int> max_independent(int _n, const vector<pair<int, int>> &edges)//注意有孤立点这种奇怪东西
{
    auto a = perfect_seq(edges);
    memset(ed + 1, 0, n * sizeof ed[0]);
    vector<int> r;
    for (int u : a) if (!ed[u])
    {
        r.push_back(u);
        for (int v : e[u]) ed[v] = 1;
    }
    for (int i = n + 1; i <= _n; i++) r.push_back(i);
    return r;
}
}
using chordal_graph::check_chordal, chordal_graph::find_cycle, chordal_graph::color,
    chordal_graph::max_independent;

```

6 计算几何

6.1 自适应 simpson 法

sim(l,r) 计算 $\int_l^r f(x) dx$

```
const db eps = 1e-7;
db sl, sr, sm, a;
db f(db x)
{
    return pow(x, a / x - x);
}
db g(db l, db r)
{
    db mid = (l + r) * 0.5;
    return (f(l) + f(r) + f(mid) * 4) / 6 * (r - l);
}
db sim(db l, db r)
{
    db mid = (l + r) * 0.5;
    sl = g(l, mid); sr = g(mid, r); sm = g(l, r);
    if (abs(sl + sr - sm) < eps) return sl + sr;
    return sim(l, mid) + sim(mid, r);
}
```

6.2 计算几何全

功能其实比较少，因为实际遇到的几何题不多。最有用的可能是闵可夫斯基和合并凸包，和常规的线段判交之类的。其余功能最好直接使用 HDU 板。

```
namespace geo
{
#define tpl template<class T>
    using ll = long long;
    using lll = __int128;
    using db = long double;
    mt19937 rnd(chrono::steady_clock::now().time_since_epoch().count());
    tpl using up = conditional_t<std::is_same_v<T, ll>, lll,
        conditional_t<std::is_same_v<T, db>, db, void>>;
    const db eps = 1e-7, pi = 3.1415926535897932384626434;
#define all(x) (x).begin(), (x).end()
    inline int sgn(const ll &x) { return x < 0 ? -1 : (x > 0); }
    inline int sgn(const lll &x) { return x < 0 ? -1 : (x > 0); }
    inline int sgn(const db &x)
    {
        if (abs(x) < eps) return 0;
        return x > 0 ? 1 : -1;
    }
    tpl struct vec/* 为叉乘, dot 为点乘, 只允许使用 long double 和 ll */
    {
        static_assert(std::is_same_v<T, ll> || std::is_same_v<T, db>);
        mutable T x, y;
        vec() { }
        vec(T a, T b) : x(a), y(b) { }
        explicit operator vec<ll>() const { return vec<ll>(x, y); }
        operator vec<db>() const { return vec<db>(x, y); }
        vec<T> operator+(const vec<T> &o) const { return vec(x + o.x, y + o.y); }
    }
}
```

```

vec<T> operator-(const vec<T> &o) const { return vec(x - o.x, y - o.y); }
vec<T> operator*(const T &k) const { return vec(x * k, y * k); }
vec<T> operator/(const T &k) const { return vec(x / k, y / k); }
T operator*(const vec<T> &o) const { return x * o.y - y * o.x; }
T dot(const vec<T> &o) const { return x * o.x + y * o.y; }
void operator+=(const vec<T> &o) { x += o.x; y += o.y; }
void operator-=(const vec<T> &o) { x -= o.x; y -= o.y; }
void operator*=(const T &k) { x *= k; y *= k; }
void operator/=(const T &k) { x /= k; y /= k; }
bool operator==(const vec<T> &o) const { return x == o.x && y == o.y; }
bool operator!=(const vec<T> &o) const { return x != o.x || y != o.y; }
db len() const { return sqrt(len2()); } //模长
T len2() const { return x * x + y * y; }
vec<db> rotate(db angle)
{
    db c = cos(angle), s = sin(angle);
    return vec<db>(x * c - y * s, x * s + y * c);
}
vec<T> rotate_90() { return vec<T>(-y, x); }
};
const vec<db> npos = vec<db>(514e194, 9810e191), apos = vec<db>(145e174, 999e180), O = vec<db>
>(0, 0);
tmpl int quad(const vec<T> &o) //坐标轴归右上象限, 返回值 [1,4]
{
    const static int d[4] = {1, 2, 4, 3};
    return d[(sgn(o.y) < 0) * 2 + (sgn(o.x) < 0)];
}
tmpl bool angle_cmp(const vec<T> &a, const vec<T> &b)
{
    int c = quad(a), d = quad(b);
    if (c != d) return c < d;
    return a * b > 0;
}
tmpl db dis(const vec<T> &a, const vec<T> &b) { return (a - b).len(); }
tmpl T dis2(const vec<T> &a, const vec<T> &b) { return (a - b).len2(); }
tmpl vec<T> operator*(const T &k, const vec<T> &o) { return vec<T>(k * o.x, k * o.y); }
tmpl bool operator<(const vec<T> &a, const vec<T> &b)
{
    int s = sgn(a * b);
    return s > 0 || s == 0 && sgn(a.len2() - b.len2()) < 0;
}
istream &operator>>(istream &cin, vec<ll> &o) { return cin >> o.x >> o.y; }
istream &operator>>(istream &cin, vec<db> &o)
{
    static string s, t;
    cin >> s >> t;
    o = vec<db>(stod(s), stod(t));
    return cin;
}
tmpl ostream &operator<<(ostream &cout, const vec<T> &o)
{
    if ((vec<db>)o == apos) return cout << "all_position";
    if ((vec<db>)o == npos) return cout << "no_position";
    return cout << '(' << o.x << ',' << o.y << ')';
}
tmpl struct line
{

```

```

    vec<T> o, d;
    line() { }
    line(const vec<T> &a, const vec<T> &b);
    line(db a, db b, db c);
    bool operator!=(const line<T> &m) { return !(*this == m); }
};

template<> line<ll>::line(const vec<ll> &a, const vec<ll> &b) :o(a), d(b - a)
{
    ll tmp = gcd(d.x, d.y);
    assert(tmp);
    if (d.x < 0 || d.x == 0 && d.y < 0) tmp = -tmp;
    d.x /= tmp; d.y /= tmp;
}

template<> line<db>::line(const vec<db> &a, const vec<db> &b) :o(a), d(b - a)
{
    int s = sgn(d.x);
    if (s < 0 || !s && d.y < 0) d.x = -d.x, d.y = -d.y;
    assert(sgn(d.x) || sgn(d.y));
}

template<> line<db>::line(db a, db b, db c) : o(abs(a) > abs(b) ? vec<db>(-c / a, 0) : vec<db>
>(0, -c / b)), d(-b, a) { }//ax+by+c=0
tpl db get_angle(const vec<T> &m, const vec<T> &n) { return atan2(m * n, m.dot(n)); }
tpl bool operator<(const line<T> &m, const line<T> &n)
{
    int s = sgn(m.d * n.d);
    return s ? s > 0 : m.d * m.o < n.d * n.o;
}

tpl bool operator==(const line<T> &m, const line<T> &n) { return sgn(m.d - n.d) == 0 && sgn
((m.o - n.o) * m.d) == 0; }

tpl ostream &operator<<(ostream &cout, const line<T> &o) { return cout << '(' << o.d.x << "k
+ " << o.o.x << " + " << o.d.y << "k+ " << o.o.y << ")"; }

tpl vec<db> intersect(const line<T> &m, const line<T> &n)
{
    if (!sgn(m.d * n.d)) return (!sgn(m.d * (n.o - m.o))) ? apos : npos;
    return (vec<db>)m.o + (n.o - m.o) * n.d / (db)(m.d * n.d) * (vec<db>)m.d;
}

tpl db dis(const line<T> &m, const vec<T> &o) { return abs(m.d * (o - m.o) / m.d.len()); }
tpl db dis(const vec<T> &o, const line<T> &m) { return abs(m.d * (o - m.o) / m.d.len()); }
tuple<vec<ll>, vec<ll>, ll, ll> proj(const line<ll> &l, const vec<ll> &p)
{
    ll x = (ll)(p.x - l.o.x) * l.d.x + (ll)(p.y - l.o.y) * l.d.y;
    ll y = (ll)l.d.x * l.d.x + (ll)l.d.y * l.d.y;
    return {l.o, l.d, x, y};
}

vec<db> proj(const line<db> &l, const vec<db> &p)
{
    return l.o + l.d * ((p - l.o).dot(l.d) / l.d.len2());
}

tpl struct circle;
template<> struct circle<db>
{
    vec<db> o;
    db r;
    circle() { }
    circle(const vec<db> &o, const db &r = 0) :o(o), r(r) { }//圆心半径构造
    circle(const vec<db> &a, const vec<db> &b) { o = (a + b) * 0.5; r = dis(b, o); }//直径构造
};

```

```

circle(const vec<db> &a, const vec<db> &b, const vec<db> &c) // 三点构造外接圆 (非最小圆)
{
    auto A = (b + c) * 0.5, B = (a + c) * 0.5;
    o = intersect(line(A, A + (c - A).rotate_90()), line(B, B + (c - B).rotate_90()));
    r = dis(o, c);
}
circle(vector<vec<db>> a)
{
    int n = a.size(), i, j, k;
    shuffle(all(a), rnd);
    *this = circle(a[0]);
    for (i = 1; i < n; i++) if (cover(a[i]) < 0)
    {
        *this = circle(a[i]);
        for (j = 0; j < i; j++) if (cover(a[j]) < 0)
        {
            *this = circle(a[i], a[j]);
            for (k = 0; k < j; k++) if (cover(a[k]) < 0)
                *this = circle(a[i], a[j], a[k]);
        }
    }
}
circle(const vector<vec<ll>> &b)
{
    vector<vec<db>> a(b.size());
    int n = a.size(), i, j, k;
    for (i = 0; i < a.size(); i++) a[i] = b[i];
    *this = circle(a);
}
int cover(const vec<db> &a) const { return sgn(r - dis(a, o)); } // 1 表示在内, 0 表示在边上, -1 表示在外
};
template<> struct circle<ll>
{
    vec<ll> a, b, c;
    short t;
    circle() : circle({0, 0}) { }
    circle(const vec<ll> &a) : a(a), t(1) { }
    circle(const vec<ll> &a, const vec<ll> &b) : a(a), b(b), t(2) { assert(a != b); }
    circle(const vec<ll> &x, const vec<ll> &y, const vec<ll> &z) : a(x), b(y), c(z), t(3)
    {
        assert(a != b && a != c && b != c);
        if ((b - a) * (c - a) < 0) swap(b, c);
    }
    circle(vector<vec<ll>> a)
    {
        int n = a.size(), i, j, k;
        shuffle(all(a), rnd);
        *this = circle(a[0]);
        for (i = 1; i < n; i++) if (cover(a[i]) < 0)
        {
            *this = circle(a[i]);
            for (j = 0; j < i; j++) if (cover(a[j]) < 0)
            {
                *this = circle(a[i], a[j]);
                for (k = 0; k < j; k++) if (cover(a[k]) < 0)
                    *this = circle(a[i], a[j], a[k]);
            }
        }
    }
}

```

```

    }
}
}
int cover(const vec<ll> &p) const//1 表示在内, 0 表示在边上, -1 表示在外
{
    if (t == 1) return -(a != p);
    if (t == 2) return -sgn((a - p).dot(b - p));
    assert(t == 3);
    vec<ll> u = a - p, v = b - p, w = c - p;
    ll u2 = (ll)u.x * u.x + (ll)u.y * u.y;
    ll v2 = (ll)v.x * v.x + (ll)v.y * v.y;
    ll w2 = (ll)w.x * w.x + (ll)w.y * w.y;
    ll vw = (ll)v.x * w.y - (ll)v.y * w.x;
    ll wu = (ll)w.x * u.y - (ll)w.y * u.x;
    ll uv = (ll)u.x * v.y - (ll)u.y * v.x;
    return sgn(u2 * vw + v2 * wu + w2 * uv);
}
};
tmpl vector<vec<db>> intersect(const circle<db> &c, const line<T> &l)
{
    vec<db> o = (vec<db>)l.o, d = (vec<db>)l.d;
    vec<db> p = o + d * ((c.o - o).dot(d) / d.len2());
    db h = c.r * c.r - (p - c.o).len2();
    if (sgn(h) < 0) return { };
    if (!sgn(h)) return {p};
    d *= sqrt(h / d.len2());
    return {p - d, p + d};
}
tmpl vector<vec<db>> intersect(const line<T> &l, const circle<db> &c) { return intersect(c, l); }
vector<vec<db>> intersect(const circle<db> &a, const circle<db> &b)
{
    vec<db> d = b.o - a.o;
    db l = d.len();
    if (!sgn(l)) return { };
    db x = (a.r * a.r - b.r * b.r + l * l) / (2 * l);
    db h = a.r * a.r - x * x;
    if (sgn(h) < 0) return { };
    vec<db> p = a.o + d * (x / l);
    if (!sgn(h)) return {p};
    d = d.rotate_90() * (sqrt(h) / l);
    return {p - d, p + d};
}
vector<pair<line<db>, vec<db>>> tangent(const circle<db> &c, const vec<db> &p)
{
    vector<pair<line<db>, vec<db>>> ans;
    vec<db> d = p - c.o;
    db z = d.len2(), h = z - c.r * c.r;
    if (sgn(h) < 0 || !sgn(z)) return ans;
    if (!sgn(h))
    {
        vec<db> t = c.o + d * (c.r / sqrt(z));
        ans.push_back({line<db>(t, t + d.rotate_90()), t});
        return ans;
    }
    vec<db> r = d.rotate_90();
    for (int s : {-1, 1})

```

```

    {
        vec<db> t = c.o + (d * (c.r * c.r) + r * (s * c.r * sqrt(h))) / z;
        ans.push_back({line<db>(p, t), t});
    }
    return ans;
}

vector<tuple<line<db>, vec<db>, vec<db>>> tangent(const circle<db> &a, const circle<db> &b)
{
    vector<tuple<line<db>, vec<db>, vec<db>>> ans;
    vec<db> d = b.o - a.o;
    db z = d.len2();
    if (!sgn(z)) return ans;
    for (int o : {-1, 1})
    {
        db r = a.r - o * b.r, h = z - r * r;
        if (sgn(h) < 0) continue;
        h = sqrt(max<db>(0, h));
        for (int s : {-1, 1})
        {
            vec<db> v = (d * r + d.rotate_90() * (s * h)) / z;
            vec<db> x = a.o + v * a.r, y = b.o + v * (o * b.r);
            ans.push_back({sgn((x - y).len2()) == 0 ? line<db>(x, x + v.rotate_90()) : line<
db>(x, y), x, y});
            if (!sgn(h)) break;
        }
    }
    return ans;
}

tpl struct segment
{
    vec<T> a, b;
    segment() { }
    segment(const vec<T> &o, const vec<T> &p) :a(o), b(p)
    {
        int s = sgn(a.x - b.x);
        if (s > 0 || !s && a.y > b.y) swap(a, b);
    }
    bool cover(const vec<T> &o) const { return sgn((o - a) * (b - a)) == 0 && sgn((o - a).dot
(o - b)) <= 0; }
};

tpl bool intersect(const segment<T> &m, const segment<T> &n)
{
    auto a = n.b - n.a, b = m.b - m.a;
    auto d = n.a - m.a;
    if (sgn(n.b.x - m.a.x) < 0 || sgn(m.b.x - n.a.x) < 0) return 0;
    if (sgn(max(n.a.y, n.b.y) - min(m.a.y, m.b.y)) < 0 || sgn(max(m.a.y, m.b.y) - min(n.a.y,
n.b.y)) < 0) return 0;
    return sgn(b * d) * sgn((n.b - m.a) * b) >= 0 && sgn(a * d) * sgn((m.b - n.a) * a) <= 0;
}

tpl bool intersect(const segment<T> &m, const line<T> &n) { return sgn(n.d * (m.a - n.o)) *
sgn(n.d * (m.b - n.o)) <= 0; }

tpl bool intersect(const line<T> &n, const segment<T> &m) { return intersect(m, n); }

tpl vector<vec<db>> intersect(const circle<db> &c, const segment<T> &s)
{
    vector<vec<db>> ans;
    if (s.a == s.b) return c.cover((vec<db>)s.a) == 0 ? vector<vec<db>>{(vec<db>)s.a} : ans;
    segment<db> t((vec<db>)s.a, (vec<db>)s.b);

```



```

    for (auto p : intersect(c, line<db>(t.a, t.b))) if (t.cover(p)) ans.push_back(p);
    return ans;
}
templ vector<vec<db>> intersect(const segment<T> &s, const circle<db> &c) { return intersect(c, s); }
templ db dis(const vec<T> &o, const segment<T> &l)
{
    if (l.a == l.b) return dis(o, l.a);
    if (sgn((l.b - l.a).dot(o - l.a)) < 0 || sgn((l.a - l.b).dot(o - l.b)) < 0) return min(
dis(o, l.a), dis(o, l.b));
    return dis(o, line(l.a, l.b));
}
templ db dis(const segment<T> &l, const vec<T> &o) { return dis(o, l); }
tuple<vec<ll>, vec<ll>, ll, ll> nearest(const segment<ll> &s, const vec<ll> &p)
{
    vec<ll> d = s.b - s.a;
    ll x = (ll)(p.x - s.a.x) * d.x + (ll)(p.y - s.a.y) * d.y;
    ll y = (ll)d.x * d.x + (ll)d.y * d.y;
    if (!y || x <= 0) return {s.a, d, 0, 1};
    if (x >= y) return {s.a, d, 1, 1};
    return {s.a, d, x, y};
}
vec<db> nearest(const segment<db> &s, const vec<db> &p)
{
    vec<db> d = s.b - s.a;
    db y = d.len2(), x = (p - s.a).dot(d);
    if (!sgn(y) || x <= 0) return s.a;
    if (x >= y) return s.b;
    return s.a + d * (x / y);
}
templ db dis(const segment<T> &a, const segment<T> &b)
{
    if (intersect(a, b)) return 0;
    return min({dis(a.a, b), dis(a.b, b), dis(b.a, a), dis(b.b, a)});
}
templ struct polygon
{
    vector<vec<T>> p;
    polygon(const vector<vec<T>> &a = { }) : p(a) { }
    db peri() const//周长
    {
        int i, n = p.size();
        if (n == 0) return 0;
        db C = (p[n - 1] - p[0]).len();
        for (i = 1; i < n; i++) C += (p[i - 1] - p[i]).len();
        return C;
    }
    db area() const { return area2() * 0.5; }//面积
    T area2() const//两倍面积
    {
        int i, n = p.size();
        if (n == 0) return 0;
        T S = p[n - 1] * p[0];
        for (i = 1; i < n; i++) S += p[i - 1] * p[i];
        return abs(S);
    }
    int cover(const vec<T> &o) const//点是否在多边形内, -1 外 0 上 1 内

```

```

{
    static uniform_int_distribution<ll> gen(1.2e9, 2e9);
    vec<T> t;
    t.x = gen(rnd); t.y = gen(rnd);
    segment<T> s(o, t);
    int i, n = p.size(), r = 0;
    for (i = 0; i < n; i++)
    {
        if (segment(p[i], p[(i + 1) % n]).cover(o)) return 0;
        r ^= intersect(s, segment(p[i], p[(i + 1) % n]));
    }
    return r ? 1 : -1;
}

};
templ struct convex : polygon<T>
{
    convex(vector<vec<T>> a)
    {
        auto &p = this->p;
        int n = a.size(), i;
        if (!n) return;
        p = a;
        for (i = 1; i < n; i++) if (p[i].x < p[0].x || p[i].x == p[0].x && p[i].y < p[0].y)
swap(p[0], p[i]);
        a.resize(0); a.reserve(n);
        for (i = 1; i < n; i++) if (p[i] != p[0]) a.push_back(p[i] - p[0]);
        sort(all(a));
        for (i = 0; i < a.size(); i++) a[i] += p[0];
        vec<T> *st = p.data() - 1;
        int tp = 1;
        for (auto &v : a)
        {
            while (tp > 1 && sgn((st[tp] - st[tp - 1]) * (v - st[tp - 1])) <= 0) --tp;
            st[++tp] = v;
        }
        p.resize(tp);
    }
    int cover(const vec<T> &o) const//点是否在凸包内, -1 外 0 上 1 内
    {
        const auto &p = this->p;
        if (sgn(o.x - p[0].x) < 0 || sgn(o.x - p[0].x) == 0 && sgn(o.y - p[0].y) < 0) return
-1;
        if (o == p[0]) return 0;
        if (p.size() == 1) return -1;
        int tmp = sgn((o - p[0]) * (p[1] - p[0]));
        if (tmp == 0) return sgn(dis2(o, p[0]) - dis2(p[1], p[0])) <= 0 ? 0 : -1;
        tmp = sgn((o - p[0]) * (p.back() - p[0]));
        if (tmp == 0) return sgn(dis2(o, p[0]) - dis2(p.back(), p[0])) <= 0 ? 0 : -1;
        if (tmp < 0 || p.size() == 2) return -1;
        int x = upper_bound(1 + all(p), o, [&](const vec<T> &a, const vec<T> &b) { return sgn
((a - p[0]) * (b - p[0])) > 0; }) - p.begin() - 1;
        tmp = sgn((o - p[x]) * (p[x + 1] - p[x]));
        if (tmp > 0) return -1;
        return tmp < 0;
    }
    convex<T> operator+(const convex<T> &A) const
    {

```

```

    auto &p = this->p;
    int n = p.size(), m = A.p.size(), i, j;
    vector<vec<T>> a(n), b(m), c;
    for (i = 0; i + 1 < n; i++) a[i] = p[i + 1] - p[i];
    a[n - 1] = p[0] - p[n - 1];
    for (i = 0; i + 1 < m; i++) b[i] = A.p[i + 1] - A.p[i];
    b[m - 1] = A.p[0] - A.p[m - 1];
    c.reserve(n + m);
    c.push_back(p[0] + A.p[0]);
    for (i = j = 0; i < n && j < m;)
    {
        int t = sgn(a[i] * b[j]);
        if (t == 0) c.push_back(c.back() + a[i] + b[j]), ++i, ++j;
        else c.push_back(c.back() + (t > 0 ? a[i++] : b[j++]));
    }
    while (i < n) c.push_back(c.back() + a[i++]);
    while (j < m) c.push_back(c.back() + b[j++]);
    c.pop_back();
    convex<T> t({ });
    t.p = c;
    return t;
}

};
tpl pair<int, int> sector(const convex<T> &c, const vec<T> &o) //返回射线 p[0]->o 所在扇形的两个凸包下标
{
    const auto &p = c.p;
    int n = p.size();
    if (!n) return {-1, -1};
    if (n == 1 || o == p[0]) return {0, 0};
    if (n == 2) return {0, 1};
    auto det = [&](int i) -> up<T> {
        return (up<T>)(p[i].x - p[0].x) * (o.y - p[0].y) - (up<T>)(p[i].y - p[0].y) * (o.x - p[0].x);
    };
    auto dot = [&](int i) -> up<T> {
        return (up<T>)(p[i].x - p[0].x) * (o.x - p[0].x) + (up<T>)(p[i].y - p[0].y) * (o.y - p[0].y);
    };
    int s = sgn(det(1));
    if (s < 0) return {0, 1};
    if (!s && sgn(dot(1)) >= 0) return {1, 1};
    s = sgn(det(n - 1));
    if (s > 0) return {n - 1, 0};
    if (!s && sgn(dot(n - 1)) >= 0) return {n - 1, n - 1};
    int l = 1, r = n - 1;
    while (l + 1 < r)
    {
        int mid = (l + r) >> 1;
        if (sgn(det(mid)) >= 0) l = mid;
        else r = mid;
    }
    if (!sgn(det(l)) && sgn(dot(l)) >= 0) return {l, l};
    if (!sgn(det(r)) && sgn(dot(r)) >= 0) return {r, r};
    return {l, r};
}

tpl pair<pair<int, vec<T>>, pair<int, vec<T>>> tangent(const convex<T> &c, const vec<T> &o)

```

```

{
    const auto &p = c.p;
    int n = p.size();
    if (!n) return {{-1, { }}, {-1, { }}};
    if (n == 1) return o == p[0] ? pair{pair{-1, vec<T>()}, pair{-1, vec<T>()}} : pair{pair
{0, p[0]}, pair{0, p[0]}};
    if (n == 2)
    {
        if (segment<T>(p[0], p[1]).cover(o)) return {{-1, { }}, {-1, { }}};
        return {{0, p[0]}, {1, p[1]}};
    }
    auto id = [&](long long x) -> int {
        x %= n;
        if (x < 0) x += n;
        return x;
    };
    auto det = [&](const vec<T> &a, const vec<T> &b, const vec<T> &d) -> up<T> {
        return (up<T>)(b.x - a.x) * (d.y - a.y) - (up<T>)(b.y - a.y) * (d.x - a.x);
    };
    auto side = [&](int i) { return sgn(det(p[i], p[(i + 1) % n], o)); };
    auto visible = [&](long long i) { return side(id(i)) < 0; };
    vector<int> cand;
    auto add = [&](int x) {
        x = id(x);
        if (find(all(cand), x) == cand.end()) cand.push_back(x);
    };
    auto [x, y] = sector(c, o);
    for (int i = -4; i <= 4; i++) add(x + i), add(y + i);
    int e = -1;
    for (int x : cand) if (side(x) < 0) { e = x; break; }
    if (e == -1) return {{-1, { }}, {-1, { }}};
    long long L, R;
    if (!visible(0) && !visible(n - 1))
    {
        long long l = 0, r = e;
        while (l < r)
        {
            long long mid = (l + r) >> 1;
            if (visible(mid)) r = mid;
            else l = mid + 1;
        }
        L = l;
        l = e; r = n - 1;
        while (l < r)
        {
            long long mid = (l + r + 1) >> 1;
            if (visible(mid)) l = mid;
            else r = mid - 1;
        }
        R = l;
    }
    else if (visible(0) && !visible(n - 1))
    {
        L = 0;
        long long l = 0, r = n - 1;
        while (l < r)
        {

```

```

        long long mid = (l + r + 1) >> 1;
        if (visible(mid)) l = mid;
        else r = mid - 1;
    }
    R = l;
}
else if (!visible(0) && visible(n - 1))
{
    R = n - 1;
    long long l = 0, r = n - 1;
    while (l < r)
    {
        long long mid = (l + r) >> 1;
        if (visible(mid)) r = mid;
        else l = mid + 1;
    }
    L = l;
}
else
{
    int f = -1;
    for (int x : cand) if (!visible(x)) { f = id(x); break; }
    for (int i = 1; f == -1 && i < n; i <= 1)
    {
        if (!visible(i)) f = i;
        else if (!visible(n - 1 - i)) f = n - 1 - i;
    }
    if (f == -1 && !visible(n / 2)) f = n / 2;
    if (f == -1) return {{-1, { }}, {-1, { }}};
    long long l = 0, r = f;
    while (l < r)
    {
        long long mid = (l + r) >> 1;
        if (!visible(mid)) r = mid;
        else l = mid + 1;
    }
    R = l - 1;
    l = f; r = n - 1;
    while (l < r)
    {
        long long mid = (l + r + 1) >> 1;
        if (!visible(mid)) l = mid;
        else r = mid - 1;
    }
    L = l + 1;
}
int R0 = id(R + 1);
int L0 = id(L);
return {{R0, p[R0]}, {L0, p[L0]}};
}
tmp1 struct half_plane//默认左侧
{
    vec<T> o, d;
    operator half_plane<ll>() const { return {(vec<ll>)o, (vec<ll>)(o + d)}; }
    operator half_plane<db>() const { return {(vec<db>)o, (vec<db>)(o + d)}; }
    half_plane() { }
    half_plane(const vec<T> &a, const vec<T> &b) :o(a), d(b - a) { }
}

```

```

bool operator< (const half_plane<T> &a) const
{
    int p = quad(d), q = quad(a.d);
    if (p != q) return p < q;
    p = sgn(d * a.d);
    if (p) return p > 0;
    return sgn(d * (a.o - o)) > 0;
}

};

template<ostream &operator<<(ostream &cout, half_plane<T> &m) { return cout << m.o << " | " << m.d; }

template<vec<db> intersect(const half_plane<T> &m, const half_plane<T> &n)
{
    if (!sgn(m.d * n.d)) return sgn(m.d * (n.o - m.o)) ? npos : apos;
    return (vec<db>)m.o + (n.o - m.o) * n.d / (db)(m.d * n.d) * (vec<db>)m.d;
}

const db inf = 1e18;
convex<db> intersect(vector<half_plane<db>> a)
{
    db I = inf;
    a.push_back({{-I, -I}, {I, -I}});
    a.push_back({{I, -I}, {I, I}});
    a.push_back({{I, I}, {-I, I}});
    a.push_back({{-I, I}, {-I, -I}});
    sort(all(a));
    int n = a.size(), i, h = 0, t = -1;
    vector<half_plane<db>> q(n);
    vector<vec<db>> p(n);
    for (i = 0; i < n; i++) if (i == n - 1 || sgn(a[i].d * a[i + 1].d))
    {
        auto x = (half_plane<db>)a[i];
        while (h < t && sgn((p[t - 1] - x.o) * x.d) >= 0) --t;
        while (h < t && sgn((p[h] - x.o) * x.d) >= 0) ++h;
        q[++t] = x;
        if (h < t) p[t - 1] = intersect(q[t - 1], q[t]);
    }
    while (h < t && sgn((p[t - 1] - q[h].o) * q[h].d) >= 0) --t;
    if (h == t) return convex<db>(vector<vec<db>>(0));
    p[t] = intersect(q[h], q[t]);
    return convex<db>(vector<vec<db>>(p.begin() + h, p.begin() + t + 1));
}

template<pair<int, int> closest_pair(const vector<vec<T>> &a)
{
    int n = a.size(), i;
    assert(n >= 2);
    vector<pair<vec<T>, int>> b(n);
    for (i = 0; i < n; i++) b[i] = {a[i], i};
    sort(all(b), [&](auto u, auto v) {
        if (u.first.x != v.first.x) return u.first.x < v.first.x;
        return u.first.y < v.first.y;
    });
    tuple<T, int, int> ans = {dis2(a[0], a[1]), 0, 1};
    set<pair<T, int>> s;
    int j = 0;
    for (auto [v, i] : b)
    {
        auto [x, y] = v;

```

```

    T d = sqrtl(get<0>(ans));
    if (d == 0) break;
    for (auto it = s.lower_bound({y - d, 0}); it != s.end() && it->first <= y + d; ++it)
        cmin(ans, tuple{dis2(a[it->second], v), i, it->second});
    s.emplace(v.y, i);
    while (b[j].first.x < v.x - d) s.erase({b[j].first.y, b[j].second}), ++j;
}
return {get<1>(ans), get<2>(ans)};
}

tmpl pair<int, int> furthest_pair(const vector<vec<T>> &a)
{
    int n = a.size(), i, j;
    assert(n >= 2);
    auto b = convex(a).p;
    int m = b.size();
    if (m == 1) return {0, 1};
    b.push_back(b[0]);
    tuple<T, int, int> ans{dis2(b[0], b[1]), 0, 1};
    for (i = 0, j = 1; i < m; i++)
    {
        while (abs((b[i + 1] - b[i]) * (b[j] - b[i])) < abs((b[i + 1] - b[i]) * (b[(j + 1) % m] - b[i]))) j = (j + 1) % m;
        cmax(ans, tuple{dis2(b[i], b[j]), i, j});
        cmax(ans, tuple{dis2(b[i + 1], b[j]), i + 1, j});
    }
    auto [_, j1, j2] = ans;
    int i1, i2;
    for (i1 = 0; i1 < n; i1++) if (a[i1] == b[j1]) break;
    for (i2 = 0; i2 < n; i2++) if (i2 != i1 && a[i2] == b[j2]) break;
    return {i1, i2};
}

tmpl array<vec<db>, 4> rectangle_cover(const vector<vec<T>> &a)
{
    const auto &p = convex(a).p;
    int n = p.size(), m = n * 4, i, j, k, l;
    if (n <= 2) return {0, 0, 0, 0};
    vector<vec<T>> b(m);
    for (i = 0; i < m; i++) b[i] = p[i % n];
    tuple<db, int, int, int, int> tmp{inf, 0, 0, 0, 0};
    for (i = j = k = l = 0; i < n * 2; i++)
    {
        cmax(j, i + 1);
        auto d = b[i + 1] - b[i];
        while (d.dot(b[j] - b[i]) < d.dot(b[j + 1] - b[i])) ++j;
        while (j > i && d.dot(b[j] - b[i]) < d.dot(b[j - 1] - b[i])) --j;
        cmax(k, j);
        while (abs(d * (b[k] - b[i])) < abs(d * (b[k + 1] - b[i]))) ++k;
        while (k > j && abs(d * (b[k] - b[i])) < abs(d * (b[k - 1] - b[i]))) --k;
        cmax(l, k);
        while (d.dot(b[l] - b[i]) > d.dot(b[l + 1] - b[i])) ++l;
        while (l > k && d.dot(b[l] - b[i]) > d.dot(b[l - 1] - b[i])) --l;
        assert(l + 1 < m);
        if (i >= n) cmin(tmp, tuple{(db)(b[j] - b[l]).dot(d) * abs((b[k] - b[i]) * d) / d.
len2(), i, j, k, l});
    }
    tie(ignore, i, j, k, l) = tmp;
    auto d = b[i + 1] - b[i], rd = d.rotate_90();

```

```

    line l1(b[i], b[i] + d), l2(b[j], b[j] + rd), l3(b[k], b[k] + d), l4(b[l], b[l] + rd);
    return {intersect(l1, l2), intersect(l2, l3), intersect(l3, l4), intersect(l4, l1)};
}
template<line<T>> convex_up(vector<line<T>> a)
{
    for (auto &t : a) t.d.y = -t.d.y;
    a = convex_down(a);
    for (auto &t : a) t.d.y = -t.d.y;
    return a;
}
template<line<T>> convex_down(vector<line<T>> a)
{
    sort(all(a), [&](const auto &u, const auto &v) {
        int t = sgn(u.d * v.d);
        return t ? t > 0 : sgn((u.o - v.o) * v.d) > 0;
    });
    vector<line<T>> b;
    int tp = -1;
    for (auto t : a)
    {
        while (tp >= 0 && sgn(b[tp].d * t.d) == 0) --tp, b.pop_back();
        while (tp >= 1 && sgn((up<T>)((b[tp].o - t.o) * b[tp].d) * (t.d * b[tp - 1].d) - (up<
T>)((b[tp - 1].o - t.o) * b[tp - 1].d) * (t.d * b[tp].d)) <= 0)
            --tp, b.pop_back();
        ++tp; b.push_back(t);
    }
    return b;
}
template<vec<T>> convex_down(vector<vec<T>> a)
{
    sort(all(a), [&](const auto &u, const auto &v) {
        int t = sgn(u.x - v.x);
        if (t) return t < 0;
        return u.y > v.y;
    });
    vector<vec<T>> b;
    int tp = -1;
    for (auto t : a)
    {
        while (tp >= 0 && sgn(b[tp].x - t.x) == 0) --tp, b.pop_back();
        while (tp >= 1 && sgn((t - b[tp]) * (t - b[tp - 1])) >= 0)
            --tp, b.pop_back();
        ++tp; b.push_back(t);
    }
    return b;
}
template<vec<T>> convex_up(vector<vec<T>> a)
{
    for (auto &t : a) t.y = -t.y;
    a = convex_down(a);
    for (auto &t : a) t.y = -t.y;
    return a;
}
template<vec<db>> to_vec(const vector<line<T>> &a)
{
    int n = a.size(), i;
    vector<vec<db>> b(n - 1);

```



```

    for (i = 0; i < n - 1; i++) b[i] = intersect(a[i], a[i + 1]);
    return b;
}
template<T> find_max(const vector<vec<T>> &a, T kx, T ky)//要求函数凸
{
    vec<T> p = {kx, ky};
    int l = 0, r = a.size() - 1, mid;
    while (l < r)
    {
        mid = (l + r) / 2;
        if (a[mid].dot(p) < a[mid + 1].dot(p)) l = mid + 1;
        else r = mid;
    }
    return max({a[l].dot(p), a[0].dot(p), a.back().dot(p)});
}
template<T> find_min(const vector<vec<T>> &a, T kx, T ky)//要求函数凸
{
    vec<T> p = {kx, ky};
    int l = 0, r = a.size() - 1, mid;
    while (l < r)
    {
        mid = (l + r) / 2;
        if (a[mid].dot(p) > a[mid + 1].dot(p)) l = mid + 1;
        else r = mid;
    }
    return min({a[l].dot(p), a[0].dot(p), a.back().dot(p)});
}
template<T> max_subset_sum(const vector<vec<T>> &a)
{
    int n = a.size(), i;
    function<convex<T>(int, int)> dfs = [&](int l, int r) {
        if (l + 1 == r) return convex<T>({vec<T>(0, 0), a[l]});
        int mid = (l + r) / 2;
        return dfs(l, mid) + dfs(mid, r);
    };
    const auto &p = dfs(0, n).p;
    T ans = 0;
    for (auto t : p) ans = max(ans, t.len2());
    return ans;
}
template<class T> struct dynamic_convex//下凸
{
    set<vec<T>, decltype([](const vec<T> &a, const vec<T> &b) {
        return sgn(a.x - b.x) < 0;
    })> s;
    void check(auto it)
    {
        decltype(it) jt, kt;
        if (it != s.begin())
        {
            jt = prev(it);
            if (jt != s.begin())
            {
                kt = prev(jt);
                while (sgn((*it - *kt) * (*jt - *kt)) >= 0)
                {
                    s.erase(jt);
                }
            }
        }
    }
};

```

```

        if (kt == s.begin()) break;
        jt = kt--;
    }
}
jt = next(it);
if (jt != s.end())
{
    kt = next(jt);
    while (kt != s.end() && sgn((*kt - *it) * (*jt - *it)) >= 0)
        s.erase(jt), jt = kt++;
}
}
void insert(const vec<T> &p)
{
    auto it = s.lower_bound(p);
    if (it == s.end() || sgn(s.begin()->x - p.x) > 0) it = s.insert(it, p);
    else if (sgn(it->x - p.x) == 0) cmin(it->y, p.y);
    else
    {
        auto l = *prev(it);
        if (sgn((p - l) * (*it - l)) > 0) it = s.insert(it, p);
    }
    check(it);
}
int cover(const vec<T> &p) //在凸壳区域以上返回 1, 在凸壳上返回 0, 其余返回 -1。
{
    if (s.size() == 0) return -1;
    if (sgn(p.x - s.begin()->x) < 0 || sgn(p.x - s.rbegin()->x) > 0) return -1;
    auto it = s.lower_bound(p);
    if (sgn(it->x - p.x) == 0) return sgn(p.y - it->y);
    auto l = *prev(it);
    return sgn((*it - l) * (p - l));
}
};
template<class T> vector<array<int, 3>> delaunay(const vector<vec<T>> &a)
{
    int n0 = a.size();
    if (n0 < 3) return { };

    vector<int> id(n0);
    iota(all(id), 0);
    sort(all(id), [&](int i, int j) {
        auto [x1, y1] = a[i];
        auto [x2, y2] = a[j];
        if (x1 != x2) return x1 < x2;
        return y1 < y2;
    });
    id.erase(unique(all(id), [&](int i, int j) {
        auto [x1, y1] = a[i];
        auto [x2, y2] = a[j];
        return x1 == x2 && y1 == y2;
    }), id.end());

    int n = id.size();
    if (n < 3) return { };

```

```

struct Q { int rot, o, p = -1; };

vector<Q> e; vector<int> del;
e.reserve(n * 16); del.reserve(n * 4);

auto mk = [&](int u, int v) {
    int xy;
    if (del.size()) xy = del.back(), del.pop_back();
    else xy = e.size(), e.resize(xy + 4);
    e[xy].rot = xy + 1; e[xy + 1].rot = xy + 2; e[xy + 2].rot = xy + 3; e[xy + 3].rot =
xy;
    e[xy].p = u; e[xy + 2].p = v; e[xy + 1].p = e[xy + 3].p = -1;
    e[xy].o = xy; e[xy + 1].o = xy + 3; e[xy + 2].o = xy + 2; e[xy + 3].o = xy + 1;
    return xy;
};
auto splice = [&](int x, int y) {
    swap(e[e[x].o].rot.o, e[e[y].o].rot.o);
    swap(e[x].o, e[y].o);
};
auto connect = [&](int x, int y) {
    int xy = mk(e[x ^ 2].p, e[y].p);
    splice(xy, e[e[x].rot ^ 2].o.rot);
    splice(xy ^ 2, y);
    return xy;
};
auto erase = [&](int xy) {
    splice(xy, e[e[xy].rot].o.rot);
    splice(xy ^ 2, e[e[xy ^ 2].rot].o.rot);
    xy = xy >> 2 << 2;
    e[xy].p = -1;
    del.push_back(xy);
};
auto solve = [&](auto &&self, int l, int r) -> pair<int, int> {
    if (r - l == 2)
    {
        int xy = mk(id[l], id[l + 1]);
        return {xy, xy ^ 2};
    }
    if (r - l == 3)
    {
        int xy = mk(id[l], id[l + 1]), yz = mk(id[l + 1], id[l + 2]);
        splice(xy ^ 2, yz);
        int s = sgn((a[id[l + 1]] - a[id[l]]) * (a[id[l + 2]] - a[id[l]]));
        if (s > 0)
        {
            connect(yz, xy);
            return {xy, yz ^ 2};
        }
        else if (s < 0)
        {
            int zx = connect(yz, xy);
            return {zx ^ 2, zx};
        }
        else return {xy, yz ^ 2};
    }
}

int m = l + r >> 1;

```

```

auto [lo, li] = self(self, l, m); auto [ri, ro] = self(self, m, r);

while (1)
{
    if (sgn((a[e[li].p] - a[e[ri].p]) * (a[e[li ^ 2].p] - a[e[ri].p])) > 0)
        li = e[e[li].rot ^ 2].o.rot;
    else if (sgn((a[e[ri ^ 2].p] - a[e[li].p]) * (a[e[ri].p] - a[e[li].p])) > 0)
        ri = e[ri ^ 2].o;
    else break;
}

int bs = connect(ri ^ 2, li);
if (e[li].p == e[lo].p) lo = bs ^ 2;
if (e[ri].p == e[ro].p) ro = bs;

while (1)
{
    int u = e[bs ^ 2].p, v = e[bs].p, lc = e[bs ^ 2].o;
    bool vl = sgn((a[v] - a[u]) * (a[e[lc ^ 2].p] - a[u])) > 0;
    if (vl)
    {
        while (1)
        {
            int t = e[lc].o, x = e[t ^ 2].p;
            if (sgn((a[v] - a[u]) * (a[x] - a[u])) <= 0 ||
                circle<T>(a[u], a[v], a[e[lc ^ 2].p]).cover(a[x]) <= 0) break;
            erase(lc);
            lc = t;
        }
    }

    int rc = e[e[bs].rot].o.rot;
    bool vr = sgn((a[v] - a[u]) * (a[e[rc ^ 2].p] - a[u])) > 0;
    if (vr)
    {
        while (1)
        {
            int t = e[e[rc].rot].o.rot, x = e[t ^ 2].p;
            if (sgn((a[v] - a[u]) * (a[x] - a[u])) <= 0 ||
                circle<T>(a[u], a[v], a[e[rc ^ 2].p]).cover(a[x]) <= 0) break;
            erase(rc);
            rc = t;
        }
    }

    if (!vl && !vr) break;

    bool cr = !vl || vr && circle<T>(a[e[lc ^ 2].p], a[e[lc].p], a[e[rc].p]).cover(a[
e[rc ^ 2].p]) > 0;

    if (cr) bs = connect(rc, bs ^ 2);
    else bs = connect(bs ^ 2, lc ^ 2);
}
return {lo, ro};
};

solve(solve, 0, n);

```

```

vector<pair<int, int>> eg; int sz = e.size();
eg.reserve(sz / 4);
for (int i = 0; i < sz; i += 4) if (e[i].p != -1)
{
    int u = e[i].p, v = e[i ^ 2].p;
    if (u == -1 || v == -1 || u == v) continue;
    if (u > v) swap(u, v);
    eg.push_back({u, v});
}
sort(all(eg)); eg.erase(unique(all(eg)), eg.end());

int m = eg.size();
vector<vector<pair<int, int>>> g(n0); vector<int> deg(n0);
for (auto [u, v] : eg) deg[u]++, deg[v]++;
for (int i = 0; i < n0; i++) g[i].reserve(deg[i]);
vector<int> from(m * 2), to(m * 2), nxt(m * 2), pos(m * 2);
for (int i = 0; i < m; i++)
{
    auto [u, v] = eg[i];
    from[i * 2] = u; to[i * 2] = v;
    from[i * 2 + 1] = v; to[i * 2 + 1] = u;
    g[u].push_back({v, i * 2});
    g[v].push_back({u, i * 2 + 1});
}
for (int u = 0; u < n0; u++)
{
    sort(all(g[u]), [&](auto x, auto y) {
        vec<T> a1 = a[x.first] - a[u], a2 = a[y.first] - a[u];
        int h1 = sgn(a1.y) > 0 || !sgn(a1.y) && sgn(a1.x) > 0;
        int h2 = sgn(a2.y) > 0 || !sgn(a2.y) && sgn(a2.x) > 0;
        if (h1 != h2) return h1 > h2;
        int s = sgn(a1 * a2);
        if (s) return s > 0;
        return a1.len2() < a2.len2();
    });
    for (int j = 0; j < (int)g[u].size(); j++) pos[g[u][j].second] = j;
}
for (int i = 0; i < m * 2; i++)
{
    auto &cur = g[to[i]];
    int sz = cur.size();
    nxt[i] = cur[(pos[i ^ 1] + sz - 1) % sz].second;
}

vector<char> vis(m * 2);
vector<array<int, 3>> trs;
vector<int> f;
trs.reserve(m); f.reserve(16);
for (int t = 0; t < m * 2; t++)
{
    if (vis[t]) continue;

    f.clear();
    int cur = t; bool flg = false;
    while (1)
    {

```

```

        vis[cur] = true;
        f.push_back(from[cur]);
        cur = nxt[cur];
        if (cur == t) { flg = true; break; }
        if ((int)f.size() > m * 2) break;
    }
    int sz = f.size();
    if (!flg || sz < 3) continue;

    up<T> ss = 0;
    for (int i = 0; i < sz; i++)
    {
        const auto &p = a[f[i]];
        const auto &o = a[f[(i + 1) % sz]];
        ss += (up<T>)p.x * o.y - (up<T>)p.y * o.x;
    }
    if (sgn(ss) <= 0) continue;
    for (int i = 1; i + 1 < sz; i++)
        if (sgn((a[f[i]] - a[f[0]]) * (a[f[i + 1]] - a[f[0]])) > 0)
            trs.push_back({f[0], f[i], f[i + 1]});
}

return trs;
}
struct union_set
{
    int n;
    vector<int> f;
    union_set() { }
    union_set(int n) :n(n), f(n)
    {
        iota(all(f), 0);
    }
    int getf(int u) { return f[u] == u ? u : f[u] = getf(f[u]); }
    bool merge(int u, int v)
    {
        u = getf(u); v = getf(v);
        if (u == v) return 0;
        f[u] = v;
        return 1;
    }
    bool connected(int u, int v) { return getf(u) == getf(v); }
};
template<class T> vector<pair<int, int>> euclidean_mst(const vector<vec<T>> &a)
{
    int n = a.size(), i;
    if (n <= 1) return { };
    vector<pair<int, int>> ans;
    ans.reserve(n - 1);
    auto tr = geo::delaunay(a);
    if (!tr.size())
    {
        vector<int> id(n);
        iota(all(id), 0);
        sort(all(id), [&](int x, int y) { return pair{a[x].x, a[x].y} < pair{a[y].x, a[y].y}; });
        for (i = 1; i < n; i++) ans.push_back({id[i - 1], id[i]});
    }
}

```

```

        return ans;
    }
    vector<tuple<T, int, int>> eg;
    eg.reserve(tr.size() * 3 + n);
    vector<int> id(n);
    iota(all(id), 0);
    sort(all(id), [&](int x, int y) { return pair{a[x].x, a[x].y} < pair{a[y].x, a[y].y}; });
    for (i = 1; i < n; i++) if (a[id[i]] == a[id[i - 1]]) eg.push_back({0, id[i - 1], id[i]});
;
    for (auto [x, y, z] : tr)
    {
        eg.push_back({dis2(a[x], a[y]), x, y});
        eg.push_back({dis2(a[y], a[z]), y, z});
        eg.push_back({dis2(a[z], a[x]), z, x});
    }
    sort(all(eg));
    union_set s(n);
    for (auto [z, x, y] : eg) if (s.merge(x, y)) ans.push_back({x, y});
    return ans;
}
#undef tpl
}
using geo::vec, geo::line, geo::circle, geo::convex, geo::polygon, geo::half_plane;
using geo::eps, geo::pi, geo::segment, geo::sgn, geo::dynamic_convex;
using Q = vec<ll>;
void read(db &x) { static string s; cin >> s; x = stod(s); }
template<typename T, typename... Args> void read(T &first, Args&... args) { read(first); read(
    args...); }

```

另附一份支持二分斜率的代码。这里维护的是 (x, y) ，其中 $y = C + x^2$ ，因此 x 整体增加会影响 y 。

```

bool FLAG = 0;
ll K;
struct Q
{
    ll x;
    mutable ll y;
    mutable decltype(set<Q>().begin()) r;
    Q operator-(const Q &o) const { return {x - o.x, y - o.y}; }
    ll operator*(const Q &o) const
    {
        return (lll)x * o.y - (lll)y * o.x;
    }
};
set<Q>::iterator END;
bool operator<(const Q &a, const Q &b)
{
    if (FLAG)
    {
        assert(b.x == 0 && b.y == 0);
        if (a.r == END) return 0;
        return (lll)a.x * K + a.y < (lll)a.r->x * K + a.r->y;
    }
    return a.x < b.x;
}
struct convex

```

```

{
    set<Q> s;
    ll tagx = 0, tagy = 0;
    void check(auto it)
    {
        decltype(it) jt, kt;
        if (it != s.begin())
        {
            jt = prev(it);
            if (jt != s.begin())
            {
                kt = prev(jt);
                while ((*it - *kt) * (*jt - *kt) <= 0)
                {
                    s.erase(jt);
                    if (kt == s.begin()) break;
                    jt = kt--;
                }
            }
            jt = next(it);
            if (jt != s.end())
            {
                kt = next(jt);
                while (kt != s.end() && (*kt - *it) * (*jt - *it) <= 0)
                    s.erase(jt), jt = kt++;
            }
        }
    }
    void insert(Q p)
    {
        p.y -= tagy + p.x * p.x;
        p.x -= tagx;
        p.y += p.x * p.x;
        auto it = s.lower_bound(p);
        if (it == s.end() || s.begin()->x - p.x > 0) it = s.insert(it, p);
        else if (it->x == p.x) cmax(it->y, p.y);
        else
        {
            auto l = *prev(it);
            if ((p - l) * (*it - l) < 0) it = s.insert(it, p);
        }
        check(it);
        it->r = next(it);
        if (it != s.begin()) prev(it)->r = it;
    }
    void add(ll X, ll Y) { tagx += X; tagy += Y; }
    ll query(ll k)
    {
        k += tagx * 2;
        FLAG = 1; K = k; END = s.end();
        auto it = s.lower_bound({0, 0});
        FLAG = 0;
        return k * (it->x + tagx) + it->y + tagy - tagx * tagx;
    }
    void fun(ll &x, ll &y) const
    {
        y -= x * x;
    }
}

```



```
        x += tagx;  
        y += tagy + x * x;  
    }  
};
```

7 杂项

7.1 枚举大小为 k 的集合

思路：通过进位创造 1，再把一串 1 移到最后。

```
for (int s = (1 << k) - 1, t; s < 1 << n; t = s + (s & -s), s = (s & ~t) >> __lg(s & -s) + 1 | t)
{
}
```

7.2 min plus 卷积

计算 $c_i = \min_{j=0}^i a_j + b_{i-j}$ 。

要求 b 是凸的，即 $b_{i+1} - b_i$ 不降。

```
template <class T> vector<T> min_plus_convolution(const vector<T> &a, const vector<T> &b)
{
    int n = a.size(), m = b.size(), i;
    vector<T> c(n + m - 1);
    function<void(int, int, int, int)> dfs = [&](int l, int r, int ql, int qr) {
        if (l > r) return;
        int mid = l + r >> 1;
        while (ql + m <= l) ++ql;
        while (qr > r) --qr;
        int qmid = -1;
        c[mid] = inf;
        for (int i = ql; i <= qr; i++) if (mid - i >= 0 && mid - i < m && cmin(c[mid], a[i] + b[
mid - i])) qmid = i;
        dfs(l, mid - 1, ql, qmid);
        dfs(mid + 1, r, qmid, qr);
    };
    dfs(0, n + m - 2, 0, n - 1);
    return c;
}
```

7.3 所有区间 GCD

需要自定义 fun，如 gcd, and, or。

```
template<class T> struct GCD
{
    vector<pair<int, T>> res;
    GCD(const vector<T> &a) :res(n)
    {
        int n = a.size(), i, j;
        vector<ll> v(n);
        vector<int> l(n);
        for (i = 0; i < n; i++)
        {
            for (v[i] = a[i], j = l[i] = i; j >= 0; j = l[j] - 1)
            {
                v[j] = fun(v[j], a[i]);
                while (l[j] && fun(a[i], v[l[j] - 1]) == fun(a[i], v[j])) l[j] = l[l[j] - 1];
                // [l[j]..j,i] 区间内的值求 fun 均为 v[j]
            }
        }
    }
}
```

```

    }
}
};

```

7.4 整体二分（区间 k -th）

$O((n + q) \log a)$, $O(n + q)$ 。

```

struct cz
{
    int x, y, kth, pos, typ;
};
cz q[M], st1[M], st2[M];
int a[N], b[N], d[N], ans[N], s[N];
int n, m, t1, t2, i, j, c, gs;
void add(int x, int y)
{
    for (; x <= n; x += x & -x) s[x] += y;
}
int sum(int x)
{
    int ans = 0;
    for (; x; x -= x & -x) ans += s[x];
    return ans;
}
void ztef(int ql, int qr, int l, int r)
{
    if (ql > qr) return;
    int mid = l + r >> 1, i, midd;
    t1 = t2 = 0;
    if (l == r)
    {
        for (i = ql; i <= qr; i++) if (q[i].typ) ans[q[i].pos] = d[l];
        return;
    }
    for (i = ql; i <= qr; i++) if (q[i].typ)
    {
        midd = sum(q[i].y) - sum(q[i].x - 1);
        if (midd >= q[i].kth) st1[++t1] = q[i]; else
        {
            st2[++t2] = q[i];
            st2[t2].kth -= midd;
        }
    }
    else if (q[i].pos <= mid)
    {
        add(q[i].x, 1);
        st1[++t1] = q[i];
    }
    else st2[++t2] = q[i];
    for (i = 1; i <= t1; i++) if (!st1[i].typ) add(st1[i].x, -1);
    for (i = 1; i <= t1; i++) q[i + ql - 1] = st1[i];
    midd = ql + t1 - 1;
    for (i = 1; i <= t2; i++) q[i + midd] = st2[i];
    ztef(ql, midd, l, mid); ztef(midd + 1, qr, mid + 1, r);
}
int main()

```

```

{
    cin >> n >> m;
    for (i = 1; i <= n; i++)
    {
        cin >> a[i];
        b[i] = a[i];
    }
    sort(b + 1, b + n + 1);
    d[gs = 1] = b[1];
    for (i = 2; i <= n; i++) if (b[i] != b[i - 1]) d[++gs] = b[i];
    for (i = 1; i <= n; i++) a[i] = lower_bound(d + 1, d + gs + 1, a[i]) - d;
    for (i = 1; i <= n; i++)
    {
        q[i].x = i; q[i].pos = a[i]; q[i].typ = 0;
    }
    for (i = 1; i <= m; i++)
    {
        cin >> q[i + n].x >> q[i + n].y >> q[i + n].kth;
        q[i + n].pos = i; q[i + n].typ = 1;
    }
    ztef(1, n + m, 1, gs);
    for (i = 1; i <= m; i++) printf("%d\n", ans[i]);
}

```

7.5 高精度

除法和取模有点问题，但 gcd 是对的。

```

struct bigint;
int cmp(const bigint &a, const bigint &b);
struct bigint
{
    using ull = unsigned long long;
    using lll = unsigned __int128;
    const static ull sign = 1llu << 63;
    const static lll p = 4'179'340'454'199'820'289;
    const static lll g = 3;
    const static ull base = 1e6;
    const static int output_base = 10;
    const static int length = round(log(bigint::base) / log(output_base));
    static_assert(output_base == 10 || output_base == 16, "output_base_must_be_10_or_16");
    static_assert(round(pow(output_base, length)) == base);
    const static int N = 1 << 23;
    static int r[N];
    static lll w[N];
    bool neg;
    vector<ull> a;
private:
    static lll ksm(lll x, ull y)
    {
        lll r = 1;
        while (y)
        {
            if (y & 1) r = r * x % p;
            x = x * x % p; y >>= 1;
        }
        return r;
    }

```

```

}
static void init(int n)
{
    static int pr = 0, pw = 0;
    if (pr == n) return;
    int b = __lg(n) - 1, i, j, k;
    for (i = 1; i < n; i++) r[i] = r[i >> 1] >> 1 | (i & 1) << b;
    if (pw < n)
    {
        for (j = 1; j < n; j = k)
        {
            k = j * 2;
            ull wn = ksm(g, (p - 1) / k);
            w[j] = 1;
            for (i = j + 1; i < k; i++) w[i] = w[i - 1] * wn % p;
        }
        pw = n;
    }
    pr = n;
}

static void dft(vector<lll> &a, int o = 0)
{
    int n = a.size(), i, j, k;
    lll y, *f, *g, *wn, *A = a.data();
    init(n);
    for (i = 1; i < n; i++) if (i < r[i]) swap(A[i], A[r[i]]);
    static const int T = 12;
    static_assert(T + 2 <= numeric_limits<lll>::max() / (p * p));
    for (k = 1; k < n; k *= 2)
    {
        wn = w + k;
        for (i = 0; i < n; i += k * 2)
        {
            f = A + i; g = A + i + k;
            for (j = 0; j < k; j++)
            {
                y = g[j] * wn[j] % p;
                g[j] = f[j] + p - y;
                f[j] += y;
            }
        }
        if (__lg(n / k) % T == 1) for (lll &x : a) x %= p;
    }
    if (o)
    {
        y = ksm(n, p - 2);
        for (lll &x : a) x = x * y % p;
        reverse(1 + all(a));
    }
}

ull &operator[](const int &x) { return a[x]; }
const ull &operator[](const int &x) const { return a[x]; }
static void plus_by(vector<ull> &a, const vector<ull> &b)
{
    int n = a.size(), m = b.size(), i, j;
    cmax(n, m);
    a.resize(++n);

```

```

    for (i = 0; i < m; i++) if ((a[i] += b[i]) >= base) a[i] -= base, ++a[i + 1];
    for (i = m; i < n && a[i] >= base; i++) a[i] -= base, ++a[i + 1];
    if (a[n - 1] == 0) a.pop_back();
}
static void minus_by(vector<ull> &a, const vector<ull> &b)
{
    int n = a.size(), m = b.size(), i, j;
    for (i = 0; i < m; i++) if (!(a[i] & sign) && a[i] >= b[i]) a[i] -= b[i];
    else --a[i + 1], a[i] += base - b[i];
    for (; i < n && (a[i] & sign); i++) a[i] += base, --a[i + 1];
    while (a.size() > 1 && !a.back()) a.pop_back();
}
static bool less(const vector<ull> &a, const vector<ull> &b)
{
    if (a.size() != b.size()) return a.size() < b.size();
    for (int i = a.size() - 1; i >= 0; i--) if (a[i] != b[i]) return a[i] < b[i];
    return 0;
}
static int cal(int x) { return 1 << __lg(max(x, 1) * 2 - 1); }
public:
bigint &operator+=(const bigint &o)
{
    if (neg == o.neg) plus_by(a, o.a);
    else if (neg)
    {
        if (less(o.a, a)) minus_by(a, o.a);
        else
        {
            neg = 0;
            auto t = o.a;
            swap(a, t);
            minus_by(a, t);
        }
    }
    else
    {
        if (less(a, o.a))
        {
            neg = 1;
            auto t = o.a;
            swap(a, t);
            minus_by(a, t);
        }
        else minus_by(a, o.a);
    }
    return *this;
}
bigint &operator--=(const bigint &o)
{
    neg ^= 1;
    *this += o;
    neg ^= 1;
    if (a == vector<ull>{0}) neg = 0;
    return *this;
}
bigint &operator*=(const bigint &o)
{

```

```

neg ^= o.neg;
int n = a.size(), m = o.a.size(), i, j;
assert(min(n, m) <= p / ((base - 1) * (base - 1)));
if (min(n, m) <= 64 && 0)
{
    vector<ull> c(n + m);
    for (i = 0; i < n; i++) for (j = 0; j < m; j++) c[i + j] += a[i] * o[j];
    for (i = 0; i < n + m - 1; i++)
    {
        c[i + 1] += c[i] / base;
        c[i] %= base;
    }
    swap(a, c);
    while (a.size() > 1 && !a.back()) a.pop_back();
    if (a == vector<ull>{0}) neg = 0;
    return *this;
}
int len = cal(n + m);
vector<lll> f(len), g(len);
copy_n(a.begin(), n, f.begin());
copy_n(o.a.begin(), m, g.begin());
dft(f); dft(g);
for (i = 0; i < len; i++) f[i] = f[i] * g[i] % p;
dft(f, 1);
a.resize(n + m);
copy_n(f.begin(), n + m - 1, a.begin());
for (i = n + m - 2; i >= 0; i--)
{
    a[i + 1] += a[i] / base;
    a[i] %= base;
}
for (i = 0; i < n + m - 1; i++)
{
    a[i + 1] += a[i] / base;
    a[i] %= base;
}
while (a.size() > 1 && !a.back()) a.pop_back();
if (a == vector<ull>{0}) neg = 0;
return *this;
}
bigint &operator/=(long long x)//to zero
{
    if (x < 0) x = -x, neg ^= 1;
    for (int i = a.size() - 1; i >= 0; i--)
    {
        a[i - 1] += a[i] % x * base;
        a[i] /= x;
    }
    a[0] /= x;
    while (a.size() > 1 && !a.back()) a.pop_back();
    if (a == vector<ull>{0}) neg = 0;
    return *this;
}
bigint operator+(bigint o) const { return o += *this; }
bigint operator-(bigint o) const { o -= *this; if (o.a != vector<ull>{0}) o.neg ^= 1; return o; }
bigint operator*(bigint o) const { return o *= *this; }

```

```

bigint operator/(long long x) const { auto res = *this; return res /= x; }
long long operator%(long long x) const
{
    bool flg = neg;
    if (x < 0) flg ^= 1, x = -x;
    ull res = 0;
    for (int i = (base % x == 0 ? 0 : a.size() - 1); i >= 0; i--) res = (res * base + a[i]) %
x;
    return (long long)res * (flg ? -1 : 1);
}
bigint(long long x = 0) : neg(0)
{
    if (x < 0) x = -x, neg = 1;
    a.push_back(x % base);
    while (x /= base) a.push_back(x % base);
}
bool operator<(const bigint &o) const { return cmp(*this, o) < 0; }
bool operator>(const bigint &o) const { return cmp(*this, o) > 0; }
bool operator<=(const bigint &o) const { return cmp(*this, o) <= 0; }
bool operator>=(const bigint &o) const { return cmp(*this, o) >= 0; }
bool operator==(const bigint &o) const { return cmp(*this, o) == 0; }
bool operator!=(const bigint &o) const { return cmp(*this, o) != 0; }
};
int cmp(const bigint &a, const bigint &b)
{
    if (a.neg != b.neg) return a.neg ? -1 : 1;
    if (a.neg) return -cmp(b, a);
    if (a.a.size() != b.a.size()) return a.a.size() < b.a.size() ? -1 : 1;
    for (int i = a.a.size() - 1; i >= 0; i--) if (a.a[i] != b.a[i]) return a.a[i] < b.a[i] ? -1 :
1;
    return 0;
}
istream &operator>>(istream &cin, bigint &x)
{
    x.neg = 0;
    x.a.clear();
    string s;
    cin >> s;
    const static int length = bigint::length;
    static int mp[128], _ = [&]() {
        for (int i = '0'; i <= '9'; i++) mp[i] = i - '0';
        for (int i = 'a'; i <= 'z'; i++) mp[i] = i - 'a' + 10;
        for (int i = 'A'; i <= 'Z'; i++) mp[i] = i - 'A' + 10;
        return 0;
    }();
    reverse(all(s));
    if (s.back() == '-') x.neg = 1, s.pop_back();
    ull base = 1;
    for (int i = 0; i < s.size(); i++)
    {
        if (i % length == 0) x.a.push_back(0), base = 1;
        x.a.back() += mp[s[i]] * base;
        base *= bigint::output_base;
    }
    return cin;
}
ostream &operator<<(ostream &cout, const bigint &x)

```



```

{
    if (x.neg) cout << "-";
    const static int length = bigint::length;
    if (bigint::output_base == 10)
    {
        cout << setfill('0') << x.a.back();
        for (int i = (int)x.a.size() - 2; i >= 0; i--) cout << setw(length) << x.a[i];
    }
    else if (bigint::output_base == 16)
    {
        cout << hex << uppercase << setfill('0') << x.a.back();
        for (int i = (int)x.a.size() - 2; i >= 0; i--) cout << setw(length) << x.a[i];
        cout << dec;
    }
    else assert(0);
    return cout;
}

bigint abs(bigint x)
{
    x.neg = 0;
    return x;
}

bigint gcd(bigint x, bigint y)
{
    x.neg = y.neg = 0;
    if (x == bigint(0)) return y;
    if (y == bigint(0)) return x;
    int c1 = 0, c2 = 0;
    while (x % 2 == 0) x /= 2, ++c1;
    while (y % 2 == 0) y /= 2, ++c2;
    cmin(c1, c2);
    if (x > y) swap(x, y);
    while (x != y)
    {
        y -= x;
        y /= 2;
        while (y % 2 == 0) y /= 2;
        if (x > y) swap(x, y);
    }
    while (c1--) y *= bigint(2);
    return y;
}

bigint::l11 bigint::w[bigint::N];
int bigint::r[bigint::N];
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
    cout << fixed << setprecision(0);
    int T; cin >> T;
    while (T--)
    {
        bigint a, b;
        cin >> a >> b;
        cout << (a * b) << '\n';
    }
}

```

7.6 分散层叠算法 (Fractional Cascading)

$O(n + q(k + \log n))$, $O(n)$ 。

给出 k 个长度为 n 的有序数组。

现在有 q 个查询：给出数 x ，分别求出每个数组中大于等于 x 的最小的数（非严格后继）。

若后继不存在，则定义为 0。你需要在线地回答这些询问。

```
int a[M][N], b[M][N << 1], c[M][N << 1][2], len[M], ans[M];
int n, m, qs, p, q, d, i, j, x, y, la;
int main()
{
    cin >> n >> m >> qs >> d;
    for (j = 1; j <= m; j++) for (i = 0; i < n; i++) cin >> a[j][i];
    for (j = 1; j <= m; j++) a[j][n] = inf + j; ++n;
    for (i = 0; i < n; i++) b[m][i] = a[m][i], c[m][i][0] = i;
    len[m] = n;
    for (j = m - 1; j; j--)
    {
        p = 0, q = 1;
        while (p < n && q < len[j + 1])
            if (a[j][p] < b[j + 1][q]) b[j][len[j]] = a[j][p], c[j][len[j]][0] = p++, c[j][len[j]
            ]++[1] = q;
            else b[j][len[j]] = b[j + 1][q], c[j][len[j]][0] = p, c[j][len[j]++][1] = q, q += 2;
        while (p < n) b[j][len[j]] = a[j][p], c[j][len[j]][0] = p++, c[j][len[j]++][1] = q;
        while (q < len[j + 1]) b[j][len[j]] = b[j + 1][q], c[j][len[j]][0] = p, c[j][len[j]++][1]
        = q, q += 2;
    }
    for (int ii = 1; ii <= qs; ii++)
    {
        cin >> x; x ^= la;
        y = lower_bound(b[1], b[1] + len[1], x) - b[1];
        ans[1] = a[1][c[1][y][0]]; y = c[1][y][1]; // 下标是 c[1][y][0]
        for (j = 2; j <= m; j++)
        {
            if (y && b[j][y - 1] >= x) --y;
            ans[j] = a[j][c[j][y][0]]; // 下标是 c[j][y][0]
            y = c[j][y][1];
        }
        la = 0;
        for (i = 1; i <= m; i++) la ^= ans[i] > inf ? 0 : ans[i];
        if (ii % d == 0) cout << la << '\n';
    }
}
```

7.7 圆上整点 (二平方和定理)

$x^2 + y^2 = n$ 的整数解的数目的四分之一 $f(n)$ 是积性数论函数，且对于素数幂有： $f(p^k) =$

$$\begin{cases} 1 & p = 2 \\ k + 1 & p \equiv 1 \pmod{4} \\ (k + 1) \bmod 2 & p \equiv 3 \pmod{4} \end{cases}$$

以下代码给出所有的非负整数解。注意非负整数解个数不等于 $f(n)$ 。

时间复杂度为 $O(n^{\frac{1}{4}} + f(n))$ ，其中 $O(n^{\frac{1}{4}})$ 是 pollard-rho 的复杂度。

$f(n)$ 的量级不好分析，但不会超过约数个数 $O(d(n)) \approx O(n^{\frac{1}{3}})$ ，且可以推测不能达到。实践上 10^{18} 以内 $f(n) \leq 3072$ 。

```

namespace pr
{
    using pa = pair<ll, int>;
    ll ksm(ll x, ll y, const ll p)
    {
        ll r = 1;
        while (y)
        {
            if (y & 1) r = (lll)r * x % p;
            x = (lll)x * x % p; y >>= 1;
        }
        return r;
    }
}

namespace miller
{
    const int p[7] = {2, 3, 5, 7, 11, 61, 24251};
    ll s, t;
    bool test(ll n, int p)
    {
        if (p >= n) return 1;
        ll r = ksm(p, t, n), w;
        for (int j = 0; j < s && r != 1; j++)
        {
            w = (lll)r * r % n;
            if (w == 1 && r != n - 1) return 0;
            r = w;
        }
        return r == 1;
    }
    bool prime(ll n)
    {
        if (n < 2 || n == 46'856'248'255'98111) return 0;
        for (int i = 0; i < 7; ++i) if (n % p[i] == 0) return n == p[i];
        s = __builtin_ctz(n - 1); t = n - 1 >> s;
        for (int i = 0; i < 7; ++i) if (!test(n, p[i])) return 0;
        return 1;
    }
}

using miller::prime;
mt19937_64 rnd(chrono::steady_clock::now().time_since_epoch().count());

namespace rho
{
    void nxt(ll &x, ll &y, ll &p) { x = ((lll)x * x + y) % p; }
    ll find(ll n, ll C)
    {
        ll l, r, d, p = 1;
        l = rnd() % (n - 2) + 2, r = 1;
        nxt(r, C, n);
        int cnt = 0;
        while (l ^ r)
        {
            p = (lll)p * llabs(l - r) % n;
            if (!p) return gcd(n, llabs(l - r));
            ++cnt;
            if (cnt == 127)
            {

```

```

        cnt = 0;
        d = gcd(llabs(l - r), n);
        if (d > 1) return d;
    }
    nxt(l, C, n); nxt(r, C, n); nxt(r, C, n);
}
return gcd(n, p);
}
vector<pa> w;
vector<ll> d;
void dfs(ll n, int cnt)
{
    if (n == 1) return;
    if (prime(n)) return w.emplace_back(n, cnt), void();
    ll p = n, C = rnd() % (n - 1) + 1;
    while (p == 1 || p == n) p = find(n, C++);
    int r = 1; n /= p;
    while (n % p == 0) n /= p, ++r;
    dfs(p, r * cnt); dfs(n, cnt);
}
vector<pa> getw(ll n)
{
    w = vector<pa>(0); dfs(n, 1);
    if (n == 1) return w;
    sort(w.begin(), w.end());
    int i, j;
    for (i = 1, j = 0; i < w.size(); i++) if (w[i].first == w[j].first) w[j].second += w[i].second; else w[++j] = w[i];
    w.resize(j + 1);
    return w;
}
void dfss(int x, ll n)
{
    if (x == w.size()) return d.push_back(n), void();
    dfss(x + 1, n);
    for (int i = 1; i <= w[x].second; i++) dfss(x + 1, n * w[x].first);
}
vector<ll> getd(ll n)
{
    getw(n); d = vector<ll>(0); dfss(0, 1);
    sort(d.begin(), d.end());
    return d;
}
}
using rho::getw, rho::getd;
using miller::prime;
}
using pr::getw, pr::getd, pr::prime;
lll roundiv(lll x, lll y)
{
    return x >= 0 ? (x + y / 2) / y : (x - y / 2) / y;
}
struct G
{
    lll x, y;
    G operator~() const { return {x, -y}; }
    lll len2() const { return x * x + y * y; }
}

```

```

G operator+(const G &o) const { return {x + o.x, y + o.y}; }
G operator-(const G &o) const { return {x - o.x, y - o.y}; }
G operator*(const G &o) const { return {x * o.x - y * o.y, x * o.y + y * o.x}; }
G operator/(const G &o) const
{
    G t = *this * ~o;
    lll l = o.len2();
    return {rounddiv(t.x, l), rounddiv(t.y, l)};
}
G operator%(const G &o) const { return *this - *this / o * o; }
};
G gcd(G a, G b)
{
    if (a.len2() > b.len2()) swap(a, b);
    while (a.len2())
    {
        b = b % a;
        swap(a, b);
    }
    return b;
}
namespace cipolla
{
    using ull = unsigned long long;
    using ulll = __uint128_t;
    ull p, w;
    struct Q
    {
        ulll x, y;
        Q operator*(const Q &o) const { return {(x * o.x + y * o.y % p * w) % p, (x * o.y + y * o
.x) % p}; }
    };
    ull ksm(ulll x, ull y)
    {
        ulll r = 1;
        while (y)
        {
            if (y & 1) r = r * x % p;
            x = x * x % p; y >>= 1;
        }
        return r;
    }
    Q ksm(Q x, ull y)
    {
        Q r = {1, 0};
        while (y)
        {
            if (y & 1) r = r * x;
            x = x * x; y >>= 1;
        }
        return r;
    }
    ull mosqrt(ull x, ull P) // 0 <= x < P
    {
        if (x == 0 || P == 2) return x;
        p = P;
        if (ksm(x, p - 1 >> 1) != 1) return -1;
    }
}

```

```

    ull y;
    mt19937_64 rnd(chrono::steady_clock::now().time_since_epoch().count());
    do y = rnd() % p, w = ((ull)y * y + p - x) % p; while (ksm(w, p - 1 >> 1) <= 1); //not
for p=2
    y = ksm({y, 1}, p + 1 >> 1).x;
    if (y * 2 > p) y = p - y; //两解取小
    return y;
}
}
using cipolla::mosqrt;
vector<pair<ll, ll>> two_sqr_sum(ll n) //只会返回非负解, 按照字典序排序
{
    if (n < 0) return { };
    if (n == 0) return {{0, 0}};
    ll m = __lg(n & -n), d = 1ll << (m / 2), i;
    n >>= m;
    auto w = getw(n);
    vector<G> r((m & 1) ? vector{G{1, 1}} : vector{G{0, 1}, G{1, 0}});
    for (auto [p, k] : w) if (p % 4 == 1)
    {
        vector<G> pw(k + 1);
        pw[0] = {1, 0};
        pw[1] = gcd(G(p, 0), G(mosqrt(p - 1, p), 1));
        assert(pw[1].len2() == p);
        for (i = 2; i <= k; i++) pw[i] = pw[i - 1] * pw[1];
        vector<G> rr; rr.reserve(r.size() * (k + 1));
        for (i = 0; i <= k; i++)
        {
            G x = pw[i] * ~pw[k - i];
            for (G y : r) rr.push_back(x * y);
        }
        swap(r, rr);
    }
    else
    {
        if (k % 2) return { };
        k /= 2;
        while (k--) d *= p;
    }
    vector<pair<ll, ll>> ans;
    ans.reserve(r.size());
    for (auto [x, y] : r) ans.push_back({abs((ll)x * d), abs((ll)y * d)});
    sort(all(ans));
    ans.resize(unique(all(ans)) - ans.begin());
    return ans;
}

```

7.8 Stern-Brocot Tree

你需要时刻保持传入的 $\frac{num}{den}$ 满足 $\gcd(num, den) = 1$ 。

```

template<typename T> struct upcast_t;
template<> struct upcast_t<int> { using type = long long; };
template<> struct upcast_t<long long> { using type = __int128; };
template<typename T> using upcast = typename upcast_t<T>::type;
template<class T> struct frac
{

```

```

T x, y;
frac<T> operator*(T k) const { return {k * x, k * y}; }
frac<T> add(const frac<T> &rhs) const { return {x + rhs.x, y + rhs.y}; }
upcast<T> cross(const frac<T> &o) const { return (upcast<T>)x * o.y - (upcast<T>)y * o.x; }
T max() const { return ::max(x, y); }
};

template<class T> int cmp(const frac<T> &a, const frac<T> &b)
{
    auto d = a.cross(b);
    return d < 0 ? -1 : !!d;
}

template<class T> struct node
{
    frac<T> l, r;
    frac<T> val() const { return l.add(r); }
    int cmp(const frac<T> &rhs) const { return ::cmp(val(), rhs); }
};

template<class T> node<T> root() { return {{0, 1}, {1, 0}}; }
template<class T> struct sbt
{
    node<T> move(const node<T> &u, char dir, T step) const { return dir == 'L' ? node{u.l, u.r.
add(u.l * step)} : node{u.l.add(u.r * step), u.r}; }
    node<T> decode(const vector<pair<char, T>> &a) const
    {
        node u = root<T>();
        for (auto [d, s] : a) u = move(u, d, s);
        return u;
    }
    vector<pair<char, T>> encode(const frac<T> &o) const
    {
        node u = root<T>();
        vector<pair<char, T>> r;
        while (1)
        {
            int d = u.cmp(o);
            if (d == 0) return r;
            char dir = d == 1 ? 'L' : 'R';
            auto kx = u.r.cross(o), ky = o.cross(u.l);
            if (d == 1)
            {
                T k = (kx - 1) / ky;
                r.push_back({'L', k});
                u = move(u, 'L', k);
            }
            else
            {
                T k = (ky - 1) / kx;
                r.push_back({'R', k});
                u = move(u, 'R', k);
            }
        }
    }
    node<T> to_node(const frac<T> &o) const
    {
        node u = root<T>();
        while (1)
        {

```

```

        int d = u.cmp(o);
        if (d == 0) return u;
        char dir = d == 1 ? 'L' : 'R';
        auto kx = u.r.cross(o), ky = o.cross(u.l);
        if (d == 1) u = move(u, 'L', (kx - 1) / ky);
        else u = move(u, 'R', (ky - 1) / kx);
    }
}

node<T> lca(const frac<T> &x, const frac<T> &y)
{
    auto a = encode(x), b = encode(y);
    vector<pair<char, T>> c;
    int sz = min(a.size(), b.size());
    for (int i = 0; i < sz; i++) if (a[i] == b[i]) c.push_back(a[i]);
    else
    {
        if (a[i].first == b[i].first) c.push_back({a[i].first, min(a[i].second, b[i].second)});
    }
    break;
}
return decode(c);
}

optional<node<T>> jump_to(const frac<T> &x, T depth) // depth(root) = 0
{
    auto a = encode(x);
    T dep = 0;
    for (auto [x, y] : a) dep += y;
    if (dep < depth) return { };
    T step = dep - depth;
    while (a.size() && a.back().second <= step) step -= a.back().second, a.pop_back();
    if (!step) return {decode(a)};
    a.back().second -= step;
    return {decode(a)};
}

optional<node<T>> jump(const frac<T> &x, T step)
{
    auto a = encode(x);
    while (a.size() && a.back().second <= step) step -= a.back().second, a.pop_back();
    if (!step) return {decode(a)};
    if (!a.size()) return { };
    a.back().second -= step;
    return {decode(a)};
}

pair<frac<T>, frac<T>> appro(const frac<T> &x, T n)
{
    if (x.x <= n && x.y <= n) return {x, x};
    auto a = encode(x);
    auto [l, r] = root<T>();
    for (auto [dir, step] : a) if (dir == 'L')
    {
        frac v = r.add(l * step);
        if (v.add(l).max() <= n) r = v;
        else
        {
            T k = x.x < x.y ? (n - r.y) / l.y : (n - r.x) / l.x;
            return {l, r.add(l * k)};
        }
    }
}

```



```

    }
    else
    {
        frac v = l.add(r * step);
        if (v.add(r).max() <= n) l = v;
        else
        {
            T k = x.x < x.y ? (n - l.y) / r.y : (n - l.x) / r.x;
            return {l.add(r * k), r};
        }
    }
    assert(0);
}
};

```

7.9 快速取模

```

__uint128_t brt = ((__uint128_t)1 << 64) / mod;
for (int i = 1; i <= n; i++)
{
    ans *= i;
    ans = ans - mod * (brt * ans >> 64);
    while (ans >= mod) ans -= mod; //可以替换为 if, 但据说会变慢。如果循环展开则需要替换
}
struct barret
{
    ll p, m; //p 表示上面的模数, m 为取模参数
    int c = 0;
    inline void init(ll t) {
        c = 48 + log2(t), p = t;
        m = (ll)((ull1(1) << c) / t);
    }
    friend inline ll operator % (ll n, const barret &d)
    { // get n % d
        ll r = n - ((ull1(n) * d.m) >> d.c) * d.p;
        while (r >= d.p) r -= d.p;
        return r;
    }
} modp;

```

7.10 IO 优化

```

class fast_iostream
{
private:
#define SIGNED
    const static int MAXBF = 1 << 20; FILE *inf, *ouf;
    char *inbuf, *inst, *ined;
    char *oubuf, *oust, *oued;
    inline void _flush() { fwrite(oubuf, 1, oued - oust, ouf); }
    inline char _getchar() {
        if (inst == ined) inst = inbuf, ined = inbuf + fread(inbuf, 1, MAXBF, inf);
        return inst == ined ? EOF : *inst++;
    }
}

```

```

inline void _putchar(char c) {
    if (oued == oust + MAXBF) _flush(), oued = oubuf;
    *oued++ = c;
}
public:
    fast_iostream(FILE *_inf = stdin, FILE *_ouf = stdout)
        : inbuf(new char[MAXBF]), inf(_inf), inst(inbuf), ined(inbuf),
          oubuf(new char[MAXBF]), ouf(_ouf), oust(oubuf), oued(oubuf) { }
    ~fast_iostream() { _flush(); delete[] inbuf; delete[] oubuf; }
    fast_iostream &operator >> (char &c) {
        while (isspace(c = _getchar()));
        return *this;
    }
    fast_iostream &operator >> (string &s) {
        static char c;
        while (isspace(c = _getchar()));
        s = c;
        while (!isspace(c = _getchar())) s += c;
        return *this;
    }
    template <class Int>
    fast_iostream &operator >> (Int &n) {
        static char c;
#ifdef SIGNED
        bool neg = 0;
        while ((c = _getchar()) < '0' || c > '9') neg |= c == '-';
#else
        while ((c = _getchar()) < '0' || c > '9');
#endif
        n = c - '0';
        while ((c = _getchar()) >= '0' && c <= '9') n = n * 10 + c - '0';
#ifdef SIGNED
        if (neg) n = -n;
#endif
        return *this;
    }
    template <class Int>
    fast_iostream &operator << (Int n) {
        using U = make_unsigned_t<Int>;
        U x;
        if (n < 0) _putchar('-'), x = U(0) - U(n); else x = n;
        static char S[20]; int t = 0;
        do { S[t++] = '0' + x % 10, x /= 10; } while (x);
        for (int i = 0; i < t; ++i) _putchar(S[t - i - 1]);
        return *this;
    }
    fast_iostream &operator << (char c) { _putchar(c); return *this; }
    fast_iostream &operator << (const char *s) {
        for (int i = 0; s[i]; ++i) _putchar(s[i]); return *this;
    }
}fio;

```

7.11 手动开栈

一种写法是文件开头放，但部分 OJ 会失效。

```
#pragma comment(linker, "/STACK:102400000,102400000")
```

另一种写法是在 main 开头写，但必须以 `exit(0)` 结束程序。
以下两个应该有一个是对的，不对会 CE。

```
{
    static int OP = 0;
    if (OP++ == 0)
    {
        int size = 256 << 20; // 256MB
        char *p = (char *)malloc(size) + size;
        __asm__("movl 0, %%esp\n" :: "r"(p));
    }
}

{
    static int OP = 0;
    if (OP++ == 0)
    {
        int size = 128 << 20; // 128MB
        char *p = new char[size] + size;
        __asm__ __volatile__ ("movq 0, %%rsp\n" "pushq $exit\n" "jmp _main\n" :: "r"(p));
    }
}
```

7.12 德扑

`solve` 返回按照出现次数排序的 `vector<int>` (0 下标处为牌型)，这样就可以字典序比较了。

```
struct Q
{
    int suit, rank;
    bool operator<(const Q &o) const { return pair{rank, suit} < pair{o.rank, o.suit}; }
    bool operator==(const Q &o) const { return pair{rank, suit} == pair{o.rank, o.suit}; }
};

auto solve = [&](vector<Q> a) {
    vector<int> res;
    vector<int> cnt(15);
    for (auto [s, r] : a) ++cnt[r];
    sort(all(a));
    int i;
    bool is_flush = 1, is_str = 0;
    for (i = 1; i < 5; i++) is_flush &= a[i].suit == a[0].suit;
    is_str = *max_element(all(cnt)) == 1 && a[0].rank + 4 == a[4].rank;
    vector<int> b(6);
    for (i = 1; i < 6; i++) b[i] = a[i - 1].rank;
    sort(1 + all(b), [&](int x, int y) {
        return pair{cnt[x], x} > pair{cnt[y], y};
    });
    if (b == vector{0, 12, 3, 2, 1, 0}) is_str = 1, b[1] = 0;
    if (is_flush && is_str) return b[0] = 9, b;
    if (cnt[b[1]] == 4) return b[0] = 8, b;
    if (cnt[b[1]] == 3 && cnt[b[4]] == 2) return b[0] = 7, b;
    if (is_flush) return b[0] = 6, b;
    if (is_str) return b[0] = 5, b;
    if (cnt[b[1]] == 3) return b[0] = 4, b;
    if (cnt[b[1]] == 2 && cnt[b[3]] == 2) return b[0] = 3, b;
    if (cnt[b[1]] == 2) return b[0] = 2, b;
```

```
    return b;
};
auto turn = [&](string s) {
    Q res = Q{"SHDC".find(s[0]), "23456789TJQKA".find(s[1])};
    return res;
};
```

7.13 随机函数

u64 → u64

```
ull rnd(ull x = 0)
{
    static ull s = chrono::high_resolution_clock::now().time_since_epoch().count() ^ 0
x9e3779b97f4a7c15;
    s += 0x60bee2bee120fc15;
    x ^= s ^ 0xa0761d6478bd642f;
    ulll m = (ulll)x * (x ^ 0xe7037ed1a0b428db);
    return (ull)m ^ (ull)(m >> 64);
}
```

7.14 约数个数表

n	n 前第一个质数	n 后第一个质数	$\max\{\omega(n)\}$	$\max\{d(n)\}$	$\pi(n)$
10^1	$n - 3$	$n + 1$	2	$d(6) = 4$	4
10^2	$n - 3$	$n + 1$	3	$d(60) = 12$	25
10^3	$n - 3$	$n + 13$	4	$d(840) = 32$	168
10^4	$n - 27$	$n + 7$	5	$d(7560) = 64$	1229
10^5	$n - 9$	$n + 3$	6	$d(83160) = 128$	9592
10^6	$n - 17$	$n + 3$	7	$d(720720) = 240$	$7.9 \cdot 10^4$
10^7	$n - 9$	$n + 19$	8	$d(8648640) = 448$	$6.7 \cdot 10^5$
10^8	$n - 11$	$n + 7$	8	$d(73513440) = 768$	$5.8 \cdot 10^6$
10^9	$n - 63$	$n + 7$	9	$d(735134400) = 1344$	$5.1 \cdot 10^7$
10^{10}	$n - 33$	$n + 19$	10	$d(6983776800) = 2304$	$4.6 \cdot 10^8$
10^{11}	$n - 23$	$n + 3$	10	$d(97772875200) = 4032$	$4.2 \cdot 10^8$
10^{12}	$n - 11$	$n + 39$	11	$d(963761198400) = 6720$	$3.8 \cdot 10^9$
10^{13}	$n - 29$	$n + 37$	12	$d(9316358251200) = 10752$	$3.5 \cdot 10^{10}$
10^{14}	$n - 27$	$n + 31$	12	$d(97821761637600) = 17280$	$3.3 \cdot 10^{11}$
10^{15}	$n - 11$	$n + 37$	13	$d(866421317361600) = 26880$	$3 \cdot 10^{12}$
10^{16}	$n - 63$	$n + 61$	13	$d(8086598962041600) = 41472$	$2.8 \cdot 10^{13}$
10^{17}	$n - 3$	$n + 3$	14	$d(74801040398884800) = 64512$	
10^{18}	$n - 11$	$n + 3$	15	$d(897612484786617600) = 103680$	
10^{19}	$n - 39$	$n + 51$	16	$d(9200527969062830400) = 161280$	

7.15 NTT 质数

$p = r \times 2^k + 1$	r	k	g (最小原根)	位数
17	1	4	3	2
97	3	5	5	2
193	3	6	5	3
257	1	8	3	3
7681	15	9	17	4
12289	3	12	11	5
40961	5	13	3	5
65537	1	16	3	5
786433	3	18	10	6
5767169	11	19	3	7
7340033	7	20	3	7
23068673	11	21	3	8
104857601	25	22	3	9
167772161	5	25	3	9
469762049	7	26	3	9
998244353	119	23	3	10
1004535809	479	21	3	10
2013265921	15	27	31	10
2281701377	17	27	3	10
3221225473	3	30	5	10
75161927681	35	31	3	11
77309411329	9	33	7	11
206158430209	3	36	22	12
2061584302081	15	37	7	13
2748779069441	5	39	3	13
6597069766657	3	41	5	13
39582418599937	9	42	5	14
79164837199873	9	43	5	14
263882790666241	15	44	7	15
1231453023109121	35	45	3	16
1337006139375617	19	46	3	16
3799912185593857	27	47	5	16
4222124650659841	15	48	19	16
7881299347898369	7	50	6	16
31525197391593473	7	52	3	17
180143985094819841	5	55	6	18
1945555039024054273	27	56	5	19
4179340454199820289	29	57	3	19

7.16 公式

向上取整的整除分块 $[i, \lfloor \frac{n-1}{\lceil \frac{n}{i} \rceil - 1} \rfloor]$

n 个点 k 个连通块的生成树方案 $n^{k-2} \prod_{i=1}^k \text{siz}_i$

(x, y) 曼哈顿距离 $\rightarrow (x+y, x-y)$ 切比雪夫距离

(x, y) 切比雪夫距离 $\rightarrow (\frac{x+y}{2}, \frac{x-y}{2})$ 曼哈顿距离

k -th $\max\{S\} = \sum_{T \subseteq S} (-1)^{|T|-k} \binom{|T|-1}{k-1} \min\{T\}$

Kummer's Theorem: $\binom{n+m}{n}$ 含 p ($p \in \text{prime}$) 的次数是 $n+m$ 在 p 进制下的进位数

$\ln(1-x^V) = -\sum_{i \geq 1} \frac{x^{Vi}}{i}$

$x^{\bar{n}} = \sum_i S_1(n, i) x^i$

$$\begin{cases} x \equiv a_1 \pmod{m_1} \\ x \equiv a_2 \pmod{m_2} \\ \dots \\ x \equiv a_n \pmod{m_n} \end{cases}$$

$V - E + F = 2$, $S = n + \frac{s}{2} - 1$. (n 为内部, s 为边上)

用途: 对于相邻的不相等的值, 在中间画一条线 (最外也画), 连通块个数 $= 1 + E - V +$ 内部框个数

注意全都是不含矩形边界上的。

卡特兰三角: 由 n 个 -1 和 m 个 1 组成一个序列, 满足所有前缀和小于 k 的方案数 ($k \leq m \leq n+k-1$): $\binom{n+m}{m} - \binom{n+m}{m-k}$

贝尔数 (划分集合方案数) EGF: $\exp(e^x - 1)$, $B_n = \sum_{i=0}^n S_2(n, i)$, 伯努利数 EGF: $\frac{x}{e^x - 1}$

$S_1(i, m)$ EGF: $\frac{(\sum_{i \geq 0} \frac{x^i}{i})^m}{m!}$, $S_2(i, m)$ EGF: $\frac{(e^x - 1)^m}{m!}$. $S_2(n, m) = [x^n] x^m (\prod_{i=1}^m (1-ix))^{-1}$, $S_2(n, m) \equiv [n-m \& \lfloor \frac{m-1}{2} \rfloor = 0] \pmod{2}$.

多项式牛顿迭代: 如果已知 $G(F(x)) \equiv 0 \pmod{x^{2n}}$, $G(F_*(x)) \equiv 0 \pmod{x^n}$, 则有 $F(x) \equiv F_*(x) - \frac{G(F_*(x))}{G'(F_*(x))} \pmod{x^{2n}}$. 求导时孤立的多项式视为常数。

$\int_0^1 t^a (1-t)^b dt = \frac{a!b!}{(a+b+1)!}$, $\sum_{i=0}^{n-1} i^k = \frac{n^{k+1}}{k+1}$

Burnside 引理: 等价类数量为 $\sum_{g \in G} \frac{X^g}{|G|}$, X^g 表示 g 变换下不动点的数量。

Polya 定理: 染色方案数为 $\sum_{g \in G} \frac{m^{c(g)}}{|G|}$, 其中 $c(g)$ 表示 g 变换下环的数量。

假设已经只保留了一个牛人酋长, 其名字为 $A = a_1 a_2 \cdots a_l$ 。

假设王国旁边开了一座赌场, 每单位时间 (就称为“秒”吧) 会有一个赌徒带着 1 铜币进入赌场。

赌场规则很简单: 支付 x 铜币赌下一秒会唱出 y , 如果猜对了就返还 nx 铜币, 否则钱就没了。

每个赌徒会如下行动: 支付 1 铜币赌下一秒会唱出 a_1 , 如果赌对了就支付得到的 n 铜币赌下一秒会唱出 a_2 , 如果还对了就支付得到的 n^2 铜币赌下一秒会唱出 a_3 , 等等, 以此类推, 最后支付 n^{l-1} 铜币赌下一秒会唱出 a_l 。

一旦连续唱出了 $a_1 a_2 \cdots a_l$, 赌场老板就会认为自己亏大了而关门, 并驱散所有赌徒。

那么关门前发生了什么呢? 以 $A = \{1, 4, 1, 5, 1, 1, 4, 1\}, n = 5$ 为例:

- 最后一位赌徒拿着 5 铜币离开; - 倒数第三位赌徒拿着 5^3 铜币离开; - 倒数第八位赌徒拿着 5^8 铜币离开; - 其他所有赌徒空手而归。

我们可以发现 1, 3 恰好是原序列的所有 border 的长度, 而且对于其他的名字也有这样的规律。

这时候最神奇的一步来了：由于这个赌博游戏是公平的，因此赌场应该期望下不赚不赔，因此关门时期望来了 $5 + 5^3 + 5^8$ 个赌徒，因此期望需要 $5 + 5^3 + 5^8$ 单位时间唱出这个名字。

同理，即可知道对于一般的 A ，答案为：

$$\sum_{a_1 a_2 \cdots a_c = a_{l-c+1} a_{l-c+2} \cdots a_l} n^c$$

8 语言基础

8.1 Makefile

```
%:%.cpp %.in
    g++ $< -o $@ -std=c++20 -DLOCAL -D_GLIBCXX_DEBUG -D_GLIBCXX_DEBUG_PEDANTIC -fsanitize=
    undefined
    ./$@ < $@.in
```

8.2 debug.h

```
template <class T, size_t size = tuple_size<T>::value> string to_dbg(T, string s = "") requires(
    not ranges::range<T>);
string to_dbg(auto x) requires requires(ostream &os) { os << x; }
{
    ostringstream os;
    os << x;
    return os.str();
}
string to_dbg(ranges::range auto x, string s = "") requires(not is_same_v<decltype(x), string>)
{
    for (auto xi : x) s += ", " + to_dbg(xi);
    return "[" + s.substr(2 * !!s.size()) + "]";
}
template <class T, size_t size> string to_dbg(T x, string s) requires(not ranges::range<T>)
{
    [&] <size_t... I>(index_sequence<I...>) { ((s += ", " + to_dbg(get<I>(x))), ...); } (
    make_index_sequence<size>());
    return "(" + s.substr(2 * !!s.size()) + ")";
}
#define dbg(...) cerr << __LINE__ << ": " #__VA_ARGS__ " " << to_dbg(tuple(__VA_ARGS__)) << "
    \n"
```

8.3 初始代码

```
#include "bits/stdc++.h"
using namespace std;
using ll = long long;
#define all(x) (x).begin(), (x).end()
int main()
{
    ios::sync_with_stdio(0); cin.tie(0);
}
```

抄写完 debug.h 后，补充于 using namespace std; 下一行：

```
#ifdef LOCAL
#include "debug.h"
#endif
```

8.4 bitset

```

#include "bits/stdc++.h"
using namespace std;
bitset<10> f(12);
char s2[] = "100101";
bitset<10> g(s2);
string s = "100101"; //reverse 了
bitset<10> h(s);
int main()
{
    for (int i = 0; i <= 9; i++) cout << f.test(i); cout << endl;
    for (int i = 0; i <= 9; i++) cout << g.test(i); cout << endl;
    for (int i = 0; i <= 9; i++) cout << h.test(i); cout << endl;
    cout << h << endl;
    foo.count(); //1的个数
    foo.flip(); //全部翻转
    foo.set(); //变1
    foo.reset(); //变0
    foo.to_string();
    foo.to_ulong();
    foo.to_ullong();
    foo._Find_first();
    foo._Find_next();
    //位运算: << 变大, >> 变小
}

```

输出:

```

0011000000
1010010000
1010010000
0000100101

```

8.5 pb_ds 和一些奇怪的用法

```

#pragma GCC optimize("Ofast")
#pragma GCC target("popcnt","sse3","sse2","sse","avx","sse4","sse4.1","sse4.2","ssse3","f16c","fma","avx2","xop","fma4")
#pragma GCC optimize("inline","fast-math","unroll-loops","no-stack-protector")
#include "bits/stdc++.h"
#include "ext/pb_ds/assoc_container.hpp"
#include "ext/pb_ds/tree_policy.hpp" //balanced tree
#include "ext/pb_ds/hash_policy.hpp" //hash table
#include "ext/pb_ds/priority_queue.hpp" //priority_queue
using namespace __gnu_pbds;
using namespace std;
template <typename T> using rbt = tree<T, null_type, less<T>, rb_tree_tag,
    tree_order_statistics_node_update>;
cc_hash_table<string, int> mp1; //拉链法
gp_hash_table<string, int> mp2; //查探法
rbt<int> s1, s2; //注意是不可重的
//null_type无映射(低版本g++为null_mapped_type)
//less<int>从小到大排序
//插入t.insert();
//删除t.erase();
//求有多少个数比 k 小:t.order_of_key(k);

```

```

//求树中第 k+1 小:t.find_by_order(k);
//a.join(b) b并入a, 前提是两棵树的 key 的取值范围不相交, b 会清空但迭代器没事, 如不满足会抛出异常。我
    听说复杂度是线性???
//a.split(v,b) key 小于等于 v 的元素属于 a, 其余的属于 b
template <typename T> using heap = __gnu_pbds::priority_queue<T, greater<T>, pairing_heap_tag>;
//join(priority_queue &other) //合并两个堆, other会被清空
//split(Pred prd, priority_queue &other) //分离出两个堆
//modify(point_iterator it, const key) //修改一个节点的值
int main()
{
    __builtin_clz();//前导 0
    __builtin_ctz();//后面的 0
    ios::sync_with_stdio(0); cin.tie(0);
    mt19937 rnd(chrono::steady_clock::now().time_since_epoch().count());
    cout << fixed << setprecision(15);
    rbtree::iterator it;
    string s = "abc", t = "dabce";
    boyer_moore_horspool_searcher S(all(s));
    if (search(all(t), S) != t.end())
    {
        cout << "find\n";
    }
    uniform_real_distribution<> a(1, 2);
    numeric_limits<int>::max();
}

```

8.6 python 使用方法

注意事项: python 容易爆栈, 且引用与赋值较为混乱。注意局部变量的 global 怎么写 (如果需要修改全局内容)。

文件操作

```

fi = open("discuss.in", "r")
fo = open("discuss.out", "w")
n = int(fi.readline())
fo.write(str(ans))

```

类的构造, 重载运算符

```

class Q:
    def __init__(self, x, y):
        self.x = x
        self.y = y

    def __add__(self, o):
        r = Q(self.x + o.x, self.y + o.y)
        return r

    def __sub__(self, o):
        r = Q(self.x - o.x, self.y - o.y)
        return r

    def __mul__(self, o):
        return self.x * o.y - self.y * o.x

    def __lt__(self, o):
        if self.x != o.x:

```

```

        return self.x < o.x
    return self.y < o.y

n, m = map(int, input().split())
c = list(map(int, input().split()))
print(*c)
a = Q(0, 0)
b = Q(1, 1)
if a < b - a:
    pass

```

初始代码

```

import sys, math
from collections import Counter, deque, defaultdict
from math import sin, cos, asin, acos, tan, atan, atan2, gcd, sqrt, hypot
from heapq import heapify, heappop, heappush
from itertools import permutations, product, combinations, combinations_with_replacement
from itertools import chain, accumulate, groupby, islice, tee, cycle
from functools import reduce
from queue import Queue
from bisect import bisect_left, bisect_right
from random import randint, sample
from copy import copy, deepcopy
from array import array
from types import GeneratorType

def bootstrap(f, stk=[]):
    def wrappedfunc(*args, **kwargs):
        if stk:
            return f(*args, **kwargs)
        else:
            to = f(*args, **kwargs)
            while True:
                if type(to) is GeneratorType:
                    stk.append(to)
                    to = next(to)
                else:
                    stk.pop()
                    if not stk:
                        break
            to = stk[-1].send(to)
        return to
    return wrappedfunc

class FastReader:
    def __init__(self):
        self.input = sys.stdin.read().splitlines()
        self.idx = 0

    def next_line(self):
        if self.idx < len(self.input):
            line = self.input[self.idx]
            self.idx += 1

```

```
        return line
    else:
        assert False

cin = FastReader()

def input():
    global cin
    return cin.next_line()

def RI():
    return int(input())

def RF():
    return float(input())

def RII():
    return [*map(int, input().split())]

def clamp(l, h, x):
    return max(min(x, h), l)

def RM(num=2):
    v = RII()
    v[:num] = [x - 1 for x in v[:num]]
    return v

def stol(s):
    return [*map(lambda x: s.find(x), list(input()))]

def vec(*dims, val=None):
    if len(dims) == 0:
        return []
    if len(dims) == 1:
        return [val] * dims[0]
    return [vec(*dims[1:], val=val) for _ in range(dims[0])]

for _ in range(RI()):
```

9 其他人的板子（补充）

9.1 最小树形图

```

struct RollbackUnionFind
{
    vector<pair<int, int>> st;
    vector<int> f;
    RollbackUnionFind(int n) : f(n, -1) { }
    int find(int u) { return f[u] < 0 ? u : find(f[u]); }
    bool merge(int u, int v)
    {
        if ((u = find(u)) == (v = find(v))) return false;
        if (f[u] < f[v]) swap(u, v);
        st.emplace_back(u, f[u]);
        f[v] += f[u];
        f[u] = v;
        return true;
    }
    void rollback(int t)
    {
        while (st.size() > t)
        {
            auto [u, v] = st.back();
            st.pop_back();
            f[f[u]] -= v;
            f[u] = v;
        }
    }
};

struct Skew
{
    int u, v;
    ll w, lazy;
    Skew *lc, *rc;
    static Skew *merge(Skew *x, Skew *y)
    {
        if (!x) return y;
        if (!y) return x;
        if (x->w > y->w) swap(x, y);
        x->push();
        x->rc = merge(x->rc, y);
        swap(x->lc, x->rc);
        return x;
    }
    Skew(tuple<int, int, ll> e) : lazy(0)
    {
        tie(u, v, w) = e;
        lc = rc = nullptr;
    }
    void add(ll x)
    {
        w += x;
        lazy += x;
    }
    void push()
    {

```

```

        if (lc) lc->add(lazy);
        if (rc) rc->add(lazy);
        lazy = 0;
    }
    Skew *pop()
    {
        push();
        return merge(lc, rc);
    }
};

pair<ll, vector<int>> directed_minimum_spanning_tree(int n, const vector<tuple<int, int, ll>> &
edges, int s) //[0,n)
{
    ll ans = 0;
    vector<Skew *> h(n), in(n);
    RollbackUnionFind f(n), r(n);
    vector<pair<Skew *, int>> c;
    for (auto [u, v, w] : edges) h[v] = Skew::merge(h[v], new Skew({u, v, w}));
    for (int i = 0; i < n; i += 1)
    {
        if (i == s) continue;
        for (int u = i;;)
        {
            while (h[u] && r.find(h[u]->u) == r.find(u)) h[u] = h[u]->pop(); // fix by codex
            if (!h[u]) return { };
            ans += (in[u] = h[u])->w;
            in[u]->add(-in[u]->w);
            int v = r.find(in[u]->u);
            if (f.merge(u, v)) break;
            int t = r.st.size();
            while (r.merge(u, v)) {
                h[r.find(u)] = Skew::merge(h[u], h[v]);
                u = r.find(u);
                v = r.find(in[v]->u);
            }
            c.emplace_back(in[u], t);
            while (h[u] && r.find(h[u]->u) == r.find(u)) h[u] = h[u]->pop();
        }
    }
    reverse(all(c));
    for (auto [p, t] : c)
    {
        int u = r.find(p->v);
        r.rollback(t);
        int v = r.find(in[u]->v);
        in[v] = exchange(in[u], p);
    }
    vector<int> res(n, -1);
    for (int i = 0; i < n; i += 1) res[i] = i == s ? i : in[i]->u;
    return {ans, res};
}

```

9.2 MTT+exp

```

#include "bits/stdc++.h"
using namespace std;

```

```

typedef long long ll;
typedef double db;
int read() {
    int res = 0;
    char c = getchar(), f = 1;
    while (c < 48 || c > 57) { if (c == '-') f = 0; c = getchar(); }
    while (c >= 48 && c <= 57) res = (res << 3) + (res << 1) + (c & 15), c = getchar();
    return f ? res : -res;
}

const int L = 1 << 19, mod = 1e9 + 7;
const db pi2 = 3.141592653589793 * 2;
int inc(int x, int y) { return x + y >= mod ? x + y - mod : x + y; }
int dec(int x, int y) { return x - y < 0 ? x - y + mod : x - y; }
int mul(int x, int y) { return (ll)x * y % mod; }
int qpow(int x, int y) {
    int res = 1;
    for (; y; y >>= 1) res = y & 1 ? mul(res, x) : res, x = mul(x, x);
    return res;
}
int inv(int x) { return qpow(x, mod - 2); }

struct cp {
    db x, y;
    cp() {}
    cp(db a, db b) { x = a, y = b; }
    cp operator+(const cp &p) const { return cp(x + p.x, y + p.y); }
    cp operator-(const cp &p) const { return cp(x - p.x, y - p.y); }
    cp operator*(const cp &p) const { return cp(x * p.x - y * p.y, x * p.y + y * p.x); }
    cp conj() { return cp(x, -y); }
} w[L];
int re[L];
int getre(int n) {
    int len = 1, bit = 0;
    while (len < n) ++bit, len <= 1;
    for (int i = 1; i < len; ++i) re[i] = (re[i >> 1] >> 1) | ((i & 1) << (bit - 1));
    return len;
}
void getw() {
    for (int i = 0; i < L; ++i) w[i] = cp(cos(pi2 / L * i), sin(pi2 / L * i));
}
void fft(cp *a, int len, int m) {
    for (int i = 1; i < len; ++i) if (i < re[i]) swap(a[i], a[re[i]]);
    for (int k = 1, r = L >> 1; k < len; k <= 1, r >>= 1)
        for (int i = 0; i < len; i += k << 1)
            for (int j = 0; j < k; ++j) {
                cp &L = a[i + j], &R = a[i + j + k], t = w[r * j] * R;
                R = L - t, L = L + t;
            }
    if (!~m) {
        reverse(a + 1, a + len);
        cp tmp = cp(1.0 / len, 0);
        for (int i = 0; i < len; ++i) a[i] = a[i] * tmp;
    }
}
void mul(int *a, int *b, int *c, int n1, int n2, int n) {
    static cp f1[L], f2[L], f3[L], f4[L];

```



```

int len = getre(n1 + n2 - 1);
for (int i = 0; i < len; ++i) {
    f1[i] = i < n1 ? cp(a[i] >> 15, a[i] & 32767) : cp(0, 0);
    f2[i] = i < n2 ? cp(b[i] >> 15, b[i] & 32767) : cp(0, 0);
}
fft(f1, len, 1), fft(f2, len, 1);
cp t1 = cp(0.5, 0), t2 = cp(0, -0.5), r = cp(0, 1);
cp x1, x2, x3, x4;
for (int i = 0; i < len; ++i) {
    int j = (len - i) & (len - 1);
    x1 = (f1[i] + f1[j].conj()) * t1;
    x2 = (f1[i] - f1[j].conj()) * t2;
    x3 = (f2[i] + f2[j].conj()) * t1;
    x4 = (f2[i] - f2[j].conj()) * t2;
    f3[i] = x1 * (x3 + x4 * r);
    f4[i] = x2 * (x3 + x4 * r);
}
fft(f3, len, -1), fft(f4, len, -1);
ll c1, c2, c3, c4;
for (int i = 0; i < n; ++i) {
    c1 = (ll)(f3[i].x + 0.5) % mod, c2 = (ll)(f3[i].y + 0.5) % mod;
    c3 = (ll)(f4[i].x + 0.5) % mod, c4 = (ll)(f4[i].y + 0.5) % mod;
    c[i] = (((c1 << 15) + c2 + c3) << 15) + c4 % mod;
}
}

void inv(int *a, int *b, int n) {
    if (n == 1) { b[0] = 1; return; }
    static int c[L];
    int l = (n + 1) >> 1;
    inv(a, b, l);
    mul(a, b, c, n, l, n);
    for (int i = 0; i < n; ++i) c[i] = mod - c[i];
    c[0] += 2;
    mul(b, c, b, n, n, n);
}

void der(int *a, int n) {
    for (int i = 1; i < n; ++i) a[i - 1] = mul(a[i], i);
    a[n - 1] = 0;
}

void its(int *a, int n) {
    for (int i = n - 1; i; --i) a[i] = mul(a[i - 1], inv(i));
    a[0] = 0;
}

void ln(int *a, int *b, int n) {
    static int c[L];
    for (int i = 0; i < n; ++i) c[i] = a[i];
    der(c, n);
    inv(a, b, n);
    mul(b, c, b, n, n, n);
    its(b, n);
}

void exp(int *a, int *b, int n) {
    if (n == 1) { b[0] = 1; return; }
    static int c[L];
    int l = (n + 1) >> 1;
    exp(a, b, l);
    ln(b, c, n);
}

```

```

    for (int i = 0; i < n; ++i) c[i] = dec(a[i], c[i]);
    ++c[0];
    mul(b, c, b, 1, n, n);
    for (int i = 0; i < n; ++i) c[i] = 0;
}

int n, k, a[L], f[L], g[L];
int main() {
    getw();
    n = read(), k = read();
    for (int i = 1; i <= k; ++i) a[i] = inv(i);
    for (int i = 2; i <= n; ++i)
        for (int j = 1; i * j <= k; ++j)
            f[i * j] = inc(f[i * j], a[j]);
    for (int i = 1; i <= k; ++i) f[i] = mod - f[i];
    for (int i = 1; i <= k; ++i) f[i] = inc(f[i], mul(n - 1, a[i]));
    exp(f, g, k + 1);
    printf("%d\n", g[k]);
}

```

9.3 半平面交

```

const int N = 305;
const db inf = 1e15, eps = 1e-10;
int sign(db x) {
    if (fabs(x) < eps) return 0;
    return x > 0 ? 1 : -1;
}

struct vec {
    db x, y;
    vec() {}
    vec(db a, db b) { x = a, y = b; }
    vec operator+(const vec &p) const {
        return vec(x + p.x, y + p.y);
    }
    vec operator-(const vec &p) const {
        return vec(x - p.x, y - p.y);
    }
    db operator*(const vec &p) const {
        return x * p.y - y * p.x;
    }
    vec operator*(const db &p) const {
        return vec(x * p, y * p);
    }
} p1[N], p2[N];

struct line {
    vec s, t;
    line() {}
    line(vec a, vec b) { s = a, t = b; }
} a[N], q[N];
db ang(vec v) {
    return atan2(v.y, v.x);
}
db ang(line l) {

```

```

    return ang(l.t - l.s);
}
bool cmp(line x, line y) {
    int s = sign(ang(x) - ang(y));
    return s ? s < 0 : sign((y.t - y.s) * (x.s - y.s)) < 0; // fix by codex
}

vec inter(line x, line y) {
    vec a = y.s - x.s, b = x.t - x.s, c = y.t - y.s;
    return y.s + c * ((a * b) / (b * c));
}
bool out(line l, vec p) {
    return sign((l.t - l.s) * (p - l.s)) < 0;
}

int n, tot = 0;
db ans = inf;
int main() {
    scanf("%d", &n);
    for (int i = 1; i <= n; ++i) scanf("%lf", &p1[i].x);
    for (int i = 1; i <= n; ++i) scanf("%lf", &p1[i].y);
    for (int i = 1; i < n; ++i) a[i] = line(p1[i], p1[i + 1]);
    a[n] = line(vec(p1[1].x, inf), vec(p1[1].x, p1[1].y));
    a[n + 1] = line(vec(p1[n].x, p1[n].y), vec(p1[n].x, inf));

    sort(a + 1, a + n + 2, cmp);
    for (int i = 1; i <= n; ++i) {
        if (!sign(ang(a[i]) - ang(a[i + 1]))) continue;
        a[++tot] = a[i];
    }
    a[++tot] = a[n + 1];

    int l = 1, r = 0;
    q[++r] = a[1], q[++r] = a[2];
    for (int i = 3; i <= tot; ++i) {
        while (l < r && out(a[i], inter(q[r], q[r - 1]))) --r;
        while (l < r && out(a[i], inter(q[l], q[l + 1]))) ++l;
        q[++r] = a[i];
    }
    while (l < r && out(q[l], inter(q[r], q[r - 1]))) --r;
    while (l < r && out(q[r], inter(q[l], q[l + 1]))) ++l;
    //.....
}

```

9.4 多项式复合 (yurzhang)

$O(n \log n \sqrt{n \log n})$, 奇慢无比, 慎用

```

#pragma GCC optimize("Ofast,inline")
#pragma GCC target("sse,sse2,sse3,ssse3,sse4,sse4.1,sse4.2,popcnt,abm,mmx,avx,avx2,tune=native")
#include <cstdio>
#include <cstring>
#include <cmath>
#include <algorithm>

#define MOD 998244353
#define G 332748118
#define N 262210

```

```

#define re register
#define gc pa==pb&&(pb=(pa=buf)+fread(buf,1,100000,stdin),pa==pb)?EOF:*pa++
typedef long long ll;
static char buf[100000], *pa(buf), *pb(buf);
static char pbuf[3000000], *pp(pbuf), st[15];
int read() {
    re int x(0); re char c(gc);
    while (c < '0' || c>'9') c = gc;
    while (c >= '0' && c <= '9')
        x = x * 10 + c - 48, c = gc;
    return x;
}
void write(re int v) {
    if (v == 0)
        *pp++ = 48;
    else {
        re int tp(0);
        while (v)
            st[++tp] = v % 10 + 48, v /= 10;
        while (tp)
            *pp++ = st[tp--];
    }
    *pp++ = 32;
}
int pow(re int a, re int b) {
    re int ans(1);
    while (b)
        ans = b & 1 ? (ll)ans * a % MOD : ans, a = (ll)a * a % MOD, b >>= 1;
    return ans;
}
int inv[N], ifac[N];
void pre(re int n) {
    inv[1] = ifac[0] = 1;
    for (re int i(2); i <= n; ++i)
        inv[i] = (ll)(MOD - MOD / i) * inv[MOD % i] % MOD;
    for (re int i(1); i <= n; ++i)
        ifac[i] = (ll)ifac[i - 1] * inv[i] % MOD;
}
int getLen(re int t) {
    return 1 << (32 - __builtin_clz(t));
}
int lmt(1), r[N], w[N];
void init(re int n) {
    re int l(0);
    while (lmt <= n)
        lmt <<= 1, ++l;
    for (re int i(1); i < lmt; ++i)
        r[i] = (r[i >> 1] >> 1) | ((i & 1) << (l - 1));
    re int wn(pow(3, (MOD - 1) / lmt));
    w[lmt >> 1] = 1;
    for (re int i((lmt >> 1) + 1); i < lmt; ++i)
        w[i] = (ll)w[i - 1] * wn % MOD;
    for (re int i((lmt >> 1) - 1); i; --i)

```

```

        w[i] = w[i << 1];
    }

void DFT(int *a, re int l) {
    static unsigned long long tmp[N];
    re int u(__builtin_ctz(lmt) - __builtin_ctz(l)), t;
    for (re int i(0); i < l; ++i)
        tmp[i] = (a[r[i] >> u]) % MOD;
    for (re int i(1); i < l; i <= 1)
        for (re int j(0), step(i << 1); j < l; j += step)
            for (re int k(0); k < i; ++k)
                t = (ll)w[i + k] * tmp[i + j + k] % MOD,
                tmp[i + j + k] = tmp[j + k] + MOD - t,
                tmp[j + k] += t;
    for (re int i(0); i < l; ++i)
        a[i] = tmp[i] % MOD;
}

void IDFT(int *a, re int l) {
    std::reverse(a + 1, a + l); DFT(a, l);
    re int bk(MOD - (MOD - 1) / l);
    for (re int i(0); i < l; ++i)
        a[i] = (ll)a[i] * bk % MOD;
}

int n, m;
int a[N], b[N], c[N];

void getInv(int *a, int *b, int deg) {
    if (deg == 1)
        b[0] = pow(a[0], MOD - 2);
    else {
        static int tmp[N];
        getInv(a, b, (deg + 1) >> 1);
        re int l(getLen(deg << 1));
        for (re int i(0); i < l; ++i)
            tmp[i] = i < deg ? a[i] : 0;
        DFT(tmp, l), DFT(b, l);
        for (re int i(0); i < l; ++i)
            b[i] = (2ll - (ll)tmp[i] * b[i] % MOD + MOD) % MOD * b[i] % MOD;
        IDFT(b, l);
        for (re int i(deg); i < l; ++i)
            b[i] = 0;
    }
}

void getDer(int *a, int *b, int deg) {
    for (re int i(0); i + 1 < deg; ++i)
        b[i] = (ll)a[i + 1] * (i + 1) % MOD;
    b[deg - 1] = 0;
}

void getComp(int *a, int *b, int k, int m, int &n, int *c, int *d) {
    if (k == 1) {
        for (re int i(0); i < m; ++i)
            c[i] = 0, d[i] = b[i];
        n = m, c[0] = a[0];
    }
}

```

```

}
else {
    static int t1[N], t2[N];
    int nl(n), nr(n), *cl, *cr, *dl, *dr;
    getComp(a, b, k >> 1, m, nl, cl = c, dl = d);
    getComp(a + (k >> 1), b, (k + 1) >> 1, m, nr, cr = c + nl, dr = d + nl);
    n = std::min(n, nl + nr - 1);
    re int _l(getLen(nl + nr));
    for (re int i(0); i < _l; ++i)
        t1[i] = i < nl ? dl[i] : 0;
    for (re int i(0); i < _l; ++i)
        t2[i] = i < nr ? cr[i] : 0;
    DFT(t1, _l), DFT(t2, _l);
    for (re int i(0); i < _l; ++i)
        t2[i] = (ll)t1[i] * t2[i] % MOD;
    IDFT(t2, _l);
    for (re int i(0); i < n; ++i)
        c[i] = ((i < nl ? cl[i] : 0) + t2[i]) % MOD;
    for (re int i(0); i < _l; ++i)
        t2[i] = i < nr ? dr[i] : 0;
    DFT(t2, _l);
    for (re int i(0); i < _l; ++i)
        t2[i] = (ll)t1[i] * t2[i] % MOD;
    IDFT(t2, _l);
    for (re int i(0); i < n; ++i)
        d[i] = t2[i];
}
}

void getComp(int *a, int *b, int *c, int deg) {
    static int ts[N], ps[N], c0[N], _t1[N], idM[N];
    if (deg == 1) {
        c[0] = a[0];
        return;
    } // fix by codex
    int M(std::max((int)ceil(sqrt(deg / log2(deg)) * 2.5), 2)), _n(deg + deg / M);
    getComp(a, b, deg, M, _n, c0, _t1);
    re int _l(getLen(_n + deg));
    for (re int i(_n); i < _l; ++i)
        c0[i] = 0;
    for (re int i(0); i < _l; ++i)
        ps[i] = i == 0;
    for (re int i(0); i < _l; ++i)
        ts[i] = M <= i && i < deg ? b[i] : 0;
    getDer(b, _t1, M);
    for (re int i(M - 1); i < deg; ++i)
        _t1[i] = 0; /// Important!!!
    getInv(_t1, idM, deg);
    for (int i = deg; i < _l; ++i)
        idM[i] = 0;
    DFT(ts, _l), DFT(idM, _l);
    for (re int t(0); t * M < deg; ++t) {
        for (re int i(0); i < _l; ++i)
            _t1[i] = i < deg ? c0[i] : 0;
        DFT(ps, _l), DFT(_t1, _l);
        for (re int i(0); i < _l; ++i)
            _t1[i] = (ll)_t1[i] * ps[i] % MOD,

```

```

        ps[i] = (ll)ps[i] * ts[i] % MOD;
    IDFT(ps, _l), IDFT(_t1, _l);
    for (re int i(deg); i < _l; ++i)
        ps[i] = 0;
    for (re int i(0); i < deg; ++i)
        c[i] = ((ll)_t1[i] * ifac[t] + c[i]) % MOD;
    getDer(c0, c0, _n);
    for (re int i(_n - 1); i < _l; ++i)
        c0[i] = 0;
    DFT(c0, _l);
    for (re int i(0); i < _l; ++i)
        c0[i] = (ll)c0[i] * idM[i] % MOD;
    IDFT(c0, _l);
    for (re int i(_n - 1); i < _l; ++i)
        c0[i] = 0;
}
}

int main() {
    n = read(), m = read();
    for (re int i(0); i <= n; ++i)
        a[i] = read();
    for (re int i(0); i <= m; ++i)
        b[i] = read();

    m = (n > m ? n : m) + 1;
    pre(m); init(m * 5);
    getComp(a, b, c, m);

    for (re int i(0); i <= n; ++i)
        write(c[i]);
    fwrite(pbuf, 1, pp - pbuf, stdout);
    return 0;
}

```

9.5 下降幂多项式乘法

$O(n \log n)$ 。

```

#include<cstdio>
#include<algorithm>
const int N = 524288, md = 998244353, g3 = (md + 1) / 3;
typedef long long LL;
int n, m, A[N], B[N], fac[N], iv[N], rev[N], C[N], g[20][N], lim, M;
int pow(int a, int b) {
    int ret = 1;
    for (; b; b >>= 1, a = (LL)a * a % md) if (b & 1) ret = (LL)ret * a % md;
    return ret;
}
void upd(int &a) { a += a >> 31 & md; }
void init(int n) {
    int l = -1;
    for (lim = 1; lim < n; lim <= 1) ++lim; M = lim + 1;
    for (int i = 1; i < lim; ++i)
        rev[i] = ((rev[i >> 1] >> 1) | ((i & 1) << 1));
}
void NTT(int *a, int f) {

```

```

for (int i = 1; i < lim; ++i) if (i < rev[i]) std::swap(a[i], a[rev[i]]);
for (int i = 0; i < M; ++i) {
    const int *G = g[i], c = 1 << i;
    for (int j = 0; j < lim; j += c << 1)
        for (int k = 0; k < c; ++k) {
            const int x = a[j + k], y = a[j + k + c] * (LL)G[k] % md;
            upd(a[j + k] += y - md), upd(a[j + k + c] = x - y);
        }
}
if (!f) {
    const int iv = pow(lim, md - 2);
    for (int i = 0; i < lim; ++i) a[i] = (LL)a[i] * iv % md;
    std::reverse(a + 1, a + lim);
}
}
int main() {
    scanf("%d%d", &n, &m); ++n, ++m;
    for (int i = 0; i < 20; ++i) {
        int *G = g[i];
        G[0] = 1;
        const int gi = G[1] = pow(3, (md - 1) / (1 << i + 1));
        for (int j = 2; j < 1 << i; ++j) G[j] = (LL)G[j - 1] * gi % md;
    }
    for (int i = 0; i < n; ++i) scanf("%d", A + i);
    for (int i = 0; i < m; ++i) scanf("%d", B + i);
    for (int i = *fac = 1; i < N; ++i)
        fac[i] = fac[i - 1] * (LL)i % md;
    iv[N - 1] = pow(fac[N - 1], md - 2);
    for (int i = N - 2; ~i; --i) iv[i] = (i + 1LL) * iv[i + 1] % md;
    init(n + m << 1);
    for (int i = 0; i < n + m - 1; ++i) C[i] = iv[i];
    NTT(A, 1), NTT(B, 1), NTT(C, 1);
    for (int i = 0; i < lim; ++i) A[i] = (LL)A[i] * C[i] % md, B[i] = (LL)B[i] * C[i] % md;
    NTT(A, 0), NTT(B, 0);
    for (int i = 0; i < lim; ++i) C[i] = 0;
    for (int i = 0; i < n + m - 1; ++i)
        C[i] = (i & 1) ? md - iv[i] : iv[i];
    for (int i = 0; i < lim; ++i) A[i] = (LL)A[i] * B[i] % md * fac[i] % md;
    for (int i = n + m - 1; i < lim; ++i) A[i] = 0;
    NTT(A, 1), NTT(C, 1);
    for (int i = 0; i < lim; ++i) A[i] = (LL)A[i] * C[i] % md;
    NTT(A, 0);
    for (int i = 0; i < n + m - 1; ++i) printf("%d%c", A[i], "\n"[i == n + m - 2]);
    return 0;
}

```

9.6 弦图找错

```

#include "bits/stdc++.h"
using namespace std;
const int MAXN = 200005;
using lint = long long;
using pi = pair<int, int>;
// the algorithm may be wrong. if you have any ideas for proving / disproving this, please
// contact me.
vector<int> gph[MAXN];

```



```

int n, m, cnt[MAXN], idx[MAXN];
int mark[MAXN], vis[MAXN], par[MAXN];
void report(int x, int y) {
    gph[x].erase(find(gph[x].begin(), gph[x].end(), y));
    gph[y].erase(find(gph[y].begin(), gph[y].end(), x));
    for (int i = 1; i <= n; i++) {
        if (binary_search(gph[i].begin(), gph[i].end(), x) &&
            binary_search(gph[i].begin(), gph[i].end(), y)) {
            mark[i] = 1;
        }
    }
    queue<int> que;
    vis[x] = 1;
    que.push(x);
    while (!que.empty()) {
        int x = que.front(); que.pop();
        for (auto &i : gph[x]) {
            if (!mark[i] && !vis[i]) {
                par[i] = x;
                vis[i] = 1;
                que.push(i);
            }
        }
    }
    assert(vis[y]);
    vector<int> v;
    while (y) {
        v.push_back(y);
        y = par[y];
    }
    printf("NO\n%d\n", v.size());
    for (auto &i : v) printf("%d_", i - 1);
}

int main() {
    scanf("%d_%d", &n, &m);
    for (int i = 0; i < m; i++) {
        int s, e; scanf("%d_%d", &s, &e);
        s++, e++;
        gph[s].push_back(e);
        gph[e].push_back(s);
    }
    for (int i = 1; i <= n; i++) sort(gph[i].begin(), gph[i].end());
    priority_queue<pi> pq;
    for (int i = 1; i <= n; i++) pq.emplace(cnt[i], i);
    vector<int> ord;
    while (!pq.empty()) {
        int x = pq.top().second, y = pq.top().first;
        pq.pop();
        if (cnt[x] != y || idx[x]) continue;
        ord.push_back(x);
        idx[x] = n + 1 - ord.size();
        for (auto &i : gph[x]) {
            if (!idx[i]) {
                cnt[i]++;
                pq.emplace(cnt[i], i);
            }
        }
    }
}

```

```

    }
}
reverse(ord.begin(), ord.end());
for (auto &i : ord) {
    int minBef = 1e9;
    for (auto &j : gph[i]) {
        if (idx[j] > idx[i]) minBef = min(minBef, idx[j]);
    }
    minBef--;
    if (minBef < n) {
        minBef = ord[minBef];
        for (auto &j : gph[i]) {
            if (idx[j] > idx[minBef] && !binary_search(gph[minBef].begin(), gph[minBef].end()
, j)) {
                report(minBef, i);
                return 0;
            }
        }
    }
}
puts("YES");
for (auto &i : ord) printf("%d_", i - 1);
}

```

9.7 最长公共子序列

复杂度 $O(\frac{nm}{\omega})$ 。

```

/*
 * Author : _Wallace_
 * Source : https://www.cnblogs.com/-Wallace-/
 * Problem : LOJ #6564. 最长公共子序列
 * Standard : GNU C++ 03
 * Optimal : -Ofast
 */
#include <algorithm>
#include <cstdio>
#include <cstring>
#include <string>

typedef unsigned long long ULL;

const int N = 7e4 + 5;
int n, m, u;

struct bitset {
    ULL t[N / 64 + 5];

    bitset() {
        memset(t, 0, sizeof(t));
    }
    bitset(const bitset &rhs) {
        memcpy(t, rhs.t, sizeof(t));
    }

    bitset &set(int p) {
        t[p >> 6] |= 1llu << (p & 63);
    }
}

```

```

        return *this;
    }
    bitset &shift() {
        ULL last = 0llu;
        for (int i = 0; i < u; i++) {
            ULL cur = t[i] >> 63;
            (t[i] <= 1) |= last, last = cur;
        }
        return *this;
    }
    int count() {
        int ret = 0;
        for (int i = 0; i < u; i++)
            ret += __builtin_popcountll(t[i]);
        return ret;
    }

    bitset &operator = (const bitset &rhs) {
        memcpy(t, rhs.t, sizeof(t));
        return *this;
    }
    bitset &operator &= (const bitset &rhs) {
        for (int i = 0; i < u; i++) t[i] &= rhs.t[i];
        return *this;
    }
    bitset &operator |= (const bitset &rhs) {
        for (int i = 0; i < u; i++) t[i] |= rhs.t[i];
        return *this;
    }
    bitset &operator ^= (const bitset &rhs) {
        for (int i = 0; i < u; i++) t[i] ^= rhs.t[i];
        return *this;
    }

    friend bitset operator - (const bitset &lhs, const bitset &rhs) {
        ULL last = 0llu; bitset ret;
        for (int i = 0; i < u; i++) {
            ULL cur = (lhs.t[i] < rhs.t[i] + last);
            ret.t[i] = lhs.t[i] - rhs.t[i] - last;
            last = cur;
        }
        return ret;
    }
} p[N], f, g;

signed main() {
    scanf("%d%d", &n, &m), u = n / 64 + 1;
    for (int i = 1, c; i <= n; i++)
        scanf("%d", &c), p[c].set(i);
    for (int i = 1, c; i <= m; i++) {
        scanf("%d", &c), (g = f) |= p[c];
        f.shift(), f.set(0);
        ((f = g - f) ^= g) &= g;
    }
    printf("%d\n", f.count());
    return 0;
}

```

另一个实现

```

#include "bits/stdc++.h"
#pragma GCC target("popcnt,bmi")

using namespace std;
using ull = uint64_t;

const int N = 70005, M = 1136;

int n, m;
ull g[N][M], f[M];

int read() {
    const int M = 1e6;
    static streambuf *in = cin.rdbuf();
#define gc (p1 == p2 && (p2 = (p1 = buf) + in -> sgetn(buf, M), p1 == p2) ? -1 : *p1++)
    static char buf[M], *p1, *p2;
    int c = gc, r = 0;

    while (c < 48)
        c = gc;

    while (c > 47)
        r = r * 10 + (c & 15), c = gc;

    return r;
}

int main() {
    cin.tie(0)->sync_with_stdio(0);
    cin >> n >> m;

    for (int i = 0; i < n; i++)
        g[read()][i / 62] |= 1ULL << (i % 62);

    int lim = (n - 1) / 62;

    for (int i = 0; i < m; i++) {
        int c = 1;
        auto can = g[read()];

        for (int j = 0; j <= lim; j++) {
            ull x = f[j], y = x | can[j];
            x += x + c + (~y & (1ULL << 62) - 1);
            f[j] = x & y, c = x >> 62;
        }
    }

    int ans = 0;

    for (int i = 0; i <= lim; i++)
        ans += __builtin_popcountll(f[i]);

    cout << ans;
}

```

9.8 区间 LIS（排列）

在线：

```

/*
References

[1] Tiskin, A. (2008).Semi-local string comparison: Algorithmic techniques
and applications. Mathematics in Computer Science, 1(4), 571-603.

[2] Claude, F., Navarro, G., & Ordóñez, A. (2015). The wavelet matrix: An
efficient wavelet tree for large alphabets. Information Systems, 47,
15-32.

*/

// #pragma GCC target("popcnt")

#include <algorithm>
#include <cassert>
#include <climits>
#include <numeric>
#include <utility>
#include <vector>

namespace noshi91 {

    namespace range_lis_query_impl {

        namespace wavelet_matrix_impl {

            using uint = unsigned int;
            using ll = long long;

            static constexpr int w = CHAR_BIT * sizeof(uint);

            int popcount(uint x) {
#ifdef __GNUC__
                return __builtin_popcount(x);
#else
                static_assert(w == 32, "");
                x -= x >> 1 & 0x55555555;
                x = (x & 0x33333333) + (x >> 2 & 0x33333333);
                x = x + (x >> 4) & 0x0F0F0F0F;
                return x * 0x01010101 >> 24 & 0x3F;
#endif
            }

            class bit_vector {
            public:
                class node_type {
                public:
                    uint bit;
                    int sum;

                    node_type() : bit(0), sum(0) { }
                };
            };
        }
    }
}

```

```

        std::vector<node_type> v;

public:
    bit_vector(const uint n) : v(n / w + 1) { }

    void set(const uint i) {
        v[i / w].bit |= uint(1) << (i % w); // fix by codex
        v[i / w].sum += 1;
    }

    void build() {
        for (int i = 1; i < int(v.size()); i++) {
            v[i].sum += v[i - 1].sum;
        }
    }

    int rank(const uint i) const {
        return v[i / w].sum - popcount(v[i / w].bit & ~uint(0) << i % w);
    }

    int one() const { return v.back().sum; }
};

class wavelet_matrix {
private:
    template <class I> static bool test(const I x, const int k) {
        return (x & I(1) << k) != I(0);
    }

    std::vector<bit_vector> mat;

public:
    template <class I>
    wavelet_matrix(const int bit_length, std::vector<I> a)
        : mat(bit_length, bit_vector(a.size())) {
        const int n = a.size();
        std::vector<I> a0;
        a0.reserve(n);
        for (int p = bit_length - 1; p >= 0; p--) {
            bit_vector &v = mat[p];
            auto itr = a.begin();
            for (int i = 0; i < n; i++) {
                if (test(a[i], p)) {
                    v.set(i);
                    *itr = a[i];
                    itr++;
                }
                else {
                    a0.push_back(a[i]);
                }
            }
            v.build();
            std::copy(a0.begin(), a0.end(), itr);
            a0.clear();
        }
    }
};

```

```

    int count_less_than(int l, int r, const ll key) const {
        int ret = r - l;
        for (int p = mat.size() - 1; p >= 0; p--) {
            const bit_vector &v = mat[p];
            const int rank_l = v.rank(l);
            const int rank_r = v.rank(r);
            if (test(key, p)) {
                l = rank_l;
                r = rank_r;
            }
            else {
                ret -= rank_r - rank_l;
                const int o = v.one();
                l += o - rank_l;
                r += o - rank_r;
            }
        }
        return ret - (r - l);
    }
};

} // namespace wavelet_matrix_impl

using wavelet_matrix_impl::wavelet_matrix;

using vi = std::vector<int>;
using iter = typename vi::iterator;

static constexpr int none = -1;

vi inverse(const vi &p) {
    const int n = p.size();
    vi q(n, none);
    for (int i = 0; i < n; i++) {
        if (p[i] != none) {
            q[p[i]] = i;
        }
    }
    return q;
}

void unit_monge_dmul(const int n, iter stack, const iter a, const iter b) {
    if (n == 1) {
        stack[0] = 0;
        return;
    }
    const iter c_row = stack;
    stack += n;
    const iter c_col = stack;
    stack += n;

    const auto map = [=](const int len, const auto f, const auto g) {
        const iter a_h = stack + 0 * len;
        const iter a_m = stack + 1 * len;
        const iter b_h = stack + 2 * len;
        const iter b_m = stack + 3 * len;
        const auto split = [=](const iter v, iter v_h, iter v_m) {

```

```

        for (int i = 0; i < n; i++) {
            if (f(v[i])) {
                *v_h = g(v[i]);
                ++v_h;
                *v_m = i;
                ++v_m;
            }
        }
    };
    split(a, a_h, a_m);
    split(b, b_h, b_m);
    const iter c = stack + 4 * len;
    unit_monge_dmul(len, c, a_h, b_h);
    for (int i = 0; i < len; i++) {
        const int row = a_m[i];
        const int col = b_m[c[i]];
        c_row[row] = col;
        c_col[col] = row;
    }
};

const int mid = n / 2;
map(mid, [mid](const int x) { return x < mid; },
    [](const int x) { return x; });
map(n - mid, [mid](const int x) { return x >= mid; },
    [mid](const int x) { return x - mid; });

class d_itr {
public:
    int delta;
    int col;
    d_itr() : delta(0), col(0) { }
};

int row = n;
const auto right = [&](d_itr &it) {
    if (b[it.col] < mid) {
        if (c_col[it.col] >= row) {
            it.delta += 1;
        }
    }
    else {
        if (c_col[it.col] < row) {
            it.delta += 1;
        }
    }
    it.col += 1;
};

const auto up = [&](d_itr &it) {
    if (a[row] < mid) {
        if (c_row[row] >= it.col) {
            it.delta -= 1;
        }
    }
    else {
        if (c_row[row] < it.col) {
            it.delta -= 1;
        }
    }
}

```



```

};
d_itr neg, pos;
while (row != 0) {
    while (pos.col != n) {
        d_itr temp = pos;
        right(temp);
        if (temp.delta == 0) {
            pos = temp;
        }
        else {
            break;
        }
    }
    row -= 1;
    up(neg);
    up(pos);
    while (neg.delta != 0) {
        right(neg);
    }
    if (neg.col > pos.col) {
        c_row[row] = pos.col;
    }
}
}

vi subunit_monge_dmul(vi a, vi b) {
    const int n = a.size();
    vi a_inv = inverse(a);
    vi b_inv = inverse(b);
    std::swap(b, b_inv);
    vi a_map, b_map;
    for (int i = n - 1; i >= 0; i--) {
        if (a[i] != none) {
            a_map.push_back(i);
            a[n - a_map.size()] = a[i];
        }
    }
    std::reverse(a_map.begin(), a_map.end());
    {
        int cnt = 0;
        for (int i = 0; i < n; i++) {
            if (a_inv[i] == none) {
                a[cnt] = i;
                cnt += 1;
            }
        }
    }
    for (int i = 0; i < n; i++) {
        if (b[i] != none) {
            b[b_map.size()] = b[i];
            b_map.push_back(i);
        }
    }
    {
        int cnt = b_map.size();
        for (int i = 0; i < n; i++) {
            if (b_inv[i] == none) {

```

```

        b[cnt] = i;
        cnt += 1;
    }
}
}
vi c([](int n) {
    int ret = 0;
    while (n > 1) {
        ret += 2 * n;
        n = (n + 1) / 2;
        ret += 4 * n;
    }
    ret += 1;
    return ret;
})(n));
unit_monge_dmul(n, c.begin(), a.begin(), b.begin());

vi c_pad(n, none);
for (int i = 0; i < int(a_map.size()); i++) {
    const int t = c[n - a_map.size() + i];
    if (t < int(b_map.size())) {
        c_pad[a_map[i]] = b_map[t];
    }
}
return c_pad;
}

vi seaweed_doubling(const vi &p) {
    const int n = p.size();
    if (n == 1) {
        return vi({none});
    }
    const int mid = n / 2;
    vi lo, hi;
    vi lo_map, hi_map;
    for (int i = 0; i < n; i++) {
        const int e = p[i];
        if (e < mid) {
            lo.push_back(e);
            lo_map.push_back(i);
        }
        else {
            hi.push_back(e - mid);
            hi_map.push_back(i);
        }
    }
    lo = seaweed_doubling(lo);
    hi = seaweed_doubling(hi);
    vi lo_pad(n), hi_pad(n);
    std::iota(lo_pad.begin(), lo_pad.end(), 0);
    std::iota(hi_pad.begin(), hi_pad.end(), 0);
    for (int i = 0; i < mid; i++) {
        if (lo[i] == none) {
            lo_pad[lo_map[i]] = none;
        }
        else {
            lo_pad[lo_map[i]] = lo_map[lo[i]];
        }
    }
}

```

```

    }
}
for (int i = 0; mid + i < n; i++) {
    if (hi[i] == none) {
        hi_pad[hi_map[i]] = none;
    }
    else {
        hi_pad[hi_map[i]] = hi_map[hi[i]];
    }
}
return subunit_monge_dmul(std::move(lo_pad), std::move(hi_pad));
}

bool is_permutation(const vi &p) {
    const int n = p.size();
    std::vector<bool> used(n, false);
    for (const int e : p) {
        if (e < 0 || n <= e || used[e]) {
            return false;
        }
        used[e] = true;
    }
    return true;
}

wavelet_matrix convert(const vi &p) {
    assert(is_permutation(p));
    int n = p.size();
    vi row;
    if (n != 0) {
        row = seaweed_doubling(vi(p.begin(), p.end()));
    }
    for (int &e : row) {
        if (e == none) {
            e = n;
        }
    }
    int bit_length = 0;
    while (n > 0) {
        bit_length += 1;
        n /= 2;
    }
    return wavelet_matrix(bit_length, std::move(row));
}

class range_lis_query {
    int n;
    wavelet_matrix wm;

public:
    range_lis_query() : range_lis_query(std::vector<int>()) { }
    explicit range_lis_query(const std::vector<int> &p)
        : n(p.size()), wm(convert(p)) { }

    int query(const int left, const int right) const {
        assert(0 <= left && left <= right && right <= n);
        return (right - left) - wm.count_less_than(left, n, right);
    }
};

```

```

    }
};

} // namespace range_lis_query_impl

using range_lis_query_impl::range_lis_query;

} // namespace noshi91

#include <iostream>
#include <vector>

int main() {
    std::ios::sync_with_stdio(false);
    std::cin.tie(nullptr);

    int N, Q;
    std::cin >> N >> Q;
    std::vector<int> P(N);
    for (int &p : P) {
        std::cin >> p;
    }

    const noshi91::range_lis_query rlq(P);

    for (int i = 0; i < Q; i++) {
        int l, r;
        std::cin >> l >> r;
        std::cout << rlq.query(l, r) << "\n";
    }

    return 0;
}

```

离线:

```

//http://10.49.18.71/submission/164696
#include<bits/stdc++.h>
using namespace std;
constexpr int M = 1e6 + 5;
int n, q, a[M], tr[M], ans[M];
vector<pair<int, int>>qry[M];
int blk, bel[M], L[M], R[M], val[M];
priority_queue<int>Q[M];
priority_queue<int, vector<int>, greater<int>>P[M];
int read() {
    int x = 0; char ch = getchar();
    while (!isdigit(ch)) ch = getchar();
    while (isdigit(ch)) x = x * 10 + ch - 48, ch = getchar();
    return x;
}
void update(int x) { while (x) tr[x]++, x -= x & -x; }
int query(int x) { int res = 0; while (x <= n) res += tr[x], x += x & -x; return res; }
int main() {
    n = read(); q = read();
    for (int i = 1; i <= n; i++) a[i] = read() + 1;
    blk = (int)ceil(sqrt(n));
    for (int i = 1; i <= n; i++) bel[i] = (i - 1) / blk + 1;
}

```

```

for (int i = 1; i <= bel[n]; i++) L[i] = R[i - 1] + 1, R[i] = R[i - 1] + blk; R[bel[n]] = n;
auto push_back = [&](int x) {
    const int p = a[x], B = bel[p];
    if (!P[B].empty()) {
        for (int i = L[B]; i <= R[B]; i++)
            if (val[i]) {
                P[B].push(val[i]);
                val[i] = P[B].top();
                P[B].pop();
            }
        while (!P[B].empty()) P[B].pop();
    }
    val[p] = x; Q[B].push(x);
    int tmp = 0; bool flag = 0;
    for (int i = p + 1; i <= R[B]; i++)
        if (tmp < val[i]) swap(tmp, val[i]), flag = 1;
    if (flag) {
        while (!Q[B].empty()) Q[B].pop();
        for (int i = L[B]; i <= R[B]; i++)
            if (val[i]) Q[B].push(val[i]);
    }
    for (int i = B + 1; i <= bel[n]; i++)
        if (!Q[i].empty() && tmp < Q[i].top()) {
            P[i].push(tmp);
            if (tmp) Q[i].push(tmp);
            tmp = Q[i].top(), Q[i].pop();
        }
    update(tmp);
};
for (int i = 1; i <= q; i++) {
    int l = read() + 1, r = read();
    ans[i] = r - l + 1;
    if (l <= r) qry[r].emplace_back(l, i);
}
for (int i = 1; i <= n; i++) {
    push_back(i);
    for (auto [x, id] : qry[i])
        ans[id] -= query(x);
}
for (int i = 1; i <= q; i++) printf("%d\n", ans[i]);
return 0;
}

```

9.9 区间 LCS

$s_{[0,a)}$ 和 $t_{[b,c)}$ 的 LCS

```

#include "bits/stdc++.h"
using namespace std;
//dengyaotriangle!

const int maxn = 1005;
const int maxq = 500005;
int n, m, q;
char a[maxn], b[maxn];
struct qryt {
    int x, nxt;

```

```

}z[maxq];
int qry[maxn][maxn];
int ans[maxq];
int r[maxn];
int bit[maxn];

int main() {
    ios::sync_with_stdio(0); cin.tie(0);
    cin >> q >> b >> a; n = strlen(a); m = strlen(b);
    //q,s,t
    for (int i = 1; i <= q; i++) {
        int a, b, c;
        cin >> a >> b >> c;
        if (a) {
            ans[i] = c - b;
            z[i].x = b; z[i].nxt = qry[a][c];
            qry[a][c] = i;
        }
    }
    for (int i = 0; i < n; i++) r[i] = i;
    for (int i = 0; i < m; i++) {
        int lp = -1;
        for (int j = 0; j < n; j++) if (a[j] == b[i]) { lp = j; break; }
        if (lp != -1) {
            for (int j = lp + 1; j < n; j++) {
                if (a[j] != b[i]) {
                    if (r[j - 1] < r[j]) swap(r[j - 1], r[j]);
                }
            }
            for (int i = n - 1; i > lp; i--) r[i] = r[i - 1];
            r[lp] = -1;
        }
        for (int i = 0; i <= n; i++) bit[i] = 0;
        for (int j = 0; j < n; j++) {
            if (r[j] != -1) {
                for (int p = n - r[j]; p <= n; p += p & -p) bit[p]++;
            }
            for (int y = qry[i + 1][j + 1]; y; y = z[y].nxt) {
                for (int p = n - z[y].x; p; p -= p & -p) ans[y] -= bit[p];
            }
        }
    }
    for (int i = 1; i <= q; i++) cout << ans[i] << '\n';
    return 0;
}

```